Introduction to Abstract Nonsense

Lecture Notes on Homological Algebra

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Date: September, 2025

Version: January 10, 2026

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Chapter 1 Setting the stage

1.1 Categories and functors

1.1.1 Categories

Definition 1.1.1 (Category)

*A **category** $\mathscr{C}$ consists of a class of objects $\mathrm{obj}(\mathscr{C})$ and a set of morphisms $\mathrm{Mor}(X, Y)$ for all $X, Y \in \mathrm{obj}(\mathscr{C})$ such that:*

(1) for any $X, Y, Z \in \mathrm{obj}(\mathscr{C})$, there exists a composition map:

$$\mathrm{Mor}(Y, Z) \times \mathrm{Mor}(X, Y) \longrightarrow \mathrm{Mor}(X, Z)$$
$$(f, g) \longmapsto f \circ g.$$

And for $X \xrightarrow{g} Y \xrightarrow{f} Z \xrightarrow{h} W$ we have $h \circ (f \circ g) = (h \circ f) \circ g$.

(2) For any $X \in \mathrm{obj}(\mathscr{C})$, there exists an identity morphism $\mathrm{id}_X \in \mathrm{Mor}(X, X)$ such that for any $f \in \mathrm{Mor}(X, Y), g \in \mathrm{Mor}(Y, X)$, $f \circ \mathrm{id}_X = f, \mathrm{id}_X \circ g = g$.

*$\mathscr{C}$ is called **small** if $\mathrm{obj}(\mathscr{C})$ is a set. We will write $X \in \mathscr{C}$, $\mathrm{End}(X) = \mathrm{Mor}(X, X)$.*

Example 1.1 Category of sets The prototypical example to keep in mind is the category of sets, denoted $\mathbf{Sets}$. The objects are sets, and the morphisms are maps of sets.

Example 1.2 Category of vector spaces The category of vector spaces over a given field k, denoted $\mathbf{Vec}_k$. The objects are k-vector spaces, and the morphisms are linear transformations.

Example 1.3 Category of abelian groups The abelian groups, along with group homomorphisms, form a category $\mathbf{Ab}$.

Example 1.4 Category of modules over a ring If A is a ring, then the A-modules form a category $\mathbf{Mod}_A$. (This category has additional structure; it will be the prototypical example of an **abelian category**.)

Example 1.5 Category of rings There is a category $\mathbf{Rings}$, where the objects are rings, and the morphisms are maps of rings in the usual sense (maps of sets which respect adition and multiplication, and which send 1 to 1)

Example 1.6 Category of topological spaces The topological spaces, along with continuous maps, form a category $\mathbf{Top}$. The isomorphisms are homeomorphisms.

Example 1.7 Category of small categories Small categories with functors form a category $\mathbf{Cat}$.

Definition 1.1.2 (Isomorphism)

We called morphism f (or g) is an isomorphism, if $f : A \to B$ such that there exists some — necessarily unique — morphism $g : B \to A$, where $f \circ g$ and $g \circ f$ are the identity on B and A respectively.

Definition 1.1.3 (Monomorphism)

*A morphism $\pi : X \to Y$ is a **monomorphism** if any two morphisms $\mu_1 : Z \to X$ and $\mu_2 : Z \to X$ such that $\pi \circ \mu_1 = \pi \circ \mu_2$ must satisfy $\mu_1 = \mu_2$. In other words, there is at most one way of filling in the dotted* *arrow such that the diagram*

$$\begin{tikzcd} Z \arrow[d, dotted, "\leq 1"'] \arrow[dr] & \\ X \arrow[r, "\pi"] & Y \end{tikzcd}$$

commutes — for any object Z, the natural map (induced by π) $\mathrm{Mor}(Z, X) \to \mathrm{Mor}(Z, Y)$ is an injection.

We give an example to show that monomorphism need not be injective.

Example 1.8 In the category of divisible groups, the map $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ is a monomorphism but not injective.

Proof Clearly, $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ is not injective.

Let $\mu_1 : G \to \mathbb{Q}$ and $\mu_2 : G \to \mathbb{Q}$ be any two group homomorphism. Say $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$. Let $\pi \circ \mu_1 = \pi \circ \mu_2$, then $\pi(\mu_1(g)) = \pi(\mu_2(g))$ for all $g \in G$, hence, $(\mu_1 - \mu_2)(g) \in \mathbb{Z}$. We claim that $\mu_1 - \mu_2 = 0$, say $h = \mu_1 - \mu_2$, then $h : G \to \mathbb{Z}$. Since G is divisible, we have $\mathrm{Im}\, h$ is divisible. The only divisible subgroup of $\mathbb{Z}$ is 0, hence $\mathrm{Im}\, h = 0$, i.e., $\mu_1 = \mu_2$. □

Definition 1.1.4 (Epimorphism)

A morphism $\pi : X \to Y$ is a ***epimorphism*** *if for any two morphisms $\mu_1 : Y \to Z$ and $\mu_2 : Y \to Z$ such that $\mu_1 \circ \pi = \mu_2 \circ \pi$ we have $\mu_1 = \mu_2$. In other words, there is at most one way of filling in the dotted arrow such that the diagram*

$$\begin{tikzcd} X \arrow[r, "\pi"] \arrow[dr] & Y \arrow[d, dotted, "\leq 1"] \\ & Z \end{tikzcd}$$

commutes — for any object Z, the natural map (induced by π) $\mathrm{Mor}(Y, Z) \to \mathrm{Mor}(X, Z)$ is an injection.

Be careful when working with categories of objects that are sets with additional structure, as epimorphisms need not be surjective.

Example 1.9 In the category **Rings**, $\mathbb{Z} \to \mathbb{Q}$ is an epimorphism, but not surjective.

Proof Say $\pi : \mathbb{Z} \to \mathbb{Q}$. Let $R \in$ **Rings**, and homomorphisms $\mu_1 : \mathbb{Q} \to R$ and $\mu_2 : \mathbb{Q} \to R$, let $\mu_1 \circ \pi = \mu_2 \circ \pi$, then $\mu_1 \circ \pi(n) = \mu_2 \circ \pi(n)$ for all $n \in \mathbb{Z}$. Since $\pi(n) = n\pi(1) = n$, we have $\mu_1(n) = \mu_2(n)$ for all $n \in \mathbb{Z}$, which implies that π is epimorphism. □

Isomorphism always implies monomorphism and epimorphism, but converse is not true. For example:

Example 1.10 $\mathbb{R}_{\mathrm{disc}} \to \mathbb{R}$ is monomorphism and epimorphism, but is not an isomorphism in **Top**, where $\mathbb{R}_{\mathrm{disc}}$ means $\mathbb{R}$ equipped with the discrete topology.

Definition 1.1.5 (Opposite category)

The ***opposite category*** *$\mathscr{C}^{\mathrm{op}}$ of $\mathscr{C}$ is defined by $\mathrm{obj}(\mathscr{C}^{\mathrm{op}}) = \mathrm{obj}(\mathscr{C})$, $\mathrm{Mor}_{\mathscr{C}^{\mathrm{op}}}(X, Y) = \mathrm{Mor}_{\mathscr{C}}(Y, X)$, for all $X, Y \in \mathscr{C}$.*

Definition 1.1.6 (Subcategory)

A category $\mathscr{C}'$ is called a ***subcategory*** *of $\mathscr{C}$, denote $\mathscr{C}' \subseteq \mathscr{C}$, if*

(i) $\mathrm{obj}(\mathscr{C}') \subseteq \mathrm{obj}(\mathscr{C})$;

(ii) $\mathrm{Mor}_{\mathscr{C}'}(X, Y) \subseteq \mathrm{Mor}_{\mathscr{C}}(X, Y)$, *for all* $X, Y \in \mathrm{obj}(\mathscr{C}')$;

(iii) the composition and identity morphism in $\mathscr{C}'$ are equal to those in $\mathscr{C}$.

1.1.2 Functors

Definition 1.1.7 (Functor)

A ***covariant functor*** *F from a category $\mathscr{C}$ to a category $\mathscr{D}$, denoted $F : \mathscr{C} \to \mathscr{D}$, is the following data.*

(1) It is a map of obejects $F : \mathrm{obj}(\mathscr{C}) \to \mathrm{obj}(\mathscr{D})$,

(2) For each $A_1, A_2 \in \mathscr{C}$ and morphism $m \in \mathrm{Mor}_{\mathscr{C}}(A_1, A_2)$, $F(m) \in \mathrm{Mor}_{\mathscr{D}}(F(A_1), F(A_2))$,

(3) F preserves identity morphisms: for $A \in \mathscr{C}$, $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$,

(4) F preserves composition: $F(m_2 \circ m_1) = F(m_2) \circ F(m_1)$.

A ***contravariant functor*** *$F : \mathscr{C} \to \mathscr{D}$ is defined by a covariant functor $\mathscr{C}^{\mathrm{op}} \to \mathscr{D}$.*

Example 1.11 Identity functors A trivial example is the **identity functors** $\mathrm{id} : \mathscr{A} \to \mathscr{A}$.

Example 1.12 Forgetful functors Consider the functor from the category of vector space (over a field k) $\mathbf{Vec}_k$ to $\mathbf{Sets}$, that associates to each vector space its underlying set. The functor sends a linear transformation to its underlying map of sets. This is an example of a **forgetful functor** is $\mathbf{Mod}_A \to \mathbf{Ab}$ from A-modules to abelian groups, remembering only the abelian group structure of the A-module.

Example 1.13 Topological examples Examples of covariant functors include the fundamental group functor π_1, which sends a topological space X with choice of a point $x_0 \in X$ to a group $\pi_1(X, x_0)$, and the i-th homology functor $\mathbf{Top} \to \mathbf{Ab}$, which sends a topological space X to its i-th homology group $H_i(X, \mathbb{Z})$. The covariance corresponds to the fact that a continuous morphism of pointed topological spaces $\varphi : X \to Y$ with $\varphi(x_0) = y_0$ induces a map of fundamental groups $\pi_1(X, x_0) \to \pi_1(Y, y_0)$, and similarly for homology groups.

Example 1.14 Hom functor Suppose A is an object in a category $\mathscr{C}$. Then there is a functor $h^A : \mathscr{C} \to \mathbf{Sets}$ sending $B \in \mathscr{C}$ to $\mathrm{Mor}(A, B)$, and sending $f : B_1 \to B_2$ to $\mathrm{Mor}(A, B_1) \to \mathrm{Mor}(A, B_2)$ described by

$$[g : A \to B_1] \mapsto [f \circ g : A \to B_1 \to B_2].$$

Example 1.15 Linear algebra example If $\mathbf{Vec}_k$ is the category of k-vector spaces, then taking duals gives a contravariant functor $(\cdot)^\vee : \mathbf{Vec}_k \to \mathbf{Vec}_k$. Indeed, to each linear transformation $f : V \to W$, we have a dual transformation $f^\vee : W^\vee \to V^\vee$, and $(f \circ g)^\vee = g^\vee \circ f^\vee$.

Example 1.16 Topological example The i-th cohomology functor $H^i(\cdot, \mathbb{Z}) : \mathbf{Top} \to \mathbf{Ab}$ is a contravariant functor.

Example 1.17 There is a contravariant functor $\mathbf{Top} \to \mathbf{Rings}$ taking a topological space X to the ring of real-valued continuous functions on X. A morphism of topological spaces $X \to Y$ (a continuous map) induces the pullback map from functions on Y to functions on X.

Example 1.18 The functor of points Suppose A is an object of a category $\mathscr{C}$. Then there is a cotravariant functor $h_A : \mathscr{C} \to \mathbf{Sets}$ sending $B \in \mathscr{C}$ to $\mathrm{Mor}(B, A)$, and sending the morphism $f : B_1 \to B_2$ to the morphism $\mathrm{Mor}(B_2, A) \to \mathrm{Mor}(B_1, A)$ via

$$[g : B_2 \to A] \mapsto [g \circ f : B_1 \to B_2 \to A].$$

This functor might reasonably be called the **functor of maps** (to A), but is actually known as the **functor of points**.

Example 1.19 Base change $R \to S$ is a morphism of rings, then $S \otimes_R \Box : \mathbf{Mod}_R \to \mathbf{Mod}_S$ is a covariant functor.

Definition 1.1.8 (Faithful, full, fully faithful and full subcategory)

*A covariant functor $F : \mathscr{C} \to \mathscr{D}$ is **faithful** if for all $A, A' \in \mathscr{C}$, the map $\mathrm{Mor}_{\mathscr{C}}(A, A') \to \mathrm{Mor}_{\mathscr{D}}(F(A), F(A'))$ is injective, and **full** if it is surjective. A functor that is full and faithful is **fully faithful**.*

*A subcategory $i : \mathscr{C} \to \mathscr{D}$ is a **full subcategory** if i is full. Thus a subcategory $\mathscr{C}'$ of $\mathscr{C}$ is full if and only if for all $A, B \in \mathrm{obj}(\mathscr{C}')$, $\mathrm{Mor}_{\mathscr{C}'}(A, B) = \mathrm{Mor}_{\mathscr{C}}(A, B)$.*

Example 1.20

(1) The forgetful functor $\mathbf{Vec}_k \to \mathbf{Sets}$ is faithful, but not full.

(2) If A is a ring, the category of f.g. A-modules is a full subcategory of the category $\mathbf{Mod}_A$ of A-modules.

Definition 1.1.9 (Essentially surjective)

*$F : \mathscr{C} \to \mathscr{D}$ is called **essentially surjective** if for all $Y \in \mathscr{D}$, there exists $X \in \mathscr{C}$ and an isomorphism $F(X) \cong Y$.*

1.1.3 Natural transformation

Definition 1.1.10 (Natural transformation)

*Suppose F and G are two covariant functors from $\mathscr{C}$ to $\mathscr{D}$. A **natural transformation of covariant functors** $F \to G$ is the data of a morphism $m_A : F(A) \to G(A)$ for each $A \in \mathscr{C}$ such that for each $f : A \to A'$ in $\mathscr{C}$, the diagram*

$$\begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[d, "m_A"'] & F(A') \arrow[d, "m_{A'}"] \\ G(A) \arrow[r, "G(f)"'] & G(A') \end{tikzcd}$$

*commutes. A **natural isomorphism** of functors is a natural transformation such that each m_A is an isomorphism. (We make analogous definition when F and G are both contravariant.)*

Remark Let $F, G, H : \mathscr{C} \to \mathscr{D}$ be functors. Let $\alpha : F \to G$, $\beta : G \to H$ be natural transformation, then $\beta \circ \alpha : F \to H$ is a natural transformation. In fact, functors $\mathscr{C} \to \mathscr{D}$ and natural transformations form a category $\mathbf{Fun}(\mathscr{C}, \mathscr{D})$.

Definition 1.1.11 (Equivalence of categories)

*The data of functors $F : \mathscr{C} \to \mathscr{D}$ and $F' : \mathscr{D} \to \mathscr{C}$ such that $F \circ F'$ is naturally isomorphic to the identity functor $\mathrm{id}_{\mathscr{D}}$ on $\mathscr{D}$ and $F' \circ F$ is naturally isomorphic to $\mathrm{id}_{\mathscr{C}}$ is said to be an **equivalence of categories**. The functors F and G are called **quasi-inverse** of each other.*

Proposition 1.1.1

A functor $F : \mathscr{C} \to \mathscr{D}$ is an equivalence of categories iff it is fully faithful and essentially surjective.

Proof If $F : \mathscr{C} \to \mathscr{D}$ is an equivalence of categories, then there exists $G : \mathscr{D} \to \mathscr{C}$ such that $F \circ G \cong \mathrm{id}_{\mathscr{D}}$ and $G \circ F \cong \mathrm{id}_{\mathscr{C}}$. We first show that F is essentially surjective. Let $Y \in \mathscr{D}$, then we have $Y \cong F(G(Y))$, hence F is essentially surjective. We next show that F is fully faithful. It suffices to show that

$$\mathrm{Mor}_{\mathscr{C}}(A, B) \xrightarrow{\sim} \mathrm{Mor}_{\mathscr{D}}(F(A), F(B))$$

is bijection for all $A, B \in \mathscr{C}$. Say $\varphi : \mathrm{Mor}_{\mathscr{C}}(A, B) \to \mathrm{Mor}_{\mathscr{D}}(F(A), F(B))$ given by $\varphi(f) = F(f)$. Define

$$\begin{aligned} \psi : \mathrm{Mor}_{\mathscr{D}}(F(A), F(B)) &\longrightarrow \mathrm{Mor}_{\mathscr{C}}(A, B) \\ g &\longmapsto G(g), \end{aligned}$$

then we have $\psi \circ \varphi(f) = G \circ F(f) = f$ and $\varphi \circ \psi(g) = F \circ G(g) = g$ for all $f \in \mathrm{Mor}_{\mathscr{C}}(A, B)$ and $g \in \mathrm{Mor}_{\mathscr{D}}(F(A), F(B))$, hence F is fully faithful.

Conversely, if F is fully faithful and essentially surjective, we want to show that F is an equivalence of categories. Define $G : \mathscr{D} \to \mathscr{C}$ as follow. Pick $B \in \mathscr{D}$, since F is essentially surjective, there exists $A \in \mathscr{C}$ such that $F(A) \cong B$ define $G(B) = A$. For each $h \in \mathrm{Mor}_{\mathscr{D}}(A, B)$, since F is fully faithful there exists unique h' such that $F(h') = h$, then we define $G(h) = h'$. It easy to see that G is a functor from $\mathscr{D}$ to $\mathscr{C}$, and $F \circ G \cong \mathrm{id}_{\mathscr{D}}$, and $G \circ F \cong \mathrm{id}_{\mathscr{C}}$, hence F is an equivalence of categories. □

Corollary 1.1.1

Let $F : \mathscr{C} \to \mathscr{D}$ be a fully faithful functor, then F induces an equivalence of categories: $\mathscr{C} \to \mathscr{D}_0$ where $\mathscr{D}_0 \subseteq \mathscr{D}$ is the full subcategory spanned by the imager of F.

1.1.4 Yoneda lemma

Suppose A is an object of category $\mathscr{C}$. For any object $C \in \mathscr{C}$, we have a set of morphisms $\mathrm{Mor}(C, A)$. If we have a morphism $f : B \to C$, we get a map of sets

$$\mathrm{Mor}(C, A) \xrightarrow{\circ f} \mathrm{Mor}(B, A),$$

by composition: given a map from C to A, we get a map from B to A by precomposing with $f : B \to C$. Hence this gives a contravariant functor $h_A = \mathrm{Mor}(\square, A) : \mathscr{C} \to \mathbf{Sets}$ (or covariant functor $h_A : \mathscr{C}^{op} \to \mathbf{Sets}$). Yoneda's Lemma states that the functor h_A determines A up to unique isomorphism.

Note *If $F, G : A \to B$ are functors of the same variance, then*

$$\mathrm{Nat}(F, G) = \{\textit{natural transformations } F \to G\}.$$

Theorem 1.1.1 (Yoneda's Lemma)

Let $\mathscr{C}$ be a category, let $A \in \mathrm{obj}(\mathscr{C})$. For every functor $G : \mathscr{C}^{op} \to \mathbf{Sets}$, the map

$$y : \mathrm{Nat}(h_A, G) \to G(A), \quad \tau \mapsto \tau_A(\mathrm{id}_A)$$

is a bijection, where $h_A = \mathrm{Mor}(\square, A)$.

Proof If $\tau : \mathrm{Mor}_{\mathscr{C}}(\square, A) \to G$ is a natual transformation, then $y(\tau) = \tau_A(\mathrm{id}_A)$ lies in the set $G(A)$, for $\tau_A : \mathrm{Mor}_{\mathscr{C}}(A, A) \to G(A)$. Thus, y is a well-defined function.

Since $\tau \in \mathrm{Nat}(h_A, G)$ is a natural transformation, for each $B \in \mathrm{obj}(\mathscr{C})$ and $\varphi \in \mathrm{Mor}_{\mathscr{C}}(B, A)$, there is a commutative diagram

$$\begin{tikzcd} \mathrm{Mor}_{\mathscr{C}}(A, A) \arrow[r, "\tau_A"] \arrow[d, "\square\circ\varphi"'] & G(A) \arrow[d, "G(\varphi)"] \\ \mathrm{Mor}_{\mathscr{C}}(B, A) \arrow[r, "\tau_B"] & G(B) \end{tikzcd}$$

so that

$$(G\varphi)\tau_A(\mathrm{id}_A) = \tau_B(\square \circ \varphi)(\mathrm{id}_A) = \tau_B(\mathrm{id}_A \circ \varphi) = \tau_B(\varphi).$$

If $\sigma : \mathrm{Mor}_{\mathscr{C}}(\square, A) \to G$ is another natural transformation, then $\sigma_B(\varphi) = (G\varphi)\sigma_A(\mathrm{id}_A)$. Hence, if $\sigma_A(\mathrm{id}_A) =$

$\tau_A(\mathrm{id}_A)$, then $\sigma_B = \tau_B$ for all $B \in \mathrm{obj}(\mathscr{C})$, and therefore $\sigma = \tau$, which implies y is an injection.

To see that y is a surjection, take $x \in G(A)$. For $B \in \mathrm{obj}(\mathscr{C})$ and $\psi \in \mathrm{Mor}(B, A)$, define

$$\tau_B(\psi) = (G\psi)(x)$$

(note that $G\psi : GA \to GB$, hence $(G\psi)(x) \in GB$). We claim that τ is a natural transformation, that is, if $\theta : C \to B$ is a morphism in $\mathscr{C}$, then the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Mor}_{\mathscr{C}}(B, A) & \xrightarrow{\tau_B} & GB \\ {\scriptstyle \Box\circ\theta}\downarrow & & \downarrow{\scriptstyle G\theta} \\ \mathrm{Mor}_{\mathscr{C}}(C, A) & \xrightarrow{\tau_C} & GC. \end{array}$$

Going clockwise, we have $(G\theta)\tau_B(\psi) = G\theta G\psi(x)$; going counterclockwise, we have $\tau_C(\Box \circ \theta)\psi = \tau_C(\psi \circ \theta) = G(\psi\theta)(x)$. Since G is a contravariant functor, $G(\psi\theta) = (G\theta)(G\psi)$, hence $(G\theta)\tau_B = \tau_C(\Box \circ \theta)$, thus, τ is a natural transformation. Now $y(\tau) = \tau_A(\mathrm{id}_A) = G(\mathrm{id}_A)(x) = x$, and so y is bijection. □

Corollary 1.1.2 (Another form of Yoneda's Lemma)

(a) Let $f : X \to Y$ be a morphism such that $h_f : h_X \to h_Y$ is a natural isomorphism. Then f is an isomorphism.

(b) Let X, Y be objects such that $h_X \cong h_Y$. Then $X \cong Y$.

Definition 1.1.12 (Representable)

*A covariant functor $F : \mathscr{C} \to$ **Sets** is said to be **representable** if there exists $A \in \mathrm{obj}(\mathscr{C})$ with natural isomorphism $F \cong \mathrm{Mor}_{\mathscr{C}}(A, \Box) = h^A$.*

*A contravariant functors $G : \mathscr{C} \to$ **Sets** is said to be **represantable** if there exists $A \in \mathrm{obj}(\mathscr{C})$ with natural isomorphism $G \cong \mathrm{Mor}_{\mathscr{C}}(\Box, A) = h_A$.*

Then we get a functor $h : \mathscr{C} \to \mathbf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathbf{Sets}), X \mapsto h_X$. Yoneda's Lemma states that this is a fully faithful functor, called **the Yoneda embedding**.

Corollary 1.1.3 (Yoneda Imbedding)

If $\mathscr{C}$ is a small category, then there is a functor $h : \mathscr{C} \to \mathbf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathbf{Sets})$ that is injective on objects and whose image is a full subcategory of $\mathbf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathbf{Sets})$.

Proof Define h on objects by $h(A) = h_A = \mathrm{Mor}_{\mathscr{C}}(\Box, A)$. If $A \neq A'$, then pairwise disjointness of Mor sets gives $\mathrm{Mor}(\Box, A) \neq \mathrm{Mor}(\Box, A')$, that is, $h_A \neq h_{A'}$, and so h is injective on objects. If $\psi : A \to B$ is a morphism in $\mathscr{C}$, then there is a natural transformation $h(\psi) : \mathrm{Mor}(\Box, A) \to \mathrm{Mor}(\Box, B)$ with $h(\psi)_C = \psi \circ \Box$, for all $C \in \mathrm{obj}(\mathscr{C})$, by Yoneda's Lemma 1.1.1. Now, we have $h(\psi\eta) = h(\psi)h(\eta)$, and so h is covariant functor. Surjectivity of the Yoneda's function y in Theorem 1.1.1 shows that every natural transformation $h_A \to h_B$ arises as $h(\psi)$ for some ψ. Therefore, the image of h is a full subcategory of the functor category $\mathbf{Sets}^{\mathscr{C}^{op}}$. □

1.2 Limites and colimits

1.2.1 Initial, final, zero object

Definition 1.2.1 (Initial object, final object, zero object)

*Let $\mathscr{C}$ be a category. An object $X \in \mathscr{C}$ is called **initial** if for all $Y \in \mathscr{C}$, there exists a unique morphism $X \to Y$.*

*An object $X \in \mathscr{C}$ is called **final** if for all $Y \in \mathscr{C}$, there exists a unique morphism $Y \to X$ (if and only if Y is initial in $\mathscr{C}^{\mathrm{op}}$).*

*If an object is both initial and final, it is called a **zero object** (unique up to isomorphism) and denoted by* 0.

Example 1.21

(i) In $\mathbf{Sets}$, the initial object is $\varnothing$ and every one-element set $\{*\}$ is a final object.

(ii) In $\mathbf{Rings}$, $\mathbb{Z}$ is an initial object, 0 is the final object.

(iii) In $\mathbf{Mod}_R$, 0 is a zero object. If $\mathbb{C}$ admits a zero object, then for all $X, Y \in \mathscr{C}$, there exists a unique morphism $X \to Y$ given by: $X \to 0 \to Y$ called the zero morphism.

1.2.2 Product, coproduct

Definition 1.2.2 (Product)

*Let $\mathscr{C}$ be a category, and let $\{A_i\}_{i \in I}$ be a family of objects in $\mathscr{C}$ indexed by a set I. A **product** is an ordered pair $(C, \{p_i : C \to A_i\}_{i\in I})$, consisting of an object C and a family $\{p_i : C \to A_i\}_{i \in I}$ of **projections**, that is a solution to the following universal mapping problem: for every object X equipped with morphisms $f_i : X \to A_i$, there exists a unique morphism $\theta : X \to C$ making the diagram commute for each i.*

$$\begin{tikzcd}
X \arrow[dr, dashed, "\exists!\theta"] \arrow[drr, "f_i"] \arrow[ddr, "f_j"'] & & \\
& C \arrow[r, "p_i"] \arrow[d, "p_j"'] & A_i \\
& A_j &
\end{tikzcd}$$

Definition 1.2.3 (Coproduct)

*Let $\mathscr{C}$ be a category, and let $\{A_i\}_{i\in I}$ be a family of obejects in $\mathscr{C}$ indexed by a set I. A **coproduct** is an ordered pair $(C, \{\alpha_i : A_i \to C\}_{i\in I})$, consisting of an object C and a family $\{\alpha_i : A_i \to C\}_{i\in I}$ of morphisms, called **injections**, that is a solution to the following universal mapping problem: for every object X equipped with morphisms $\{f_i : A_i \to X\}_{i\in I}$, there exists a unique morphism $\theta : C \to X$*

making the diagram commute for each i.

$$
\begin{tikzcd}
& A_i \arrow[d, "\alpha_i"'] \arrow[ddr, "f_i"] & \\
A_j \arrow[r, "\alpha_j"] \arrow[drr, "f_j"'] & C \arrow[dr, dashed, "\exists!\theta"] & \\
& & X
\end{tikzcd}
$$

Coproduct is denoted by $\coprod$.

Example 1.22

(i) In $\mathbf{Sets}$, $\prod_{i\in I} X_i$ is the usual product, $\coprod_{i\in I} X_i$ is the disjoint union.

(ii) In $\mathbf{Mod}_R$, $\prod_{i\in I} M_i$ is the usual product, $\coprod_{i\in I} X_i = \bigoplus_{i\in I} X_i$ is the direct sum.

1.2.3 Fibered product, fibered coproduct

Definition 1.2.4 (Fibered product)

*Suppose morphsims $\alpha : X \to Z$ and $\beta : Y \to Z$ (in any category). Then the **fibered product** (= pullback) is an object $X \times_Z Y$ along with morphisms $\mathrm{pr}_X : X \times_Z Y \to X$ and $\mathrm{pr}_Y : X \times_Z Y \to Y$, where the two compositions $\alpha \circ \mathrm{pr}_X = \beta \circ \mathrm{pr}_Y$, such that given any object W with maps to X and Y whose compositions to Z agree (commute), these maps factor through some unique $W \to X \times_Z Y$:*

$$
\begin{tikzcd}
W \arrow[dr, dashed, "\exists!"] \arrow[drr] \arrow[ddr] & & \\
& X \times_Z Y \arrow[r, "\mathrm{pr}_Y"'] \arrow[d, "\mathrm{pr}_X"] & Y \arrow[d, "\beta"] \\
& X \arrow[r, "\alpha"'] & Z
\end{tikzcd}
$$

Definition 1.2.5 (Fibered coproduct)

*Suppose morphisms $\alpha : Z \to X$ and $\beta : Z \to Y$ (in any category). Then the **fibered coproduct** (= pushout) is an object $X \cup_Z Y$ along with morphisms $\iota_X : X \to X \cup_Z Y$ and $\iota_Y : Y \to X \cup_Z Y$, where the two compositions $\iota_X \circ \alpha = \iota_Y \circ \beta$, such that given any object W with maps from X and Y whos compositions from Z agree, these maps factor through some unique $X \cup_Z Y \to W$:*

$$
\begin{tikzcd}
Z \arrow[r, "\beta"] \arrow[d, "\alpha"'] & Y \arrow[d, "\iota_Y"'] \arrow[ddr] & \\
X \arrow[r, "\iota_X"] \arrow[drr] & X \cup_Z Y \arrow[dr, dashed, "\exists!"] & \\
& & W.
\end{tikzcd}
$$

Example 1.23

(i) In $\mathbf{Sets}$, fibered product exists: $X \times_Z Y = \{(x, y) \in X \times Y : f(x) = g(y)\}$, where $f : X \to Z$ and $g : Y \to Z$. This induces in $\mathbf{Top}$ and $\mathbf{Grp}$ fibered product exists.

(ii) In $\mathbf{Sets}$, fibered coproducts exist: $X \cup_Z Y = X \coprod Y/\sim$, where $\sim$ is given by $f(z) = g(z)$ ($f : Z \to X$ and $g : Z \to Y$).

1.2.4 Limit, colimit

Definition 1.2.6 (Comma category)

Let $F : \mathscr{C} \to \mathscr{D}$ be a functor and $Y \in \mathscr{D}$. The ***comma category*** *$(F \downarrow Y)$ is defined as follows:*

(i) An object is a pair (X, h), where $X \in \mathrm{obj}(\mathscr{C})$, $h \in \mathrm{Mor}_{\mathscr{D}}(FX, Y)$.

(ii) A morphism between (X_1, h_1) and (X_2, h_2) consists of a morphism $f : X_1 \to X_2$ in $\mathscr{C}$ and a comma diagram

$$\begin{tikzcd} FX_1 \arrow[rr, "Ff"] \arrow[dr, "h_1"'] & & FX_2 \arrow[dl, "h_2"] \\ & Y & \end{tikzcd}$$

Dually, we can define $(Y \downarrow F)$ by reversing the arrow of h.

Definition 1.2.7 (Diagonal functor)

Let $\mathscr{I}$ and $\mathscr{C}$ be arbitrary categories. The $\mathscr{I}$-ary diagonal functor of $\mathscr{C}$ is the functor $\Delta : \mathscr{C} \to \mathbf{Fun}(\mathscr{I}, \mathscr{C})$ sending each object X to the constant functor ΔX (the functor having value X for each object of $\mathscr{I}$ and value id_X for each arrow of $\mathscr{I}$), and each arrow $f : X \to Y$ of $\mathscr{C}$ to the natural transformation $\Delta f : \Delta X \to \Delta Y$ which has the same value f at each object i of $\mathscr{I}$.

Here are some applications of comma categories:

Example 1.24

(i) Let I be a fixed set, $(X_i)_{i \in I}$ a family of objects in $\mathscr{C}$, it can be viewed as an object X in $\mathscr{C}^I = \mathbf{Fun}(I, \mathscr{C})$. Let $\Delta : \mathscr{C} \to \mathscr{C}^I$ be the diagonal functor. Then $\prod_{i \in I} X_i$ is a final object in $(\Delta \downarrow X)$; $\coprod_{i \in I} X_i$ is an initial object in $(X \downarrow \Delta)$.

Proof We just prove $\prod_{i \in I} X_i$ is a final object in $(\Delta \downarrow X)$. The natural transformation $\Delta(\prod_{i \in I} X_i) \to X$ is the family of maps $\{\mathrm{pr}_j : \prod_{i \in I} X_i \to X_j\}_j$. Let (c, θ) be any object in $(\Delta \downarrow X)$, from natural transformation $\Delta c \to X$, we get the following diagram.

$$\begin{tikzcd} c \arrow[dr, dashed] \arrow[drr, bend left=10] \arrow[ddr] & & \\ & \prod_{k \in I} X_k \arrow[r] \arrow[d] & X_j \\ & X_i & \end{tikzcd}$$

By the universal property of product, there exists unique morphism from c to $\prod_{k \in I} X_k$. This induced a unique natural transformation $\Delta c \to \prod_{k \in I} X_k$. So $\prod_{i \in I} X_i$ is a final object in $(\Delta \downarrow X)$. $\square$

(ii) Let $\mathscr{C}, \mathscr{I}$ be categories and $F \in \mathscr{C}^{\mathscr{I}} = \mathbf{Fun}(\mathscr{I}, \mathscr{C})$ be a functor from $\mathscr{I}$ to $\mathscr{C}$. Let $\Delta : \mathscr{C} \to \mathscr{C}^{\mathscr{I}}$ be the diagonal functor (= constant functor). Then the category $(\Delta \downarrow F)$ is defined as follows:

- An object consists of a pair (X, h) where $X \in \mathscr{C}$.
- A morphism $h : \Delta X \to F$ natural transformation, i.e., a family of morphisms $\{h_i : X \to F(i)\}_{i \in \mathscr{I}}$, such that $\forall i \xrightarrow{f} j$, $h_j = F(f) \circ h_i$.

$$\begin{tikzcd} \Delta X(i) = X \arrow[r, leftrightarrow, "\sim"] \arrow[d, "h_i"'] & \Delta X(j) = X \arrow[d, "h_j"] \\ F(i) \arrow[r, "F(f)"] & F(j) \end{tikzcd}$$

Dually we have $(F \downarrow \Delta)$.

Definition 1.2.8 (Limit,colimit)

*A **limit** (= projective/inverse limit) of F is a final object of $(\Delta \downarrow F)$. A **colimit** (= inductive/direct limit) of F is an initial object of $(F \downarrow \Delta)$. When exists they are unique up to isomorphism, denoted by $\lim_{i\in\mathscr{I}} F(i)$ and $\operatorname{colim}_{i\in\mathscr{I}} F(i)$ respectively.*

Example 1.25 (Important examples!)

(i) For $\mathscr{I} = (\bullet \to \bullet \leftarrow \bullet)$, $F : \mathscr{I} \to \mathscr{C}$ is given by a pair of morphism in $\mathscr{C}$: $f : X \to Z$, $g : Y \to Z$. A limit of F is a pullback of (f, g). Dually, for $\mathscr{I} = (\bullet \leftarrow \bullet \to \bullet)$, a colimit is a pushout.

(ii) If I is a directed poset, viewed as a category, then given $F : I^{\mathrm{op}} \to \mathscr{C}$ is same as given an inverse system $(X_i)_{i\in I}$. We also write $\lim\limits_{i\in I} F(i) = \varprojlim\limits_{i\in I} X_i$.

Given $G : I \to \mathscr{C}$ is same as given a direct system $(X_i)_{i\in I}$, we also write $\operatorname{colim}_{i\in I} G(i) = \varinjlim\limits_{i\in I} X_i$. (We can generalize I to filtered small category)

Definition 1.2.9 (Equalizer, coequalizer, kernel, and cokernel)

*For $\mathscr{I} = (\bullet \rightrightarrows \bullet)$, $F : \mathscr{I} \to \mathscr{C}$ is represented by a pair of morphisms $f, g : X \to Y$. Then $\lim F$ (resp., $\operatorname{colim} F$) is called the **equalizer** (resp., **coequalizer**) of (f, g), denoted $\operatorname{Eq}(f, g)$ (resp., $\operatorname{Coeq}(f, g)$).*

*If $\mathscr{C}$ admits a zero object and $g = 0$, then $\operatorname{Ker} f := \operatorname{Eq}(f, 0)$, $\operatorname{Coker} f := \operatorname{Coeq}(f, 0)$ are called the **kernel** and **cokernel** of f.*

Proposition 1.2.1

Let $f, g : X \to Y$, then canonical morphisms $e : \operatorname{Eq}(f, g) \to X$ is monomorphism and $c : Y \to \operatorname{Coeq}(f, g)$ is epimorphism.

Proof Consider the diagram

$$\begin{tikzcd}
Z \arrow[d, dashed, "\exists!"'] \arrow[dr] \arrow[drr, bend left] & & \\
E = \operatorname{Eq}(f,g) \arrow[r, "e"'] & X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y
\end{tikzcd}$$

the canonical morphism e is a monomorphism.

Dually,

$$X \overset{f}{\underset{g}{\rightrightarrows}} Y \xrightarrow{c} C = \operatorname{Coeq}(f, g),$$

c is an epimorphism. □

Definition 1.2.10

*Let $\mathscr{C}, \mathscr{I}$ be categories, we say $\mathscr{C}$ **admits limits of shape** $\mathscr{I}$ if for any functor $F : \mathscr{I} \to \mathscr{C}$ has a limit, i.e., it induces a functor $\lim : \mathscr{C}^{\mathscr{I}} \to \mathscr{C}$.*

*We say $\mathscr{C}$ **admits finite (resp., small) limits**, if $\mathscr{C}$ admits limits of shape $\mathscr{I}$ for all finite (resp., small) category $\mathscr{I}$.*

*We say $\mathscr{C}$ is **complete** if it admits small limits. Dually we have the notion of a **cocomplete category**.*

Example 1.26

(i) **Sets** admits small limit and small colimit. Let $F : \mathscr{I} \to \mathbf{Sets}$ be a functor, $\mathscr{I}$ a small category. Then $\lim F$ is represented by $L = \{(X_i)_{i\in I} : Ff(X_i) = X_j, \forall f : i \to j\}$.
colim F is represented by $Q = \coprod_{i\in I} F(i)/\sim$, where $\sim$ is the equivalent relation generated by $x \sim Ff(x)$ for $f : i \to j, x \in F(i)$.

(ii) Similarly $\mathbf{Mod}_R$ admits small limits and small colimits.

Definition 1.2.11

*Let $\mathscr{C}$, $\mathscr{D}$ be two categories and $F : \mathscr{C} \to \mathscr{D}$ be a functor. We say F **preserves all limits of shape** $\mathscr{I}$, if for any functor $S : \mathscr{I} \to \mathscr{C}$ with (X, h) a limit of S, where $h : \Delta X \to S$ is a functor, then*

$$\Delta(FX) = F \circ \Delta X \xrightarrow{Fh} FS,$$

is a limit of $FS : \mathscr{I} \to \mathscr{D}$.
*We can say F preserves products, equalizers, pullbacks, etc. A **continuous functor** is one that preserves all small limits and a **cocontinuous functor** is one that preserves all small colimits.*

1.3 Adjoint functors

Definition 1.3.1 (Adjoint)

*Let $\mathscr{C}$, $\mathscr{D}$ be two categories, an **adjunction** is a triple (F, G, τ) where $F : \mathscr{C} \to \mathscr{D}$, $G : \mathscr{D} \to \mathscr{C}$ are functors, and τ is a natural isomorphism:*

$$\tau_{\square,\square} : \mathrm{Mor}_{\mathscr{D}}(F\square, \square) \xrightarrow{\sim} \mathrm{Mor}_{\mathscr{C}}(\square, G\square).$$

*i.e., $\tau_{X,Y} : \mathrm{Mor}_{\mathscr{D}}(FX, Y) \to \mathrm{Mor}_{\mathscr{C}}(X, GY)$, $\forall X \in \mathscr{C}, Y \in \mathscr{D}$ are functorially bijective. We say that F is **left adjoint** to G, G is **right adjoint** to F and write $\tau : F \dashv G$.*

Example 1.27

(i) Let $U : \mathbf{Grp} \to \mathbf{Sets}$ be the forgetful functor, the free group functor (send any set to a free group) is the left adjoint to U.

(ii) Let $\mathscr{I}, \mathscr{C}$ be two categories. If $\mathscr{C}$ admits limits of shape $\mathscr{I}$, then the functor $\lim : \mathscr{C}^{\mathscr{I}} \to \mathscr{C}$ is right adjoint to the diagonal functor $\Delta : \mathscr{C} \to \mathscr{C}^{\mathscr{I}}$.
Similarly, if $\mathscr{C}$ admits colimits of shape $\mathscr{I}$, the colim : $\mathscr{C}^{\mathscr{I}} \to \mathscr{C}$ is left adjoint to $\Delta : \mathscr{C} \to \mathscr{C}^{\mathscr{I}}$.
Proof We just proof $\Delta \dashv \lim$. Let $\alpha : \Delta X \to G$ be a natural transformation. Since $\lim G$ is the final object of $(\Delta \downarrow G)$, then there exists unique morphism $\alpha' : X \to \lim G$. Then we defined $\mathrm{Mor}_{\mathscr{C}^{\mathscr{I}}}(\Delta X, G) \to \mathrm{Mor}_{\mathscr{I}}(X, \lim G)$.
Conversely, let $\beta : X \to \lim G$ be a morphism in $\mathrm{Mor}_{\mathscr{C}}(X, \lim G)$. Say $\theta : \Delta \lim G \to G$. Define natural transformation $\beta' : \Delta X \to G$ by setting $\beta'_i = \theta_i \circ \beta$. Then we defined $\mathrm{Mor}_{\mathscr{I}}(X, \lim G) \to \mathrm{Mor}_{\mathscr{C}^{\mathscr{I}}}(\Delta X, G)$. $\square$

(iii) Suppose M, N, and P are A-modules (where A is a ring). There is a bijection $\mathrm{Hom}_A(M \otimes_A N, P) \leftrightarrow \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P))$. This isomorphisms shows $\square \otimes_A N$ is left adjoint to $\mathrm{Hom}_A(N, \square)$.

(iv) The inclusion functor $i : \mathbf{Ab} \hookrightarrow \mathbf{Grp}$ has a left adjoint $a : \mathbf{Grp} \to \mathbf{Ab}$ ($G \mapsto G^{\mathrm{ab}}$, $G^{\mathrm{ab}} = G/[G, G]$, $[G, G]$ is generated by $[g, h] = ghg^{-1}h^{-1}$) since $\mathrm{Hom}_{\mathbf{Ab}}(G^{\mathrm{ab}}, A) \cong \mathrm{Hom}_{\mathbf{Grp}}(G, A)$.

Proposition 1.3.1

Let $\tau : F \dashv G$. Then G is determined by F up to natural isomorphism.

Proof Assume $\tau' : F \dashv G'$, $F : \mathscr{C} \to \mathscr{D}$, $G, G' : \mathscr{D} \to \mathscr{C}$. Consider the natural isomorphism $\tau' \circ \tau^{-1}$:

$$\forall X \in \mathscr{C}, Y \in \mathscr{D}, (\tau' \circ \tau^{-1})_{X,Y} : \mathrm{Mor}_{\mathscr{C}}(X, GY) \xrightarrow{\sim} \mathrm{Mor}_{\mathscr{C}}(X, G'Y).$$

By Yoneda's Lemma 1.1.2, this is given by an isomorphism $GY \cong GY'$ which is natural in Y. Then $G \cong G'$. □

Proposition 1.3.2

Let $F : \mathscr{C} \to \mathscr{D}$ be a functor. Then F admits a right adjoint if and only if for every object Y is $\mathscr{D}$, the functor $h_Y \circ F = \mathrm{Mor}_{\mathscr{D}}(F\square, Y)$ is representable.

Proof If $F \dashv G$, then $\forall Y \in D$, $\mathrm{Mor}_{\mathscr{D}}(F\square, Y) \cong \mathrm{Mor}_{\mathscr{C}}(\square, GY) = h_{GY}$ is represented by GY.

Conversely, we construct an adjunction $\varphi : F \dashv G$ as follows: $\forall Y \in \mathscr{D}$, we choose an object $GY \in \mathscr{C}$ with an isomorphism $\mathrm{Mor}_{\mathscr{D}}(F\square, Y) \cong h_{GY}(\square)$. By Yoneda's Lemma 1.1.2, such GY is unique up to a natural isomorphism. Let $g : Y \to Y'$, then g induces a natural morphism $\mathrm{Mor}_{\mathscr{D}}(F\square, Y) \to \mathrm{Mor}_{\mathscr{D}}(F\square, Y')$, i.e., a natural transformation $h_{GY} \to h_{GY'}$. By Yoneda's Lemma 1.1.1, we get a morphism $GY \to GY'$, define Gg be this morphism. Hence, we get a right adjoint G to F. □

Lemma 1.3.1

Let $F : \mathscr{C} \to \mathscr{D}$, $F' : \mathscr{D} \to \mathscr{E}$, $G : \mathscr{D} \to \mathscr{C}$, $G' : \mathscr{E} \to \mathscr{D}$ be functors and let $\tau : F \dashv G$, $\tau' : F' \dashv G'$ be adjunctions. Then $\tau\tau' : F'F \dashv GG'$.

Proof Consider $\mathrm{Mor}_{\mathscr{E}}(F'FX, Y)$, by $F \dashv G$ and $F' \dashv G'$, we have

$$\mathrm{Mor}_{\mathscr{E}}(F'FX, Y) \cong \mathrm{Mor}_{\mathscr{D}}(FX, G'Y) \cong \mathrm{Mor}_{\mathscr{C}}(X, GG'Y),$$

for all $X \in \mathscr{C}$ and $Y \in \mathscr{E}$. Next we need to check the naturality of this map. The entire process is straightforward, and the proof will not be provided here. □

Lemma 1.3.2

Let $F : \mathscr{C} \to \mathscr{D}$, $G : \mathscr{D} \to \mathscr{C}$ be functors and let $\tau : F \dashv G$ be an adjunction. Then $\eta : \mathrm{id}_{\mathscr{C}} \to GF$ with

$$\eta_X = \tau_{X,FX}(\mathrm{id}_{FX}) : X \to GFX, \forall X \in \mathscr{C}$$

is a natural transformation; similarly, $\varepsilon : FG \to \mathrm{id}_{\mathscr{D}}$ with

$$\varepsilon_Y = \tau^{-1}_{GY,Y}(\mathrm{id}_{GY}) : FGY \to Y, \forall Y \in \mathscr{D}$$

is a natural transformation. The compositions:

$$F \xrightarrow{F\eta} FGF \xrightarrow{\varepsilon F} F, \qquad G \xrightarrow{\eta G} GFG \xrightarrow{G\varepsilon} G$$

are identities, i.e.,

$$\varepsilon F \circ F\eta = \mathrm{id}_F, \qquad G\varepsilon \circ \eta G = \mathrm{id}_G.$$

The data $(F, G, \varepsilon, \eta)$ with $\varepsilon F \circ F\eta = \mathrm{id}_F$, $G\varepsilon \circ \eta G = \mathrm{id}_G$ uniquely determined $\tau : F \dashv G$.

Proof Consider $\mathrm{id}_{FX} : FX \xrightarrow{\sim} FX$, by $F \dashv G$, we have $\tau_{X,FX}(\mathrm{id}_{FX}) : X \to GFX$ for all $X \in \mathscr{C}$. Define $\eta_X = \tau_{X,FX}(\mathrm{id}_{FX})$ for all $X \in \mathscr{C}$. It is easy to check that $\eta : \mathrm{id}_{\mathscr{C}} \to GF$ is a natural transformation.

Similarly, consider $\mathrm{id}_{GY} : GY \xrightarrow{\sim} GY$, by $F \dashv G$, we have $\tau^{-1}_{GY,Y}(\mathrm{id}_{GY}) : FGY \to Y$ for all $Y \in \mathscr{D}$. Define $\varepsilon_Y = \tau^{-1}_{GY,Y}(\mathrm{id}_{GY})$ for all $Y \in \mathscr{D}$. It is easy to check $\varepsilon : FG \to \mathrm{id}_{\mathscr{D}}$ is a natural transformation.

We next show that $\varepsilon F \circ F\eta = \mathrm{id}_F$. It suffices to show that $\mathrm{id}_{FX} = \varepsilon_{FX} \circ F(\eta_X)$, for all $X \in \mathscr{C}$. By naturality of $\tau : F \dashv G$, we have the following commutative diagram.

$$\begin{array}{ccc} \mathrm{Mor}_{\mathscr{C}}(GFX, GFX) & \xrightarrow{\square\circ\eta_X} & \mathrm{Mor}_{\mathscr{C}}(X, GFX) \\ {\scriptstyle \tau^{-1}_{X,GFX}}\big\downarrow & & \big\downarrow{\scriptstyle \tau^{-1}_{X,FX}} \\ \mathrm{Mor}_{\mathscr{D}}(FX, FGFX) & \xrightarrow{\square\circ F(\eta_X)} & \mathrm{Mor}_{\mathscr{D}}(FX, FX) \end{array}$$

Hence

$$\varepsilon_{FX} \circ F(\eta_X) = \tau^{-1}_{GFX,FX}(\mathrm{id}_{GFX}) \circ F(\eta_X) = \tau^{-1}_{X,FX}(\mathrm{id}_{GFX} \circ \eta_X),$$

and therefore

$$\tau_{X,FX}(\varepsilon_{FX} \circ F(\eta_X)) = \mathrm{id}_{GFX} \circ \eta_X = \mathrm{id}_{GFX} \circ \tau_{X,FX}(\mathrm{id}_{FX}) = \tau_{X,FX}(\mathrm{id}_{FX}).$$

Since $\tau_{X,FX}$ is bijection, we have $\varepsilon_{FX} \circ F(\eta_X) = \mathrm{id}_{FX}$, as we desired. Hence $\varepsilon F \circ F\eta = \mathrm{id}_F$.

We now show that $G\varepsilon \circ \eta G = \mathrm{id}_G$. It suffices to show that $\mathrm{id}_{GY} = G(\varepsilon_Y) \circ \eta_{GY}$, for all $Y \in \mathscr{D}$. By naturality of $\tau : F \dashv G$, we have the following commutative diagram.

$$\begin{array}{ccc} \mathrm{Mor}_{\mathscr{D}}(FGY, FGY) & \xrightarrow{\varepsilon_Y\circ\square} & \mathrm{Mor}_{\mathscr{D}}(FGY, Y) \\ {\scriptstyle \tau_{GY,FGY}}\big\downarrow & & \big\downarrow{\scriptstyle \tau_{GY,Y}} \\ \mathrm{Mor}_{\mathscr{C}}(GF, GFGY) & \xrightarrow{G(\varepsilon_Y)\circ\square} & \mathrm{Mor}_{\mathscr{C}}(GY, GY) \end{array}$$

Hence

$$G(\varepsilon_Y) \circ \eta_{GY} = G(\varepsilon_Y) \circ \tau_{GY,FGY}(\mathrm{id}_{FGY}) = \tau_{GY,Y}(\varepsilon_Y \circ \mathrm{id}_{FGY}) = \tau_{GY,Y}(\varepsilon_Y) = \mathrm{id}_{GY},$$

i.e., $G\varepsilon \circ \eta G = \mathrm{id}_G$.

Finally, we show that we can recover τ from $\varepsilon F \circ F\eta = \mathrm{id}_F$ and $G\varepsilon \circ \eta G = \mathrm{id}_G$. Let $f \in \mathrm{Mor}_{\mathscr{D}}(FX, Y)$ and $g \in \mathrm{Mor}_{\mathscr{C}}(X, GY)$, define $\tau_{X,Y}(f) = Gf \circ \eta_X$ and $\tau^{-1}_{X,Y}(g) = \varepsilon_Y \circ Fg$. Now we check that $\tau_{X,Y}(\tau^{-1}_{X,Y}(g)) = g$ and $\tau^{-1}_{X,Y}(\tau_{X,Y}(f)) = f$:

$$\tau_{X,Y}(\tau^{-1}_{X,Y}(g)) = \tau_{X,Y}(\varepsilon_Y \circ Fg) = G(\varepsilon_Y \circ Fg) \circ \eta_X = G(\varepsilon_Y) \circ GFg \circ \eta_X,$$

note that $GFg \circ \eta_X = \eta_{GY} \circ g$ (naturality of η) and $G(\varepsilon_Y) \circ \eta_{GY} = \mathrm{id}_{GY}$, we have

$$\tau_{X,Y}(\tau^{-1}_{X,Y}(g)) = G(\varepsilon_Y) \circ \eta_{GY} \circ g = \mathrm{id}_{GY} \circ g = g.$$

Similarly, we have $\tau^{-1}_{X,Y}(\tau_{X,Y}(f)) = f$. Check that τ is natural transformation is straightforward. $\square$

Definition 1.3.2 (Unit, counit)

*Let $F : \mathscr{C} \to \mathscr{D}$, $G : \mathscr{D} \to \mathscr{C}$ be functors and let $\tau : F \dashv G$ be an adjunction. Define **unit** $\eta : \mathrm{id}_{\mathscr{C}} \to GF$ by setting $\eta_X = \tau_{X,FY}(\mathrm{id}_{FX}), \forall X \in \mathscr{C}$, and **counit** $\varepsilon : FG \to \mathrm{id}_{\mathscr{D}}$ by setting $\varepsilon_Y = \tau^{-1}_{GY,Y}(\mathrm{id}_{GY}), \forall Y \in \mathscr{D}$.*

Proposition 1.3.3

Let $F : \mathscr{C} \to \mathscr{D}$, $G : \mathscr{D} \to \mathscr{C}$, be functors and let $\tau : F \dashv G$ be adjunction. Then

(i) F is fully faithful if and only if unit $\eta : \mathrm{id}_{\mathscr{C}} \to GF$ is a natural isomorphism.

(ii) G is fully faithful if and only if counit $\varepsilon : FG \to \mathrm{id}_{\mathscr{D}}$ is a natural isomorphism.

Proof We just prove (i), (ii) is similarly. Let $f : X \to X'$ be a morphism in $\mathscr{C}$, we have a commutative diagram:

$$\begin{tikzcd}
\mathrm{Mor}_{\mathscr{D}}(FX, FX) \arrow[r, "\tau_{X,FX}"] \arrow[d, "F(f)\circ\square"'] & \mathrm{Mor}_{\mathscr{C}}(X, GFX) \arrow[d, "GF(f)\circ\square"] \\
\mathrm{Mor}_{\mathscr{D}}(FX, FX') \arrow[r, "\tau_{X,FX'}"] & \mathrm{Mor}_{\mathscr{C}}(X, GFX')
\end{tikzcd}$$

Then

$$\tau_{X,FX'}(F(f)\circ\square)(\mathrm{id}_{FX}) = \tau_{X,FX'}(F(f)) = GF(f)\circ\tau_{X,FX}(\mathrm{id}_{FX}) = GF(f)\circ\eta_X = \eta_{X'}\circ f,$$

where the last equal sign given by the naturality of η. Then the composition

$$\mathrm{Mor}_{\mathscr{C}}(X, X') \xrightarrow{F} \mathrm{Mor}_{\mathscr{D}}(FX, FX') \xrightarrow{\tau_{X,FX'}} \mathrm{Mor}_{\mathscr{C}}(X, GFX')$$

is induced by $\eta_{X'}\circ\square$.

If F is fully faithful, then F is bijection (as map of Mor sets). Since $\tau_{X,FX'}$ is bijection, we know that $\eta_{X'}$ is bijection for all $X'\in\mathscr{C}$.

Conversely, if η is natural isomorphism, we have $F = \tau_{X,FX'}^{-1}\circ\eta_{X'}$ is bijection (as map of Mor sets) for all $X, X'\in\mathscr{C}$. Hence F is fully faithful. $\square$

Corollary 1.3.1

Let $F:\mathscr{C}\to\mathscr{D}$, $G:\mathscr{D}\to\mathscr{C}$, be functors and let $\tau: F\dashv G$ be adjunctions. The following are equivalent:

(i) F is an equivalence of categories.

(ii) G is an equivalence of categories.

(iii) F and G are fully faithful.

(iv) The unit η and counit ε are natural isomorphisms under these conditions, F and G are quasi-inverse to each other.

Proof Clear. $\square$

Example 1.28 In Example 1.27 (iv): $i:\mathbf{Ab}\to\mathbf{Grp}$ is fully faithful, the counit $\varepsilon: a\circ i\to\mathrm{id}_{\mathbf{Ab}}$ is natural isomorphism (not that $a\dashv i$). The unit $\eta:\mathrm{id}_{\mathbf{Grp}}\to i\circ a$ is not a natural isomorphism.

Proposition 1.3.4 (Important!)

Let $F:\mathscr{C}\to\mathscr{D}$, $G:\mathscr{D}\to\mathscr{C}$, be functors and let $\tau: F\dashv G$ be adjunction. Then

(i) $F^{\mathscr{I}}:\mathscr{C}^{\mathscr{I}}\to\mathscr{D}^{\mathscr{I}}$ is adjoint to $G^{\mathscr{I}}:\mathscr{D}^{\mathscr{I}}\to\mathscr{C}^{\mathscr{I}}$ for any category $\mathscr{I}$.

(ii) G preserves limits and F preserves colimits whenever exists.

Proof

(i) Check it directly.

(ii) We just need to G preserves limits, the case for F is similar. Let $S:\mathscr{I}\to\mathscr{D}$ be a functor such that $\lim S$ exists. We need to check $G(\lim S)\cong\lim G^{\mathscr{I}}(S)$.

Since $\Delta \dashv \lim$, $F \dashv G$, $F^{\mathscr{I}} \dashv G^{\mathscr{I}}$, we get $\forall X \in \mathscr{C}$,

$$\begin{aligned}
\mathrm{Mor}(X, \lim G^{\mathscr{I}}(S)) &\cong \mathrm{Mor}(\Delta X, G^{\mathscr{I}}(S)) && \Delta \dashv \lim \\
&\cong \mathrm{Mor}(F^{\mathscr{I}}\Delta X, S) && F^{\mathscr{I}} \dashv G^{\mathscr{I}} \\
&\overset{(*)}{=} \mathrm{Mor}(\Delta F X, S) && \\
&\cong \mathrm{Mor}(FX, \lim S) && \Delta \dashv \lim \\
&\cong \mathrm{Mor}(X, G(\lim S)) && F \dashv G,
\end{aligned}$$

where $(*)$ follows from the commutative diagram:

$$\begin{tikzcd}
\mathscr{C} \arrow[r, "\Delta"] \arrow[d, "F"'] & \mathscr{C}^{\mathscr{I}} \arrow[d, "F^{\mathscr{I}}"] \\
\mathscr{D} \arrow[r, "\Delta"] & \mathscr{D}^{\mathscr{I}}
\end{tikzcd}$$

By Yoneda's Lemma 1.1.2, we have $G(\lim S) \cong \lim G^{\mathscr{I}}(S)$.

□

1.4 Additive and abelian categories

1.4.1 Additive categories

Recall that a **magma** is a set equipped with a binary operation. The magma is said to be **unital** if it has an identity element.

Proposition 1.4.1

Let $\mathscr{A}$ be a category with each $\mathrm{Mor}_{\mathscr{A}}(X,Y)$ equipped with a structure of unital magma such that composition is bilinear. Let A_1 and A_2 be objects of $\mathscr{A}$. Then the following conditions are euivalent.

(1) $A_1 \prod A_2$ exists.

(2) $A_1 \coprod A_2$ exists.

Under these assumptions, the morphism $\varphi_{A_1,A_2} : A_1 \coprod A_2 \to A_1 \prod A_2$ described by the matrix $\begin{bmatrix} \mathrm{id}_{A_1} & 0 \\ 0 & \mathrm{id}_{A_2} \end{bmatrix}$ is an isomorphism. Moreover, if $Y \prod Y$ and $Y \prod Y \prod Y$ exist, then $\mathrm{Mor}_{\mathscr{A}}(X,Y)$ is a commutative monoid, and for $f, f' : X \to Y$, $f + f'$ is given by the composition

$$X \xrightarrow{(f,f')} Y \prod Y \xrightarrow{\varphi^{-1}} Y \coprod Y \xrightarrow{(\mathrm{id}_Y, \mathrm{id}_Y)} Y, \tag{1.1}$$

We denote the operation on $\mathrm{Mor}_{\mathscr{A}}(X,Y)$ by "+" and the identity element by $0 = 0_{XY}$. The bilinearity of composition means the following: for $f, f' : X \to Y$ and $g, g' : Y \to Z$, we have

(a) $g \circ (f + f') = g \circ f + g \circ f'$, $(g + g') \circ f = g \circ f + g' \circ f$;

(b) $g \circ 0_{XY} = 0_{XZ}$, $0_{YZ} \circ f = 0_{XZ}$.

Condition (b) that the zero morphism $z_{XY} : X \to Y$ (that factors through every zero object) is the unit of $\mathrm{Mor}_{\mathscr{A}}(X,Y)$. Indeed, $z_{XY} = 0_{YY} \circ z_{XY} = 0_{XY}$.

Proof By duality we may assume that $A_1 \prod A_2$ exists. Let $i_1 = (\mathrm{id}_{A_1}, 0) : A_1 \to A_1 \prod A_2$ and let $i_2 = (0, \mathrm{id}_{A_2}) : A_2 \to A_1 \prod A_2$. We show that $(A_1 \prod A_2, i_1, i_2)$ exhibits $A_1 \prod A_2$ as a coproduct of A_1 and

A_2. Let B be an object of $\mathscr{A}$ equipped with $h_1 : A_1 \to B$ and $h_2 : A_2 \to B$.

$$\begin{tikzcd}
 & A_2 \arrow[d, "i_2"'] \arrow[ddr, "h_2"] & \\
A_1 \arrow[r, "i_1"] \arrow[drr, "h_1"'] & A_1 \prod A_2 \arrow[dr, dashed, "\exists!?"] & \\
 & & B
\end{tikzcd}$$

We put $h =< h_1, h_2 > (p_1, p_2) = h_1p_1 + h_2p_2 : A_1 \prod A_2 \to B$. Here $p_1 : A_1 \prod A_2 \to A_1$ and $p_2 : A_1 \prod A_2 \to A_2$ are the projections. Then $h \circ i_1 = h_1p_1i_1 + h_2p_2i_1 = h_1 + 0 = h_1$ and similarly $hi_2 = h_2$. If $h' : A_1 \times A_2 \to B$ is a morphism such that $h'i_1 = h_1$, $h'i_2 = h_2$. Then

$$h' = h'(i_1p_1 + i_2p_2) = h_1p_1 + h_2p_2 = h$$

by Lemma 1.4.1 below. Hence $A_1 \prod A_2$ is coproduct of A_1 and A_2, and therefore the morphism φ_{A_1,A_2} : $A_1 \coprod A_2 \to A_1 \prod A_2$ described by the matrix $\begin{bmatrix} \mathrm{id}_{A_1} & 0 \\ 0 & \mathrm{id}_{A_2} \end{bmatrix}$ is an isomorphism.

Assume that $Y \times Y$ exists. Then in (1.1), $(\mathrm{id}_Y, \mathrm{id}_Y) = (p_1 + p_2)\varphi_{YY}$. for $f, f' : X \to Y$, $f + f'$ is given by the composition

$$X \xrightarrow{(f,f')} Y \prod Y \xrightarrow{\varphi^{-1}} Y \coprod Y \xrightarrow{(\mathrm{id}_Y,\mathrm{id}_Y)} Y, \tag{1.2}$$

where the first map is induced by the universal property of product $Y \prod Y$ from $f, f' : X \to Y$, and similarly for the third map. Note that the diagram

$$\begin{tikzcd}
X \arrow[r, "{(f,f')}"] \arrow[dr, "{(f',f)}"'] & Y \prod Y \arrow[r, "\varphi_{YY}^{-1}"] \arrow[d, "\sim"] & Y \coprod Y \arrow[r, "{(\mathrm{id}_Y,\mathrm{id}_Y)}"] \arrow[d, "\sim"'] & Y \\
 & Y \prod Y \arrow[r, "\varphi_{YY}^{-1}"'] & Y \coprod Y \arrow[ur, "{(\mathrm{id}_Y,\mathrm{id}_Y)}"'] &
\end{tikzcd}$$

commutes, which implies $f + f' = f' + f$. Here the square in the middle commutes since the following square commutes

$$\begin{tikzcd}
Y \coprod Y \arrow[d, "\sim"'] & Y \prod Y \arrow[l, "\varphi_{YY}"'] \arrow[d, "\sim"] \\
Y \coprod Y & Y \prod Y \arrow[l, "\varphi_{YY}"']
\end{tikzcd}$$

Assume moreover that $Y \prod Y \prod Y$ exists. For $f, f', f : X \to Y$, $f + (f' + f'')$ is given by the composition

$$\begin{tikzcd}
X \arrow[r, "{(f,f'+f'')}"] \arrow[d, "{(f,(f',f''))}"'] & Y \prod Y \arrow[r, "\varphi_{YY}^{-1}"] & Y \coprod Y \arrow[r] & Y \\
Y \prod (Y \prod Y) \arrow[r, "\mathrm{id}_Y \prod \varphi_{YY}^{-1}"] & Y \prod (Y \coprod Y) \arrow[u] \arrow[r, "\varphi_{Y,Y\coprod Y}^{-1}"] & Y \coprod (Y \coprod Y) \arrow[u] &
\end{tikzcd}$$

The diagram

$$\begin{tikzcd}
Y \prod (Y \prod Y) \arrow[r, "\mathrm{id}_Y \prod \varphi_{Y,Y}^{-1}"] \arrow[d, "\sim"'] & Y \prod (Y \coprod Y) \arrow[r, "\varphi_{Y,Y\coprod Y}^{-1}"] & Y \coprod (Y \coprod Y) \arrow[d, "\sim"] \\
(Y \prod Y) \prod Y \arrow[r, "\varphi_{Y,Y}^{-1} \prod \mathrm{id}_Y"] & (Y \coprod Y) \prod Y \arrow[r, "\varphi_{Y\coprod Y,Y}^{-1}"] & (Y \coprod Y) \coprod Y
\end{tikzcd}$$

commutes, because the inverse of the composition of the upper horizontal arrows and the inverse of the composition of the lower horizontal arrows both given by the matrix $\begin{bmatrix} \mathrm{id}_Y & 0 & 0 \\ 0 & \mathrm{id}_Y & 0 \\ 0 & 0 & \mathrm{id}_Y \end{bmatrix}$. Therefore, $f + (f' + f'') = (f + f') + f''$. $\square$

Lemma 1.4.1

Under the above notion, $i_1 p_1 + i_2 p_2 = \mathrm{id}_{A_1 \prod A_2}$.

Proof Consider the following diagram.

$$\begin{tikzcd}
A_1 \prod A_2 \arrow[dr, dashed, "i_1p_1+i_2p_2"] \arrow[drr, bend left, "p_2"] \arrow[ddr, bend right, "p_1"'] & & \\
& A_1 \prod A_2 \arrow[r, "p_2"'] \arrow[d, "p_1"] & A_2 \\
& A_1 &
\end{tikzcd}$$

We have $p_1(i_1p_1 + i_2p_2) = p_1i_1p_1 + p_1i_2p_2 = p_1 + 0 = p_1$ and similarly $p_2(i_1p_1 + i_2p_2) = p_2$. By the universal property of product, we know that $i_1p_1 + i_2p_2 = \mathrm{id}_{A_1 \prod A_2}$. $\square$

Remark Under the assumptions (1) and (2) of Proposition 1.4.1, the composition of

$$C \xrightarrow{(a,b)} A_1 \times A_2 \xrightarrow{<c,d>} B$$

is $< c, d > (a, b) = ca + db$. In other words, composition is given by matrix multiplication, with (a, b) consider as a column vector and $< c, d >$ consider as a row vector.

The proof of Proposition 1.4.1 gives the following:

Corollary 1.4.1

If there exists an object $Z \in \mathscr{C}$ together with morphisms $i_1 : X \to Z$, $i_2 : Y \to Z$, $p_1 : Z \to X$, $p_2 : Z \to Y$ satisfying

$$\begin{aligned} p_1 i_1 &= \mathrm{id}_X, & p_1 i_2 &= 0 \\ p_2 i_1 &= 0, & p_2 i_2 &= \mathrm{id}_Y \\ i_1 p_1 + i_2 p_2 &= \mathrm{id}_Z, \end{aligned} \tag{1.3}$$

Then the objects $X \prod Y$, $X \coprod Y$, and Z are naturally isomorphic.

Proposition 1.4.2

Let $\mathscr{A}$ be a category admitting a zero object, finite products, and finite coproducts, such that the natural map from the coproduct to the product

$$\varphi : Y \coprod Y \longrightarrow Y \prod Y \tag{1.4}$$

is an isomorphism for every $Y \in \mathrm{obj}(\mathscr{A})$. Then there exists a unique way to equip every $\mathrm{Mor}_{\mathscr{A}}(X, Y)$ with a unital magma such that composition is bilinear. Moreover, $\mathrm{Mor}_{\mathscr{A}}(X, Y)$ is a commutative monoid.

Proof We first explain how to equip $\mathrm{Mor}(X, Y)$ a structure of magma operation using (1.4). Let $f, f' : X \to Y$, we define $f + f'$ to be the composite

$$X \xrightarrow{(f,f')} Y \prod Y \xrightarrow{\varphi^{-1}} Y \coprod Y \xrightarrow{(\mathrm{id}_Y, \mathrm{id}_Y)} Y,$$

where the first map is induced by the universal property of product $Y \prod Y$ from $f, f' : X \to Y$, and similarly for the third map.

Note that the diagram

$$\begin{tikzcd}
X \arrow[r, "{(f,0)}"] \arrow[dr, "f"'] & Y\prod Y \arrow[r, "\varphi^{-1}_{Y,Y}"] & Y\coprod Y \arrow[dr, "{(\mathrm{id}_Y,\mathrm{id}_Y)}"] & \\
& Y \arrow[u, "{(\mathrm{id}_Y,0)}"] \arrow[r, "{(\mathrm{id}_Y,0)^{-1}}"'] & Y\coprod 0 \arrow[u, "\mathrm{id}_Y\coprod 0"] \arrow[r, "{(\mathrm{id}_Y,0)}"'] & Y
\end{tikzcd}$$

commutes, so that $f+0=f$. Similarly $0+f=f$. This construction equips $\mathrm{Mor}_{\mathscr{A}}(X,Y)$ with the structure of a unital magma. It is clear from the construction that $(g+g')f=gf+g'f$. The diagram

$$\begin{tikzcd}
X \arrow[r, "{(f,f')}"] \arrow[dr, "{(gf,gf')}"'] & Y\prod Y \arrow[r, "\varphi^{-1}_{Y,Y}"] \arrow[d, "g\times g"] & Y\coprod Y \arrow[r, "{(\mathrm{id}_Y,\mathrm{id}_Y)}"] \arrow[d, "g\coprod g"] & Y \arrow[d, "g"] \\
& Z\prod Z \arrow[r, "\varphi^{-1}_{ZZ}"'] & Z\coprod Z \arrow[r, "{(\mathrm{id}_Z,\mathrm{id}_Z)}"'] & Z
\end{tikzcd}$$

commutes, so that $g\circ(f+f')=g\circ f+g\circ f'$. Moreover, $0f=0$ and $g0=0$. Thus composition law on $\mathscr{A}$ is bilinear. Now, in category $\mathscr{A}$, each $\mathrm{Mor}_{\mathscr{A}}(X,Y)$ equipped with a structure of unital magma such that composition is bilinear, $Y\prod Y\cong Y\coprod Y$, and $\mathscr{A}$ admitting finite products and finite coproducts.

The uniqueness and the fact that $\mathrm{Mor}_{\mathscr{A}}(X,Y)$ is a commutative monoid follows from Proposition 1.4.1.

□

Note *In a category satisfying the above assumptions, coproducts are also called* ***direct sums*** *and we often write* $\oplus$ *in stead of* $\coprod$.

Definition 1.4.1 (Pre-additive category)

A ***pre-additive category*** $\mathscr{C}$ *is a category such that every* Mor-set $\mathrm{Mor}(X,Y)$ *in* $\mathscr{C}$ *has the structure of an abelian group, and composition of morphisms is bilinear, in the sense that composition of morphisms distributes over the group operation. In formulas:*

$$f\circ(g+h)=f\circ g+f\circ h,\qquad \textit{resp. } (g+h)\circ f=g\circ f+g\circ f,$$

where $+$ *is the group operation and* $f:X\to Y$, $g,h:W\to X$ *(resp.,* $g,h:Y\to Z$*).*

Remark If $\mathscr{C}$ is a pre-additive category, then for any $X\in\mathscr{C}$, $\mathrm{End}_{\mathscr{C}}(X):=\mathrm{Mor}_{\mathscr{C}}(X,X)$ becomes a ring. In fact, pre-additive categories can be seen as a generalisation of rings. Given a ring R, we form the pre-additive category with one single object $\{*\}$ and $\mathrm{Mor}(*,*)=R$ which is an abelian group and the composition of morphisms is bilinear.

Lemma 1.4.2

Let $\mathscr{C}$ *be a pre-additive category and* $X\in\mathrm{obj}(\mathscr{C})$. *The following are equivalent:*

(1) X *is an initial object;*

(2) X *is a final object;*

(3) $\mathrm{id}_X=0$ *in* $\mathrm{Mor}_{\mathscr{C}}(X,X)$.

Proof We prove (1) $\Leftrightarrow$ (3) here, the case for (2) $\Rightarrow$ (3) is similar.

(1) $\Rightarrow$ (3): If X is initial, then $\mathrm{Mor}_{\mathscr{C}}(X,Y)$ is a singleton for any Y; in particular $\mathrm{Mor}_{\mathscr{C}}(X,X)$ is a singleton and must be equal to $\{\mathrm{id}_X\}$. On the other hand, since $\mathscr{C}$ is pre-additive, $\mathrm{Mor}_{\mathscr{C}}(X,X)$ is an abelian group, thus $\mathrm{id}_X=0$.

(3) $\Rightarrow$ (1) We need to show that $\mathrm{Mor}_{\mathscr{C}}(X,Y)$ is singleton for any Y. Since $\mathrm{Mor}_{\mathscr{C}}(X,Y)$ is an abelian group,

it is equivalent to show that $f = 0$ for any $f \in \mathrm{Mor}_{\mathscr{C}}(X, Y)$. Consider the composition $X \xrightarrow{\mathrm{id}_X} X \xrightarrow{f} Y$. Since $\mathrm{id}_X = 0$ in $\mathrm{Mor}_{\mathscr{C}}(X, X)$, we have $\mathrm{id}_X = \mathrm{id}_X + \mathrm{id}_X$. So

$$f = f \circ \mathrm{id}_X = f \circ (\mathrm{id}_X + \mathrm{id}_X) = f \circ \mathrm{id}_X + f \circ \mathrm{id}_X = f + f,$$

which implies that $f = 0$. $\square$

Definition 1.4.2 (Additive category)

*An **additive category** is a pre-additive category $\mathscr{A}$ admitting all finite products.*

Note *In an additive category, the morphism are often called **homomorphisms**, and* Mor *is denoted by* Hom*. In fact, this notation* Hom *is a good indication that you're working in an additive category.*

Proposition 1.4.3

A category $\mathscr{A}$ is additive if and only if

(1) it admits a zero object;

(2) it admits finite products;

(3) it admits finite coproduct;

(4) the natural map $\varphi : Y \coprod Y \to Y \prod Y$ is an isomorphism for every $Y \in \mathrm{obj}(\mathscr{A})$;

(5) the monoid $\mathrm{Mor}_{\mathscr{A}}(X, Y)$ is an abelian group.

Proof $\Leftarrow$ is clear by Proposition 1.4.2.

$\Rightarrow$ First, take the product over empty set, we obtain a final object in $\mathscr{A}$ and Lemma 1.4.2 shows that this is furthermore a zero object. This proves (1). By the definition of pre-additive category and additive category, apply Proposition 1.4.1, $\mathscr{A}$ admits finite products and finite coproduct. (2), (3), (4) are given by Proposition 1.4.1. (5) is given by definition of pre-additive category. $\square$

Example 1.29

(1) The category $\mathbf{Mod}_R$ is additive. The category of finitely generated R-modules is full subcategory of $\mathbf{Mod}_R$, it is also an additive category.

(2) The category $\mathbf{Grp}$ is not preadditive, because $\mathrm{Mor}(G_1, G_2)$ is in general not an abelian group. The category $\mathbf{CRings}$ is not preadditive neither.

(3) Let $\mathscr{A}$ be an additive category and $\mathscr{I}$ be a category. Then the functor category $\mathscr{A}^{\mathscr{I}}$ is also an additive category.

Lemma 1.4.3

(1) In a category with a zero object, every monomorphism $f : X \to Y$ has zero kernel, and every epimorphism has zero cokernel.

(2) In an additive category $\mathscr{A}$, every morphism of zero kernel $f : X \to Y$ is a monomorphism and every morphism of zero cokernel is an epimorphism.

Proof

(1) Show that $0 \to X$ satisfies the universal property of $\mathrm{Ker}(f : X \to Y)$. Consider the following diagram.

$$\begin{tikzcd}
Z \arrow[d, dashed, "\exists!?"'] \arrow[dr, "g"] \arrow[drr, bend left, "0"] & & \\
0 \arrow[r, "0"] & X \arrow[r, "f"] & Y
\end{tikzcd}$$

where $f \circ g = 0$. Note that $f \circ 0 = f \circ 0$ and f is monic, we have $g = 0_{Z \to X}$. Since 0 is final object,

$Z \to 0$ must be 0, so $\mathrm{Ker}(f : X \to Y) = (0 : 0 \to X)$.

The case of cokernel is similar.

(2) We only check the statement in the first case. We need to show if

$$W \overset{g_1}{\underset{g_2}{\rightrightarrows}} X \xrightarrow{f} Y$$

is such that $fg_1 = fg_2$, then $g_1 = g_2$. Since $\mathscr{A}$ is additive, we have $f(g_1 - g_2) = 0$. Consider the following diagram.

$$\begin{tikzcd} W \arrow[d, dashed, "\exists!"'] \arrow[dr, "g_1-g_2"] \arrow[drr, bend left, "0"] & & \\ 0 = \mathrm{Ker}\, f \arrow[r, "0"'] & X \arrow[r, "f"'] & Y \end{tikzcd}$$

By the universal property of kernel, there exists a unique morphism $\theta : W \to 0 = \mathrm{Ker}\, f$ such that $g_1 - g_2 = 0 \circ \theta = 0$, hence f is monic.

□

1.4.2 Additive functors

Proposition 1.4.4

Let $F : \mathscr{A} \to \mathscr{B}$ be a functor between additive categories. Then the following conditions are equivalent:

(1) F preserves products of pairs of objects;

(2) F preserve coproducts of pair of objects;

(3) for every pair A, A' of $\mathscr{A}$, the map

$$\mathrm{Mor}_{\mathscr{A}}(A, A') \longrightarrow \mathrm{Mor}_{\mathscr{B}}(FA, FA')$$

induced by F is a group homomorphism.

Proof We only prove (1) $\Leftrightarrow$ (3), (1) $\Leftrightarrow$ (3) is similar.

(1) $\Rightarrow$ (3). Let $f, g : A \to A'$. Then $f + g$ is given by the composition

$$A \xrightarrow{(f,g)} A' \prod A' \xrightarrow{\varphi^{-1}} A' \coprod A' \xrightarrow{(\mathrm{id},\mathrm{id})} A'.$$

One checks that applying F to it, we obtain

$$FA \xrightarrow{(Ff,Fg)} FA' \prod FA' \xrightarrow{\varphi^{-1}} FA' \coprod FA' \xrightarrow{(\mathrm{id},\mathrm{id})} FA',$$

so that $F(f + g) = Ff + Fg$.

(3) $\Rightarrow$ (1). We must show that $(Fp_1, Fp_2) : F(A_1 \prod A_2) \to FA_1 \prod FA_2$ is an isomorphism, where $p_j : A_1 \prod A_2 \to A_j$ is the natural projection. We have morphisms

$$Fp_j : F(A_1 \prod A_2) \longrightarrow FA_j, \qquad Fi_j : FA_j \longrightarrow F(A_1 \prod A_2),$$

and by assumption they also satisfy the resolutions listed in (1.3), by Corollary 1.4.1, $F(A_1 \prod A_2) \cong FA_1 \prod FA_2$. □

Definition 1.4.3 (Additive functor)

*A functor is called **additive** if it satisfies the conditions in Proposition 1.4.4*

Example 1.30 Let $\mathscr{A}$ be an additive category. The functor

$$F : \mathscr{A} \longrightarrow \mathscr{A}$$
$$A \longmapsto A \oplus A$$

is additive. However, the functor $G : \mathscr{A} \to \mathscr{A}$, sending A to $A \oplus B$ is not additive unless $B = 0$.

Proof Let $f : A \to A'$, then $F(f) : A \oplus A \to A' \oplus A'$. In fact, $F(f)$ can be expressed by a matrix:

$$F(f) = \begin{bmatrix} f & 0 \\ 0 & f \end{bmatrix}.$$

Let $g : A \to A'$, we want to show that $F(f + g) = F(f) + F(g)$. Note that

$$F(f+g) = \begin{bmatrix} f+g & 0 \\ 0 & f+g \end{bmatrix}$$

and

$$F(f) + F(g) = \begin{bmatrix} f & 0 \\ 0 & f \end{bmatrix} + \begin{bmatrix} g & 0 \\ 0 & g \end{bmatrix} = \begin{bmatrix} f+g & 0 \\ 0 & f+g \end{bmatrix}.$$

Hence F is an additive functor.

Next, we want to show that G is not an additive functor unless $B = 0$. Let $f, g : A \to A'$, then

$$G(f+g) = \begin{bmatrix} f+g & 0 \\ 0 & \mathrm{id}_B \end{bmatrix}.$$

and

$$G(f) + G(g) = \begin{bmatrix} f & 0 \\ 0 & \mathrm{id}_B \end{bmatrix} + \begin{bmatrix} g & 0 \\ 0 & \mathrm{id}_B \end{bmatrix} = \begin{bmatrix} f+g & 0 \\ 0 & \mathrm{id}_B + \mathrm{id}_B \end{bmatrix}.$$

Hence $G(f+g) = G(f) + G(g)$ iff $\mathrm{id}_B = \mathrm{id}_B + \mathrm{id}_B$ iff $\mathrm{id}_B = 0_B$ iff $B = 0$, i.e., G is not an additive functor unless $B = 0$. □

Example 1.31 Be attention that the functor

$$\mathbf{Mod}_R \longrightarrow \mathbf{Mod}_R$$
$$M \longmapsto M \otimes_R M$$

is not additive unless $R = 0$, because it does not preserve products.

1.4.3 Abelian categories

Definition 1.4.4 (Coimage, image)

Let $\mathscr{A}$ be an additive category admitting kernels and cokernels. Let $f : A \to B$ be a morphism. We define the coimage and image of f to be

$$\mathrm{Coim}(f) := \mathrm{Coker}(\mathrm{Ker}\, f \to A), \qquad \mathrm{Im}(f) := \mathrm{Ker}(B \to \mathrm{Coker}\, f).$$

Remark Recall that Coker is coequalizer, hence $A \to \mathrm{Coim}(f)$ is an epimorphism. Dually, recall that Ker is equalizer, hence $\mathrm{Im}(f) \to B$ is monomorphism.

Lemma 1.4.4

In an additive category $\mathscr{A}$, every morphism $f : A \to B$ factors uniquely as

$$A \twoheadrightarrow \mathrm{Coim}(f) \longrightarrow \mathrm{Im}(f) \hookrightarrow B. \tag{1.5}$$

Proof First, since the composition

$$\mathrm{Ker}(f) \longrightarrow A \longrightarrow B$$

is zero, the definition of $\mathrm{Coim}(f)$ implies the following commutative diagram,

$$\begin{tikzcd}
\mathrm{Ker}\, f \arrow[r] \arrow[drr, bend right, "0"'] & A \arrow[r, two heads] \arrow[dr, "f"'] & \mathrm{Coim}(f) \arrow[d, dashed] \\
& & B
\end{tikzcd}$$

i.e., f factors uniquely as $A \twoheadrightarrow \mathrm{Coim}(f) \to B$. Next, since the composition

$$A \twoheadrightarrow \mathrm{Coim}(f) \longrightarrow B \longrightarrow \mathrm{Coker}(f)$$

is zero and since $A \twoheadrightarrow \mathrm{Coim}(f)$ is epic, the composition $\mathrm{Coim}(f) \to B \to \mathrm{Coker}(f)$ is also equal to 0. Thus we have

$$\begin{tikzcd}
\mathrm{Coim}(f) \arrow[d, dashed] \arrow[dr] \arrow[drr, bend left, "0"] & & \\
\mathrm{Ker}(B \to \mathrm{Coker}(f)) \arrow[r] & B \arrow[r] & \mathrm{Coker}(f)
\end{tikzcd}$$

i.e., the morphism $\mathrm{Coim}(f) \to B$ factors through $\mathrm{Ker}(B \to \mathrm{Coker}(f))$, namely $\mathrm{Im}(f)$. $\square$

Definition 1.4.5 (Abelian category)

An ***abelian category*** *is an additive category $\mathscr{A}$ satisfying the following axioms:*

(AB1) $\mathscr{A}$ admits kernels and cokernels.

(AB2) For each morphism $f : A \to B$, the morphism $\mathrm{Coim}(f) \to \mathrm{Im}(f)$ is an isomorphism.

Remark Since an additive category $\mathscr{A}$ admits all finite products and finite coproducts, hence combined with (AB1) which implies that $\mathscr{A}$ admits equalizers and coequalizers. More precisely, $\mathrm{Eq}(f, g) = \mathrm{Ker}(f - g)$ and $\mathrm{Coeq}(f, g) = \mathrm{Coker}(f - g)$ (this is easy to check).

The following lemma tells us additive category $\mathscr{A}$ with (AB1) admits all finite limits and finite colimits.

Lemma 1.4.5

The following are equivalent for the category $\mathscr{C}$:

(1) Equalizers and finite products exist in $\mathscr{C}$.

(2) Pullbacks and a final object exist in $\mathscr{C}$.

Moreover, if the equivalent conditions hold, then $\mathscr{C}$ admits all finite limits.

Proof (1) $\Rightarrow$ (2). Assume that equalizers and finite products exist in $\mathscr{C}$. A product of the empty family of objects is a final object of $\mathscr{C}$, so it remains to show that pullbacks exist.

Let $f_1 : X_1 \to Z$ and $f_2 : X_2 \to Z$ be morphisms in $\mathscr{C}$; then $X_1 \prod X_2$ exists in $\mathscr{C}$ with natural projections $p_i : X_1 \prod X_2 \to X_i$. Consider the following diagram.

$$\begin{tikzcd}
& & X_2 \arrow[dr, "f_2"] & \\
\mathrm{Eq}(f_1p_1, f_2p_2) \arrow[r, "e"] & X_1 \prod X_2 \arrow[ur, "p_2"] \arrow[dr, "p_1"'] & & Z \\
& & X_1 \arrow[ur, "f_1"'] &
\end{tikzcd}$$

Say $E := \mathrm{Eq}(f_1p_1, f_2p_2)$. We claim that E is a pullback of (f_1, f_2).

Given an object Q of $\mathscr{C}$ with compatible morphisms $q_i : Q \to X_i$ such that $f_1 q_1 = f_2 q_2$. Then we have the following commutative diagram,

$$\begin{tikzcd}
Q \arrow[dr, dashed, "\exists!"] \arrow[ddrr, dashed, "\exists!"] \arrow[rrrd, bend left, "q_2"] \arrow[dddr, bend right, "q_1"'] & & & \\
& E \arrow[rr, "p_2 \circ e"] \arrow[dd, "p_1 \circ e"] \arrow[dr] & & X_2 \arrow[dd, "f_2"] \\
& & X_1 \coprod X_2 \arrow[ur] \arrow[dl] & \\
& X_1 \arrow[rr, "f_1"'] & & Z
\end{tikzcd}$$

where dashed arrow $Q \to X_1 \coprod X_2$ is given by the universal property of product and dashed arrow $Q \to E$ is given by the universal property of equalizer. Hence E is pullback of (f_1, f_2).

(2) $\Rightarrow$ (1). Let Z be the final object in $\mathscr{C}$. Let $X, Y \in \operatorname{obj}(\mathscr{C})$, then $X \times_Z Y$ exists, note that $X \times_Z Y$ can be seen as product, so $\mathscr{C}$ admits finite product.

Consider the following commutative diagram,

$$\begin{tikzcd}
Q \arrow[dr, dashed] \arrow[drr, bend left, "q"] \arrow[ddr, bend right, "q"'] & & \\
& X \times_Y X \arrow[r, "p"'] \arrow[d, "p"] & X \arrow[d, "g"] \\
& X \arrow[r, "f"'] & Y
\end{tikzcd}$$

where Q is any object with $g \circ q = f \circ q$ and dashed arrow is given by universal property of pullback. Then $X \times_Y X \to X$ can be seen as equalizer $\operatorname{Eq}(f, g) \to X$. Hence $\mathscr{C}$ admits equalizer.

To prover the final statement, let $F : \mathscr{I} \to \mathscr{C}$ with $\mathscr{I}$ a finite category. We consider the (finite) products

$$A = \prod_{i \in \operatorname{obj}(\mathscr{I})} F(i), \quad B = \prod_{f : i \to j} F(j)$$

and morphisms $a, b : A \to B$, where

- a is the product (over all morphisms f) of the natural projections $a_f : A \to F(j)$;
- b is the product of $b_f : A \longrightarrow F(i) \xrightarrow{F(f)} F(j)$.

Say $\pi_f : A \to F(i)$ be natural projection, then $b_f = F(f) \circ \pi_f$.

$$\begin{tikzcd}
& F(i) \arrow[r, "F(f)"] & F(j) \\
\operatorname{Eq}(a, b) \arrow[r, "e"'] & A \arrow[u, "\pi_f"] \arrow[ur, "a_f"] \arrow[r, dashed, shift left, "a"] \arrow[r, dashed, shift right, "b"'] & B \arrow[u, "p_f"']
\end{tikzcd}$$

Since $a \circ e = b \circ e$, we have $a_f \circ e = b_f \circ e$, this gives a natural transformation $\Delta \operatorname{Eq}(a, b) \to F$. Let (X, h) be any object in $(\Delta \downarrow F)$. By the universal property of product, there exists unique morphism $\theta : X \to A$ such that $h_i = \pi_f \circ \theta$ and $h_j = a_f \circ \theta$ for all $f : i \to j$. By the natural transformation $h : \Delta X \to F$, we have $h_j = F(f) \circ h_i$ for all $f : i \to j$. Hence

$$h_j = a_f \circ \theta = F(f) \circ \pi_f \circ \theta = b_f \circ \theta$$

for all $f : i \to j$, hence $a \circ \theta = b \circ \theta$. By the universal property of equalizer, there exists unique morphism $X \to \operatorname{Eq}(a, b)$. Hence $(\operatorname{Eq}(a, b), \Delta \operatorname{Eq}(a, b) \to F)$ is final object of $(\Delta \downarrow F)$, which implies $\mathscr{C}$ admits all finite limits. □

Corollary 1.4.2

Additive category $\mathscr{A}$ with (AB1) admits all finite limits and finite colimits.

Proof $\mathscr{A}$ admits finite colimits is given by the dual of Lemma 1.4.5, admits finite limits is Lemma 1.4.5. □

Example 1.32 Let $\mathscr{A}$ be an abelian category, then $\mathscr{A}^{\mathrm{op}}$ is an abelian category.

Example 1.33 Let $\mathscr{A}$ be an abelian category and $\mathscr{I}$ be any category, then the functor category $\mathscr{A}^{\mathscr{I}}$ is an abelian category.

Example 1.34 ($\mathbf{Mod}_R$ is an abelian category) $\mathbf{Mod}_R$ is an abelian category. The subcategory $\mathbf{f.g.Mod}_R$, the category of finite generated R-modules is not an abelian category in general. In fact, if $R \in \mathbf{f.g.Mod}_R$ is not noetherian, there exists an ideal $I \subseteq R$ which is not finitely generated, then the canonical map $R \to R/I$ has no kernel in $\mathbf{f.g.Mod}_R$.

The full subcategory of $\mathbf{Mod}_R$ consisting of all noetherian (or artinian) modules is an abelian category.

Example 1.35 ((Pre)sheaves) Let X be a topological space, which induces a category $\mathbf{Open}(X)$, whose objects consist of open subsets of X and morphisms are inclusions. (More generally we can start with a site, i.e., a category with a Grothendieck topology.) Now define **the abelian category of presheaves of abelian groups** on X as:

$$\mathbf{PreShv}(X) = \mathbf{Fun}(\mathbf{Open}(X)^{\mathrm{op}}, \mathbf{Ab}).$$

The category of sheaves of abelian groups $\mathbf{Shv}(X)$ is the full subcategory of $\mathbf{PreShv}$ consists of functors $F : \mathbf{Open}(X)^{\mathrm{op}} \to \mathbf{Ab}$ satisfying the following condition: for any open subset U and any open cover $\{U_i \hookrightarrow U\}_{i\in I}$ the restriction map $F(U) \to F(U_i)$ induces a bijection:

$$F(U) \xleftrightarrow{\sim} \mathrm{Eq}\left(\prod_{i\in I} F(U_i) \rightrightarrows \prod_{i,j\in I} F(U_i \cap U_j)\right),$$

where two arrows are induced by $F(U_i) \to F(U_i \cap U_j)$ and $F(U_j) \to F(U_i \cap U_j)$. Identity axiom ensures that $\mathscr{F}(U) \to \prod \mathscr{F}(U_i)$ is injective. Gluability axiom implies that the equalizer of the two maps $\prod \mathscr{F}(U_i) \rightrightarrows \prod \mathscr{F}(U_i \cap U_j)$ (two maps are $(s_i) \mapsto (s_i|_{U_i \cap U_j})$ and $(s_i) \mapsto (s_j|_{U_i \cap U_j})$) must coincide exactly with the image of $\mathscr{F}(U)$ (recall that $\mathrm{Eq}(f, g) = \mathrm{Ker}(f - g)$ in abelian categories).

The category $\mathbf{Shv}(X)$ is an abelian category, and the forgetful functor: $\mathbf{Shv}(X) \to \mathbf{PreShv}(X)$ admits a left adjoint, the sheafification functor, i.e., sheafification is left adjoint to forgetful functor.

Remark ⋆⋆ **Define the Shv on a site. (Unimportant, just casual conversation.)**

A **site** is given by a category $\mathscr{C}$ and a set $\mathrm{Cov}(\mathscr{C})$ of families of morphisms with fixed target $\{U_i \to U\}_{i\in I}$, called **coverings of** $\mathscr{C}$, satisfying the following axioms:

(1) If $V \to U$ is an isomorphism then $\{V \to U\} \in \mathrm{Cov}(\mathscr{C})$.

(2) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$ and for each i we have $\{V_{ij} \to U_i\}_{j\in J_i} \in \mathrm{Cov}(\mathscr{C})$, then $\{V_{ij} \to U\}_{i\in I, j\in J_i} \in \mathrm{Cov}(\mathscr{C})$.

(3) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$ and $V \to U$ is a morphism of $\mathscr{C}$ then $U_i \times_U V$ exists for all i and $\{U_i \times_U V \to V\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$.

Define **the category of presheaves of sets** on X as:

$$\mathbf{PreShv}(X) = \mathbf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathbf{Sets}).$$

Let $\mathscr{C}$ be a site, and let $\mathscr{F}$ be a presheaf of sets on $\mathscr{C}$. We say $\mathscr{F}$ is a **sheaf** if for every coving

$\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$ the diagram

$$\mathscr{F}(U) \longrightarrow \prod_{i\in I} \mathscr{F}(U_i) \overset{\mathrm{pr}_0^*}{\underset{\mathrm{pr}_1^*}{\rightrightarrows}} \prod_{(i_0,i_1)\in I\times I} \mathscr{F}(U_{i_0} \times_U U_{i_1}) \tag{1.6}$$

represents the first arrow as the equalizer of pr_0^* and pr_1^*.

A **topos** is the category $\mathbf{Shv}(\mathscr{C})$ of sheaves on a site $\mathscr{C}$.

(1) Let $\mathscr{C}$, $\mathscr{D}$ be sites. A **morphism of topoi** f from $\mathbf{Shv}(\mathscr{D})$ to $\mathbf{Shv}(\mathscr{C})$ is given by a pair of functors $f_* : \mathbf{Shv}(\mathscr{D}) \to \mathbf{Shv}(\mathscr{C})$ and $f^{-1} : \mathbf{Shv}(\mathscr{C}) \to \mathbf{Shv}(\mathscr{D})$ such that

(a) we have

$$\mathrm{Mor}_{\mathbf{Shv}(\mathscr{D})}(f^{-1}\mathscr{G}, \mathscr{F}) = \mathrm{Mor}_{\mathbf{Shv}(\mathscr{C})}(\mathscr{G}, f_*\mathscr{F})$$

bifunctorially, and

(b) the functor f^{-1} commutes with finite limits, i.e., is left exact.

(2) Let $\mathscr{C}, \mathscr{D}, \mathscr{E}$ be sites. Given morphisms of topoi $f : \mathbf{Shv}(\mathscr{D}) \to \mathbf{Shv}(\mathscr{C})$ and $g : \mathbf{Shv}(\mathscr{E}) \to \mathbf{Shv}(\mathscr{D})$ the composition $f \circ g$ is the morphism of topoi defined by the functors $(f \circ g)_* = f_* \circ g_*$ and $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

Proposition 1.4.5

Some properties in abelian category $\mathscr{A}$:

(1) If a morphism is both monomorphism and an epimorphism, then it is an isomorphism.

(2) Every monomorphism is the kernel of its cokernel.

(3) Every epimorphism is the cokernel of its kernel.

(4) Every morphism $f : A \to B$ can be decomposed into

$$A \xrightarrow{g} \mathrm{Im}(f) \xrightarrow{h} B$$

where g is an epimorphism and h is a monomorphism.

Proof

(1) By (1.5), it suffices to show that $A \twoheadrightarrow \mathrm{Coim}(f)$ and $\mathrm{Im}(f) \hookrightarrow B$ are isomorphisms. We prove only the first. By definition $\mathrm{Coim}(f) \cong \mathrm{Coker}(\mathrm{Ker}(f) \to A)$. Since f is monomorphism, we know $\mathrm{Ker}(f) = 0$ and $\mathrm{Coker}(\mathrm{Ker}(f) \to A) = \mathrm{Coker}(0 : 0 \to A)$. Then it is direct to check that $\mathrm{id}_A : A \to A$ is the cokernel of the zero morphism.

(2) Consider a monomorphism $0 \longrightarrow X \xrightarrow{f} Y$. By definition, $\mathrm{Im}(f)$ is the kernel of $Y \to \mathrm{Coker}(f)$. But the morphism $X \to \mathrm{Im}(f)$ is monic and epic, hence is an isomorphism by (1).

(3) Same as (2).

(4) By Lemma 1.4.4, and recall (AB2) in Definition 1.4.5.

□

Example 1.36 (Filtered modules over k) Another important non-abelian category is the category $\mathbf{FilMod}_k$ of filtered modules over k. Precisely, an object is a k-vector space M with a decreasing filtration $\{F_nM\}_{n\in\mathbb{Z}}$. A morphism between two objects is a morphism $f : M \to M'$ such that $f(F_nM) \subseteq F_nM'$. Take $M = k[X]$ with the filtration given by

$$F_nM = \begin{cases} k[X] & n \leq 0 \\ (X^n) & n \geq 0. \end{cases}$$

We can define another filtration on $k[X]$ by putting $F_nM' := (X^{2n})$. Clearly the identity morphism $M' \to M$ is not an isomorphism. Hence $\mathbf{FilMod}_k$ is not abelian.

1.5 Exact functors

1.5.1 Exactness

Definition 1.5.1 (Exact functors)

Let $\mathscr{C}$ be a category admitting finite limits (resp., finite colimits). We say that functor $F : \mathscr{C} \to \mathscr{D}$ is ***left exact (resp., right exact)*** *if it preserves finite limits (resp., finite colimits). For $\mathscr{C}$ admitting finite limits and finite colimits, we say that F is* ***exact*** *if it is both left exact and right exact.*

Lemma 1.5.1

For $\mathscr{A}$ an abelian category and a sequence $A \xrightarrow{f} B \xrightarrow{g} C$ in $\mathscr{A}$, suppose $g \circ f = 0$, we have a natural decomposition:

$$A \longrightarrow \operatorname{Coim} f \longrightarrow \operatorname{Ker} g \hookrightarrow B \twoheadrightarrow \operatorname{Coker} f \longrightarrow \operatorname{Im} g \longrightarrow C.$$

Proof By Lemma 1.4.4, $f : A \to B$ factors into

$$A \twoheadrightarrow \operatorname{Coim} f \xrightarrow{\sim} \operatorname{Im} f \hookrightarrow B,$$

since $\mathscr{A}$ is abelian category, we have $\operatorname{Coim} f \cong \operatorname{Im} f$, so $f : A \to B$ factors through $\operatorname{Coim} f$. Note that $g \circ f = 0$ and $A \twoheadrightarrow \operatorname{Coim} f$ is epic, the composition of $\operatorname{Coim} f \to B \to C$ is zero. Consider the following commutative diagram,

$$\begin{tikzcd}
A \arrow[d, two heads] \arrow[drr, "f"] \arrow[drrr, bend left, "0"] & & & \\
\operatorname{Coim} f \arrow[d, dashed] \arrow[dr, hook] \arrow[drr, "0"] & & & \\
\operatorname{Ker} g \arrow[r, hook] & B \arrow[r, "g"'] & C &
\end{tikzcd}$$

where dashed arrow given by the universal property of $\operatorname{Ker} g \hookrightarrow B$. Hence f factors into

$$A \twoheadrightarrow \operatorname{Coim} f \longrightarrow \operatorname{Ker} g \hookrightarrow B.$$

By Lemma 1.4.4, $g : B \to C$ factors into

$$B \twoheadrightarrow \operatorname{Coim} g \xrightarrow{\sim} \operatorname{Im} g \hookrightarrow C,$$

so $g : B \to C$ factors through $\operatorname{Im} g$. Note that $g \circ f = 0$ and $\operatorname{Im} g \to C$ is monic, the composition of $A \to B \to \operatorname{Im} g$ is zero. Consider the following commutative diagram,

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[drr, "0"] \arrow[ddrr, bend right, "0"'] & B \arrow[r, two heads] \arrow[dr, two heads] \arrow[ddr, "g"] & \operatorname{Coker} f \arrow[d, dashed] \\
& & \operatorname{Im} g \arrow[d, hook] \\
& & C
\end{tikzcd}$$

where dashed arrow given by the universal property of $B \twoheadrightarrow \operatorname{Coker} f$. Hence g factors into

$$B \twoheadrightarrow \operatorname{Coker} f \longrightarrow \operatorname{Im} g \hookrightarrow C.$$

□

Lemma 1.5.1 yields the definition of exact sequence:

Definition 1.5.2 (Exact sequence)

*Let $\mathscr{A}$ be an abelian category. We say that a sequence $A \xrightarrow{f} B \xrightarrow{g} C$ with $g \circ f = 0$ in $\mathscr{A}$ is **exact at** B if the induced morphism $\operatorname{Coim} f \to \operatorname{Ker} g$ is an isomorphism (or equivalently, $\operatorname{Coker} f \to \operatorname{Im} g$ is an isomorphism).*

We say that a sequence

$$A^0 \longrightarrow A^1 \longrightarrow \cdots \longrightarrow A^n$$

*is **exact** if it is exact at each A^i, for $1 \leq i \leq n-1$.*

Proposition 1.5.1

Some properties of exact sequence (in abelian category):

(1) Sequence

$$0 \longrightarrow A \longrightarrow 0$$

is exact if and only if $A = 0$.

(2) Sequence

$$0 \longrightarrow A \xrightarrow{f} B$$

is exact if and only if f is a monomorphism.

(3) Sequence

$$A \xrightarrow{f} B \longrightarrow 0$$

is exact if and only if f is a epimorphism.

(4) Sequence

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow 0$$

is exact if and only if f is an isomorphism.

(5) Sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is exact if and only if f is kernel of g, and sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact if and only if g is cokernel of f.

Remark In additive category, Proposition 1.5.1 also holds.

Proof

(1) If sequence

$$0 \longrightarrow A \longrightarrow 0$$

exact, then $\operatorname{Ker}(A \to 0) = \operatorname{Im}(0 \to A)$. We shall to show that $\operatorname{Ker}(A \to 0) = \mathrm{id}_A : A \to A$ and $\operatorname{Im}(0 \to A) = 0 : 0 \to A$.

Note that $\mathrm{id}_A : A \to A$ is the solution of the following universal problem,

$$\begin{tikzcd}
X \arrow[d, dashed, "f"'] \arrow[dr, dashed, "f"] \arrow[drr, "0"] & & \\
A \arrow[r, "\mathrm{id}_A"] \arrow[rr, bend right, "0"'] & A \arrow[r, "0"] & 0,
\end{tikzcd}$$

hence, $\mathrm{Ker}(A \to 0) = \mathrm{id}_A : A \to A$. Since $\mathrm{Im}(0 \to A) = \mathrm{Ker}(\mathrm{Coker}(0 \to A))$, consider the following diagram,

$$\begin{tikzcd}
0 \arrow[r, "0"] \arrow[drr, "0"'] & A \arrow[r, "\mathrm{id}_A"] \arrow[dr, dashed, "g"] & A \arrow[d, dashed, "g"] \\
& & Y
\end{tikzcd}$$

$\mathrm{id}_A : A \to A$ is the solution of above universal problem, hence $\mathrm{Coker}(0 \to A) = \mathrm{id}_A : A \to A$. Now, it suffice to show that $\mathrm{Ker}(\mathrm{id} : A \to A) = 0 : 0 \to A$. Consider the following universal problem,

$$\begin{tikzcd}
Z \arrow[d, "0"'] \arrow[dr, dashed, "0"] \arrow[drr, "0"] & & \\
0 \arrow[r, "0"'] \arrow[rr, bend right] & A \arrow[r, "\mathrm{id}"'] & A,
\end{tikzcd}$$

$0 : 0 \to A$ is the solution, and therefore $\mathrm{Im}(0 \to A) = 0 : 0 \to A$. Since $\mathrm{Ker}(A \to 0) = \mathrm{Im}(0 \to A)$, we have $A = 0$.

Conversely, if $A = 0$, obviously, $\mathrm{Ker}(A \to 0) = \mathrm{Im}(0 \to A)$.

(2) If sequence

$$0 \longrightarrow A \xrightarrow{f} B$$

is exact, then $\mathrm{Im}(0 \to A) = \mathrm{Ker}\, f$. By part (1), $\mathrm{Im}(0 \to A) = 0 : 0 \to A$, hence, $\mathrm{Ker}(f) = 0 : 0 \to A$. Let two morphisms $\mu_1 : X \to A$ and $\mu_2 : X \to A$ with $f \circ \mu_1 = f \circ \mu_2$. Since in Abelian category, $f \circ (\mu_1 - \mu_2) = 0$ and $\mu_1 - \mu_2 : X \to A$, consider the following diagram,

$$\begin{tikzcd}
X \arrow[d, dashed, "0"'] \arrow[dr, dashed, "\mu_1 - \mu_2"] \arrow[drr, "0"] & & \\
0 \arrow[r, "0"'] \arrow[rr, bend right] & A \arrow[r, "f"'] & B,
\end{tikzcd}$$

by the universal property of $\mathrm{Ker}\, f$, $\mu_1 - \mu_2 = 0$, that is, f is monomorphism.

Conversely, if f is monomorphism, we shall to show that $\mathrm{Ker}(f) = 0 : 0 \to A$. It suffice to show that $0 : 0 \to A$ is the solution of the following universal problem.

$$\begin{tikzcd}
X \arrow[d, dashed] \arrow[dr, dashed, "g"] \arrow[drr, "0"] & & \\
0 \arrow[r, "0"'] \arrow[rr, bend right] & A \arrow[r, "f"'] & B
\end{tikzcd}$$

Since, f is monomorphism, g is the unique morphism such that $f \circ g = 0$, note that $0 = f \circ 0$, $g = 0$. Hence, $0 : 0 \to A$ is the solution of the universal problem, and therefore $\mathrm{Ker}(f) = 0 : 0 \to A = \mathrm{Im}(0 \to A)$, which implies the sequence

$$0 \longrightarrow A \xrightarrow{f} B$$

is exact.

(3) If sequence

$$A \xrightarrow{f} B \longrightarrow 0$$

is exact, then $\operatorname{Im} f = \operatorname{Ker}(B \to 0) = \mathrm{id}_B : B \to B$. Let two morphisms $\eta_1 : B \to Y$ and $\eta_2 : B \to Y$ with $\eta_1 \circ f = \eta_2 \circ f$, since in Abelian category, we have $(\eta_1 - \eta_2) \circ f = 0$. Note that $\operatorname{Im} f = \operatorname{Ker}(\operatorname{Coker}(f)) = \mathrm{id}_B : B \to B$, $\operatorname{Coker}(f) \circ \mathrm{id}_B = \operatorname{Coker}(f) = 0 : B \to 0$. Now consider the following diagram,

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[rr, bend left, "0"] \arrow[drr, "0"'] & B \arrow[r, "0"] \arrow[dr] & 0 \arrow[d, dashed] \\
 & & Y
\end{tikzcd}$$

$\eta_1 - \eta_2$ agree with above diagram, hence, $\eta_1 = \eta_2$, which implies that f is epimorphism.

Conversely, if f is epimorphism, we shall to show that $\operatorname{Im}(f) = \operatorname{Ker}(\operatorname{Coker}(f)) = \operatorname{Ker}(B \to 0) = \mathrm{id}_B : B \to B$. It suffice to show that $\mathrm{id}_B : B \to B$ is the solution of the following universal problem.

$$\begin{tikzcd}
Z \arrow[d, dashed] \arrow[dr] \arrow[drr, "0"] & & \\
B \arrow[r] \arrow[rr, bend right, "0"'] & B \arrow[r] & \operatorname{Coker}(f)
\end{tikzcd}$$

We need to show that $\operatorname{Coker}(f) \circ \mathrm{id}_B = \operatorname{Coker}(f) = 0 : B \to 0$. Consider the following universal problem,

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[rr, bend left, "0"] \arrow[drr, "0"'] & B \arrow[r] \arrow[dr] & 0 \arrow[d, dashed] \\
 & & W,
\end{tikzcd}$$

since f is epimorphism, $B \to 0$ and $B \to W$ must be same, hence, $W = 0$, and the unique morphism is $0 : 0 \to 0$. Hence, $\operatorname{Coker}(f) = 0 : B \to 0$. It is clearly $\mathrm{id}_B : B \to B$ is the solution of the universal problem

$$\begin{tikzcd}
Z \arrow[d, dashed, "g"'] \arrow[dr, "g"] \arrow[drr, "0"] & & \\
B \arrow[r] \arrow[rr, bend right, "0"'] & B \arrow[r] & \operatorname{Coker}(f),
\end{tikzcd}$$

hence, $\operatorname{Im}(f) = \mathrm{id}_B : B \to B = \operatorname{Ker}(B \to 0)$.

(4) By part (2) and (3), immediately.

(5) If sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is exact, then $\operatorname{Ker} g = \operatorname{Im} f$. We shall to show that $g \circ f = 0$. By Lemma 1.4.4, we have $f = me$, where epimorphism $e : A \to \operatorname{Im} f$ and monomorphism $m : \operatorname{Im} f \to B$. Since $\operatorname{Ker} g = \operatorname{Im} f$, we have $e : A \to \operatorname{Ker} g$ and $m : \operatorname{Ker} g \to B$. By the definition of kernal, we have $g \circ m = 0$. Hence,

$g \circ f = g \circ m \circ e = 0$.
By Lemma 1.4.4, $f = me$, where $m : \operatorname{Im} f \to B$ monomorphism and $e : A \to \operatorname{Im} f$. We claim that e is an isomorphism. It suffice to show that e is monomorphism. Let $\eta_1 : X \to A$ and $\eta_2 : X \to A$ morphisms with $e \circ \eta_1 = e \circ \eta_2$, then $m \circ e \circ \eta_1 = m \circ e \circ \eta_2$, since $f = me$ is monomorphism, $\eta_1 = \eta_2$. Hence, e is isomorphism. Consider the following universal problem.

$$\begin{tikzcd}
& X \arrow[dl, dashed] \arrow[d, dashed, "t"] \arrow[dr, "h"] \arrow[drr, "0"] & & \\
A \arrow[r, "e"] \arrow[rrr, bend right, "0"'] & \operatorname{Im} f \arrow[r, "m"] \arrow[rr, bend right, "0"'] & B \arrow[r, "g"] & C
\end{tikzcd}$$

Since $\operatorname{Im} f = \operatorname{Ker} g$, by the universal property of kernel, exists unique $t : X \to \operatorname{Im} f$. Let $\theta : X \to A$ by setting $\theta = e^{-1} \circ t$, then above diagram commutes. Note that $f = me$ is monomorphism, θ must be unique. Hence, $f = \operatorname{Ker} g$.
Conversely, if $f : A \to B = \operatorname{Ker}(g)$, then $g \circ f = 0$. By Lemma 1.4.4, $f = me$ uniquely, where $e : A \to \operatorname{Im} f$ epimorphism and $m : \operatorname{Im} f \to B$ monomorphism. Consider the following diagram,

$$\begin{tikzcd}
A \arrow[d, dashed, "e"] \arrow[dr, "f"] \arrow[drr, "0"] & & \\
\operatorname{Im} f \arrow[r, "m"'] \arrow[rr, bend right, "0"'] & B \arrow[r, "g"'] & C,
\end{tikzcd}$$

since e is epimorphism, $g \circ f = g \circ m \circ e = 0$ implies $g \circ m = 0$. Hence, e is the solution of above universal problem, and therefore $\operatorname{Im} f = \operatorname{Ker} g$. Let X be any object, $u : X \to A$ and $v : X \to A$ are two morphisms with $f \circ u = f \circ v$, then $f \circ (u - v) = 0$. Consider the following diagram,

$$\begin{tikzcd}
0 \arrow[d, dashed, "\exists!"'] \arrow[dr, "0"] \arrow[drr, "0"] & & \\
A \arrow[r, "f"] \arrow[rr, bend right, "0"'] & B \arrow[r, "g"] & C,
\end{tikzcd}$$

since $f = \operatorname{Ker}(g)$, by the universal property of kernel, $u - v = 0$, that is, $u = v$. Hence, f is monomorphism.
If sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact, then $\operatorname{Im} f = \operatorname{Ker} g$ and g is epimorphism. We shall to show that $g = \operatorname{Coker} f$. Note that $\operatorname{Im} f = \operatorname{Ker}(\operatorname{Coker}(f))$, we have $\operatorname{Coker}(\operatorname{Ker}(\operatorname{Coker}(f))) = \operatorname{Coker}(f) = \operatorname{Coker}(\operatorname{Ker} g)$. It suffice to show that $g = \operatorname{Coker}(\operatorname{Ker}(g))$. By Lemma 1.4.4, $g = me$, where $m = \operatorname{Ker}(C \to \operatorname{Coker} g)$ monomorphism and $e = \operatorname{Coker}(\operatorname{Ker}(g) \to B)$ epimorphism. Since g is epimorphism, $\operatorname{Coker}(g) = 0 : C \to 0$, and therefore $m = \operatorname{Ker}(0 : C \to 0) = \mathrm{id}_C : C \to C$. Hence, $g = \operatorname{Coker}(\operatorname{Ker}(g))$.
Conversely, If $g = \operatorname{Coker} f$, then $\operatorname{Ker}(g) = \operatorname{Ker}(\operatorname{Coker}(f)) = \operatorname{Im}(f)$. We shall to show that g is epimorphism. Let morphisms $\mu_1 : C \to X$ and $\mu_2 : C \to X$ with $\mu_1 \circ g = \mu_2 \circ g$, then $(\mu_1 - \mu_2) \circ g = 0$. Since $g = \operatorname{Coker}(f)$, by the universal property of cokernel, $\mu_1 - \mu_2 = 0$, that is, $\mu_1 = \mu_2$. Hence, g is epimorphism. Consequently, sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact.

□

In what follows, we will mainly consider additive functors between abelian categories.

Proposition 1.5.2

Let $F : \mathscr{A} \to \mathscr{B}$ be an additive functor between abelian categories. Then the following conditions are equivalent:

(1) F is left exact.

(2) F preserves kernels; namely, for every exact sequence $0 \to X \to Y \to Z$ in $\mathscr{A}$, $0 \to FX \to FY \to FZ$ is an exact sequence in $\mathscr{B}$;

(3) For every short exact sequence $0 \to X \to Y \to Z \to 0$ in $\mathscr{A}$, $0 \to FX \to FY \to FZ$ is exact in $\mathscr{B}$.

Proof (1) $\Rightarrow$ (2). Consider the following exact sequence.

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z$$

By Proposition 1.5.1 (5), we know that $f : X \to Y = \operatorname{Ker} g \to Y$. Since F is left exact, F preserves finite limits, note that $\operatorname{Ker} g \to B$ is limit, we have $F(f) : FX \to FY = \operatorname{Ker} Fg \to FB$. By Proposition 1.5.1 (5) again, we know that the following sequence is exact.

$$0 \longrightarrow FX \xrightarrow{Ff} FY \xrightarrow{Fg} FZ$$

(2) $\Rightarrow$ (3). Obviously!

(2) $\Rightarrow$ (1). This is because F preserves finite products and that a finite limit can be described as the equalizer of a suitable morphism between two finite products (see the proof of Lemma 1.4.5).

(3) $\Rightarrow$ (2). Let

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z$$

be an exact sequence in $\mathscr{A}$. Spilt the morphism $g : Y \to Z$ to be $Y \to \operatorname{Im} g \to 0$ and $0 \to \operatorname{Im} g \to Z \to \operatorname{Coker} g \to 0$.

$$\begin{tikzcd}
0 \arrow[dr] & & 0 & & \\
& \operatorname{Im} g \arrow[ur] \arrow[dr, hook] & & & \\
Y \arrow[ur, two heads] \arrow[rr, "g"] & & Z \arrow[dr] & & \\
& & & \operatorname{Coker} g \arrow[dr] & \\
& & & & 0
\end{tikzcd}$$

Then

$$0 \longrightarrow X \longrightarrow Y \longrightarrow \operatorname{Im} g \longrightarrow 0$$

and

$$0 \longrightarrow \operatorname{Im} g \longrightarrow Z \longrightarrow \operatorname{Coker} g \longrightarrow 0$$

are exact. By the hypothesis, we have

$$0 \longrightarrow FX \xrightarrow{Ff} FY \xrightarrow{a} F(\operatorname{Im} g) \tag{1.7}$$

and

$$0 \longrightarrow F(\operatorname{Im} g) \xrightarrow{b} FZ \longrightarrow F(\operatorname{Coker} g) \tag{1.8}$$

exacts. Next we want to show that the kernel of $FY \to F(\operatorname{Im} g) \to FZ$ is the image of $FX \to FY$. By (1.7), we know that $\operatorname{Im} Ff \cong \operatorname{Ker} a$. We claim that $\operatorname{Ker} a \cong \operatorname{Ker} b \circ a$. Consider the following diagram.

$$\begin{tikzcd}
\operatorname{Ker} a \arrow[d, dashed, "\exists!"'] \arrow[dr] \arrow[drr, "0"] \arrow[drrr, "0"] & & & \\
\operatorname{Ker} b \circ a \arrow[r] & FY \arrow[r, "a"'] & F(\operatorname{Im} g) \arrow[r, hook, "b"'] & FZ
\end{tikzcd}$$

By the universal property of $\operatorname{Ker} b \circ a$, there exists unique morphism form $\operatorname{Ker} a$ to $\operatorname{Ker} b \circ a$. From (1.8), we know that $\operatorname{Ker} b = 0$. Consider the following diagram.

$$\begin{tikzcd}
\operatorname{Ker} b \circ a \arrow[dd, dashed, "\exists!"'] \arrow[ddr] \arrow[drr, dashed, "\exists!"] \arrow[ddrrr, "0"] & & & \\
& & \operatorname{Ker} b = 0 \arrow[d] & \\
\operatorname{Ker} a \arrow[r] & FY \arrow[r, "a"'] & F(\operatorname{Im} g) \arrow[r, hook, "b"'] & FZ
\end{tikzcd}$$

Dashed arrow $\operatorname{Ker} b \circ a \to \operatorname{Ker} b$ given by the universal property of $\operatorname{Ker} b$, then we have $\operatorname{Ker} b \circ a \to FY \to F(\operatorname{Im} g)$ is zero, by the universal property of $\operatorname{Ker} a$, there exists unique morphism from $\operatorname{Ker} b \circ a$ to $\operatorname{Ker} a$. Hence $\operatorname{Ker} b \circ a \cong \operatorname{Ker} a$, and therefore $\operatorname{Im} Ff \cong \operatorname{Ker} b \circ a$. Thus the sequence

$$0 \longrightarrow FX \longrightarrow FY \longrightarrow FZ$$

is exact. $\square$

Example 1.37 (An additive functor F preserving monomorphisms need not be left exact) Let $n \geq 2$. Consider the functor $\mathbf{Mod}_{\mathbb{Z}} \to \mathbf{Mod}_{\mathbb{Z}}$ which sends M to its submodule nM. Clearly this functor preserves monomorphisms: if $M \hookrightarrow N$, then $aM \hookrightarrow aN$. But it is not left exact: the exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

is sent to $0 \to n\mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$, which is not exact.

Corollary 1.5.1

Let $F : \mathscr{A} \to \mathscr{B}$ be an additive functor between abelian categories. Then the following conditions are equivalent:

(1) F is exact.

(2) For every exact sequence $X \to Y \to Z$ in $\mathscr{A}$, $FX \to FY \to FZ$ is exact in $\mathscr{B}$.

(3) For every short exact sequence $0 \to X \to Y \to Z \to 0$ in $\mathscr{A}$, $0 \to FX \to FY \to FZ \to 0$ is a short exact sequence in $\mathscr{B}$.

(4) F is left exact and preserves epimorphisms.

(5) F is right exact and preserves monomorphisms.

Proof Obvious. $\square$

Proposition 1.5.3 (Important!)

If a functor $F : \mathscr{A} \to \mathscr{B}$ is a right (resp., left) adjoint, i.e., admits a left (resp., right) adjoint, then F is left (resp., right) exact.

Proof By Proposition 1.3.4, right adjoints preserves limits and left adjoints preserves colimits, we know that right adjoints are left exact and left adjoints are right exact. □

Example 1.38 We consider the category $\mathbf{Ab}$ in the following examples.

(1) The functor $h_{\mathbb{Z}} = \mathrm{Hom}_{\mathbf{Ab}}(\square, \mathbb{Z})$ is left exact but not right exact. For example, consider the exact sequence $0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$, we obtain

$$0 \longrightarrow \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \longrightarrow \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}) \xrightarrow{\square \circ \times 2} \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}).$$

It is easy to see that $\mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0$ and the second morphism is not surjective: the identity map is not in the image.

(2) The functor $\square \otimes_{\mathbb{Z}} (\mathbb{Z}/2\mathbb{Z})$ is right exact but not left exact. Still consider the above short exact sequence which implies

$$\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{F}_2 \xrightarrow{\times 2 \otimes \mathrm{id}} \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{F}_2 \longrightarrow \mathbb{F}_2 \otimes_{\mathbb{Z}} \mathbb{F}_2 \longrightarrow 0$$

here the first morphism is clearly zero morphism, hence not injective.

(3) If $\mathscr{A}$ admits limits (resp., colimits) indexed by a category $\mathscr{I}$, since $\Delta \dashv \lim$ (resp., $\mathrm{colim} \dashv \Delta$), then $\lim : \mathscr{A}^{\mathscr{I}} \to \mathscr{A}$ (resp., $\mathrm{colim} : \mathscr{A}^{\mathscr{I}} \to \mathscr{A}$) is left (resp., right) exact.

1.5.2 The Mittag-Leffler condition

Example 1.39 $\varprojlim : \mathbf{Mod}_R^{\mathscr{I}} \to \mathbf{Mod}_R$ is left exact, but not right exact in general.

Consider $R = k[X]$ with k a field. Denote $J = (X)$ be the principle ideal generated by X. One has

$$\varprojlim_n R/J^n \cong k[[X]],$$

the ring of formal series with coefficients in k. On the other hand, if $m < n$, the monomorphism $J^n \hookrightarrow J^m$ defines a projective system, i.e., $\cdots \hookrightarrow J^n \hookrightarrow \cdots \hookrightarrow J^1 \hookrightarrow R$, whose projective limit is 0 (intersection of all J^n). Thus, if we consider the exact sequence of R-modules

$$0 \longrightarrow J^n \longrightarrow R \longrightarrow R/J^n \longrightarrow 0$$

and taking the projective limit, we get the sequence

$$0 \longrightarrow 0 \longrightarrow k[X] \longrightarrow k[[X]] \longrightarrow 0$$

which is no more exact, neither in the category $\mathbf{Mod}_R$ nor in the category $\mathbf{Mod}_k$.

Note *Let I be a directed set.*

Definition 1.5.3 (Mittag-Leffler)

*Let $(A_i, \varphi_{ij} : A_j \to A_i)$ be a directed inverse system of sets over I. Then we say (A_i, φ_{ij}) is **Mittag-Leffler** if for each $i \in I$, the family $\varphi_{ij}(A_j) \subseteq A_i$ for $j \geq i$ stabilizes. Explicitly, this means that for each $i \in I$, there exists $j \geq i$ such that for $k \geq j$ we have $\varphi_{ik}(A_k) = \varphi_{ij}(A_j)$.*

*If (A_i, φ_{ij}) is a directed inverse system of modules over a ring R, we say that it is **Mittag-Leffler** if the underlying inverse system of sets is Mittag-Leffler.*

Example 1.40 If (A_i, φ_{ij}) is a directed inverse system of sets or of modules and the maps φ_{ij} are surjective, then clearly the system is Mittag-Leffler (since each $\varphi_{ij}(A_j) = A_i$). Conversely, suppose (A_i, φ_{ij}) is Mittag-Leffler.

Let $A_i' \subseteq A_i$ be the stable image of $\varphi_{ij}(A_j)$ for $j \geq i$. Then $\varphi_{ij}|_{A_j'} : A_j' \to A_i'$ is surjective for all $j \geq i$ (note that $\varphi_{ij} \circ \varphi_{jk} = \varphi_{ik}$, let k sufficient large, then we have $\varphi_{ij}(\varphi_{jk}(A_k)) = \varphi_{ij}(A_j') = A_i'$), and therefore $\varprojlim A_i = \varprojlim A_i'$. Hence the limit of the Mittag-Leffler system (A_i, φ_{ij}) can also be written as the limit of a directed inverse system over I with surjective maps.

Remark In the category **Sets**,

$$\left\{(a_i)_{i\in\mathscr{I}} \in \prod_i A_i : F(m)(a_j) = a_k \text{ for all } m \in \mathrm{Mor}_{\mathscr{I}}(j,k) \subseteq \mathrm{Mor}(\mathscr{I})\right\},$$

along with the obvious projection maps to each A_i, is the limit $\varprojlim_{\mathscr{I}} A_i$.

Lemma 1.5.2

Let (A_i, φ_{ij}) be a directed inverse system over I. Suppose I is countable. If (A_i, φ_{ij}) is Mittag-Leffler and the A_i are nonempty, then $\varprojlim A_i$ is nonempty.

Proof Let $i_1, i_2, i_3, \ldots$ be an enumeration of the elements of I (this sequence may not ordered). Define inductively a sequence of elements $j_n \in I$ for $n = 1, 2, 3, \ldots$ by the conditions: $j_1 = i_1$, and $j_n \geq i_n$ and $j_n \geq j_m$ for $m < n$. Then the subsequence $\{j_n\}_{n\geq 1}$ is increasing and forms a cofinal subset of I. Hence we may assume $I = 1, 2, 3, \ldots$ (with the natural total order). So by Example 1.40 we are reduced to showing that the limit of an inverse system of nonempty sets with surjective maps index by the positive integers is nonempty.

Since A_1 is nonempty, pick $a_1 \in A_1$. Since $A_1 \twoheadleftarrow A_2$ is surjective, there exists $a_2 \in A_2$ such that $\varphi_{12}(a_2) = a_1$. Inductively, we have a sequence $(a_i)_{i\in I} \in \varprojlim A_i$, which implies that $\varprojlim A_i$ is nonempty. □

Theorem 1.5.1

Let

$$0 \longrightarrow A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i \longrightarrow 0$$

be an exact sequence of directed inverse systems of abelian groups over I. Suppose I is countable. If (A_i) is Mittag-Leffler, then

$$0 \longrightarrow \varprojlim A_i \longrightarrow \varprojlim B_i \longrightarrow \varprojlim C_i \longrightarrow 0$$

is exact.

Proof Taking limits of directed inverse system is left exact, hence we only need to prove surjective of $\varprojlim B_i \to \varprojlim C_i$. So let $(c_i) \in \varprojlim C_i$. For each $i \in I$, let $E_i = g_i^{-1}(c_i)$, which is nonempty since $g_i : B_i \to C_i$ is surjective. The system of maps $\varphi_{ij} : B_j \to B_i$ restrict to maps $E_j \to E_i$ which make (E_i) into an inverse system of nonempty sets. It is enough to show that (E_i) is Mittag-Leffler. For then Lemma 1.5.2 would show $\varprojlim E_i$ is nonempty, and taking any element of $\varprojlim E_i$ would give an element of $\varprojlim B_i$ mapping to (c_i).

By the injection $f_i : A_i \to B_i$ we will regard A_i as a subset of B_i. Since (A_i) is Mittag-Leffler, if $i \in I$ then there exists $j \geq i$ such that $\varphi_{ik}(A_k) = \varphi_{ij}(A_j)$ for $k \geq j$. We claim that also $\varphi_{ik}(E_k) = \varphi_{ij}(E_j)$ for $k \geq j$. Always $\varphi_{ik}(E_k) \subseteq \varphi_{ij}(E_j)$ for $k \geq j$ (since $\varphi_{ik} = \varphi_{ij} \circ \varphi_{jk}$). For the reverse inclusion let $e_j \in E_j$, and we need to find $e_k \in E_k$ such that $\varphi_{ik}(e_k) = \varphi_{ij}(e_j)$. Let $e_k' \in E_k$ be any element, and set $e_j' = \varphi_{jk}(e_k')$. Then $g_j(e_j - e_j') = c_j - c_j = 0$, hence $e_j - e_j' = a_j \in A_j$. Since $\varphi_{ik}(A_k) = \varphi_{ij}(A_j)$, there exists $a_k \in A_k$ such that $\varphi_{ik}(a_k) = \varphi_{ij}(a_j)$. Hence

$$\varphi_{ik}(e_k' + a_k) = \varphi_{ij} \circ \varphi_{jk}(e_k' + a_k) = \varphi_{ij}(e_j) + \varphi_{ij}(a_j) = \varphi_{ij}(e_j),$$

so we can take $e_k = e_k' + a_k$ (since $g_k(e_k' + a_k) = c_k + 0 = c_k$, hence $e_k' + a_k \in g_k^{-1}(c_k) = E_k$). □

1.5.3 Exactness of filtered colimits

Example 1.41 colim is not left exact in general. We give an example for cokernels. Consider the following diagram:

$$\begin{tikzcd}
0 \arrow[r] & 4\mathbb{Z} \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "="] & \mathbb{Z}/4\mathbb{Z} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & 2\mathbb{Z} \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d] & \mathbb{Z}/2\mathbb{Z} \arrow[r] \arrow[d] & 0 \\
& 2\mathbb{Z}/4\mathbb{Z} & 0 & 0 &
\end{tikzcd}$$

The snake lemma gives

$$2\mathbb{Z}/4\mathbb{Z} \longrightarrow 0 \longrightarrow 0 \longrightarrow 0$$

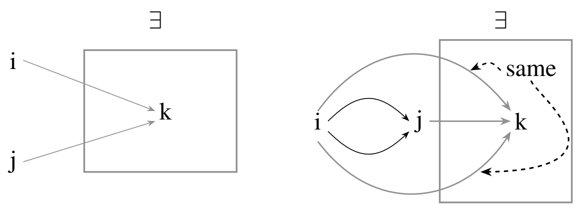
which shows that taking cokernel is not left exact.

Definition 1.5.4 (Filtered category)

*A non-empty category $\mathscr{I}$ is said to be **filtered** if*

(1) For $i, j \in \mathrm{obj}(\mathscr{I})$, there exists morphisms $i \to k$, $j \to k$ in $\mathscr{I}$.

(2) For morphisms $u : i \to j$ and $v : i \to j$, there exists a morphism $w : j \to k$ such that $wu = wv$.



(i) (ii)

Figure 1.1: A filtered category (pictorial definition)

Remark The category $(\bullet \to \bullet \leftarrow \bullet)$ is filtered, but $(\bullet \rightrightarrows \bullet)$ is not.

Example 1.42 Suppose $\mathscr{I}$ is filtered. Recall the symbol $\coprod$ for disjoint union of sets. Then any diagram in **Sets** indexed by $\mathscr{I}$ has the following, with the obvious maps to it, as a colimit:

$$\left\{ (a_i, i) \in \coprod_{i \in \mathscr{I}} A_i \right\} \Big/ \left(\begin{array}{l} (a_i, i) \sim (a_j, j) \text{ if and only if there are } f : A_i \to A_k \text{ and} \\ g : A_j \to A_k \text{ in the diagram for which } f(a_i) = g(a_j) \text{ in } A_k \end{array} \right)$$

Proof Say

$$\mathscr{S} = \left\{ (a_i, i) \in \coprod_{i \in \mathscr{I}} A_i \right\} \Big/ \left(\begin{array}{l} (a_i, i) \sim (a_j, j) \text{ if and only if there are } f : A_i \to A_k \text{ and} \\ g : A_j \to A_k \text{ in the diagram for which } f(a_i) = g(a_j) \text{ in } A_k \end{array} \right).$$

Then for $A_j \to A_k$, the following diagram commutes,

$$\begin{tikzcd}
A_j \arrow[r] \arrow[d, "e_j"'] & A_k \arrow[dl, "e_k"] \\
\mathscr{S},
\end{tikzcd}$$

where e_j, e_k are embedding.

Cosider ther following diagram.

$$\begin{tikzcd}
\mathscr{S} \arrow[rr, dashed, "\theta"] & & W \\
& A_j \arrow[ul, "e_j"'] \arrow[ur, "\alpha_j"] \arrow[d, "g"] & \\
& A_k \arrow[uul, bend left, "e_k"] \arrow[uur, bend right, "\alpha_k"'] &
\end{tikzcd}$$

Define $\theta : W \to \mathscr{S}$ by setting $\theta(a_n, n) = \alpha_n(a_n)$. We need to show that θ is well-defined. Let $(a_i, i) \sim (a_j, j) \in \coprod_{i\in\mathscr{I}}$, then there are $f : A_i \to A_k$ and $g : A_j \to A_k$ in the diagram for which $f(a_i) = g(a_j)$. Note that $\alpha_k \circ f = \alpha_i$ and $\alpha_k \circ g = \alpha_j$, $\alpha_i(a_i) = \alpha_k \circ f(a_i) = \alpha_k \circ g(a_j) = \alpha_j(a_j)$, hence $\theta(a_i) = \theta(a_j)$. Hence, θ is well-defined and such that above diagram commutes.

To show the uniqueness of θ, suppose there is ψ such that above diagram is commutes. Let $(a_n, n) \in \coprod_{i\in\mathscr{I}} A_i$, then $\psi(a_n, n) = \psi \circ e_n(a_n) = \alpha_n(a_n) = \theta(a_n, n)$. Hence, $\psi = \theta$. □

Proposition 1.5.4

In the category $\mathbf{Mod}_R$ for a ring R, small filtered colimits are exact.

Proof Since we know that colim is right exact, it suffices to check that it preserves monomorphisms. Let $f : F \to G$ be a monomorphism in $\mathbf{Mod}_R^{\mathscr{I}}$ (equivalently, $f_i : F(i) \to G(i)$ in injective for any $i \in \mathscr{I}$) and $\operatorname{colim} f : \operatorname{colim} F \to \operatorname{colim} G$ be the morphism in $\mathbf{Mod}_R$. Recall that $\operatorname{colim} F$ is represented by $\bigoplus_{i\in\mathrm{obj}(\mathscr{I})} F(i)/S$, where S is the submodule generated by all elements

$$\iota_j(x_j) - \iota_i((Fu)(x_j)), \tag{1.9}$$

for all $i, j \in \mathscr{I}$ and all $u : j \to i$, where $\iota_i : F(i) \to \bigoplus_i F(i)$ denotes the natural inclusion. In particular, every element $[x] \in \operatorname{colim} F$ can be represented by an element of the form $\sum_{j\in I} \iota_j(x_j)$, where I is a finite subset of $\mathrm{obj}(\mathscr{I})$. But since $\mathscr{I}$ is directed, i.e., satisfies the condition (1) in Definition 1.5.4, we check that $[x]$ has a representative of the form $\iota_i(x_i)$ for some single $i \in \mathrm{obj}(\mathscr{I})$. Indeed, we may choose i such that for any $j \in S$ there is a morphism $u_j : j \to i$, then $[x]$ can be represented by $\sum_{j\in S} \iota_j\left((Fu_j)(x_j)\right)$ which is an element in $F(i)$.

Let $[x] \in \operatorname{Ker}(\operatorname{colim} f)$ and let $x_i \in F(i)$ be a representative of $[x]$ (here we omit the inclusion ι_i to simplify the notation). Since $\operatorname{colim} f([x]) = [f_i(x_i)] = 0$, we can write $f_i(x_i)$ as a finite sum of elements of the form (1.9); moreover, since $\mathscr{I}$ is filtered, there exists $u : i \to k$ such that $(Gu)(f_i(x_i)) = 0$, see [Jos79, Lemma 5.30]. The commutative diagram below

$$\begin{tikzcd}
F(i) \arrow[r, "f_i"] \arrow[d, "Fu"'] & G(i) \arrow[d, "Gu"] \\
F(k) \arrow[r, "f_k"] & G(k)
\end{tikzcd}$$

shows that $f_k((Fu)(x_i)) = 0$. But f_k is injective, so $(Fu)(x_i) = 0$, [Jos79, Lemma 5.30], hence $[x] = 0$. □

Remark In Grothendieck's Tōhoku article [Gro57], listed four additional axioms (and their duals) that an abelian category $\mathscr{A}$ might satisfy. Some of them are:

(AB3) For every family (A_i) of objects of $\mathscr{A}$, the coproduct of A_i exists in $\mathscr{A}$ (i.e. $\mathscr{A}$ is cocomplete). (Since $\mathscr{A}$ admits coequalizers, $\mathscr{A}$ admits all colimits.)

(AB5) $\mathscr{A}$ satisfies (AB3), and filtered colimits of exact sequences are exact.

Hence the categories $\mathbf{Mod}_R$ and $\mathbf{Sets}$ satisfy (AB3), (AB5). (But note that $\mathbf{Sets}$ is not additive.)

The proof of the following result is omitted.

Proposition 1.5.5

For any small filtered category $\mathscr{I}$, any finite category $\mathscr{J}$ and any functor $F : \mathscr{I} \times \mathscr{J} \to \mathbf{Sets}$, the map

$$\mathrm{colim}_{i\in\mathscr{I}} \lim_{j\in\mathscr{J}} F(i,j) \longrightarrow \lim_{j\in\mathscr{J}} \mathrm{colim}_{i\in\mathscr{I}} F(i,j)$$

is a bijection.

1.5.4 Freyd-Mitchell Embedding Theorem

Theorem 1.5.2 (Freyd-Mitchell Embedding Theorem)

*Let $\mathscr{A}$ be a small abelian category, i.e., $\mathrm{obj}(\mathscr{A})$ is a set. Then there exists a small ring and a **fully faithful exact** functor $F : \mathscr{A} \to \mathbf{Mod}_R$.*

Remark For a fully faithful exact functor $F : \mathscr{A} \to \mathscr{B}$ between abelian categories, we have the following facts:

(1) if $A \in \mathscr{A}$ is such that $FA = 0$, then $A = 0$;

(2) if $f : A \to A'$ is a morphism in $\mathscr{A}$ such that $Ff : FA \to FA'$ is the zero morphism, then f is the zero morphism.

Remark The functor F yields an equivalence between $\mathscr{A}$ and a full subcategory of $\mathbf{Mod}_R$ in such a way that kernels and cokernels computed in $\mathscr{A}$ correspond to the ordinary kernels and cokernels computed in $\mathbf{Mod}_R$. The theorem thus essentially says that the objects of $\mathscr{A}$ can be thought of as R-modules, and the morphisms as R-linear maps, with kernels, cokernels, exact sequences and sums of morphisms being determined as in the case of modules.

However, projective and injective objects in $\mathscr{A}$ do not necessarily correspond to projective and injective R-modules. There are more morphisms to test in $\mathscr{A}$.

For practical use, we also need the following complement of Freyd-Mitchell Embedding Theorem 1.5.2.

Lemma 1.5.3

Let $\mathscr{A}$ be an abelian category and $S \subseteq \mathrm{obj}(\mathscr{A})$ be a non-empty (small) set of objects. There is a full small abelian subcategory $\mathscr{B}$ of $\mathscr{A}$ containing S.

Proof Define inductively a sequence $\{\mathscr{A}_n\}_{n\geq 0}$ of subcategories of $\mathscr{A}$ as follows. First, $\mathscr{A}_0$ is the category whose objects are the objects in S and whose morphisms are as in $\mathscr{A}$. Given $\mathscr{A}_n$, $\mathscr{A}_{n+1}$ is the full subcategory of $\mathscr{A}$ consisting of the objects in $\mathscr{A}_n$ together with

- a single representative for the kernel and cokernel in $\mathscr{A}$ of every morphism in $\mathscr{A}_n$;
- a single representatives for every finite product in $\mathscr{A}$ of objects in $\mathscr{A}_n$.

If $\mathscr{A}_n$ is small, then so is $\mathscr{A}_{n+1}$. Since $\mathscr{A}_0$ is small, so is $\mathscr{A}_n$ for all $n \geq 0$. Consequently, $\mathscr{B} = \bigcup_{n=0}^{\infty} \mathscr{A}_n$ is small, and $\mathscr{B}$ is a full abelian subcategory of $\mathscr{A}$. $\square$

With Freyd-Mitchell Embedding Theorem 1.5.2 and Lemma 1.5.3, we may reduce many proofs in general abelian category to the category $\mathbf{Mod}_R$ for some ring R.

1.5.5 Diagrams

Let $\mathscr{A}$ be an abelian category.

Lemma 1.5.4 (Snake lemma)

Consider a commutative diagram in abelian category $\mathscr{A}$ with exact rows:

$$\begin{tikzcd} & X' \arrow[r, "f"] \arrow[d, "u'"'] & X \arrow[r, "g"] \arrow[d, "u"] & X'' \arrow[r] \arrow[d, "u''"] & 0 \\ 0 \arrow[r] & Y' \arrow[r, "f'"] & Y \arrow[r, "g'"] & Y'' & \end{tikzcd}$$

There exists an exact sequence

$$\operatorname{Ker}(u') \longrightarrow \operatorname{Ker}(u) \longrightarrow \operatorname{Ker}(u'') \xrightarrow{\delta} \operatorname{Coker}(u') \longrightarrow \operatorname{Coker}(u) \longrightarrow \operatorname{Coker}(u'')$$

If, moreover, f is monic, then so is $\operatorname{Ker}(u') \to \operatorname{Ker}(u)$; *if g' is epic, then so is* $\operatorname{Coker}(u) \to \operatorname{Coker}(u'')$.

Proof By diagram chasing (or use spectral sequence of modules), we check that the statement is true if $\mathscr{A}$ is replaced by $\mathbf{Mod}_R$. Then let $\mathscr{B}$ be a small, full abelian subcategory of $\mathscr{A}$ containing the X, X', X'', Y, Y', Y''. Then the diagram is commutative and the rows are exact in $\mathscr{B}$. Let $F : \mathscr{B} \to \mathbf{Mod}_R$ be an exact fully faithful embedding. In $\mathscr{B}$ we have the desired long exact sequence in $\mathbf{Mod}_R$:

$$\begin{tikzcd} F\operatorname{Ker}(u') \arrow[r] & F\operatorname{Ker}(u) \arrow[r] & F\operatorname{Ker}(u'') \arrow[d, "F\delta"] & & \\ & & F\operatorname{Coker}(u') \arrow[r] & F\operatorname{Coker}(u) \arrow[r] & F\operatorname{Coker}(u'') \end{tikzcd}$$

Since $F : \mathscr{B} \to \mathbf{Mod}_R$ is fully faithful, this gives morphisms in $\mathscr{B}$, in particular the existence of δ in $\mathscr{B}$, and the corresponding long sequence is exact. □

Remark We give a proof of statement is true in $\mathbf{Mod}_R$, by using spectral sequence of modules:

Proof Consider the diagram

$$\begin{tikzcd} 0 \arrow[r] & D \arrow[r] & E \arrow[r] & F \arrow[r] & 0 \\ 0 \arrow[r] & A \arrow[r] \arrow[u, "\alpha"] & B \arrow[r] \arrow[u, "\beta"] & C \arrow[r] \arrow[u, "\gamma"] & 0 \end{tikzcd}$$

where the rows are exact in the middle (at A, B, C, D, E, F) and the squares commute. (Normally the Snake Lemma is described with the vertical arrows pointing downwards, but I want to fit this into my spectral sequence conventions.) We wish to show that there is an exact sequence

$$0 \longrightarrow \operatorname{Ker}\alpha \longrightarrow \operatorname{Ker}\beta \longrightarrow \operatorname{Ker}\gamma \longrightarrow \operatorname{Coker}\alpha \longrightarrow \operatorname{Coker}\beta \longrightarrow \operatorname{Coker}\gamma \longrightarrow 0.$$

We have the fact that there is a filtration

$$E_\infty^{0,k} \xhookrightarrow{E_\infty^{1,k-1}} F_1 \xhookrightarrow{E_\infty^{2,k-2}} \cdots \xhookrightarrow{E_\infty^{k-1,1}} F_{k-1} \xhookrightarrow{E_\infty^{k,0}} H^k(E^\bullet) \tag{1.10}$$

where the quotients are displayed above each inclusion (i.e., $F_1/E_\infty^{0,k} \cong E_\infty^{1,k-1}, \ldots, F_i/F_{i-1} \cong E_\infty^{i,k-i}, \ldots,$ $H^k(E^\bullet)/F_{k-1} \cong E_\infty^{k,0}$).

We plug this into our spectral sequence machinery. We first compute the cohomology using the rightward orientation. Then because the rows are exact, $E_1^{p,q} = 0$, so the spectral sequence has already converged: $E_\infty^{p,q} = 0$.

We next compute this "0" in another way, by computing the spectral sequence using the upward orientation. Then ${}_\uparrow E_1^{\bullet,\bullet}$ (with its differentials) is:

$$0 \longrightarrow \operatorname{Coker}\alpha \longrightarrow \operatorname{Coker}\beta \longrightarrow \operatorname{Coker}\gamma \longrightarrow 0$$

$$0 \longrightarrow \operatorname{Ker}\alpha \longrightarrow \operatorname{Ker}\beta \longrightarrow \operatorname{Ker}\gamma \longrightarrow 0.$$

Then ${}_{\uparrow}E_2^{\bullet,\bullet}$ is of the form:

$$\begin{tikzcd}
0 \arrow[drr] & 0 \arrow[drr] & & & & & \\
& 0 \arrow[drr] & ?? \arrow[drr] & ? \arrow[drr] & ? & 0 & \\
& 0 & ? & ? \arrow[drr] & ?? \arrow[drr] & 0 & \\
& & & & & 0 & 0
\end{tikzcd}$$

We see that after ${}_{\uparrow}E_2$, all the terms will stabilize except for the double question marks — all maps to and form the single question marks are to and from 0-entires. And after ${}_{\uparrow}E_3$, even these two double-question-mark terms will stabilize. But in the end our complex must be the 0 complex (since ${}_{\rightarrow}E_\infty^{p,q} = 0$, by (1.10), we know that $H^\bullet(E^\bullet) = 0$). This means that in ${}_{\uparrow}E_2$, all the entries must be zero, except for the two double question marks (these two ?? are $\operatorname{Ker}(\operatorname{Coker}\alpha \to \operatorname{Coker}\beta)$ and $\operatorname{Coker}(\operatorname{Ker}\beta \to \operatorname{Ker}\gamma)$), and these two must be isomorphic. This means that $0 \to \operatorname{Ker}\alpha \to \operatorname{Ker}\beta \to \operatorname{Ker}\gamma$ and $\operatorname{Coker}\alpha \to \operatorname{Coker}\beta \to \operatorname{Coker}\gamma \to 0$ are both exact (that comes from the vanishing of the vanishing of the single question marks), and

$$\operatorname{Coker}(\operatorname{Ker}\beta \to \operatorname{Ker}\gamma) \cong \operatorname{Ker}(\operatorname{Coker}\alpha \to \operatorname{Coker}\beta)$$

is an isomorphism (that comes from the equality of the double question marks). Taken together, we have proved the exactness of (**??**), and hence the Snake Lemma! (**Notice:** in the end we didn't really care about the double complex. We just used it as a prop to prove the Snake Lemma.)

Spectral sequences make it easy to see how to generalize results further. For example, if $A \to B$ is no longer assumed to be injective, then ${}_{\rightarrow}E_1^{\bullet,\bullet}$ (with its differentials) is:

$$\begin{tikzcd}
0 & 0 & 0 \\
\operatorname{Ker}(A \to B) \arrow[u] & 0 \arrow[u] & 0 \arrow[u] \\
0 \arrow[u] & 0 \arrow[u] & 0. \arrow[u]
\end{tikzcd}$$

Hence ${}_{\rightarrow}E_\infty^{0,0} = H^0(E^\bullet) = \operatorname{Ker}(A \to B)$, by (1.10). Note that ${}_{\uparrow}E_\infty^{0,0} = H^0(E^\bullet)$ ((1.10) again), ${}_{\uparrow}E_2^{\bullet,\bullet}$ is of the form:

$$\begin{tikzcd}
0 \arrow[drr] & 0 \arrow[drr] & & & & & \\
0 \arrow[drr] & 0 \arrow[drr] & \operatorname{Ker}(\operatorname{Coker}\alpha \to \operatorname{Coker}\beta) \arrow[drr] & 0 \arrow[drr] & 0 & & \\
& 0 & \operatorname{Ker}(A \to B) \arrow[drr] & 0 \arrow[drr] & \operatorname{Coker}(\operatorname{Ker}\beta \to \operatorname{Ker}\gamma) \arrow[drr] & 0 & \\
& & & & 0 & 0 & 0.
\end{tikzcd}$$

Since $E_\infty^{p,q} = 0$ for all $p \neq 0$ or $q \neq 0$ and $\operatorname{Ker}(\operatorname{Ker}\alpha \to \operatorname{Ker}\beta) = \operatorname{Ker}(A \to B)$, we have exact sequence

$$\operatorname{Ker}\alpha \longrightarrow \operatorname{Ker}\beta \longrightarrow \operatorname{Ker}\gamma \longrightarrow \operatorname{Coker}\alpha \longrightarrow \operatorname{Coker}\beta \longrightarrow \operatorname{Coker}\gamma \longrightarrow 0.$$

$\square$

Corollary 1.5.2

Let $0 \to X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$ be a short exact sequence in $\mathscr{A}$. Then the following conditions are equivalent:

(S1) f admits a retraction, i.e. there exists $r : X \to X'$ such that $rf = \mathrm{Id}_{X'}$.

(S2) g admits a section, i.e. there exists $s : X'' \to X$ such that $gs = \mathrm{Id}_{X''}$.

(S3) The sequence is isomorphic to the short exact sequence $0 \to X' \to X' \times X'' \to X'' \to 0$ where i and p are the canonical morphisms, i.e. there is an isomorphism $\alpha : X \to X' \times X''$ and a commutative diagram:

$$\begin{tikzcd}
0 \arrow[r] & X' \arrow[r, "f"] \arrow[d, equal] & X \arrow[r, "g"] \arrow[d, "\alpha"] & X'' \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & X' \arrow[r, "i"] & X' \times X'' \arrow[r, "p"] & X'' \arrow[r] & 0
\end{tikzcd}$$

Proof By Freyd-Mitchell Embedding Theorem 1.5.2 and Lemma 1.5.3, it suffices to show this in the category $\mathbf{Mod}_R$. We only prove (S2) $\Leftrightarrow$ (S3), (S1) $\Leftrightarrow$ (S3) is similar.

(S2) $\Rightarrow$ (S3). Consider $\mathrm{id}_X - s \circ g : X \to X$. Note that $g(\mathrm{id}_X - s \circ g) = g - g = 0$, we have $\mathrm{Im}(\mathrm{id}_X - s \circ g) \subseteq \mathrm{Ker}\, g$. Since $\mathrm{Ker}\, g = \mathrm{Im}\, f$, $\mathrm{Im}(\mathrm{id}_X - s \circ g) \subseteq \mathrm{Im}\, f$. Since f is monic, each element in $\mathrm{Im}(\mathrm{id}_X - s \circ g)$ has a unique preimage in X', define $r : X \to X'$ by setting $r = f^{-1} \circ (\mathrm{id}_X - s \circ g)$ (r is well-defined, since each element in $\mathrm{Im}(\mathrm{id}_X - s \circ g)$ has a unique preimage in X'). Now define $\alpha : X \to X' \times X''$ by setting $x \mapsto (r(x), g(x))$. Then the following diagram commutes.

$$\begin{tikzcd}
0 \arrow[r] & X' \arrow[r, "f"] \arrow[d, equal] & X \arrow[r, "g"] \arrow[d, "\alpha"] & X'' \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & X' \arrow[r, "i"] & X' \times X'' \arrow[r, "p"] & X'' \arrow[r] & 0
\end{tikzcd}$$

Then by Snake Lemma 1.5.4, we done.

(S3) $\Rightarrow$ (S2): Define $s = \alpha^{-1} \circ i_{X''}$ where $i_{X''} : X'' \to X' \times X''$ be natural inclusion, we done. □

Definition 1.5.5 (Split)

A short exact sequence satisfying (S1) (or (S2) or (S3) in Corollary 1.5.2) is said to be split.

Example 1.43 The sequence $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ is not split.

Corollary 1.5.3 (Five Lemma)

Consider a commutative diagram in abelian category $\mathscr{A}$

$$\begin{tikzcd}
X^0 \arrow[r] \arrow[d, "u^0"'] & X^1 \arrow[r] \arrow[d, "u^1"] & X^2 \arrow[r] \arrow[d, "u^2"] & X^3 \arrow[r] \arrow[d, "u^3"] & X^4 \arrow[d, "u^4"] \\
Y^0 \arrow[r] & Y^1 \arrow[r] & Y^2 \arrow[r] & Y^3 \arrow[r] & Y^4
\end{tikzcd}$$

with exact rows.

(1) If u^1 and u^3 are epimorphisms and u^4 is monomorphism, then u^2 is epimorphism.

(2) If u^1 and u^3 are monomorphisms and u^0 is epimorphism, then u^2 is monomorphism.

(3) If u^1 and u^3 are isomorphisms, u^0 is epimorphism, and u^4 is monomorphism, then u^2 is isomorphism.

Proof By Freyd-Mitchell Embedding Theorem 1.5.2 and Lemma 1.5.3, it suffices to show this in the category

$\mathbf{Mod}_R$. In $\mathbf{Mod}_R$, Five Lemma is obviously. □

Remark We give a proof of Five Lemma in $\mathbf{Mod}_R$ by using spectral sequence:

Proof Consider the following commutative diagram.

$$\begin{tikzcd} B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5 \\ A_1 \arrow[r] \arrow[u, "f_1"] & A_2 \arrow[r] \arrow[u, "f_2"'] & A_3 \arrow[r] \arrow[u, "f_3"'] & A_4 \arrow[r] \arrow[u, "f_4"'] & A_5 \arrow[u, "f_5"'] \end{tikzcd}$$

where the top and bottom rows are exact sequences. We want to show:

(1) If f_2 and f_4 are surjective and f_5 is injective, then f_3 is surjective.

(2) If f_2 and f_4 are injective and f_1 is surjective, then f_3 is injective.

(3) If f_2 and f_4 are isomorphisms, f_1 is surjective, and f_5 is injective, then f_3 is isomorphism.

(2) is the dual statement of (1), we just prove (1), and (2) is similarly.

Suppose f_2 and f_4 are surjective and f_5 is injective, we want to show that f_3 is surjective.

We first compute the cohomology of the total complex using the rightward orientation. Then ${}_{\rightarrow}E_1^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd} \mathrm{Ker}(B_1 \to B_2) & 0 & 0 & 0 & \mathrm{Coker}(B_4 \to B_5) \\ \mathrm{Ker}(A_1 \to A_2) \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & \mathrm{Coker}(A_4 \to A_5). \arrow[u] \end{tikzcd}$$

Then ${}_{\rightarrow}E_2^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd} 0 & & & & 0 & & \\ 0 & & & & 0 & & \\ & ? \arrow[uul] & 0 & 0 & 0 & ? \arrow[uul] & \\ & ? \arrow[uul] & 0 & 0 & 0 & ? \arrow[uul] & \\ & & 0 \arrow[uul] & & & & 0 \arrow[uul] \\ & & 0 \arrow[uul] & & & & 0 \arrow[uul] \end{tikzcd}$$

and therefore the sequence will converge by ${}_{\rightarrow}E_2$. Hence by (1.10),

$$E_2^{0,3} \xhookrightarrow{E_2^{1,2}} F_1 \xhookrightarrow{E_2^{2,1}} F_2 \xhookrightarrow{E_2^{3,0}} H^3(E^\bullet),$$

we have $H^3(E^\bullet) = 0$.

We next compute this using the upward orientation. The ${}_{\uparrow}E_1$ looks like this:

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{Coker}\, f_1 \arrow[r] & 0 \arrow[r] & \mathrm{Coker}\, f_3 \arrow[r] & 0 \arrow[r] & \mathrm{Coker}\, f_5 \arrow[r] & 0 \\ 0 \arrow[r] & \mathrm{Ker}\, f_1 \arrow[r] & \mathrm{Ker}\, f_2 \arrow[r] & \mathrm{Ker}\, f_3 \arrow[r] & \mathrm{Ker}\, f_4 \arrow[r] & 0 \arrow[r] & 0. \end{tikzcd}$$

And ${}_{\uparrow}E_2$ looks like this:

$$\begin{tikzcd}[column sep=small]
0 \arrow[drr] & & 0 \arrow[drr] & & 0 \arrow[drr] & & & & \\
0 \arrow[drr] & 0 \arrow[drr] & \operatorname{Coker} f_1 \arrow[drr] & 0 \arrow[drr] & \operatorname{Coker} f_3 \arrow[drr] & 0 & \operatorname{Coker} f_5 \arrow[drr] & 0 & \\
0 & 0 & ? \arrow[drr] & ? \arrow[drr] & ? \arrow[drr] & ?? \arrow[drr] & 0 & 0 & 0 \\
& & & & 0 & 0 & 0 & 0. &
\end{tikzcd}$$

Hence ${}_{\uparrow}E_{\infty}^{2,1} = \operatorname{Coker} f_3$. By (1.10), i.e. filtration

$${}_{\uparrow}E_2^{3,0} \xhookrightarrow{{}_{\uparrow}E_2^{2,1}} {}_{\uparrow}F_1 \xhookrightarrow{{}_{\uparrow}E_2^{1,2}} {}_{\uparrow}F_2 \xhookrightarrow{{}_{\uparrow}E_2^{0,3}} H^3(E^{\bullet}),$$

we have $H^3(E^{\bullet})/?? \cong \operatorname{Coker} f_3$, since $H^3(E^{\bullet}) = 0$, $\operatorname{Coker} f_3 = 0$, i.e., f_3 is surjective.

Now, we prove (3). Suppose f_2 and f_4 are isomorphism, and f_5 is injective, we want to show f_3 is isomorphism.

We first compute the cohomology of the total complex using the rightward orientation. Same as the proof of (1), we have $H^3(E^{\bullet}) = 0$ and $H^2(E^{\bullet}) = 0$.

We next compute this using the upward orientation. The ${}_{\uparrow}E_1$ looks like this:

$$\begin{tikzcd}
0 \arrow[r] & 0 \arrow[r] & \operatorname{Coker} f_3 \arrow[r] & 0 \arrow[r] & \operatorname{Coker} f_5 \\
\operatorname{Ker} f_1 \arrow[r] & 0 \arrow[r] & \operatorname{Ker} f_3 \arrow[r] & 0 \arrow[r] & 0.
\end{tikzcd}$$

Hence the spectral sequence converges at page 1. By (1.10), i.e. filtration

$${}_{\uparrow}E_1^{3,0} \xhookrightarrow{{}_{\uparrow}E_1^{2,1}} \bullet \xhookrightarrow{{}_{\uparrow}E_1^{1,2}} \bullet \xhookrightarrow{{}_{\uparrow}E_1^{0,3}} H^3(E^{\bullet})$$

and

$${}_{\uparrow}E_1^{2,0} \xhookrightarrow{{}_{\uparrow}E_1^{1,1}} \bullet \xhookrightarrow{{}_{\uparrow}E_1^{0,2}} H^2(E^{\bullet}),$$

we have $H^3(E^{\bullet}) \cong {}_{\uparrow}E_1^{2,1} = \operatorname{Coker} f_3$ and $H^2(E^{\bullet}) \cong \operatorname{Ker} f_3$, hence $\operatorname{Coker} f_3 = 0$ and $\operatorname{Ker} f_3 = 0$, i.e., f_3 is isomorphism, we are done! $\square$

Chapter 2 Derived Functors

2.1 Complexes

2.1.1 Basic definition

Definition 2.1.1 (Category of complexes)

Let $\mathscr{A}$ be an additive category. A ***(cochain) complex*** *in $\mathscr{A}$ consists of $X^\bullet = (X^n, d^n)_{n\in\mathbb{Z}}$ (or simply X), where*

- *X^n is an object of $\mathscr{A}$,*
- *$d^n : X^n \to X^{n+1}$ is a morphism in $\mathscr{A}$, called differential such that*
$$d^{n+1} \circ d^n = 0.$$

A ***morphism of (cochain) complexes*** *$X^\bullet \to Y^\bullet$ is a collection of morphisms $(f^n)_{n\in\mathbb{Z}}$, where $f^n : X^n \to Y^n$ in $\mathscr{A}$ such that $d_Y^n f^n = f^{n+1} d_X^n$. Namely*

$$\begin{tikzcd}
\cdots \arrow[r] & X^{n-1} \arrow[r, "d_X^{n-1}"] \arrow[d, "f^{n-1}"'] & X^n \arrow[r, "d_X^n"] \arrow[d, "f_n"] & X^{n+1} \arrow[r] \arrow[d, "f^{n+1}"] & \cdots \\
\cdots \arrow[r] & Y^{n-1} \arrow[r, "d_Y^{n-1}"] & Y^n \arrow[r, "d_Y^n"] & Y^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

Similarly, we may define a ***chain complex*** *in $\mathscr{A}$ and morphisms between them.*
We let $C^\bullet(\mathscr{A})$ denote the category of (cochain) complexes in $\mathscr{A}$ (for simply, we write $C(\mathscr{A})$), then we can see $\mathscr{A}$ is an additive category:

- *$C(\mathscr{A})$ admits a zero object: the zero complex 0 with $0^n = 0$ is a zero object.*
- *Finite products exists in $C(\mathscr{A})$: $\forall X^\bullet, Y^\bullet \in C(\mathscr{A}), (X^\bullet \oplus Y^\bullet)^n := X^n \oplus Y^n$, and give the obvious differential.*
- *$\forall X^\bullet, Y^\bullet \in C(\mathscr{A}), \mathrm{Hom}_{C(\mathscr{A})}(X^\bullet, Y^\bullet)$ is a subgroup of $\prod_{n\in\mathbb{Z}} \mathrm{Hom}(X^n, Y^n)$ and hence abelian and the composition:*
$$\mathrm{Hom}_{C(\mathscr{A})}(Y^\bullet, Z^\bullet) \times \mathrm{Hom}_{C(\mathscr{A})}(X^\bullet, Y^\bullet) \to \mathrm{Hom}_{C(\mathscr{A})}(X^\bullet, Z^\bullet)$$
is bilinear $\forall X^\bullet, Y^\bullet, Z^\bullet \in C(\mathscr{A})$.

If $\mathscr{A}$ is an abelian category, then so is $C(\mathscr{A})$: since
$$(\mathrm{Ker}\, f)^n = \mathrm{Ker}\, f^n, \qquad (\mathrm{Coker}\, f)^n = \mathrm{Coker}\, f^n.$$
A sequence $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ is ***exact*** *in $C(\mathscr{A})$ if and only if $0 \to X^n \to Y^n \to Z^n \to 0$ is exact, $\forall n \in \mathbb{Z}$.*
We shall also notice there is an exact fully faithful functor
$$\mathscr{A} \longrightarrow C(\mathscr{A}), \quad X \longmapsto (\cdots \to 0 \to X \to 0 \to \cdots),$$
where X is at degree 0 in the right complex.

Definition 2.1.2 (Cocyclic, coboundary, and cohomology)

Let $\mathscr{A}$ be an abelian category, for $X^\bullet \in C(\mathscr{A})$, we define
$$Z^n(X^\bullet) := \mathrm{Ker}\, d^n, \qquad B^n(X^\bullet) := \mathrm{Im}\, d^{n-1}$$

and

$$H^n(X^\bullet) := \mathrm{Coker}(B^n(X^\bullet) \to Z^n(X^\bullet)) =: Z^n(X^\bullet)/B^n(X^\bullet)$$

and call them the ***cocyclic****,* ***coboundary****, and* ***cohomology*** *object of degree n.*

Remark By Definition 2.1.2, $H^n(X^\bullet) = 0$ if and only if $X^{n-1} \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1}$ is exact. And Z^n, B^n, H^n can be regarded as additive functors from $C(\mathscr{A})$ to $\mathscr{A}, \forall n \in \mathbb{Z}$. We shall also remark Z^n is left exact.

Example 2.1 (De Rham complex) Let M be a smooth manifold of dimension n. The **de Rham complex** $\Omega^\bullet(M)$ of M is a complex of $\mathbb{R}$-vector space:

$$\cdots \longrightarrow 0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^n(M) \longrightarrow 0 \longrightarrow \cdots,$$

where $\Omega^i(M)$ consists of differential i-forms on M and d is the exterior derivative. The i-th cohomology of $\Omega^\bullet(M)$ is denoted by $H^i_{dR}(M)$.

The circle S^1 is a 1-dimensional manifold. Therefore, we only need to consider 0-forms and 1-forms; higher order forms are all zero. Thus $H^k_{dR}(S^1) = 0$ for all $k \geq 2$. We first calculate $H^0_{dR}(S^1) = Z^0(S^1)/B^0(S^1)$. $Z^0(S^1) = \mathrm{Ker}\, d^0$, let $f \in Z^0(S^1)$, then f is a 0-form (smooth function on S^1) and $df = 0$. Note that the differential of a function is zero if and only if the functions is constant on each connected component. Since S^1 is connected, a function f with $df = 0$ must be a constant function. Therefore, $Z^0(S^1) \cong \mathbb{R}$. By definition, $B^0(S^1)$ is the image of the exterior derivative of (-1)-forms. Since (-1)-forms do not exist, $B^0(S^1) = 0$. So $H^0_{dR}(S^1) \cong \mathbb{R}/0 \cong R$, and $h^0(S^1) := \dim H^0_{dR}(S^1) = 1$ (the circle has only one connected component). We next calculate $H^1_{dR}(S^1)$. Since S^1 is 1-dimensional, any 2-form is automatically zero. Hence $Z^1(S^1) = \Omega^1(S^1)$. $B^1(S^1)$ is the set of all 1-forms of the form df where f is a smooth function on S^1. A 1-form ω is exact if and only if its integral over any closed path (loop) is zero. For S^1, this means $\int_{S^1} \omega = 0$. We can prove that a 1-form $\omega = g(\theta)d\theta$ is exact if and only if $\int_0^{2\pi} g(\theta)d\theta = 0$. Now define a morphism $\int_0^{2\pi} : \Omega^1(S^1) \to \mathbb{R}$, note that $\int_0^{2\pi} d\theta = 2\pi \neq 0$, $\int_0^{2\pi}$ is a surjective, so we have $\mathbb{R} \cong \Omega^1(S^1)/B^1(S^1) = H^1_{dR}(S^1)$. The dimension $h^1(S^1) = \dim H^1_{dR}(S^1) = 1$, which indicates that S^1 has one "1-dimensional hole".

Example 2.2 (Koszul chain complex) Let $R \in$ **CRings**, $\underline{x} = (x_1, x_2, \ldots, x_n) \in R^n$. The Koszul chain complex $K_\bullet(\underline{x}, R) = (0 \to K_n \to K_{n-1} \to \cdots \to K_0 \to 0)$ is defined as follows

- $K_l = 0$ if $l \notin \{0, \ldots, n\}$.
- $K_0 = R$, and for $1 \leq l \leq n$, $K_l = \bigoplus_{i_1 \ldots i_l} Re_{i_1 \ldots i_l}$ is the free R-module of $\binom{n}{l}$ with basis:
$$\{e_{i_1 \ldots i_l} : 1 \leq i_1 < \cdots < i_l \leq n\}.$$
- The differential map $d : K_l \to K_{l-1}$ is defined as follows: if $l = 1$, $d(e_i) = x_i$; if $l \geq 2$, then
$$d(e_{i_1 \ldots i_l}) = \sum_{r=1}^{l} (-1)^{r-1} x_{i_r} e_{i_1 \ldots \hat{i_r} \ldots i_l};$$
For example, when $n = 2$, we get
$$0 \longrightarrow R \xrightarrow{\begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}} R \oplus R \xrightarrow{(x_1, x_2)} R \longrightarrow 0$$

Its homology $H_0(K_\bullet)$ is just $R/(x_1, x_2)$, $H_2(K_\bullet) = \mathrm{Ann}_A((x_1, x_2))$, and

$$H_1(K_\bullet) = \frac{\{(r_1, r_2) \in R^2 : x_1 r_1 + x_2 r_2 = 0\}}{\{(-rx_2, rx_1) : r \in R\}}.$$

If $R = k[x_1, x_2]$ is a polynomial ring, then the Koszul complex is exact at degrees $n = 1, 2$; however, if

$R = k[x_1, x_2]/(x_1^2, x_2^2)$, it is not exact.

Definition 2.1.3 (Acyclic, quasi-isomorphism)

*A complex $X^\bullet$ is said to be **acyclic** (= exact) if $H^n(X^\bullet) = 0$, $\forall n \in \mathbb{Z}$. A morphism of complexes $f : X^\bullet \to Y^\bullet$ is **quasi-isomorphism** if $H^n(f) : H^n(X^\bullet) \to H^n(Y^\bullet)$ is an isomorphism, $\forall n \in \mathbb{Z}$.*

2.1.2 Operations on complexes

(1) **Shifting:** $\forall X^\bullet \in C(\mathscr{A}), m \in \mathbb{Z}$, let $X^\bullet[m] \in C(\mathscr{A})$ be the complex with $X^\bullet[m]^n = X^{n+m}$ and $X^\bullet[m]^n \to X^\bullet[m]^{n+1}$ is given by $(-1)^m d^{n+m}$.
For $X \in \mathscr{A}$, we can view X an object in $C(\mathscr{A})$ by the embedding functor discussed above, then $X[m] = (\cdots \to 0 \to X \to 0 \to \cdots)$, where X is at degree $-m$.

(2) **Truncation:** $\forall n \in \mathbb{Z}$, $X^\bullet \in C(\mathscr{A})$. We define

$$\tau^{\leq n}X^\bullet := (\cdots \to X^{n-2} \to X^{n-1} \to \operatorname{Ker} d^n \to 0 \to 0 \to \cdots),$$
$$\widetilde{\tau}^{\leq n}X^\bullet := (\cdots \to X^{n-2} \to X^{n-1} \to X^n \to \operatorname{Im} d^n \to 0 \to \cdots),$$
$$\tau^{\geq n}X^\bullet := (\cdots \to 0 \to 0 \to \operatorname{Coker} d^{n-1} \to X^{n+1} \to X^{n+2} \to \cdots),$$
$$\widetilde{\tau}^{\geq n}X^\bullet := (\cdots \to 0 \to \operatorname{Im} d^{n-1} \to X^n \to X^{n+1} \to X^{n+2} \to \cdots).$$

They induce natural morphisms

$$\tau^{\leq n}X^\bullet \longrightarrow \widetilde{\tau}^{\leq n}X^\bullet \longrightarrow X^\bullet \longrightarrow \widetilde{\tau}^{\geq n}X^\bullet \longrightarrow \tau^{\geq n}X^\bullet$$

and exact sequence (by taking quotients):

$$0 \longrightarrow \widetilde{\tau}^{\leq n-1}X^\bullet \longrightarrow \tau^{\leq n}X^\bullet \longrightarrow H^n(X^\bullet)[-n] \longrightarrow 0,$$

$$0 \longrightarrow H^n(X^\bullet)[-n] \longrightarrow \tau^{\geq n}X^\bullet \longrightarrow \widetilde{\tau}^{\geq n+1}X^\bullet \longrightarrow 0,$$

$$0 \longrightarrow \tau^{\leq n}X^\bullet \longrightarrow X^\bullet \longrightarrow \widetilde{\tau}^{\geq n+1}X^\bullet \longrightarrow 0,$$

$$0 \longrightarrow \widetilde{\tau}^{\leq n-1}X^\bullet \longrightarrow X^\bullet \longrightarrow \tau^{\geq n}X^\bullet \longrightarrow 0.$$

(3) **Direct sum:** Give $X, Y \in C(\mathscr{A})$, we let $X \oplus Y = (X^n \oplus Y^n, d_X^n \oplus d_Y^n)$.

(4) **Tensor products:** Let $R \in \mathbf{CRings}$ and consider the category $\mathbf{Mod}_R$. The tensor product of two complexes $X, Y \in C(\mathbf{Mod}_R)$ is defined as:

$$(X \otimes Y)^n = \bigoplus_{i+j=n} X^i \otimes Y^j, \quad d^n(a \otimes b) = d_X a \otimes b + (-1)^{\deg a} a \otimes d_Y b.$$

Precisely, the restriction of d to $X^i \otimes Y^j$ is given by:

$$a \otimes b \longmapsto d_X^i a \otimes b + (-1)^i a \otimes d_Y^{n-i} b.$$

One checks that this is indeed a complex.

2.1.3 Long exact sequence

Theorem 2.1.1 (Fundamental Theorem of Homological Algebra)
If $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ is a short exact sequence of cochain complexes, then there are natural morphisms: $\delta : H^n(Z^\bullet) \to H^{n+1}(X^\bullet)$ and a long exact sequence:

$$\cdots \longrightarrow H^{n-1}(Z^\bullet) \xrightarrow{\delta}$$

$$\longrightarrow H^n(X^\bullet) \longrightarrow H^n(Y^\bullet) \longrightarrow H^n(Z^\bullet) \xrightarrow{\delta}$$

$$\longrightarrow H^{n+1}(X^\bullet) \longrightarrow \cdots$$

Similarly, if $0 \to X_\bullet \to Y_\bullet \to Z_\bullet \to 0$ is a short exact sequence of cochain complexes, then there are natural morphisms: $\delta : H_n(Z_\bullet) \to H_{n-1}(X_\bullet)$ and a long exact sequence:

$$\cdots \longrightarrow H_{n+1}(Z_\bullet) \xrightarrow{\delta}$$

$$\longrightarrow H_n(X_\bullet) \longrightarrow H_n(Y_\bullet) \longrightarrow H_n(Z_\bullet) \xrightarrow{\delta}$$

$$\longrightarrow H_{n-1}(X_\bullet) \longrightarrow \cdots$$

Proof We only show the statement for cohomology. Consider the diagram:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & Z^{n-1}(X^\bullet) \arrow[r] \arrow[d] & Z^{n-1}(Y^\bullet) \arrow[r] \arrow[d] & Z^{n-1}(Z^\bullet) \arrow[d] & \\
0 \arrow[r] & X^{n-1} \arrow[r] \arrow[d] & Y^{n-1} \arrow[r] \arrow[d] & Z^{n-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & X^n \arrow[r] \arrow[d] & Y^n \arrow[r] \arrow[d] & Z^n \arrow[r] \arrow[d] & 0 \\
 & X^n/d_X^{n-1}X^{n-1} \arrow[r] \arrow[d] & Y^n/d_Y^{n-1}Y \arrow[r] \arrow[d] & Z^n/d_Z^{n-1}Z^{n-1} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

By the Snake Lemma 1.5.4, the top and bottom rows are exact. Consider the diagram:

$$\begin{tikzcd}
 & X^n/d_X^{n-1}X^{n-1} \arrow[r] \arrow[d] & Y^n/d_Y^{n-1}Y^{n-1} \arrow[r] \arrow[d] & Z^n/d_Z^{n-1}Z^{n-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & Z^{n+1}(X^\bullet) \arrow[r] & Z^{n+1}(Y^\bullet) \arrow[r] & Z^{n+1}(Z^\bullet) &
\end{tikzcd}$$

By the Snake Lemma 1.5.4 again, we obtain

$$H^n(X^\bullet) \longrightarrow H^n(Y^\bullet) \longrightarrow H^n(Z^\bullet) \longrightarrow H^{n+1}(X^\bullet) \longrightarrow H^{n+1}(Y^\bullet) \longrightarrow H^{n+1}(Z^\bullet).$$

Pasting these sequence again, we get the result. $\square$

Let

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & Z^\bullet & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & X'^\bullet & \longrightarrow & Y'^\bullet & \longrightarrow & Z'^\bullet & \longrightarrow & 0 \end{array}$$

be a commutative diagram of complexes, with exact rows. Then it induces a commutative diagram of long exact sequence:

$$\begin{array}{ccccccccccccc} \cdots \longrightarrow & H^n(X^\bullet) & \longrightarrow & H^n(Y^\bullet) & \longrightarrow & H^n(Z^\bullet) & \longrightarrow & H^{n+1}(X^\bullet) & \longrightarrow & H^{n+1}(Y^\bullet) & \longrightarrow & H^{n+1}(Z^\bullet) & \longrightarrow \cdots \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & \\ \cdots \longrightarrow & H^n(X'^\bullet) & \longrightarrow & H^n(Y'^\bullet) & \longrightarrow & H^n(Z'^\bullet) & \longrightarrow & H^{n+1}(X'^\bullet) & \longrightarrow & H^{n+1}(Y'^\bullet) & \longrightarrow & H^{n+1}(Z'^\bullet) & \longrightarrow \cdots \end{array}$$

Corollary 2.1.1

Let

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & Z^\bullet & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle v} & & \downarrow{\scriptstyle w} & & \\ 0 & \longrightarrow & X'^\bullet & \longrightarrow & Y'^\bullet & \longrightarrow & Z'^\bullet & \longrightarrow & 0 \end{array}$$

be a commutative diagram of complexes with exact rows. If two of the morphisms u, v, w are quasi-isomorphisms, then so is the third one.

Proof Apply Five Lemma 1.5.3 to the commutative diagram above. □

2.1.4 Mapping cones

Definition 2.1.4 (Mapping cones)

*Let $f : X^\bullet \to Y^\bullet$ be a morphism in $C(\mathscr{A})$. The **mapping cone of** f is the complex: for all $n \in \mathbb{Z}$, $\mathrm{Cone}(f)^n = X^{n+1} \oplus Y^n$, with differential*

$$d^n := \begin{pmatrix} -d_X^{n+1} & 0 \\ f^{n+1} & d_Y^n \end{pmatrix} : X^{n+1} \oplus Y^n \longrightarrow X^{n+2} \oplus Y^{n+1}.$$

Remark $\mathrm{Cone}(f)$ is a complex since

$$d^n \circ d^{n-1} = \begin{pmatrix} -d_X^{n+1} & 0 \\ f^{n+1} & d_Y^n \end{pmatrix} \begin{pmatrix} -d_X^n & 0 \\ f^n & d_Y^{n-1} \end{pmatrix} = \begin{pmatrix} d_X^{n+1} d_X^n & 0 \\ -f^{n+1} d_X^n + d_Y^n f^n & d_Y^n d_Y^{n-1} \end{pmatrix} = 0.$$

Remark **Topological remark:** Let K be a simplicial complex (or more generally a cell complex). The topological cone CK of K is obtained by adding a new vertex s to K and "coning off" the simplices (cells) to get a new $(n+1)$-simplex for every old n-simplex of K. (See Figure 2.1.) The simplicial (cellular) chain complex $C_\bullet(s)$ of the one-point space $\{s\}$ is R in degree 0 and zero elsewhere. $C_\bullet(s)$ is a subcomplex of the simplicial (cellular) chain complex $C_\bullet(CK)$ of the topological cone CK. The quotient $C_\bullet(CK)/C_\bullet(s)$ is the chain complex $\mathrm{Cone}(C_\bullet K)$ of the identity map of $C_\bullet(K)$. The algebraic fact that $\mathrm{Cone}(C_\bullet K)$ is split exact (null homotopic) reflects the fact that the topological cone CK is contractible.

More generally, if $f : K \to L$ is a simplicial map (or a cellular map), the topological mapping cone $\mathrm{Cone}\, f$ of f is obtained by glueing CK and L together identifying the subcomplex K of CK with its image in L (Figure 2.1). This is a cellular complex, which is simplicial if f is an inclusion of simplicial complexes. Write $C_\bullet(\mathrm{Cone}\, f)$ for the cellular chain complex of the topological map ping cone $\mathrm{Cone}\, f$. The quotient chain complex $C_\bullet(\mathrm{Cone}\, f)/C_\bullet(s)$ may be identified with cone (f_*), the mapping cone of the chain map

$f_* : C_\bullet(K) \to C_\bullet(L)$.

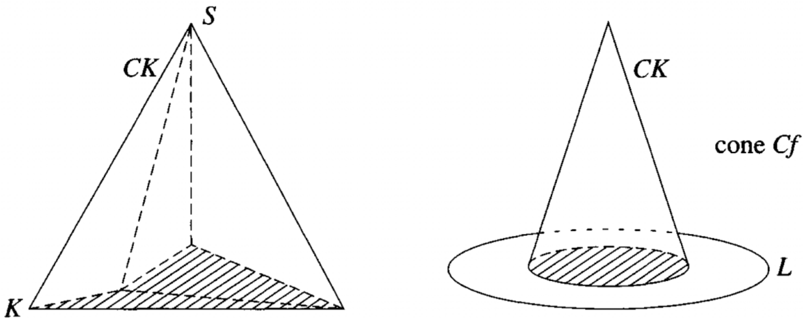


Figure 2.1: The topological cone CK and mapping cone $\operatorname{Cone} f$.

By construction, we have a short exact sequence in $C(\mathscr{A})$:

$$0 \longrightarrow Y^\bullet \xrightarrow{i} \operatorname{Cone}(f) \xrightarrow{p} X^\bullet[1] \longrightarrow 0,$$

which induces a long exact sequence:

$$\cdots \longrightarrow H^{n-1}X^\bullet[1] \xrightarrow{\delta} H^nY^\bullet \xrightarrow{H^ni} H^n(\operatorname{Cone} f) \longrightarrow H^nX^\bullet[1] \longrightarrow \cdots$$

Lemma 2.1.1

Via the isomorphism $H^{n-1}(X^\bullet[1]) \cong H^n(X^\bullet)$, the connecting morphism δ can be identified with $H^n : H^nX^\bullet \to H^nY^\bullet$. The long exact sequence thus has the form

$$\cdots \longrightarrow H^nX^\bullet \xrightarrow{H^nf} H^nY^\bullet \xrightarrow{H^ni} H^n(\operatorname{Cone} f) \longrightarrow H^{n+1}X^\bullet \longrightarrow \cdots$$

Proof Recall that the connecting morphism is constructed using the snake lemma applied to the commutative diagram.

$$\begin{tikzcd}
 & \frac{Y^{n-1}}{dY^{n-2}} \arrow[r] \arrow[d, "d_Y^{n-1}"] & \frac{\operatorname{Cone}(f)^{n-1}}{d\operatorname{Cone}(f)^{n-2}} \arrow[r] \arrow[d, "d_{\operatorname{Cone}(f)}^{n-1}"] & \frac{X^n}{dX^{n-1}} \arrow[r] \arrow[d, "d_X^n"] & 0 \\
0 \arrow[r] & Z^n(Y) \arrow[r] & Z^n(\operatorname{Cone}(f)) \arrow[r] & Z^{n+1}(X) &
\end{tikzcd}$$

Say $C = \operatorname{Cone}(f)$. Recall the construction of the snake lemma (in the category $\mathbf{Mod}_R$): for $x \in \operatorname{Ker} d_X^n$, take $\begin{pmatrix} x \\ 0 \end{pmatrix} + dC^{n-2}$ to be a lifting. Then $\delta(x)$ is by definition the image in $\operatorname{Coker}(d_Y^{n-1})$ of the element $d_C^{n-1}\begin{pmatrix} x \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ f^n(x) \end{pmatrix}$, which belongs to $Z^n(Y^\bullet)$. □

Corollary 2.1.2

A morphism $f : X^\bullet \to Y^\bullet$ in $C(\mathscr{A})$ is a quasi-isomorphism if and only if $\operatorname{Cone}(f)$ is acyclic.

Proof If $\operatorname{Cone}(f)$ is acyclic, then $H^n(\operatorname{Cone}(f)) = 0$ for all n. By Lemma 2.1.1, we know that $H^nf : H^nX^\bullet \to H^nY^\bullet$ is an isomorphism for all n, and therefore f is a quasi-isomorphism.

Conversely, if f is a quasi-isomorphism, we have $\operatorname{Im}(H^nf) = \operatorname{Ker}(H^nY^\bullet \to H^n(\operatorname{Cone}(f))) = H^nY^\bullet$ and $\operatorname{Ker}(H^{n+1}f) = \operatorname{Im}(H^n(\operatorname{Cone}(f)) \to H^{n+1}X^\bullet) = 0$. Hence $H^nY^\bullet \to H^n(\operatorname{Cone}(f))$ and $H^n(\operatorname{Cone}(f)) \to H^{n+1}X^\bullet$ are zero morphisms. By Lemma 2.1.1, we have $\operatorname{Im}(H^nY^\bullet \to H^n(\operatorname{Cone}(f))) = \operatorname{Ker}(H^n(\operatorname{Cone}(f)) \to H^{n+1}X^\bullet)$, note that $\operatorname{Im}(H^nY^\bullet \to H^n(\operatorname{Cone}(f))) = 0$ and $\operatorname{Ker}(H^n(\operatorname{Cone}(f)) \to$

$H^{n+1}X^\bullet) = H^n(\mathrm{Cone}(f))$, so $H^n(\mathrm{Cone}(f)) = 0$, which implies that $\mathrm{Cone}(f)$ is acyclic. $\square$

Proposition 2.1.1

Let $0 \longrightarrow X^\bullet \xrightarrow{f} Y^\bullet \xrightarrow{g} Z^\bullet \longrightarrow 0$ be a short exact sequence in $C(\mathscr{A})$. Then the morphism $\varphi = (0, g) : \mathrm{Cone}(f) \to Z^\bullet$ is a quasi-isomorphism.

Proof Define $\psi : \mathrm{Cone}(\mathrm{id}_X) \to \mathrm{Cone}(f)$ by setting $\psi^n : X^{n+1} \oplus X^n \to X^{n+1} \oplus Y^n, (a, b) \mapsto (a, f^n(b))$, i.e., $\psi^n = (\mathrm{id}_{X^{n+1}}, f^n)$. It is easy to check that ψ is a well-defined chain morphism. Then we have a short exact sequence,

$$0 \longrightarrow \mathrm{Cone}(\mathrm{id}_{X^\bullet}) \xrightarrow{\psi} \mathrm{Cone}(f) \xrightarrow{\varphi} Z^\bullet \longrightarrow 0.$$

By Fundamental Theorem of Homological Algebra 2.1.1, we have an exact sequence

$$\begin{tikzcd}
H^{n+1}\mathrm{Cone}(f) \arrow[dr] & & H^{n+1}\mathrm{Cone}(\mathrm{id}_{X^\bullet}) \arrow[ll, bend right] \\
H^n\mathrm{Cone}(f) \arrow[dr] & H^{n+1}Z^\bullet & H^n\mathrm{Cone}(\mathrm{id}_{X^\bullet}) \arrow[ll, dotted, bend right] \\
H^{n-1}\mathrm{Cone}(f) \arrow[dr] & H^nZ^\bullet \arrow[uur] & H^{n-1}\mathrm{Cone}(\mathrm{id}_{X^\bullet}) \arrow[ll, dotted, bend right] \\
& H^{n-1}Z^\bullet. \arrow[uur] &
\end{tikzcd}$$

By Corollary 2.1.2, $\mathrm{Cone}(\mathrm{id}_{X^\bullet})$ is acyclic, and therefore $H^n\,\mathrm{Cone}(\mathrm{id}_{X^{bullet}}) = 0$ for all n, by the long exact sequence we know that $\mathrm{Cone}(f) \to Z^\bullet$ is quasi-isomorphism. $\square$

2.1.5 Chain homotopies

Definition 2.1.5 (Homotopy)

(1) *A morphism $f : X^\bullet \to Y^\bullet$ in $C(\mathscr{A})$ is* ***homotopic to zero*** *if $\forall n \in \mathbb{Z}$, there exists a morphism $s^n : X^n \to Y^{n-1}$, such that $f^n = s^{n+1}d_X^n + d_Y^{n-1}s^n$:*

$$\begin{tikzcd}
X^{n-1} \arrow[r] & X^n \arrow[r, "d_X^n"] \arrow[dl, "s^n"'] \arrow[d, "f^n"] & X^{n+1} \arrow[dl, "s^{n+1}"] \\
Y^{n-1} \arrow[r, "d_Y^{n-1}"'] & Y^n \arrow[r] & Y^{n+1}
\end{tikzcd}$$

Two morphism $f, g : X^\bullet \to Y^\bullet$ are ***homotopic*** *if $f - g$ is homotopic to zero, denoted by $f \sim g$ (or equivalent, $f - g \sim 0$).*

(2) *An* ***object*** *$X^\bullet \in C(\mathscr{A})$* ***is homotopic to zero*** *if id_X is homotopic to zero.*

(3) *A morphism $f : X^\bullet \to Y^\bullet$ is a* ***cochain homotopy equivalence****, if $\exists g : Y^\bullet \to X^\bullet$, such that $fg \sim \mathrm{id}_Y$, $gf \sim \mathrm{id}_X$.*

Proposition 2.1.2

If $f : X^\bullet \to Y^\bullet$ is homotopic to zero, then $H^n f = 0 : H^n(X^\bullet) \to H^n(Y^\bullet), \forall n \in \mathbb{Z}$. If $f \sim g$, then $H^n f = H^n g, \forall n \in \mathbb{Z}$.

Proof It is enough to prove the first assertion, so suppose $f^n = s^{n+1}d_X^n + d_Y^{n-1}s^n$ for all n. Every element of $H^n(X^\bullet)$ is represented by an n-cocycle x. Then $f^n(x) = s^{n+1}d_X^n(x) + d_Y^{n-1}s^n(x) = d_Y^{n-1}s^n(x)$, hence $f^n(x)$ is an n-coboundary in Y. Thus $f^n(x) = 0$ in $H^n(Y^\bullet)$, which implies that $H^n f = 0$ for all n. $\square$

Definition 2.1.6 (Homotopy category $K(\mathscr{A})$)

The homotopy category $K(\mathscr{A})$ is defined by:

$$\operatorname{obj}(K(\mathscr{A})) = \operatorname{obj}(C(\mathscr{A}))$$

$$\operatorname{Hom}_{K(\mathscr{A})}(X^\bullet, Y^\bullet) = \operatorname{Hom}_{C(\mathscr{A})}(X^\bullet, Y^\bullet)/\operatorname{Ht}(X^\bullet, Y^\bullet),$$

where $\operatorname{Ht}(X^\bullet, Y^\bullet) := \{f : X^\bullet \to Y^\bullet : f \text{ is homotopic to } 0\}$.

2.1.6 Double complexes

Double complexes, Total complexes

Definition 2.1.7 (Double complexes)

A double complex $(X^{\bullet,\bullet}, d_X)$ in $\mathscr{A}$ consists of

- *objects $(X^{i,j})_{i,j\in\mathbb{Z}}$ of $\mathscr{A}$;*
- *Differentials:*

$$\begin{tikzcd}
 & {} & {} & \\
{} \arrow[r] & X^{i,j+1} \arrow[u] \arrow[r, "\mathrm{d_I}^{i,j+1}"] & X^{i+1,j+1} \arrow[u] \arrow[r] & {} \\
{} \arrow[r] & X^{i,j} \arrow[u, "\mathrm{d_{II}}^{i,j}"] \arrow[r, "\mathrm{d_I}^{i,j}"] & X^{i+1,j} \arrow[u, "\mathrm{d_{II}}^{i+1,j}"'] \arrow[r] & {} \\
 & {} \arrow[u] & {} \arrow[u] &
\end{tikzcd}$$

such that $\mathrm{d_I}^2 = \mathrm{d_{II}}^2 = 0$ *and* $\mathrm{d_I}^{i,j+1} \circ \mathrm{d_{II}}^{i,j} = \mathrm{d_{II}}^{i+1,j} \circ \mathrm{d_I}^{i,j}$.

One defines naturally the notion of a morphism of double complexes and obtains the abelian category $C^2(\mathscr{A})$. There are two functors ${}_{\rightarrow}F, {}_{\uparrow}F : C^2(\mathscr{A}) \to C(C(\mathscr{A}))$ which associates to a double complex X the complex whose objects are the rows (resp., the columns) of X. They are both isomorphisms of categories.

We set

$$\mathrm{H_I}^{i,j}(X^{\bullet,\bullet}) = \operatorname{Ker}(\mathrm{d_I}^{i,j})/\operatorname{Im}(\mathrm{d_I}^{i-1,j}), \quad \mathrm{H_{II}}^{i,j}(X^{\bullet,\bullet}) = \operatorname{Ker}(\mathrm{d_{II}}^{i,j})/\operatorname{Im}(\mathrm{d_{II}}^{i,j-1}).$$

Definition 2.1.8 (Total complex)

(1) Let $X^{\bullet,\bullet} \in C^2(\mathscr{A})$, we define two complexes in $\mathscr{A}$:

$$(\mathrm{Tot}_\oplus)^n := \bigoplus_{i+j=n} X^{i,j} \qquad \textit{(if coproducs exists),}$$

$$(\mathrm{Tot}_\prod)^n := \prod_{i+j=n} X^{i,j} \qquad \textit{(if products exists);}$$

with differentials: for $i + j = n$, the composition

$$\begin{tikzcd}
X^{i,j} \arrow[d, hook, "\iota^{i,j}"'] \arrow[dr, "d_{\mathrm{Tot}_\oplus} \circ \iota^{i,j}"] & \\
(\mathrm{Tot}_\oplus X^{\bullet,\bullet})^n \arrow[r, dashed, "d_{\mathrm{Tot}_\oplus}"'] & (\mathrm{Tot}_\oplus X^{\bullet,\bullet})^{n+1}
\end{tikzcd}$$

is given by $\mathrm{d_I}^{i,j} + (-1)^i \mathrm{d_{II}}^{i,j}$*, the composition*

$$\begin{tikzcd} (\mathrm{Tot}_{\prod} X^{\bullet,\bullet})^{n-1} \arrow[r, dashed, "d_{\mathrm{Tot}_{\prod}}"] \arrow[dr, "\mathrm{pr}^{i,j} \circ d_{\mathrm{Tot}_{\prod}}"'] & (\mathrm{Tot}_{\prod} X^{\bullet,\bullet})^n \arrow[d, "\mathrm{pr}^{i,j}"] \\ & X^{i,j} \end{tikzcd}$$

is given by $\mathrm{d_I}^{i-1,j} + (-1)^i \mathrm{d_{II}}^{i,j-1}$*. We call them the* ***direct sum total complex*** *and* ***product total complex*** *of* $X^{\bullet,\bullet}$*.*

(2) $X^{\bullet,\bullet} \in C^2(\mathscr{A})$ *is called* ***biregular****, if* $\forall n \in \mathbb{Z}$*,* $X^{i,j} = 0$ *for all but finitely many pairs* (i,j) *with* $i+j=n$*. Let* $C^2_{\mathrm{reg}}(\mathscr{A}) \subseteq C^2(\mathscr{A})$ *be the full subcategory of biregular complexes. For* $X^{\bullet,\bullet} \in C^2_{\mathrm{reg}}(\mathscr{A})$*, we have* $\mathrm{Tot}_{\oplus} X^{\bullet,\bullet} \cong \mathrm{Tot}_{\prod} X^{\bullet,\bullet}$*, which is denoted by* $\mathrm{Tot}\, X^{\bullet,\bullet}$*. This give us an additive functor* $\mathrm{Tot} : C^2_{\mathrm{reg}}(\mathscr{A}) \to C(\mathscr{A})$*,* $X^{\bullet,\bullet} \mapsto \mathrm{Tot}\, X^{\bullet,\bullet}$*. We call it the* ***total complex*** *of* $X^{\bullet,\bullet}$*.*

Example 2.3 Let $f : X^\bullet \to Y^\bullet$ be a morphism in $C(\mathscr{A})$. Consider the double complex $Z^{\bullet,\bullet}$ with

$$Z^{-1,\bullet} = X^\bullet, \quad Z^{0,\bullet} = Y^\bullet, \quad Z^{i,\bullet} = 0 \quad i \neq -1, 0$$

and differentials $d^{-1,j} = f^j : Z^{-1,j} \to Z^{0,j}$ and $d^{i,j} = 0, i \neq -1$. Then $\mathrm{Tot}\, Z^\bullet \cong \mathrm{Cone}(f)$.

Bifunctors

Let $\mathscr{A}, \mathscr{A}', \mathscr{A}''$ be additive categories. Let $F : \mathscr{A} \times \mathscr{A}' \to \mathscr{A}''$ be a functor that is additive in each variable. Then F extends to a functor

$$C^2(F) : C(\mathscr{A}) \times C(\mathscr{A}') \longrightarrow C^2(\mathscr{A}'')$$

which is additive in each variable. For $X^\bullet \in C(\mathscr{A}), Y^\bullet \in C(\mathscr{A})$, the double complex $C^2(F)(X,Y)$ is defined by $C^2(F)(X,Y)^{i,j} = F(X^i, Y^j)$, with

$$\mathrm{d_I}^{i,j} = F(d^i_X, \mathrm{id}_{Y^j}), \quad \mathrm{d_{II}}^{i,j} = F(\mathrm{id}_{X^i}, d^j_Y)$$

Example 2.4 Let R be a ring. The functor $\square \otimes_R \square : \mathbf{Mod}_R \times \mathbf{Mod}_R \to \mathbf{Mod}_R$ is additive in each variable. Thus it extends to

$$\square \otimes_R \square : C(\mathbf{Mod}_R) \times C(\mathbf{Mod}_R) \longrightarrow C^2(\mathbf{Mod}_R).$$

When R is commutative, one check that the composite

$$C(\mathbf{Mod}_R) \times C(\mathbf{Mod}_R) \longrightarrow C^2(\mathbf{Mod}_R) \longrightarrow C(\mathbf{Mod}_R)$$

is the tensor product we defined in §2.1.2 (4).

2.2 Projective and injective resolution

2.2.1 Projective objects, Injective objects

Definition 2.2.1 (Projective objects, injective objects)

*Let $\mathscr{C}$ be a category. An object P of $\mathscr{C}$ is said to be **projective** if given a morphism $f : P \to Y$ and an epimorphism $e : X \to Y$, there exists $g : P \to X$ such that $f = eg$:*

$$\begin{tikzcd} & P \arrow[dl, dashed, "g"'] \arrow[d, "f"] & \\ X \arrow[r, two heads, "e"'] & Y \arrow[r] & 0 \end{tikzcd}$$

*Dually, an object I of $\mathscr{C}$ is said to be **injective** if given as morphism $f : X \to I$ and a monomorphism $u : X \to Y$, there exists $g : Y \to I$, such that $f = gu$:*

$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r, hook, "u"] \arrow[d, "f"'] & Y \arrow[dl, dashed, "g"] \\ & I & \end{tikzcd}$$

Remark We see I is injective in $\mathscr{C}$ if and only if I^{op} is projective in $\mathscr{C}^{\mathrm{op}}$.

Proposition 2.2.1

Let $\mathscr{A}$ be the abelian category and $P \in \mathscr{A}$. The following conditions are equivalent:

(1) P is projective.

(2) $\mathrm{Hom}_{\mathscr{A}}(P, \square) : \mathscr{A} \to \mathbf{Ab}$ is exact.

(3) Every short exact sequence $0 \to A \to B \to P \to 0$ in $\mathscr{A}$ splits.

Similarly, for $I \in \mathscr{A}$, the following conditions are equivalent:

(1) I is injective.

(2) $\mathrm{Hom}_{\mathscr{A}}(\square, I) : \mathscr{A} \to \mathbf{Ab}$ is exact.

(3) Every short exact sequence $0 \to I \to A \to B \to 0$ in $\mathscr{A}$ splits.

Proof We only prove the case of projective objects, the case of injective objects is similar. By Lemma 1.5.3 and Freyd-Mitchell Embedding Theorem 1.5.2, we may assume that $\mathscr{A} = \mathbf{Mod}_R$ for some ring R.

(1) $\Rightarrow$ (2). Note that $\mathrm{Hom}(P, \square)$ is left exact, so it suffices to show that $h^P = \mathrm{Hom}(P, \square)$ is right exact. Let $f : A \to B$ surjective, we want to show $h^P(f) =: f_* : h^P(A) \to h^P(B)$ is surjective. Pick $h \in h^P(B)$, since P is projective, h factors through A, i.e., there exists a $g \in h^P(A)$ such that $h = f \circ g = f_*(g)$, which implies that f_* is surjective. Hence h^P is exact.

(2) $\Rightarrow$ (3). Since $\mathrm{Hom}(P, \square)$ is exact, we get an exact sequence

$$0 \longrightarrow h^P(A) \longrightarrow h^P(B) \longrightarrow h^P(P) \longrightarrow 0.$$

Say $f : B \to P$. Consider $\mathrm{id}_P \in h^P(P)$, since $f_* : h^P(B) \to h^P(P)$ is surjective, there exists $g \in h^P(B)$ such that $f_*(g) = f \circ g = \mathrm{id}_P$. Recall Definition 1.5.5, $0 \to A \to B \to P \to 0$ splits.

(3) $\Rightarrow$ (1). Let $s : A \twoheadrightarrow B$ is a surjective, and $f : P \to B$ be a morphism. We want to show that f factors through A. In $\mathbf{Mod}_R$, we have $A \times_B P = \{(a, b) \in A \times P : s(a) = f(b)\}$. Since s is surjective, $\mathrm{pr}_2 : A \times_B P \to P$ is surjective, then we have an exact sequence

$$0 \longrightarrow \mathrm{Ker}\,\mathrm{pr}_2 \hookrightarrow A \times_B P \xrightarrow{\mathrm{pr}_2} P \longrightarrow 0$$

by condition, this exact sequence splits, so there exists $\iota : P \to A \times_B P$ such that $\mathrm{pr}_2 \circ \iota = \mathrm{id}_P$. Then

$$f = f \circ \mathrm{id}_P = f \circ \mathrm{pr}_2 \circ \iota = s \circ (\mathrm{pr}_1 \circ \iota),$$

we are done. □

2.2.2 Projective modules

Lemma 2.2.1

Let R be a ring. Then free R-modules are projective.

Proof Direct check, omitted. □

Proposition 2.2.2

Let $P \in \mathbf{Mod}_R$. The following conditions are equivalent:

(1) P is projective.

(2) P is a direct summand of a free R-module.

Proof (1) $\Rightarrow$ (2). Let $F = \bigoplus_{x \in P} Re_x$, then F is a free module with surjective $F \twoheadrightarrow P$. Consider the following exact sequence.

$$0 \longrightarrow \mathrm{Ker}(F \twoheadrightarrow P) \hookrightarrow F \twoheadrightarrow P \longrightarrow 0$$

Since P is projective, by Proposition 2.2.1, $F \cong P \oplus \mathrm{Ker}(F \twoheadrightarrow P)$, we are done.

(2) $\Rightarrow$ (1). Suppose $F = P \oplus Q$, let $\mathrm{pr} : F \to P$ be natural projection. Let $f : A \twoheadrightarrow B$ be a surjective homomorphism and $g : P \to B$ be a morphism. We want to show that g factor through A. Consider the following diagram.

$$\begin{tikzcd}
& F \arrow[d, "\mathrm{pr}"] \arrow[ddl, dashed, "h"'] \\
& P \arrow[d, "g"] \arrow[dl, dashed, "h|_P"'] \\
A \arrow[r, two heads, "f"'] & B
\end{tikzcd}$$

Since F is projective, $g \circ \mathrm{pr}$ factors through A, i.e., $f \circ h = g \circ \mathrm{pr}$. Restrict h to P, then we have $f \circ h|_P = g$, which implies that P is projective. □

Example 2.5 Let R be a PID. Then every submodule of a free R-module is still free. So over R, an R-module is projective if and only if it is free (recall Proposition 2.2.2, a projective module is a direct summand of a free R-module, so it is a submodule of a free R-module).

2.2.3 Injective modules

Proposition 2.2.3 (Baer's criterion)

Let $R \in \mathbf{Rings}$, $I \in \mathbf{Mod}_R$. Then I is injective if and only if for any ideal $\mathfrak{a} \subseteq R$, every morphism $\mathfrak{a} \to I$ can be extended to $R \to I$.

Proof If I is injective. We may regard $\mathfrak{a} \subseteq R$ as an R-module, then there is an injection $\mathfrak{a} \hookrightarrow R$, by the definition, every morphism $\mathfrak{a} \hookrightarrow I$ can be extended to $R \to I$.

Conversely, let $f : A \to I$ and $u : A \hookrightarrow B$ be given R-module homomorphisms where u is an injective. We may view u as the inclusion $A \subseteq B$. Consider

$$\mathcal{S} = \{(h : M \to I) : A \subseteq M \subseteq B, h|_A = f\},$$

$\mathcal{S}$ is nonempty since $f \in \mathcal{S}$. We define a partial order on $\mathcal{S}$ by $(h_1 : M_1 \to I) \leq (h_2 : M_2 \to I)$ if $M_1 \subseteq M_2$ and $h_2|_{M_1} = h_1$. Let $\mathcal{C} = \{(h_\alpha : M_\alpha \to I)\}$ be a chain in $\mathcal{S}$. We can construct an upper bound $(h : \bigcup_\alpha M_\alpha \to I)$ where $h(x) = h_\alpha(x)$ for any α where $x \in M_\alpha$. This is well-defined and is an R-module homomorphism. By Zorn's Lemma, $\mathcal{S}$ contains a maximal element, say $(H : M \to I)$. We intend to prove $M = B$. If not, take $b \in B \setminus M$, and consider $M + Rb$. The set

$$\mathfrak{a} := \{r \in R : rb \in M\}$$

is an ideal of R and $f_0 : \mathfrak{a} \to I$ given by $f_0(r) := H(rb)$ is an R-module homomorphism. By assumption it extends to $\widetilde{f}_0 : R \to I$, since this is R-map, if we let $u := \widetilde{f}_0(1)$, then

$$\widetilde{f}_0(r) = r\widetilde{f}_0(1) = ru, \quad \forall r \in R.$$

Now define

$$\varphi : M + Rb \longrightarrow I, \quad a + rb \longmapsto H(a) + ru.$$

This is well-defined because: if $rb \in M \cap Rb$, then $r \in \mathfrak{a}$, thus

$$\varphi(rb) = H(rb) = f_0(r) = \widetilde{f}_0(r) = ru.$$

Note that $\varphi|_A = h|_A = f$, $\varphi \in \mathcal{S}$, this gives a contradiction to the choice of M. □

Divisible

Recall that $M \in \mathbf{Mod}_R$ is called **divisible** if for any $m \in M$, $r \in R \setminus \{0\}$, there exists $x \in M$ such that $m = rx$.

Proposition 2.2.4

Let R be an integral domain. Then any injective module over R is divisible. If R is a PID or a DD (Dedekind domain, integrally close Noetherian domain of dimension one), the converse also holds.

Proof See [Jos79, Lemma 3.33, Corollary 3.35, Theorem 4.24]. □

Example 2.6 $\mathbb{Q}/\mathbb{Z}, \mathbb{Q}$ are injective $\mathbb{Z}$-modules since they are divisible and $\mathbb{Z}$ is a PID. Consider $R = \mathbb{Z}[X] \subset K$ where K is the fractional field, then K/R is a divisible R-module but not injective.

Flatness

Definition 2.2.2 (Flat)

*$M \in \mathbf{Mod}_R$ is called **flat** if $\square \otimes_R M : \mathbf{Mod}_R \to \mathbf{Ab}$ is exact.*

Proposition 2.2.5 (Projective R-modules are flat)

(1) Let $\{M_i\}_{i \in I}$ be a family of R-modules, $M = \bigoplus_{i \in I} M_i$ is flat if and only if each M_i is flat.

(2) Projective R-modules are flat.

Proof For (1). M is flat if and only if for any injective $f : A \to B$, $\mathrm{id}_M \otimes f : M \otimes_R A \to M \otimes_R B$ is injective. Note that $M \otimes_R A \cong \bigoplus_{i \in I}(M_i \otimes_R A)$ and $M \otimes_R B \cong \bigoplus_{i \in I}(M_i \otimes_R B)$, $\mathrm{id}_M \otimes f : M \otimes_R A \to M \otimes_R B$ is injective if and only if each $\mathrm{id}_{M_i} \otimes f : M_i \otimes_R A \to M_i \otimes_R B$ is injective, which implies each M_i is flat.

$$\begin{tikzcd} M \otimes_R A \arrow[r, "\sim"] \arrow[d, "\mathrm{id}_M \otimes f"'] & \bigoplus_{i\in I}(M_i \otimes_R A) \arrow[d, "\bigoplus_{i\in I}(\mathrm{id}_{M_i} \otimes f)"] \\ M \otimes_R B \arrow[r, "\sim"] & \bigoplus_{i\in I}(M_i \otimes_R B) \end{tikzcd}$$

For (2). Let P be a projective module. By part (1), we know that free R-modules are flat. Recall Proposition 2.2.2, any projective module is a direct summand of a free R-algebra, we may write $F = P \oplus C$, where F is free. By part (1) again, P is projective. □

Example 2.7 (A flat module may not be a projective R-module.) $\mathbb{Q}$ is flat over $\mathbb{Z}$, but $\mathbb{Q}$ is not projective.

Proof Let $f : A \to B$ be any injective of $\mathbb{Z}$-module, we want to show that $A \otimes_{\mathbb{Z}} \mathbb{Q} \to B \otimes_{\mathbb{Z}} \mathbb{Q}$ is injective. Note that $\mathbb{Q} = \mathbb{Z}_{(0)}$, localization is an exact functor, so $A \otimes_{\mathbb{Z}} \mathbb{Q} \to B \otimes_{\mathbb{Z}} \mathbb{Q}$ is injective.

We next show that $\mathbb{Q}$ is not projective. If $\mathbb{Q}$ is projective, then $\mathbb{Q}$ is a direct summand of a free $\mathbb{Z}$-module, note that $\mathbb{Z}$ is a PID and every submodule of a free module over a PID is free, $\mathbb{Q}$ is free $\mathbb{Z}$-module, this is impossible! So $\mathbb{Q}$ is not projective. □

Proposition 2.2.6 (Flatness is preserved by base change)

Let $R \to S$ be a ring homomorphism. Let B be a flat R-module. Then $S \otimes_R B$ is S-flat.

Proof Obviously. □

Definition 2.2.3 (Pontrjagin dual)

*The **Pontrjagin dual** B^* of a left R-module B is the right R-module $\mathrm{Hom}_{\mathbb{Z}}(B, \mathbb{Q}/\mathbb{Z})$; an element r of R acts via $(fr)(b) = f(rb)$. It is also called the **character module of** B.*

Lemma 2.2.2

A sequence of right R-modules

$$A_1 \longrightarrow A_2 \longrightarrow A_3$$

is exact if and only if the sequence of character modules

$$A_3^* \longrightarrow A_2^* \longrightarrow A_1^*$$

is exact.

Proof Recall that $\mathbb{Q}/\mathbb{Z}$ is injective, by Proposition 2.2.1, $\square^*$ is exact. So if $A_1 \to A_2 \to A_3$ is exact, then $A_3^* \to A_2^* \to A_1^*$ is exact.

Conversely, if $A_3^* \xrightarrow{g^*} A_2^* \xrightarrow{f^*} A_1^*$, we want to show that $A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3$ is exact. We first show that $\mathrm{Im}\, f \subseteq \mathrm{Ker}\, g$. Pick $\beta \in A_3^*$, since $f^* \circ g^* = 0$, we have $\beta \circ (g \circ f) = 0$ for all $\beta \in A_3^*$. If $g \circ f \neq 0$, then there exists $x \in A_1$ such that $y = (g \circ f)(x) \neq 0$. Then there exists a group homomorphism $\alpha \in A_3^*$ such that $\alpha(y) \neq 0$, contradicts to the fact that $\beta \circ (g \circ f) = 0$ for all $\beta \in A_3^*$. So $\mathrm{Im}\, f \subseteq \mathrm{Ker}\, g$. We next show that $\mathrm{Ker}\, g \subseteq \mathrm{Im}\, f$. If there exists $x \in \mathrm{Ker}\, g$ such that $x \notin \mathrm{Im}\, f$. Consider $A_2/\mathrm{Im}\, f$, let $\overline{x}$ be the image of x in $A_2/\mathrm{Im}\, f$, then $\overline{x} \neq 0$. Then there exists a group homomorphism $\overline{\varphi} : A_2/\mathrm{Im}\, f \to \mathbb{Q}/\mathbb{Z}$, such that $\overline{\varphi}(\overline{x}) \neq 0$. Let $\pi : A_2 \to \mathbb{Q}/\mathbb{Z}$ be a natural projection. Define $\varphi = \overline{\varphi} \circ \pi$, then for all $z \in \mathrm{Im}\, f$, $\pi(z) = 0$, so $\varphi(z) = 0$. It follows that $\varphi \circ f = f^*(\varphi) = 0$, hence $\varphi \in \mathrm{Ker}\, f^* = \mathrm{Im}\, g^*$, hence $\varphi = g^*(\psi) = \psi \circ g$ for some $\psi \in A_3^*$. Consider $\varphi(x)$,

$$\varphi(x) = (\psi \circ g)(x) = \psi(g(x)) = \psi(0) = 0,$$

since $x \in \operatorname{Ker} g$. Since $\overline{\varphi}(\overline{x}) \neq 0$, we have $\varphi(x) = \overline{\varphi}(\overline{x}) \neq 0$, a contradiction! Hence $\operatorname{Ker} f^* = \operatorname{Im} g^*$, as we desired. $\square$

Proposition 2.2.7

A R-module B is flat if and only if B^* is an injective R-module.

Proof If B is a flat R-module. Let $f : M \to N$ be an injective of R-modules. Since $\square \otimes_R B$ is exact, we have an exact sequence

$$0 \longrightarrow M \otimes_R B \xrightarrow{f \otimes \mathrm{id}_B} N \otimes_R B.$$

Now, we apply the functor $(\square)^* = \operatorname{Hom}_{\mathbb{Z}}(\square, \mathbb{Q}/\mathbb{Z})$, since $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module (Example 2.6), the functor $(\square)^*$ is exact, then we have an exact sequence

$$(N \otimes_R B)^* \xrightarrow{(f\otimes \mathrm{id}_B)^*} (M \otimes_R B)^* \longrightarrow 0.$$

Recall that $\square \otimes_R B \dashv \operatorname{Hom}_{\mathbb{Z}}(B, \square)$, we get the following commutative diagram

$$\begin{tikzcd}
\operatorname{Hom}_{\mathbb{Z}}(N \otimes_R B, \mathbb{Q}/\mathbb{Z}) \arrow[r, equal] \arrow[d, "\sim"'] & (N \otimes_R B)^* \arrow[r, two heads] & (M \otimes_R B)^* \arrow[r, equal] & \operatorname{Hom}_{\mathbb{Z}}(M \otimes_R B, \mathbb{Q}/\mathbb{Z}) \arrow[d, "\sim"] \\
\operatorname{Hom}_R(N, \operatorname{Hom}_{\mathbb{Z}}(B, \mathbb{Q}/\mathbb{Z})) \arrow[r, equal] & \operatorname{Hom}_R(N, B^*) \arrow[r] & \operatorname{Hom}_R(M, B^*) \arrow[r, equal] & \operatorname{Hom}_R(M, \operatorname{Hom}_{\mathbb{Z}}(B, \mathbb{Q}/\mathbb{Z})).
\end{tikzcd}$$

So $\operatorname{Hom}_R(N, B^*) \to \operatorname{Hom}_R(M, B^*)$ must be surjective. Since $f : M \to N$ was an arbitrary injection, the functor $\operatorname{Hom}_R(\square, B^*)$ maps injections to surjections, it follows that $\operatorname{Hom}_R(\square, B^*)$ is right exact, by Proposition 2.2.1, B^* is injective.

Conversely, if B^* is an injective R-module, then $\operatorname{Hom}_R(\square, B^*)$ is exact. Let $f : M \to N$ be an injective of R-modules, we want to show that $f \otimes \mathrm{id}_B : M \otimes_R B \to N \otimes_R B$. If $f \otimes \mathrm{id}_B : M \otimes_R B \to N \otimes_R B$ is not injective, then there exists a nonzero elements $x \in M \otimes_R B$ such that $(f \otimes \mathrm{id}_B)(x) = 0$. Consider $(x) < M \otimes_R B$ as a subgroup (regard as sub-$\mathbb{Z}$-module), we may define a homomorphism $h : (x) \to \mathbb{Q}/\mathbb{Z}$ by setting $x \mapsto \frac{1}{\operatorname{ord}(x)} + \mathbb{Z}$. Since $\mathbb{Q}/\mathbb{Z}$ is injective $\mathbb{Z}$-module, h can be extended to $\widetilde{h} : M \otimes_R B \to \mathbb{Q}/\mathbb{Z}$. By $\square \otimes_R B \dashv \operatorname{Hom}_{\mathbb{Z}}(\square, B^*)$, there is an isomorphism

$$\tau : \operatorname{Hom}_{\mathbb{Z}}(M \otimes_R B, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} \operatorname{Hom}_R(M, \operatorname{Hom}_{\mathbb{Z}}(B, \mathbb{Q}/\mathbb{Z})) = \operatorname{Hom}_R(M, B^*),$$

denote $\widehat{h} = \tau(\widetilde{h})$. Since B^* is injective, we have an surjection

$$\square \circ f : \operatorname{Hom}_R(N, B^*) \longrightarrow \operatorname{Hom}_R(M, B^*).$$

Hence there exists $\widehat{g} \in \operatorname{Hom}_R(N, B^*)$ such that $\widehat{g} \circ f = \widehat{h}$. Apply τ^{-1} both side,

$$\tau^{-1}(\widehat{g} \circ f) = g \circ (f \otimes \mathrm{id}_B) = h,$$

where $g \in (N \otimes_R B)^*$ is the corresponding element of $\widetilde{h}$. Consider $h(x)$, by the construction of h, we know that $h(x) \neq 0$, but

$$g \circ (f \otimes \mathrm{id}_B)(x) = g((f \otimes \mathrm{id}_B)(x)) = g(0) = 0 = h(x),$$

a contradiction! So $\square \otimes_R B$ is exact, i.e., B is flat. $\square$

Remark We have $B \hookrightarrow B^{**}$, but need not be an isomorphism.

Proposition 2.2.8 (Flatness test)

An R-module B is flat if and only if for every ideal $\mathfrak{a} \subseteq R$ the natural map

$$\mathfrak{a} \otimes_R B \longrightarrow \mathfrak{a}B$$

is an isomorphism of abelian group.

Proof Consider the natural morphism $\varphi : \mathfrak{a} \otimes_R B \to R \otimes_R B = B$, its image is exactly $\mathfrak{a}B \subseteq B$. Assume B is flat, since $\mathfrak{a} \hookrightarrow R$ is injective, φ is also injective, hence $\varphi : \mathfrak{a} \otimes_R B \to \mathfrak{a}B$ is isomorphism.

Conversely, assume the condition satisfied. By Proposition 2.2.7, it suffices to show B^* is injective R-module. By Baer's criterion 2.2.3, we only need to show the lifting property for inclusion of the form $\mathfrak{a} \subseteq R$ where $\mathfrak{a}$ us a right ideal of R, i.e., need to show

$$\mathrm{Hom}_R(R, B^*) \longrightarrow \mathrm{Hom}_R(\mathfrak{a}, B^*) \longrightarrow 0$$

is surjective. By the hypothesis, $\mathfrak{a} \otimes_R B \to \mathfrak{a}B$ is an isomorphism, note that $\mathfrak{a}B$ is a submodule of $B \cong R \otimes_R B$, then we get an injective $\mathfrak{a} \otimes_R B \hookrightarrow R \otimes_R B$. By the adjunction isomorphism

$$\mathrm{Hom}_R(\mathfrak{a}, \mathrm{Hom}_{\mathbb{Z}}(B, \mathbb{Q}/\mathbb{Z})) \cong \mathrm{Hom}_{\mathbb{Z}}(\mathfrak{a} \otimes_R B, \mathbb{Q}/\mathbb{Z})$$

and

$$\mathrm{Hom}_R(R, \mathrm{Hom}_{\mathbb{Z}}(B, \mathbb{Q}/\mathbb{Z})) \cong \mathrm{Hom}_{\mathbb{Z}}(R \otimes_R B, \mathbb{Q}/\mathbb{Z}),$$

recall that $\mathbb{Q}/\mathbb{Z}$ is $\mathbb{Z}$-injective, we have surjective

$$\mathrm{Hom}_R(R, B^*) \longrightarrow \mathrm{Hom}_R(\mathfrak{a}, B^*) \longrightarrow 0.$$

□

Theorem 2.2.1 (Lazard Theorem)

Every flat R-module is a filtered colimit of free R-modules.

Proof See [Sta25, Theorem 058G]. □

Definition 2.2.4 (Torsion-free)

*An R-module M is said to be **torsion-free** if $rm = 0$ implies that either r is a zero-divisor in R or $m = 0$.*

Corollary 2.2.1

Let R be a PID. $M \in \mathbf{Mod}_R$ is flat if and only if it is torsion free.

Proof Let $\mathfrak{a} \subseteq R$ be an ideal, then $\mathfrak{a} = (r)$ for some $r \in R$. Then any element of $\mathfrak{a} \otimes_R M$ is of the form $r \otimes m$ for some $m \in B$. M is flat (Flatness test 2.2.8) if and only if the morphism

$$\mathfrak{a} \otimes_R M \longrightarrow M$$

is injective for all $\mathfrak{a}$ if and only if M is torsion-free. □

Summary

Let R be a PID. Then

$$\text{projective} \Longleftrightarrow \text{free} \Longrightarrow \text{flat} \Longleftrightarrow \text{torsion free},$$

$$\text{injective} \Longleftrightarrow \text{divisible}.$$

For counterexample of the converse of the arrows above, let $R = \mathbb{Z}$, the $\mathbb{Q}, \mathbb{Q}/\mathbb{Z}$ are injective, $\mathbb{Q}$ is flat over $\mathbb{Z}$ but not projective (Example 2.7), $\mathbb{Q}/\mathbb{Z}$ is even not flat. In $\mathbf{f.g.Mod}_R$ where R is a local ring, the category of finitely generated R-modules, then

$$\text{projective} \Longleftrightarrow \text{free} \Longleftrightarrow \text{flat} \Longleftrightarrow \text{torsion free},$$

you can see [Mat70, (3.G)] for the proof.

Finitely presented modules

Definition 2.2.5 (Finitely presented modules)

An R-module M is said to be ***finitely presented****, if there exists an exact sequence*

$$R^m \longrightarrow R^n \longrightarrow\!\!\!\!\!\rightarrow M \longrightarrow 0.$$

We study the relationship between flat modules and projective modules. We already see that: projective $\Rightarrow$ flat.

Lemma 2.2.3

Let R, S be rings. If M is a finitely presented (f.p.) left R-module and N is a (R, S)-bimodule, then

$$\sigma : N^* \otimes_R M \longrightarrow \mathrm{Hom}_R(M, N)^* \tag{2.1}$$

is an isomorphism.

Proof As M is f.p., then there are $m, n \in \mathbb{Z}$ such that

$$R^m \longrightarrow R^n \longrightarrow\!\!\!\!\!\rightarrow M \longrightarrow 0.$$

is an exact sequence of left R-modules.

If $M = R$, then as $N^* \otimes_R R \cong N^*$ and $\mathrm{Hom}_R(R, N) \cong N$ it follows that $\mathrm{Hom}_R(R, N)^* \cong N^* \otimes_R R$. Therefore

$$\begin{aligned} N^* \otimes_R R^m &\cong \bigoplus_{i=1}^m (N^* \otimes_R R) \\ &\cong \bigoplus_{i=1}^m \mathrm{Hom}_R(R, N)^* = \bigoplus_{i=1}^m \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_R(R, N), \mathbb{Q}/\mathbb{Z}) \\ &\cong \mathrm{Hom}_{\mathbb{Z}}\left(\bigoplus_{i=1}^m \mathrm{Hom}_R(R, N), \mathbb{Q}/\mathbb{Z}\right) \\ &\cong \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_R(R^m, N), \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}_R(R^m, N)^*. \end{aligned}$$

Now we apply $N^* \otimes_R \square$ to the exact sequence (2.2.3), then we get the following commutative diagram.

$$\begin{tikzcd} N^* \otimes_R R^m \arrow[r] \arrow[d, "\sim"'] & N^* \otimes_R R^n \arrow[r] \arrow[d, "\sim"'] & N^* \otimes_R M \arrow[r] \arrow[d, "\sigma"] & 0 \\ \mathrm{Hom}_R(R^m, N)^* \arrow[r] & \mathrm{Hom}_R(R^n, N)^* \arrow[r] & \mathrm{Hom}_R(M, N)^* \arrow[r] & 0, \end{tikzcd}$$

where the exactness of the bottom row follows by applying $\mathrm{Hom}_R(\square, N)$ and $\square^*$. By Five Lemma 1.5.3, $\sigma : N^* \otimes_R M \to \mathrm{Hom}_R(M, N)^*$ is an isomorphism. $\square$

Proposition 2.2.9

Any finitely presented flat R-module B is projective.

Proof Let $M \xrightarrow{f} N \longrightarrow 0$ be an exact sequence. We want to show

$$\mathrm{Hom}_R(B, M) \xrightarrow{f_*} \mathrm{Hom}_R(B, N) \longrightarrow 0$$

is exact. By Lemma 2.2.2, we have

$$0 \longrightarrow N^* \xrightarrow{f^*} M^*$$

is exact. Applying $\square \otimes_R B$, we find the commutative diagram.

$$\begin{tikzcd} 0 \arrow[r] & N^* \otimes_R B \arrow[r] \arrow[d, "\sim"] & M^* \otimes_R B \arrow[d, "\sim"] \\ 0 \arrow[r] & \operatorname{Hom}_R(B, N)^* \arrow[r] & \operatorname{Hom}_R(B, M)^* \end{tikzcd}$$

As M is flat than the top row is exact, and the vertical maps are isomorphisms by Lemma 2.2.3, so the bottom row is also exact. Thus by Lemma 2.2.2 we deduce that

$$\operatorname{Hom}_R(B, M) \xrightarrow{f_*} \operatorname{Hom}_R(B, N) \longrightarrow 0$$

is an exact sequence. Hence, B is projective. □

2.2.4 Projective resolution

Definition 2.2.6 (Projective resolution)

*Let $\mathscr{A}$ be an abelian category and $M \in \mathscr{A}$. A **left resolution of** M is a chain complex $P_\bullet$ with $P_i = 0$ for $i < 0$ together with a map $\varepsilon : P_0 \to M$ such that the complex*

$$\cdots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \longrightarrow 0$$

*is exact. It is a **projective resolution** if each P_i is projective.*

Remark Let $P_\bullet$ be a chain complex of projective objects with $P_i = 0$ for $i < 0$. Then $\varepsilon : P_\bullet \to M$ defines a resolution of M if and only if $\varepsilon : P_\bullet \to M[0]$ is a quasi-isomorphism.

Proof Chain map $\varepsilon : P_\bullet \to M[0]$ is quasi-isomorphism if and only if $H^n\varepsilon : H^n(P_\bullet) \to H^n(M)$ is isomorphism if and only if $\varepsilon : P_0/\operatorname{Im} d_1 \to M$ and $H^n\varepsilon : \operatorname{Ker} d_n/\operatorname{Im} d_{n+1} \to 0, \forall n \neq 0$ are isomorphisms if and only if the sequence

$$\cdots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \cong P_0/\operatorname{Im} d_1 \longrightarrow 0$$

is surjective. □

Definition 2.2.7 (Enough projective objects)

*Let $\mathscr{A}$ be an abelian category. We say that $\mathscr{A}$ has **enough projective objects** if for any $A \in \mathscr{A}$, there exists an epimorphism $P \twoheadrightarrow A$ with P projective.*

Lemma 2.2.4

Every $M \in \mathbf{Mod}_R$ has a projective resolution. More generally, if an abelian category $\mathscr{A}$ has enough projective objects, then every object M in $\mathscr{A}$ has a projective resolution.

Proof Take a surjective morphism $\varepsilon : P_0 \twoheadrightarrow M$ with P_0 projective (for example, let P_0 be the free module generated by a generator of M). Set $M_0 = \operatorname{Ker}\varepsilon$. Inductively, given M_{n-1}, take a projective module P_n and a surjective morphism $\varepsilon_n : P_n \twoheadrightarrow M_{n-1}$. Set $M_n = \operatorname{Ker}\varepsilon_n$ and let d_n be the composition of $P_n \twoheadrightarrow M_{n-1} \hookrightarrow P_{n-1}$. Then $(P_\bullet, d_\bullet)$ is a projective resolution of M.

If $\mathscr{A}$ has enough projective objects, use the proof as above, we get $\forall M \in \mathscr{A}$, M admits a projective resolution. □

Example 2.8 Let $R = k[x_1, x_2]$, then the Koszul complex (Example 2.2)

$$K_\bullet((x_1,x_2),R): \qquad 0 \longrightarrow R \xrightarrow{\begin{pmatrix}-x_2\\ x_1\end{pmatrix}} R\oplus R \xrightarrow{(x_1,x_2)} R \longrightarrow k \longrightarrow 0$$

provides a resolution of $k = H_0(K_\bullet) = R/(x_1,x_2)$.

Example 2.9 Let $R = k[x]/(x^2)$. The following complex (of finite length) gives a projective resolution of k:

$$\cdots \longrightarrow R \xrightarrow{\times x} R \xrightarrow{\times x} R \longrightarrow k \longrightarrow 0.$$

Remark We will see later that two projective resolutions in Example 2.8 and Example 2.9 are homotopy equivalent. This proves that a projective resolution need not be (left) bounded.

Theorem 2.2.2 (Comparison Theorem)

Let $\varepsilon : P_\bullet \to M$ be a chain complex with each P_i projective and $f : M \to N$ a map in abelian category $\mathscr{A}$. Then for every resolution $\eta : Q_\bullet \to N$ of N, there is a chain map $\check{f} : P_\bullet \to Q_\bullet$ lifting f such that the following diagram commutes.

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d, dashed, "\check{f}_2"'] & P_1 \arrow[r, "d_1"] \arrow[d, dashed, "\check{f}_1"] & P_0 \arrow[r, "\varepsilon"] \arrow[d, dashed, "\check{f}_0"] & M \arrow[r] \arrow[d, "f"] & 0 \\
\cdots \arrow[r] & Q_2 \arrow[r, "d'_2"'] & Q_1 \arrow[r, "d'_1"'] & Q_0 \arrow[r, "\eta"'] & N \arrow[r] & 0,
\end{tikzcd}$$

Moreover, the chain map $\check{f}$ is unique up to homotopic.

Proof Let $\check{f}_{-1} = f$, we construct $\check{f}_n, n \geq -1$ by induction. For each n, $\check{f}_{n-1}\circ d_n = d'_n \circ \check{f}_n$. Then $\check{f}_n$ induces $\check{f}'_n : Z_n(P_\bullet) \to Z_n(Q_\bullet)$. We have two commutative diagrams with the rows exact:

$$\begin{tikzcd}
0 \arrow[r] & Z_n(P_\bullet) \arrow[r, hook] \arrow[d, "\check{f}'_n"] & P_n \arrow[r, "d_n"] \arrow[d, "\check{f}_n"] & P_{n-1} \arrow[d, "\check{f}_{n-1}"] \\
0 \arrow[r] & Z_n(Q_\bullet) \arrow[r, hook] & Q_n \arrow[r, "d'_n"'] & Q_{n-1}
\end{tikzcd}$$

Consider the diagram:

$$\begin{tikzcd}
P_{n+1} \arrow[r, "d_{n+1}"] \arrow[d, dashed, "\exists \check{f}_{n+1}"'] & Z_n(P_\bullet) \arrow[d, "\check{f}'_n"] & \\
Q_{n+1} \arrow[r, "d'_{n+1}"'] & Z_n(Q_\bullet) \arrow[r] & 0
\end{tikzcd}$$

Since P_{n+1} is projective, we get $\check{f}_{n+1} : P_{n+1} \to Q_{n+1}$ such that $d'_{n+1}\circ \check{f}_{n+1} = \check{f}_n \circ d_{n+1} = \check{f}_n \circ d$. This complete our construction of $\check{f}$.

Suppose $g : P_\bullet \to Q_\bullet$ is another lift of f. Set $h = \check{f} - g$, we construct $s_n : P_n \to Q_{n+1}$ of h by induction (recall Def.2.1.5). Let $s_{-1} = 0$. If $n = 0$, note that $\eta\circ h_0 = \eta\circ(\check{f}_0 - g_0) = f\circ\varepsilon - f\circ\varepsilon = 0$, h_0 sends P_0 to $Z_0(Q_\bullet) = d'_1(Q_1)$:

$$\begin{tikzcd}
& P_0 \arrow[dl, dashed, "\exists s_0"'] \arrow[d, "h_0"] & \\
Q_1 \arrow[r] & Z_0(Q_\bullet) \arrow[r] & 0
\end{tikzcd}$$

Since P_0 is projective, we get $s_0 : P_0 \to Q_1$ such that $h_0 = d'_1\circ s_0 = d'\circ s_0 + s_{-1}\circ d$. Suppose given s_i,

$i < n$ and $h_{n-1} = d's_{n-1} + s_{n-2}d$.

$$\begin{tikzcd} P_n \arrow[r, "d_n"] \arrow[d, "h_n"'] & P_{n-1} \arrow[r, "d_{n-1}"] \arrow[d, "h_{n-1}"] \arrow[dl, "s_{n-1}"] & P_n \arrow[dl, "s_n"] \\ Q_n \arrow[r, "d'_n"'] & Q_{n-1} & \end{tikzcd}$$

Consider $h_n - s_{n-1}d : P_n \to Q_n$. We have $d'(h_n - s_{n-1}d) = d'h_n - (h_{n-1} - s_{n-2}d)d = 0$. Then $h_n - s_{n-1}d$ sends P_n to $Z_n(Q_\bullet) = d'(Q_{n+1})$:

$$\begin{tikzcd} & P_n \arrow[dl, dashed, "\exists s_n"'] \arrow[d, "h_n - s_{n-1}d"] & \\ Q_{n+1} \arrow[r] & Z_n(Q_\bullet) \arrow[r] & 0 \end{tikzcd}$$

Since P_n is projective, we get $s_n : P_n \to Q_{n+1}$ such that $h_n = d's_n + s_{n-1}d$. □

Corollary 2.2.2

Projective resolutions are unique up to chain homotopy equivalence.

Proof Pick $\varepsilon : P_\bullet \to M, \eta : Q_\bullet \to M$ be two projective resolutions, $N = M$, $f = \mathrm{id}_M$ in Comparison Theorem 2.2.2, we get a chain map $g : P_\bullet \to Q_\bullet$ lifting id_M. Similarly, we have a chain map $h : Q_\bullet \to P_\bullet$ lifting id_M. So $h \circ g : P_\bullet \to P_\bullet$ lifting id_M. Since $h \circ g$ is unique up to homotopic, we have $h \circ g \sim \mathrm{id}_{P\bullet}$, recall Def.2.1.5, we know that g is chain homotopy equivalence. □

Here is another way to construct projective resolutions. It is called the Horseshoe Lemma because we are required to fill in the horseshoe-shaped diagram

Lemma 2.2.5 (Horseshoe Lemma)

Suppose given a commutative diagram

$$\begin{tikzcd} & & & & 0 \arrow[d] & \\ \cdots \arrow[r] & P_2' \arrow[r] & P_1' \arrow[r] & P_0' \arrow[r, "\varepsilon'"] & A' \arrow[r] \arrow[d, "i_A"] & 0 \\ & & & & A \arrow[d, "\pi_A"] & \\ \cdots \arrow[r] & P_2'' \arrow[r] & P_1'' \arrow[r] & P_0'' \arrow[r, "\varepsilon''"] & A'' \arrow[r] \arrow[d] & 0 \\ & & & & 0 & \end{tikzcd}$$

where the column is exact and the rows are projective resolutions. Set $P_n = P_n' \oplus P_n''$. Then the P_n assemble to form a projective resolution P of A, and right-hand column lifts to an exact sequence of complexes

$$0 \longrightarrow P_\bullet' \xrightarrow{i} P_\bullet \xrightarrow{\pi} P_\bullet'' \longrightarrow 0,$$

where $i_n : P_n' \to P_n$ and $\pi_n : P_n \to P_n''$ are the natural inclusion and projection, respectively.

Proof Lift ε'' to a map $\widetilde{\varepsilon''} : P_0'' \to A$, the direct sum of this with the map $i_A \circ \varepsilon' : P_0' \to A$ gives a map

$\varepsilon = (i_A \circ \varepsilon', \widetilde{\varepsilon''}) : P_0 \to A$. Then we get a commutative diagram below.

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \operatorname{Ker}\varepsilon' \arrow[r] \arrow[d] & P_0' \arrow[r, "\varepsilon'"] \arrow[d, "i_0"'] \arrow[dr, dashed, "i_A\circ\varepsilon'"] & A' \arrow[r] \arrow[d, "i_A"] & 0 \\
0 \arrow[r] & \operatorname{Ker}\varepsilon \arrow[r] \arrow[d] & P_0 \arrow[r, dashed, "\varepsilon"] \arrow[d, "\pi_0"'] & A \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \operatorname{Ker}\varepsilon'' \arrow[r] \arrow[d] & P_0'' \arrow[r, "\varepsilon''"'] \arrow[d] \arrow[ur, dashed, "\widetilde{\varepsilon''}"] & A'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd} \tag{2.2}$$

The right two columns of (2.2) are short exact sequences, by Snake Lemma 1.5.4, the left column is exact and $\operatorname{Coker}\varepsilon = 0$. So P_0 maps onto A. This finishes the initial step and brings us to the situation

$$\begin{tikzcd}
 & & 0 \arrow[d] & \\
\cdots \arrow[r] & P_1' \arrow[r, "d'"] & \operatorname{Ker}\varepsilon' \arrow[r] \arrow[d] & 0 \\
 & & \operatorname{Ker}\varepsilon \arrow[d] & \\
\cdots \arrow[r] & P_1'' \arrow[r, "d''"] & \operatorname{Ker}\varepsilon'' \arrow[r] \arrow[d] & 0 \\
 & & 0 &
\end{tikzcd}$$

The filling in of the "horseshoe" now proceeds by induction. $\square$

2.2.5 Injective resolution

Proposition 2.2.10

Let $F : \mathscr{A} \to \mathscr{B}$ be additive functors between abelian categories.

(1) If $F \dashv G$ and G is right exact, then F maps projective objects to projective objects.

(2) If $G \dashv F$ and G is left exact, then F maps injective objects to injective objects.

Proof We only need to show (2). The proof of (1) is similar.

Let $I \in \mathscr{A}$ be injective. Recall Prop.2.2.1, it suffices to show that $\operatorname{Hom}_{\mathscr{B}}(\square, F(I))$ is exact. Let $f : B \to B'$ be a monomorphism in $\mathscr{B}$. Since $G \dashv F$, we have the commutative diagram:

$$\begin{tikzcd}
\operatorname{Hom}_{\mathscr{B}}(B', F(I)) \arrow[r, "f^*"] \arrow[d, "\sim"'] & \operatorname{Hom}_{\mathscr{B}}(B, F(I)) \arrow[d, "\sim"] \\
\operatorname{Hom}_{\mathscr{A}}(G(B'), I) \arrow[r, "(Gf)^*"] & \operatorname{Hom}_{\mathscr{A}}(G(B), I)
\end{tikzcd}$$

Since G is left exact and I is injective, we know $(Gf)^*$ is an epimorphism. From the commutative diagram above, f^* is an epimorphism. Then $\operatorname{Hom}_{\mathscr{B}}(\square, F(I))$ is exact. $\square$

Definition 2.2.8 (Enough injective objects)

*Let $\mathscr{A}$ be an abelian category. We say that $\mathscr{A}$ has **enough injective objects** if for any $A \in \mathscr{A}$, there exists a monomorphism $A \hookrightarrow I$ with I injective.*

Theorem 2.2.3 (Enough injective in Mod_R)

Let $R \in \mathbf{CRings}$, then every R-module M is a submodule of an injective module. That is, $\mathbf{Mod}_R$ has enough injective objects.

Proof **Step 1.** Assume $R = \mathbb{Z}$, $M \in \mathbf{Mod}_{\mathbb{Z}}$, choose a free $\mathbb{Z}$-module F and a surjection $F \twoheadrightarrow M$ with kernel K, write $F = \mathbb{Z}^{(I)}$ and $D = \mathbb{Q}^{(I)}$, then $M \cong F/K \subseteq D/K$ and D/K is divisible (since the quotient of abelian divisible group is also divisible), by Prop.2.2.4, D/K is injective.

Step 2. For general R, the functor

$$\mathrm{Hom}_{\mathbb{Z}}(R, \square) : \mathbf{Mod}_{\mathbb{Z}} \longrightarrow \mathbf{Mod}_R$$

admits a left adjoint, namely the restriction of scalars functor $\mathbf{Mod}_R \to \mathbf{Mod}_{\mathbb{Z}}$, which is exact. Thus $\mathrm{Hom}_{\mathbb{Z}}(R, \square)$ maps injective $\mathbb{Z}$-modules to injective R-modules by Prop.2.2.10 (2).

Let M be a left R-module. We embed the underlying $\mathbb{Z}$-module of M into an injective $\mathbb{Z}$-module I. Then $\mathrm{Hom}_{\mathbb{Z}}(R, I)$ is an injective R-module and we have injective homomorphisms

$$M \cong \mathrm{Hom}_R(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, I).$$

This finishes the proof. $\square$

Definition 2.2.9 (Injective resolution)

*Let $\mathscr{A}$ be an abelian category and $M \in \mathscr{A}$. A **right resolution of** A is a cochain complex $(I^\bullet, d^\bullet)$ with $I^i = 0$ for $i < 0$ and a morphism $A \to I^0$ such that the complex*

$$0 \longrightarrow A \longrightarrow I^0 \longrightarrow I^1 \longrightarrow \cdots$$

*is exact. It is called an **injective resolution** if each I^i is injective ($\Leftrightarrow A[0] \to I^\bullet$ is a quasi-isomorphism).*

Example 2.10 Let $\mathscr{A} = \mathbf{Mod}_R$, then $\forall M \in \mathscr{A}$ admits an injective resolution by Thm.2.2.3.

Theorem 2.2.4 (Comparison Theorem)

Let $A \to I^\bullet$ be a complex morphism with I^i injective and $f : A' \to A$ a morphism in $\mathscr{A}$. Then for any resolution $A' \to J^\bullet$, there is a cochain morphism $\check{f} : J^\bullet \to I^\bullet$ lifting f, which is unique up to homotopic. In particular, injective resolutions are unique up to cochain homotopy.

Proof Similar to the proof of Thm.2.2.2 and Cor.2.2.2. $\square$

Theorem 2.2.5 ($\mathrm{Shv}(X)$ has enough injective objects)

Let X be a topological space. Then $\mathbf{Shv}(X)$, the category of sheaves of abelian groups on X has enough injective objects.

Proof We give a sketch of proof.

Step 1. $\forall x \in X$, consider the taking stalk functor:

$$(\square)_x : \mathbf{Shv}(X) \longrightarrow \mathbf{Ab}, \quad \mathscr{F} \longmapsto \mathscr{F}_x := \varinjlim_{U \ni x} \mathscr{F}(U).$$

This is an exact functor. Moreover, $\mathscr{F} = 0 \Longleftrightarrow \forall x \in X, \mathscr{F}_x = 0$.

Step 2. $\forall x \in X, A \in \mathbf{Ab}$. Consider the skyscraper sheaf $x_* A \in \mathbf{Shv}(X)$:

$$(x_*A)(U) = \begin{cases} A, & x \in U; \\ 0, & x \notin U. \end{cases}$$

And for $U \subseteq U'$,

$$\mathrm{res}_{U',U} : (x_*A)(U') \longrightarrow (x_*A)(U) = \begin{cases} \mathrm{id}_A, & x \in U \\ 0, & x \notin U. \end{cases}$$

Then $x_*\Box : \mathbf{Ab} \to \mathbf{Shv}(X)$ is a functor and $(\Box)_* \dashv x_*\Box$. From Prop.2.2.10 (2), $x_*\Box$ maps injective abelian groups to injective objects in $\mathbf{Shv}(X)$.

Step 3. $\forall \mathscr{F} \in \mathbf{Shv}(X), \forall x \in X$, take injection $\mathscr{F}_x \hookrightarrow \mathscr{I}_x$ with $\mathscr{I}_x$ injective. Then we get a natural morphism $\mathscr{F} \to x_*\mathscr{F}_x \to x_*\mathscr{I}_X$ which induces

$$\mathscr{F} \longrightarrow \mathscr{I} := \prod_{x \in X} x_*\mathscr{I}.$$

Then $\mathscr{I}$ is injective since the product of injective sheaves is injective, and the map is an injection since it is so at each stalk. $\square$

Remark

(1) In general, $\mathbf{Shv}(X)$ does not have enough projective objects. In fact, if X is locally connected, then $\mathbf{Shv}(X)$ has enough projective objects if and only if X is an Alexandrov space, i.e. any (infinite) intersection of open subsets is open.

(2) More generally, any Grothendieck abelian category(e.g., $\mathbf{Mod}_R$, $\mathbf{Shv}(X)$) has enough injective objects, cf. [Gro57].

2.3 δ-functors and derived functors

Philosophy

Let $\mathscr{A}$ and $\mathscr{B}$ be abelian categories and let $F : \mathscr{A} \to \mathscr{B}$ be a left exact functor. For any short exact sequence

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

in $\mathscr{A}$, we have, by the left exactness of F, an exact sequence

$$0 \longrightarrow FX \longrightarrow FY \longrightarrow FZ$$

in $\mathscr{B}$. Under suitable conditions, we can define additive functors $R^nF : \mathscr{A} \to \mathscr{B}, i \geq 1$, call the **right derived functors of** F, such that the exact sequence in $\mathscr{B}$ extends to a long exact sequence

$$0 \longrightarrow FX \longrightarrow FY \longrightarrow FZ \longrightarrow$$

$$\longrightarrow R^1FX \longrightarrow R^1FY \longrightarrow R^1FZ \longrightarrow \cdots$$

$$\cdots \longrightarrow R^nFX \longrightarrow R^nFY \longrightarrow R^nFXZ \longrightarrow \cdots.$$

Roughly speaking, the right derived functors measure the lack of right exactness of F. The functors can be assembled into one single functor $RF : D^+(\mathscr{A}) \to D^+(\mathscr{B})$ between derived categories.

2.3.1 δ-functors

Definition 2.3.1 (Covariant (co)homological δ-functor)

*A **covariant cohomological δ-functor** (T^i, δ^i) from $\mathscr{A}$ to $\mathscr{B}$ is a collection of additive functors $T^n : \mathscr{A} \to \mathscr{B}$ for $n \geq 0$, together with morphisms $\delta^n : T^n(C) \to T^{n+1}(A)$ defined for each short exact sequence $0 \to A \to B \to C \to 0$ in $\mathscr{A}$, satisfying the following condition:*

(1) For any short exact sequence as above, there is a long exact sequence:

$$\cdots \longrightarrow 0 \longrightarrow T^0(A) \longrightarrow T^0(B) \longrightarrow$$

$$\longrightarrow T^0(C) \xrightarrow{\delta^0} T^1(A) \longrightarrow T^1(B) \longrightarrow \cdots$$

$$\cdots \longrightarrow T^n(C) \xrightarrow{\delta^n} T^{n+1}(A) \longrightarrow T^{n+1}(B) \longrightarrow \cdots$$

(2) For any morphism of short exact sequences

$$\begin{tikzcd} 0 \arrow[r] & A' \arrow[r] \arrow[d] & B' \arrow[r] \arrow[d] & C' \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0 \end{tikzcd}$$

there is a commutative diagram:

$$\begin{tikzcd} T^n(C') \arrow[r, "\delta^n"] \arrow[d] & T^{n+1}(A') \arrow[d] \\ T^n(C) \arrow[r, "\delta^n"] & T^{n+1}(A) \end{tikzcd}$$

A morphism $(T'^i, \delta'^i) \to (T^i, \delta^i)$ of δ-functors is a system of natural transformations $T'^n \to T^n$ that commute with δ.

*Similarly, we can define a **covariant homological δ-functor** from $\mathscr{A}$ to $\mathscr{B}$ be a collection of additive functors $T = (T_n)_{n\geq 0}$, $T_n : \mathscr{A} \to \mathscr{B}$, together with morphisms $\delta_n : T_n(C) \to T_{n-1}(A)$ defined for any short exact sequence $0 \to A \to B \to C \to 0$ with similar imposed condition as above.*

Remark By definition, T^0 is left exact and T_0 is right exact.

Example 2.11 Cohomologies gives a cohomological δ functor $H^\bullet$ from $C^{\geq 0}(\mathscr{A})$ to $\mathscr{A}$.

Example 2.12 For $n \in \mathbb{N}$, define $T_k : \mathbf{Ab} \to \mathbf{Ab}$, $k \geq 0$ as

$$T_0(A) = A/nA, \quad T_1(A) = A[n] := \{a \in A : na = 0\}, \quad T_k = 0, k \geq 2.$$

This forms a homological δ functor from $\mathbf{Ab}$ to $\mathbf{Ab}$. Actually, for any short exact sequence $0 \to A \to B \to C \to 0$, applying the Snake Lemma to

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, "\times n"] & B \arrow[r] \arrow[d, "\times n"] & C \arrow[r] \arrow[d, "\times n"] & 0 \\ 0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0 \end{tikzcd}$$

we get an exact sequence

$$0 \longrightarrow A[n] \longrightarrow B[n] \longrightarrow C[n] \longrightarrow$$

$$\longrightarrow A/nA \longrightarrow B/nB \longrightarrow C/nC \longrightarrow 0.$$

Definition 2.3.2 (Universal)

A cohomological δ-functor (T^i, δ^i) is ***universal*** *if for any other δ-functor (T'^i, δ'^i), and any natural transformation of functors $\alpha : T^0 \to T'^0$, there is a unique morphism of δ-functors $(T^i, \delta^i) \to (T'^i, \delta'^i)$ extending α. Similarly, we can define universal homological δ-functors.*

Remark Definition 2.3.2 implies, a universal δ-functor with $T^0 = F$ (resp., $T_0 = F$) if exists, is unique up to unique isomorphism.

Definition 2.3.3 (Effaceable, coeffaceable)

An additive functor $F : \mathscr{A} \to \mathscr{B}$ is ***effaceable*** *if for each object $A \in \mathscr{A}$, there exists a monomorphism $u : A \to I$ for some I such that $F(u) = 0$. Dually, it is* ***coeffaceable*** *if for each $A \in \mathscr{A}$, there exists an epimorphism $u : P \to A$ for some P such that $F(u) = 0$.*

The key fact about universal δ-functors is the following.

Theorem 2.3.1

Let (T^i, δ^i) be a cohomological δ-functor from $\mathscr{A}$ to $\mathscr{B}$. If T^i is effaceable for each $i > 0$, then T is universal. If (T_i, δ_i) is a homological δ-functor such that T_i is coeffaceable for each $i > 0$, then T is universal.

Proof Let (S^i, δ'^i) be a δ-functor from $\mathscr{A}$ to $\mathscr{B}$ and let $\alpha^0 : T^0 \to S^0$ be a morphism of functors. We want to show that there exists a unique morphism of δ-functors $\{\alpha^n : T^n \to S^n\}_{n\geq 0}$. We construct α^n by induction on n. Suppose we have already constructed a unique sequence of transformation of functors α^i for $i \leq n$ compatible with the maps δ in degree $\leq n$.

Let $A \in \operatorname{obj}(\mathscr{A})$. By assumption we may choose an embedding $u : A \to I$ such that $T^{n+1}(u) = 0$. Let $M = \operatorname{Coker}(u : A \to I)$ such that we have a short exact sequence $0 \to A \to I \to M \to 0$. The long exact sequence attached to the δ-functor T gives

$$T^n(I) \longrightarrow T^n(M) \longrightarrow T^{n+1}(A) \longrightarrow 0.$$

By induction, we already have morphisms α^n,

$$\begin{tikzcd} T^n(I) \arrow[r] \arrow[d, "\alpha^n_I"] & T^n(M) \arrow[r] \arrow[d, "\alpha^n_M"] & T^{n+1}(A) \arrow[r] \arrow[d, dashed] & 0 \\ S^n(I) \arrow[r] & S^n(M) \arrow[r] & S^{n+1}(A) & \end{tikzcd}$$

and we define $\alpha^{n+1}_A : T^{n+1}(A) \to S^{n+1}(A)$ to be the unique morphism making the above diagram commute (by the universal property of epimorphism).

First, we check that α^{n+1}_A does not depend on the choice of effacement. Indeed, let $u' : A \to I'$ be another effacement and let $M' = \operatorname{Coker}(u' : A \to I')$. It induces a morphism

$$\alpha^{n+1}_{A,u'} : T^{n+1}(A) \longrightarrow S^{n+1}(A)$$

and we need to show $\alpha^{n+1}_{A,u} = \alpha^{n+1}_{A,u'}$. In general, we cannot lift id_A to a morphism $I \to I'$. But the universal property of a pushout provides a diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & I \arrow[d] \\ I' \arrow[r] & I'', \end{tikzcd}$$

where $I'' := I \sqcup_A I'$. It is clear that the map $u'' : A \to I''$ is also a monomorphism, and an effacement. So the exact sequence $0 \to A \to I'' \to M'' \to 0$ also allows to define a morphism

$$\alpha^{n+1}_{A,u''} : T^{n+1}(A) \longrightarrow S^{n+1}(A).$$

It suffices to check

$$\alpha^{n+1}_{A,u} = \alpha^{n+1}_{A,u''} = \alpha^{n+1}_{A,u'}.$$

Consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, equal] & I \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & A \arrow[r] & I'' \arrow[r] & M'' \arrow[r] & 0
\end{tikzcd}$$

(where $M \to M''$ is induced by universal property of $\operatorname{Coker} u$) which induces

$$\begin{tikzcd}
\cdots \arrow[rr] & & S^n M \arrow[rr] \arrow[dd] & & S^{n+1} A \arrow[dd, equal] \\
0 \arrow[r] & T^n M \arrow[ur, "\alpha^n_{M,u}"] \arrow[rr] \arrow[dd] & & T^{n+1} A \arrow[ur, "\alpha^{n+1}_{A,u}"] \arrow[dd, equal] & \\
\cdots \arrow[rr] & & S^n M'' \arrow[rr] & & S^{n+1} A \\
0 \arrow[r] & T^n M'' \arrow[ur, "\alpha^n_{M'',u''}"] \arrow[rr] & & T^{n+1} A \arrow[ur, "\alpha^{n+1}_{A,u''}"] &
\end{tikzcd}$$

showing that $\alpha^{n+1}_{A,u} = \alpha^{n+1}_{A,u''}$. Similarly, $\alpha^{n+1}_{A,u'} = \alpha^{n+1}_{A,u''}$ which shows that α^{n+1}_{A} is well-defined, independent of the choice of effacement.

Next, we also need to check α^{n+1} is a natural transformation from $T^{n+1} \to S^{n+1}$. Namely, if $A \to A'$ is a morphism in $\mathscr{A}$, then there is a commutative diagram

$$\begin{tikzcd}
T^{n+1} A \arrow[r] \arrow[d] & S^{n+1} A \arrow[d] \\
T^{n+1} A' \arrow[r] & S^{n+1} A'.
\end{tikzcd}$$

The idea of the proof is similar to the above, so we omit it. $\square$

2.3.2 Left derived functors

Let $\mathscr{A}, \mathscr{B}$ be abelian categories, $F : \mathscr{A} \to \mathscr{B}$ is a (covariant) right exact additive functor. Assume that $\mathscr{A}$ has enough projective objects.

Definition 2.3.4 (Left derived functors)

*The **left derived functors** $(L_i F) : \mathscr{A} \to \mathscr{B}$, $i > 0$ of F are defined as follows: $\forall A \in \mathscr{A}$, take a projective resolution $P_\bullet \to A$ and set $L_i F(A) := H_i(F(P_\bullet))$.*

Since F is right exact, we have $L_0 F(A) = F(A)$ (denote $\varepsilon : P_0 \twoheadrightarrow A$, by exactness, $\operatorname{Im} F d_1 = \operatorname{Ker} F\varepsilon$, so $L_0 F(A) = H_0(F(P_\bullet)) = F(P_0)/\operatorname{Ker} F\varepsilon \cong F(A)$).

The following lemma shows the functor $L_i F$ is well-defined.

Lemma 2.3.1

(1) $\forall A \in \mathscr{A}$, $L_iF(A) \in \mathscr{B}$ is well-defined up to natural isomorphism, i.e., if $Q_\bullet \to A$ is another projective resolution, then there exists a canonical isomorphism:

$$L_iF(A) = H_i(F(P_\bullet)) \cong H_i(F(Q_\bullet)).$$

(2) If $f : A \to A'$ is a morphism, then there exists a natural morphism $L_iF(f) : L_iF(A) \to L_iF(A')$.

Proof

(1) Comparison Theorem 2.2.2 shows there exists a morphism of complexes $f : P_\bullet \to Q_\bullet$, $g : Q_\bullet \to P_\bullet$ which are chain homotopy equivalence. Lifting id_A, $gf \sim \mathrm{id}_{P_\bullet}$ and $fg \sim \mathrm{id}_{Q_\bullet}$, so we get

$$FgFf \sim \mathrm{id}_{FP_\bullet}, \quad FfFg \sim \mathrm{id}_{FQ_\bullet}.$$

By Proposition 2.1.2, this implies $H_i(F(P_\bullet)) \cong H_i(F(Q_\bullet))$.

(2) Let $P_\bullet \to A$, $P'_\bullet \to A'$ be the chosen projective resolutions. Comparison Theorem 2.2.2 shows that there exists a lifting $\widetilde{f} : P_\bullet \to P'_\bullet$ which induces $f_* : H_i(F(P_\bullet)) \to H_i(F(P'_\bullet))$. Any other lift is chain homotopic to $\widetilde{f}$, this implies $f_* = L_iF(f)$ is independent of the choice of $\widetilde{f}$.

□

Corollary 2.3.1

If A is projective, then A is F-acyclic, i.e., $L_iF(A) = 0$, $\forall i > 0$.

Lemma 2.3.2

Each L_iF is an additive functor from $\mathscr{A}$ to $\mathscr{B}$.

Proof Let $f, g : A \to B$ in $\mathscr{A}$, $\varepsilon : P_\bullet \to A$ and $\eta : Q_\bullet \to B$ are projective resolutions, by Comparison Theorem 2.2.2, there exists liftings $f_\bullet, g_\bullet : P_\bullet \to Q_\bullet$. It is easy to check that $f_\bullet + g_\bullet$ is the lifting of $f + g$. So

$$\begin{aligned} L_iF(f+g) =& H_i(F(f_\bullet + g_\bullet)) \\ =& H_i(F(f_\bullet) + F(g_\bullet)) \\ =& H_i(F(f_\bullet)) + H_i(F(g_\bullet)) \\ =& L_iF(f) + L_iF(g), \end{aligned}$$

i.e., L_iF is an additive functor. □

Theorem 2.3.2

The left derived functors $(L_iF)_{i\geq 0}$ form a universal homological δ-functor.

Proof Let $A \in \mathscr{A}$, then there exists a projective object such that $u : P \twoheadrightarrow A$. By Lemma 2.3.1 (2), there exists a natural morphism $L_iF(u) : L_iF(P) \to L_iF(A)$. Since P is projective, by Corollary 2.3.1, we know that $L_iF(P) = 0$, so $L_iF(u) = 0$, hence L_iF is coeffaceable. We next show that (L_iF) form a δ-functor.

Given a short exact sequence $0 \to A' \to A \to A'' \to 0$, choose projective resolutions $P'_\bullet \to A'$ and $P''_\bullet \to A''$. By the Horseshoe Lemma 2.2.5, there is a projective resolution $P_\bullet \to A$ fitting into a short exact sequence

$$0 \longrightarrow P'_\bullet \longrightarrow P_\bullet \longrightarrow P''_\bullet \longrightarrow 0$$

of projective complexes in $\mathscr{A}$. Since the P''_n are projective, each sequence $0 \to P'_n \to P_n \to P''_n \to 0$ is split exact. As F is additive, the induced sequence

$$0 \longrightarrow F(P'_n) \longrightarrow F(P_n) \longrightarrow F(P''_n) \longrightarrow 0$$

is also split exact in $\mathscr{B}$. Therefore,

$$0 \longrightarrow F(P'_\bullet) \longrightarrow F(P_\bullet) \longrightarrow F(P''_\bullet) \longrightarrow 0$$

is a short exact sequence of chain complexes. By Fundamental Theorem of Homological Algebra 2.1.1, we get

$$\cdots \longrightarrow H_iF(P'_\bullet) \longrightarrow H_iF(P_\bullet) \longrightarrow H_iF(P''_\bullet) \xrightarrow{\quad\delta_i\quad}$$

$$\xrightarrow{\quad\delta_i\quad} H_{i-1}F(P'_\bullet) \longrightarrow H_{i-1}F(P_\bullet) \longrightarrow H_{i-1}F(P''_\bullet) \longrightarrow \cdots,$$

that is,

$$\cdots \longrightarrow L_iF(A') \longrightarrow L_iF(A) \longrightarrow L_iF(A'') \xrightarrow{\quad\delta_i\quad}$$

$$\xrightarrow{\quad\delta_i\quad} L_{i-1}F(A') \longrightarrow L_{i-1}F(A) \longrightarrow L_{i-1}F(A'') \longrightarrow \cdots$$

this verifies the first condition of a δ-functor.

To see the naturality of the δ_i, assume we are given a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & A' \arrow[r, "i_A"] \arrow[d, "f'"'] & A \arrow[r, "\pi_A"] \arrow[d, "f"'] & A'' \arrow[r] \arrow[d, "f''"] & 0 \\ 0 \arrow[r] & B' \arrow[r, "i_B"] & B \arrow[r, "\pi_B"] & B'' \arrow[r] & 0 \end{tikzcd}$$

in $\mathscr{A}$, and projective resolution of the corners:

$$\varepsilon' : P'_\bullet \to A', \quad \varepsilon'' : P''_\bullet \to A'', \quad \eta' : Q'_\bullet \to B', \quad \eta'' : Q''_\bullet \to B''.$$

Use the Horseshoe Lemma 2.2.5 to get projective resolutions

$$\varepsilon : P_\bullet \longrightarrow A, \quad \eta : Q_\bullet \longrightarrow B,$$

where $P_n \cong P'_n \oplus P''_n$ and $Q_n \cong Q'_n \oplus Q''_n$. Use the Comparison Theorem 2.2.2 to obtain chain maps

$$\widetilde{f}'_\bullet : P'_\bullet \longrightarrow Q'_\bullet, \quad \widetilde{f}''_\bullet : P''_\bullet \longrightarrow Q''_\bullet$$

lifting the maps f' and f'', respectively. We shall show that there is also a chain map $\widetilde{f}_\bullet : P_\bullet \to Q_\bullet$ lifting f, and giving a commutative diagram of chain complexes with exact rows

$$\begin{tikzcd} 0 \arrow[r] & P'_\bullet \arrow[r] \arrow[d, "\widetilde{f}'_\bullet"'] & P_\bullet \arrow[r] \arrow[d, dashed, "\widetilde{f}_\bullet"] & P''_\bullet \arrow[r] \arrow[d, "\widetilde{f}''_\bullet"] & 0 \\ 0 \arrow[r] & Q'_\bullet \arrow[r] & Q_\bullet \arrow[r] & Q''_\bullet \arrow[r] & 0 \end{tikzcd}$$

The naturality of the connecting morphism in the long exact sequence then translates into the naturality of δ. We will construct inductively such an $\widetilde{f}_n : P_n \to Q_n$ of the form

$$\widetilde{f}_n = \begin{pmatrix} \widetilde{f}'_n & \gamma_n \\ 0 & \widetilde{f}''_n \end{pmatrix} : \begin{matrix} P'_n \\ \oplus \\ P''_n \end{matrix} \longrightarrow \begin{matrix} Q'_n \\ \oplus \\ Q''_n \end{matrix}$$

for suitable maps $\gamma_n : P''_n \to Q'_n$.

Let us look at $\widetilde{f}_0 : P_0' \oplus P_0'' \to Q_0' \oplus Q_0''$, as follows

$$\begin{array}{ccc} P_0' \oplus P_0'' & \xrightarrow{\varepsilon} & A \\ {\scriptstyle \widetilde{f}_0}\big\downarrow & & \big\downarrow{\scriptstyle f} \\ Q_0' \oplus Q_0'' & \xrightarrow{\eta} & B, \end{array}$$

where $\widetilde{f}_0 = \begin{pmatrix} \widetilde{f}_0' & \gamma_0 \\ 0 & \widetilde{f}_0'' \end{pmatrix}$, and we need $f\varepsilon = \eta\widetilde{f}_0$. On P_0', this is clear because it becomes $f'\varepsilon' = \eta'\widetilde{f}_0'$ (see (2.2)) and by construction $\widetilde{f}'$ lifts f'. On P_0'' this requires

$$f \circ (\varepsilon|_{P_0''}) = \eta \circ (\gamma_0 + \widetilde{f}_0'') = \eta'\gamma_0 + \eta|_{Q_0''}\widetilde{f}_0''. \tag{2.3}$$

Let $\beta := f \circ (\varepsilon|_{P_0''}) - eta|_{Q_0''}\widetilde{f}_0'' : P_0'' \to B$; if β takes values in B', then the projectivity of P_0'' will provide us a map γ_0, satisfying (2.3)

$$\begin{tikzcd} & P_0'' \arrow[dl, dashed, "\gamma_0"'] \arrow[d, "\beta"] & \\ Q_0' \arrow[r, "\eta'"] & B' \arrow[r] & 0 \end{tikzcd}$$

So we only need to check $\pi_B \circ \beta = 0$, which is true:

$$\pi_B \circ \beta = \pi_B \circ f \circ \varepsilon|_{P_0''} - \pi_B \circ \eta|_{Q_0''} \circ \widetilde{f}_0'' = f''\pi_A\varepsilon|_{P_0''} - \pi_B\eta|_{Q_0''}\widetilde{f}_0'' = f''\varepsilon - \eta''\widetilde{f}_0'' = 0.$$

The general case can be done in a similar way. Then we proved (L_iF) form a δ-functor. By Theorem 2.3.1, we are done. $\square$

Remark Dimension shifting. If $0 \to M \to P \to A \to 0$ is exact with P projective or F-acyclic. Then the long exact sequence

$$\cdots \longrightarrow L_iF(P) \longrightarrow L_iF(A) \xrightarrow{\quad\delta_i\quad}$$

$$\xrightarrow{\quad\delta_i\quad} L_{i-1}F(M) \longrightarrow L_{i-1}F(P) \longrightarrow \cdots$$

shows that

$$L_iF(A) \cong L_{i-1}F(M)$$

for $i > 2$ and that $L_1F(A)$ is the kernel of $F(M) \to F(P)$. More generally, if

$$0 \longrightarrow M_m \longrightarrow P_m \longrightarrow \cdots \longrightarrow P_0 \longrightarrow A \longrightarrow 0$$

is exact with the P_i projective or F-acyclic, then $L_iF(A) \cong L_{i-m-1}F(M_m)$ for $i \geq m+2$, and $L_{m+1}F(A)$ is the kernel of $F(M_m) \to F(P_m)$.

2.3.3 Right derived functors

Let $\mathscr{A}, \mathscr{B}$ be abelian categories and $F : \mathscr{A} \to \mathscr{B}$ be a left exact additive functor. Assume that $\mathscr{A}$ has enough injective objects.

Definition 2.3.5 (Right derived functors)

*The **right derived functors** $(R^iF) : \mathscr{A} \to \mathscr{B}$, $i \geq 0$ of F are defined as follows: $\forall A \in \mathscr{A}$, choose an injective resolution: $A \to I^\bullet$ and set $R^iF(A) = H^i(F(I^\bullet))$.*

Theorem 2.3.3

The objects $R^iF(A)$ are independent of the choice of injective resolutions. $R^iF(I^\bullet) = 0$ for $i \geq 1$ if I injective. The collection $(R^iF)_{i\geq 0}$ form a universal cohomological δ-functor.

Proof Consider $F^{\mathrm{op}} : \mathscr{A}^{\mathrm{op}} \to \mathscr{B}^{\mathrm{op}}$ which is right exact. The category $\mathscr{A}^{\mathrm{op}}$ has enough projective objects. $I^\bullet$ becomes a projective resolution of A in $\mathscr{A}^{\mathrm{op}}$. Then $R^iF(A) \cong (L_iF^{\mathrm{op}})^{\mathrm{op}}(A)$. So all the results about right exact functors apply to left exact functors. □

Example 2.13 Let X be a topological space. $\Gamma = \Gamma(X, \square) : \mathbf{Shv}(X) \to \mathbf{Ab}$ is left exact. The right derived functors of Γ are the so-called cohomology functors on X: $\mathscr{F} \in \mathbf{Shv}(X)$, $H^i(X, \mathscr{F}) := R^i\Gamma(X, \mathscr{F})$.

Let $F : \mathscr{A} \to \mathscr{B}$ be a contravariant left exact functor then $F' : \mathscr{A}^{\mathrm{op}} \to \mathscr{B}$ is a covariant left exact functor. If $\mathscr{A}$ has enough projective objects, we can define right derived functor: $R^iF(A) = H^iF(P_\bullet)$ where $P_\bullet \to A$ is a projective resolution which is a universal δ-functor. Similarly, if $\mathscr{A}$ has enough injective objects, one can define left derived functors for contravariant right exact functors.

2.4 Tor

2.4.1 Tor_n

Given a right R-module A and a left R-module B, we have two ways to construct the left derived functors $\mathrm{Tor}_i(A, B)$:

- as $L_i(\square \otimes_R B)(A)$, i.e., take a projective resolution $\varepsilon : P_\bullet \to A$, and
$$\mathrm{Tor}_i^R(A, B) = H_i(P_\bullet \otimes_R B);$$
- as $L_i(A \otimes_R \square)(B)$, i.e., take a projective resolution $\eta : Q_\bullet \to B$, and
$$\overline{\mathrm{Tor}}_i^R(A, B) = H_i(A \otimes_R Q_\bullet).$$

Theorem 2.4.1

We have natural isomorphisms $\mathrm{Tor}_i^R(A, B) \cong \overline{\mathrm{Tor}}_i^R(A, B)$.

In fact, $\mathrm{Tor}_0^R(A, B) = H_0(P_\bullet \otimes_R B) = \frac{P_0 \otimes_R B}{\mathrm{Ker}(\varepsilon \otimes \mathrm{id}_B)} \cong A \otimes_R B$, similarly, we have $\overline{\mathrm{Tor}}_0^R(A, B) = H_0(A \otimes_R Q_\bullet) \cong A \otimes_R B$. Hence in the case $i = 0$, we have $\mathrm{Tor}_0^R(A, B) \cong \overline{\mathrm{Tor}}_0^R(A, B)$. Let's looked at the case $i = 1$. Let A_1, B_1 be defined so that $0 \to A_1 \to P_0 \to A \to 0$, and $0 \to B_1 \to Q_0 \to B \to 0$. Since projective R-modules are flat (Proposition 2.2.5), $\square \otimes_R Q_0$ and $P_0 \otimes_R \square$ are exact. We tensor these two complexes into a commutative diagram as below:

$$\begin{tikzcd}
 & & 0 \arrow[d] & \overline{\mathrm{Tor}}_1^R(A,B) \arrow[d, hook] & \\
 & A_1 \otimes B_1 \arrow[r] \arrow[d] & P_0 \otimes B_1 \arrow[r] \arrow[d] & A \otimes B_1 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & A_1 \otimes Q_0 \arrow[r] \arrow[d] & P_0 \otimes Q_0 \arrow[r] \arrow[d] & A \otimes Q_0 \arrow[r] \arrow[d] & 0 \\
 & A_1 \otimes B \arrow[r] \arrow[d] & P_0 \otimes B \arrow[r] \arrow[d] & A \otimes B \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0, &
\end{tikzcd}$$

where right exact column is given by following: we have long exact sequence

$$\cdots \longrightarrow \overline{\mathrm{Tor}}_1^R(A, Q_0) \longrightarrow \overline{\mathrm{Tor}}_1^R(A,B) \longrightarrow A \otimes B_1 \longrightarrow A \otimes Q_0 \longrightarrow A \otimes B \longrightarrow 0$$

since Q_0 is projective, by Corollary 2.3.1, $\overline{\mathrm{Tor}}_1^R(A, Q_0) = 0$. By Snake Lemma 1.5.4, we have

$$0 \longrightarrow \overline{\mathrm{Tor}}_1^R(A,B) \longrightarrow A_1 \otimes B \longrightarrow P_0 \otimes B \longrightarrow A \otimes B \longrightarrow 0.$$

On the other hand, $\{\mathrm{Tor}_i^R(\square, B)\}$ form a δ-functor and, when applied to $0 \to A_1 \to P_0 \to A \to 0$ induces a long exact sequence

$$0 = \mathrm{Tor}_1^R(P_0, B) \longrightarrow \mathrm{Tor}_1^R(A,B) \longrightarrow A_1 \otimes B \longrightarrow P_0 \otimes B \longrightarrow A \otimes B \longrightarrow 0$$

All the maps being natural ones, we deduce the isomorphism

$$\mathrm{Tor}_1^R(A,B) \cong \overline{\mathrm{Tor}}_1^R(A,B).$$

One can treat the general case using dimension shifting trick to reduce to $i = 1$ case. We will rather give a more general proof.

Proof of Theorem 2.4.1

Proof Tensoring $P_\bullet$ and $Q_\bullet \to B \to 0$ gives a map of double complexes

$$P_\bullet \otimes Q_\bullet \longrightarrow P_\bullet \otimes B,$$

Taking Tot, we obtain

$$f : \mathrm{Tot}(P_\bullet \otimes Q_\bullet) \longrightarrow \mathrm{Tot}(P_\bullet \otimes B) = P_\bullet \otimes B.$$

We claim that it is a quasi-isomorphism, hence induces isomorphism

$$H_*(\mathrm{Tot}(P_\bullet \otimes Q_\bullet)) \xrightarrow{\sim} \mathrm{Tor}_*^R(A,B). \tag{2.4}$$

Recall that a morphism f of complexes is a quasi-isomorphism if and only if the mapping cone $\mathrm{Cone}(f)$ is acyclic, see Corollary 2.1.2. It is easy to check that

$$\mathrm{Cone}(f) \cong \mathrm{Tot}(P_\bullet \otimes Q'_\bullet)$$

where $Q'_\bullet$ is the augmented complex $Q_\bullet \to B \to 0$, with $B = Q'_{-1}$. The double complex $P_\bullet \otimes Q'_\bullet$ visually looks as:

$$\begin{tikzcd}
\vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
P_0 \otimes Q_1 \arrow[d] & P_1 \otimes Q_1 \arrow[l] \arrow[d] & P_2 \otimes Q_1 \arrow[l] \arrow[d] & \cdots \arrow[l] \\
P_0 \otimes Q_0 \arrow[d] & P_1 \otimes Q_0 \arrow[l] \arrow[d] & P_2 \otimes Q_0 \arrow[l] \arrow[d] & \cdots \arrow[l] \\
P_0 \otimes B & P_1 \otimes B \arrow[l] & P_2 \otimes B \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

Since P_i are projective, hence flat, we see that each column is exact. By Lemma 2.4.1 below, we deduce that $P_\bullet \otimes Q'_\bullet$ is acyclic, proving (2.4).

Similarly, one checks that the morphism $\mathrm{Tot}(P_\bullet \otimes Q_\bullet) \to A \otimes Q_\bullet$ induces isomorphisms

$$H_*(\mathrm{Tot}(P_\bullet \otimes Q_\bullet)) \xrightarrow{\sim} H_*(A \otimes Q_\bullet) =: \overline{\mathrm{Tor}}_i^R(A, B).$$

The result follows. □

Lemma 2.4.1 (Acyclic Assembly Lemma)

Let $C_{\bullet,\bullet}$ be a double complex of $\mathbf{Mod}_R$.

- $\mathrm{Tot}^{\prod}(C_{\bullet,\bullet})$ *is an acyclic chain complex, assuming either of the following:*
  - *(1) $C_{\bullet,\bullet}$ is an upper half-plane complex with exact columns.*
  - *(2) $C_{\bullet,\bullet}$ is a right half-plane complex with exact rows.*
- $\mathrm{Tot}^{\oplus}(C_{\bullet,\bullet})$ *is an acyclic chain complex, assuming either of the following:*
  - *(3) $C_{\bullet,\bullet}$ is an upper half-plane complex with exact rows.*
  - *(4) $C_{\bullet,\bullet}$ is a right half-plane complex with exact columns.*

Here the convention for a double chain complex is as follows:

$$\begin{tikzcd}
 & {} \arrow[d] & {} \arrow[d] & \\
{} & C_{-1,1} \arrow[l] \arrow[d] & C_{0,1} \arrow[l] \arrow[d] & {} \arrow[l] \\
{} & C_{-1,0} \arrow[l] \arrow[d] & C_{0,0} \arrow[l] \arrow[d] & {} \arrow[l] \\
 & {} & {} &
\end{tikzcd}$$

Proof See [Wei94, Lem. 2.7.3]. □

Corollary 2.4.1

If R is commutative, then there is a canonical isomorphism $\mathrm{Tor}_i^R(A, B) \cong \mathrm{Tor}_i^R(B, A)$.

Proof Choose a projective resolution $P_\bullet \to A$, then

$$\mathrm{Tor}_i^R(A, B) = H_i(P_\bullet \otimes B).$$

On the other hand, since R is commutative, there is a natural isomorphism of complexes:

$$P_\bullet \otimes_R B \cong B \otimes_R P_\bullet$$

so that $H_i(P_\bullet \otimes B) \cong H_i(B \otimes P_\bullet)$, the latter is isomorphic to $\overline{\mathrm{Tor}}_i^R(B, A)$ by definition. Then by Theorem

2.4.1, we have

$$\operatorname{Tor}_i^R(B,A) \cong \overline{\operatorname{Tor}}_i^R(B,A) \cong H_i(B\otimes P_\bullet) \cong H_i(P_\bullet \otimes B) \cong \operatorname{Tor}_i^R(A,B).$$

□

2.4.2 Tor and colimits

First recall that if $F:\mathscr{A}\to\mathscr{B}$ is a left adjoint, then

(a) F preserves colimits whenever exist, see Proposition 1.3.4.

(b) F is right exact, see Proposition 1.5.3.

Proposition 2.4.1

Assume $\mathscr{A}$ has enough projective objects and arbitrary direct sum exists in $\mathscr{A}$. Let $F:\mathscr{A}\to\mathscr{B}$ be a left adjoint. Then for any set $\{A_i\}_{i\in I}$ of objects in $\mathscr{A}$:

$$L_*F\left(\bigoplus_{i\in I} A_i\right) = \bigoplus_{i\in I} L_*F(A_i).$$

Proof If $P_i \twoheadrightarrow A_i$ are projective resolutions, then so is $\bigoplus_i P_i \twoheadrightarrow \bigoplus_i A_i$. Hence

$$L_*F(\oplus A_i) = H_*(F(\oplus P_i)) = H_*(\oplus F(P_i)) \overset{(*)}{\cong} \oplus H_*(F(P_i)) = \oplus L_*F(A_i).$$

Here the isomorphism $(*)$ holds because homology commutes with arbitrary direct sum of chain complex. □

Corollary 2.4.2

In the category $\mathbf{Mod}_R$, we have $\operatorname{Tor}_*^R(\bigoplus_i A_i, B) \cong \bigoplus_i \operatorname{Tor}_*^R(A_i,B)$.

We also have the following.

Proposition 2.4.2

In $\mathbf{Mod}_R$, Tor_* commutes with filtered colimits.

Proof Let $\{A_i\}_{i\in I}$ be a direct system of R-modules and let $A = \varinjlim_i A_i$. By Lemma 2.4.2 below, we can find a projective resolution P_i of A_i, which form a direct system and such that the direct limit $P := \varinjlim_i P_i$ is a projective resolution of A. Thus

$$\operatorname{Tor}_*(\varinjlim A_i, B) = H_*((\varinjlim P_i)\otimes B) \overset{(1)}{\cong} H_*(\varinjlim(P_i\otimes B)) \overset{(2)}{\cong} \varinjlim H_*(P_i\otimes B) = \varinjlim \operatorname{Tor}_*(A_i,B).$$

Here the isomorphism (1) holds because $\square\otimes B$ is left adjoint, and left adjoint functor preserves colimits; (2) holds because filtered colimits are exact (see Prop. 1.5.4). □

Lemma 2.4.2

Let $\{A_i\}_{i\in I}$ be a direct system of left R-modules, and $A=\varinjlim_i A_i$. Then there exist projective resolutions $P_{i,\bullet}$ of A_i forming a direct system such that $P_\bullet := \varinjlim_i P_{i,\bullet}$ is a projective resolution of A.

Proof For each $i\in I$, let $P_{0,i}$ be the free module $R^{(A_i)}$ (i.e., the free module with a basis indexed by elements in A_i) and let $P_0 = R^{(A)}$. The maps $A_i\to A_j$ induce maps $P_{0,i}\to P_{0,j}$ and P_0 may be identified with the limit $\varinjlim_i P_{0,i}$ (since free functor is left adjoint to forgetful functor, so it is commute with colimits). Let $K_i = \operatorname{Ker}(P_{i,0}\to A_i)$ and $K=\operatorname{Ker}(P_0\to A)$, since filtered colimits are exact, we have $\varinjlim_i K_i \cong \operatorname{Ker}(\varinjlim_i P_{i,0}\to \varinjlim_i A_i)\cong K$. We now repeat the argument with A_i replaced by K_i. The complex $P_\bullet$ is thus constructed.

$$\begin{tikzcd}
0 \arrow[r] & K \arrow[r, hook] & P_{0,\bullet} \arrow[r, two heads] & A_\bullet \arrow[r] & 0 & \\
& 0 \arrow[r] & K_i \arrow[r, hook] \arrow[ul] \arrow[d] & P_{0,i} \arrow[r, two heads] \arrow[ul] \arrow[d] & A_i \arrow[r] \arrow[ul] \arrow[d] & 0 \\
& 0 \arrow[r] & K_j \arrow[r, hook] \arrow[uul] & P_{0,j} \arrow[r, two heads] \arrow[uul] & A_j \arrow[r] \arrow[uul] & 0
\end{tikzcd}$$

□

Remark Be attention that we have no analogue of Lemma 2.2.5 for injective resolution.

2.4.3 Tor and flatness

Proposition 2.4.3

The following are equivalent for every left R-module B.

(1) B is flat.

(2) $\mathrm{Tor}_n^R(A, B) = 0$ for all $n \neq 0$ and all right R-module A.

(3) $\mathrm{Tor}_1^R(A, B) = 0$ for all right R-module A.

Proof (1) ⇒ (2): Let $P_\bullet \to A$ be a projective resolution of A, then $\mathrm{Tor}_*^R(A, B)$ is computed via $H_*(P_\bullet \otimes_R B)$. Since B is flat, $P_\bullet \otimes_R B$ is also acyclic, hence $\mathrm{Tor}_n^R(A, B) = 0$ for all $n \neq 0$.

(2) ⇒ (3): Obviously.

(3) ⇒ (1): Let $0 \to X \to Y \to Z \to 0$ be any exact sequence of R-Mods, by applying $\mathrm{Tor}^R(\square, B)$, we have long exact sequence

$$\cdots \longrightarrow \mathrm{Tor}_1^R(X, B) \longrightarrow \mathrm{Tor}_1^R(Y, B) \longrightarrow \mathrm{Tor}_1^R(Z, B) \longrightarrow X \otimes_R B \longrightarrow Y \otimes_R B \longrightarrow Z \otimes_R B \longrightarrow 0.$$

By assumption $\mathrm{Tor}_1^R(A, B) = 0$ for all $A \in \mathbf{Mod}_R$, we have $0 \to X \otimes_R B \to Y \otimes_R B \to Z \otimes_R B$ is exact, which implies that B is flat. □

Lemma 2.4.3 (Flat resolution lemma)

The groups $\mathrm{Tor}_*^R(A, B)$ may be computed using resolutions by flat modules. That is, if $F_\bullet \to A$ is a resolution of A with the F_n being flat modules, then

$$\mathrm{Tor}_*^R(A, B) \cong H_*(F_\bullet \otimes_R B).$$

Similarly, if $F'_\bullet \to B$ is a resolution of B by flat modules, then

$$\mathrm{Tor}_*^R(A, B) \cong \overline{\mathrm{Tor}}_*^R(A, B) \cong H_*(A \otimes_R F'_\bullet).$$

Proof We use induction and dimension shifting to prove $\mathrm{Tor}_n^R(A, B) \cong H_*(F_\bullet \otimes B)$. The assertion is true for $n = 0$ because $\square \otimes_R B$ is right exact. Let $K = \mathrm{Ker}(F_0 \to A) \cong \mathrm{Im}(F_1 \to F_0)$ be such that $0 \to K \to F_0 \to A \to 0$ is exact, so that $\cdots \to F_2 \to F_1 \to K \to 0$ is a resolution of K. Then

$$\begin{aligned}
\mathrm{Tor}_1^R(A, B) &= \mathrm{Ker}(K \otimes B \to F_0 \otimes B) && \text{(by Proposition 2.4.3)} \\
&= \mathrm{Ker}\left(\frac{K \otimes B}{\mathrm{Im}(d_2^F \otimes \mathrm{id}_B)} \to F_0 \otimes B\right) && \text{(given by the flat resolution of } K \text{ tensoring } B) \\
&= \mathrm{Ker}(F_1 \otimes B \to F_0 \otimes B)/\,\mathrm{Im}(d_2^F \otimes \mathrm{id}_B) && \text{(direct check)} \\
&= H_1(F_\bullet \otimes B).
\end{aligned}$$

For $n \geq 2$, suppose Lemma holds $< n$, we want to show that $\operatorname{Tor}_n^R(A,B) = H_n(F_\bullet \otimes_R B)$. By dimension shifting, we have

$$\operatorname{Tor}_i^R(A,B) \cong \operatorname{Tor}_{i-1}^R(K,B)$$

for any i. So it suffices to show that $\operatorname{Tor}_{n-1}^R(K,B) = H_n(F_\bullet \otimes_R B)$. As $\cdots \to F_2 \to F_1 \to K \to 0$ is a resolution of K, by induction hypothesis, we have $\operatorname{Tor}_{n-1}^R(K,B) = H_{n-1}(F_\bullet[1] \otimes_R B) = H_n(F_\bullet \otimes_R B)$, we are done. □

Proposition 2.4.4 (Flat base change for Tor)

Suppose $R \to S$ is a ring homomorphism, such that S is flat as an R-module. Then for any R-module A, and S-module C and any n,

$$\operatorname{Tor}_n^R(A,C) \cong \operatorname{Tor}_n^S(A \otimes_R S, C).$$

Remark Geometric version: Suppose $f : X \to Y$ is a flat morphism. Let $\mathscr{F}$ be an $\mathscr{O}_X$-module and $\mathscr{G}$ be an $\mathscr{O}_Y$-module, then

$$\operatorname{Tor}_n^{\mathscr{O}_Y}(\mathscr{G}, f_*\mathscr{F}) \cong \operatorname{Tor}_n^{\mathscr{O}_X}(f^*\mathscr{G}, \mathscr{F})$$

Proof First note that for any projective right R-module P, $P \otimes_R S$ is a projective S-module (this does not use the assumption on R), since we have isomorphisms

$$\operatorname{Hom}_S(P \otimes_R S, \square) \cong \operatorname{Hom}_R(P, \operatorname{Hom}_S(S, \square)) \cong \operatorname{Hom}_R(P, \square).$$

If $P_\bullet \to A$ is a projective resolution of A, then since S is R-flat, $P_\bullet \otimes_R S$ is a projective resolution of $A \otimes_R S$, which implies

$$\operatorname{Tor}_n^S(A \otimes_R S, C) \cong H_n((P_\bullet \otimes_R S) \otimes_S C) \cong H_n(P_\bullet \otimes_R C) \cong \operatorname{Tor}_n^R(A,C).$$

□

Corollary 2.4.3

*If $R \in$ **CRings**, S is a flat R-algebra, then $\forall A, B \in \mathbf{Mod}_R$, we have*

$$\operatorname{Tor}_n^R(A,B) \otimes_R S \cong \operatorname{Tor}_n^S(A \otimes_R S, B \otimes_R S), \quad \forall n \geq 0.$$

Proof By Proposition 2.4.4, we have

$$\operatorname{Tor}_n^R(A, B \otimes_R S) \cong \operatorname{Tor}_n^S(A \otimes_R S, B \otimes_R S).$$

It suffices to show that $\operatorname{Tor}_n^R(A, B \otimes_R S) \cong \operatorname{Tor}_n^R(A,B) \otimes_R S$.

Let $P_\bullet \to A$ be a projective resolution, then

$$\operatorname{Tor}_n^R(A,B) = H_n(P_\bullet \otimes_R B),$$

by tensoring S, we have

$$\operatorname{Tor}_n^R(A,B) \otimes_R S = H_n(P_\bullet \otimes_R B) \otimes_R S \cong H_n((P_\bullet \otimes_R B) \otimes_R S),$$

where last "$\cong$" is given by the flatness of S. Note that $H_n((P_\bullet \otimes_R B) \otimes_R S) = \operatorname{Tor}_n^R(A, B \otimes_R S)$, we have

$$\operatorname{Tor}_n^R(A, B \otimes_R S) \cong \operatorname{Tor}_n^R(A,B) \otimes_R S,$$

as we desired. □

In the following we assume R is commutative, so that the $\operatorname{Tor}_*^R(A,B)$ are actually R-modules.

Lemma 2.4.4

If $\mu : A \to A$ is multiplication by an element $r \in R$, then so are the induced maps

$$\mu_n : \mathrm{Tor}_n^R(A, B) \longrightarrow \mathrm{Tor}_n^R(A, B)$$

for all n and B.

Proof Pick a projective resolution $P_\bullet \to A \to 0$. Multiplication by r gives an R-module chain map $\widetilde{\mu} : P_\bullet \to P_\bullet$ lifting μ, and $\widetilde{\mu} \otimes \mathrm{id}_B$ is multiplication by r on $P_\bullet \otimes B$. The induced map μ_n on the subquotient $\mathrm{Tor}_n(A, B)$ of $P_n \otimes B$ is therefore also multiplication by r. $\square$

Corollary 2.4.4

If A is an $R/(r)$-module, then for every R-module B, the modules $\mathrm{Tor}_n^R(A, B)$ are $R/(r)$-modules, i.e., annihilated by the ideal (r).

Proof Apply Lemma 2.4.4. $\square$

Lemma 2.4.5

Assume R is Noetherian. Then for any finitely generated R-modules A, B, $\mathrm{Tor}_n^R(A, B)$ is finitely generated R-module.

Proof Since R is Noetherian and A is f.g. R-module, we may choose a projective resolution $P_\bullet \to A \to 0$ with each P_n f.g.. Since B is also f.g., so is each $P_n \otimes_R B$. $\mathrm{Tor}_n^R(A, B)$ is a subquotient of $P_n \otimes_R B$, so is also f.g. over R, because R is Noetherian. $\square$

2.4.4 External products for Tor

Definition 2.4.1 (External products)

Let A, A', B, B' be R-modules (with R commutative). We may construct a map

$$\mathrm{Tor}_i^R(A, B) \otimes_R \mathrm{Tor}_j^R(A', B') \longrightarrow \mathrm{Tor}_{i+j}^R(A \otimes_R A', B \otimes_R B')$$

*which is called **external product**, as follows.*

- *There are natural maps $H_i(C_\bullet) \otimes H_j(C'_\bullet) \to H_{i+j}(\mathrm{Tot}(C_\bullet \otimes C'_\bullet))$ for every pair of complexes C, C': one maps the tensor product $c \otimes c'$ of two cycles $c \in C_i$ and $c' \in C'_j$ to the class of $c \otimes c' \in C_i \otimes C'_j \hookrightarrow \mathrm{Tot}(C_\bullet \otimes C'_\bullet)_{i+j}$. Choose projective resolutions $P_\bullet \twoheadrightarrow A$, $P'_\bullet \twoheadrightarrow A'$, such that*

  $$\mathrm{Tor}_i^R(A, B) = H_i(P_\bullet \otimes B), \qquad \mathrm{Tor}_j^R(A', B') = H_j(P'_\bullet \otimes B').$$

  Then we obtain a map

  $$\mathrm{Tor}_i^R(A, B) \otimes \mathrm{Tor}_j^R(A', B') \longrightarrow H_{i+j}(\mathrm{Tot}((P_\bullet \otimes B) \otimes (P'_\bullet \otimes B'))).$$

- *Choose a projective resolution $P''_\bullet \to A \otimes A'$. The Comparison Theorem 2.2.2 gives a chain map $\mathrm{Tot}(P_\bullet \otimes P'_\bullet) \to P''_\bullet$ which is unique up to homotopic. This yields a natural map*

  $$H_n(\mathrm{Tot}((P_\bullet \otimes B) \otimes (P'_\bullet \otimes B'))) \cong H_n(\mathrm{Tot}(P_\bullet \otimes P'_\bullet \otimes B \otimes B')) \longrightarrow H_n(P''_\bullet \otimes B \otimes B').$$

2.4.5 Examples

Example 2.14 Let $R = k[x,y]$. The Koszul complex $K_\bullet((x,y),R)$ gives a projective resolution of k:

$$0 \longrightarrow R \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} R \oplus R \xrightarrow{(x\ y)} R \longrightarrow 0$$

Thus, for any R-module M, $\mathrm{Tor}_i^R(k, M)$ may be computed by taking the homology of

$$K_\bullet((x,y),M): \qquad 0 \longrightarrow M \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} M \oplus M \xrightarrow{(x\ y)} M \longrightarrow 0 \tag{2.5}$$

(1) Take $M = k$, viewed as an R-module via $R \twoheadrightarrow R/(x,y) \cong k$, then the complex (2.5) becomes

$$0 \longrightarrow k \xrightarrow{0} k \oplus k \xrightarrow{0} k \longrightarrow 0$$

and $\mathrm{Tor}_0^R(k,k) = k$, $\mathrm{Tor}_1^R(k,k) = k \oplus k$, $\mathrm{Tor}_2^R(k,k) \cong k$.

(2) Take $M = \mathfrak{m} = (x,y)$, we could compute $\mathrm{Tor}_i^R(k,\mathfrak{m})$ using (2.5). Alternatively, we may use the short exact sequence $0 \to \mathfrak{m} \to R \to k \to 0$ which induces

$$0 \to \mathrm{Tor}_1(k,k) \to \mathrm{Tor}_0(k,\mathfrak{m}) \to \mathrm{Tor}_0(k,R) \to \mathrm{Tor}_0(k,k) \to 0$$
$$\mathrm{Tor}_i(k,k) \cong \mathrm{Tor}_{i-1}(k,\mathfrak{m}), \quad \forall i \geq 2.$$

Note that $\mathrm{Tor}_0(k,R) \to \mathrm{Tor}_0(k,k)$ is an isomorphism because both of them are isomorphic to k and the morphism is surjective. Thus we deduce from (1) that

$$\mathrm{Tor}_0(k,\mathfrak{m}) = 0, \quad \mathrm{Tor}_1(k,\mathfrak{m}) \cong k, \quad \mathrm{Tor}_i(k,\mathfrak{m}) = 0, \forall i \geq 2.$$

Tor for $\mathbb{Z}$-modules

Lemma 2.4.6

For any $\mathbb{Z}$-module B, we have

$$\mathrm{Tor}_0^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) = B/nB, \qquad \mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) = B[n] = \{b \in B : nb = 0\}.$$

Proof We have a projective resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0,$$

which implies that $\mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B)$ is computed via the homology of the complex

$$0 \longrightarrow B \xrightarrow{\times n} B \longrightarrow 0.$$

The result then follows. □

Example 2.15 We have $\mathrm{Tor}_0^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/d\mathbb{Z}$, where $d = \gcd(m,n)$; and

$$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong m_1\mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z},$$

where $m = dm_1$.

We say a group is **torsion**, if every element has finite order.

Proposition 2.4.5

For all $\mathbb{Z}$-modules A and B:

(1) $\mathrm{Tor}_1^{\mathbb{Z}}(A,B)$ is a torsion $\mathbb{Z}$-module;

(2) $\mathrm{Tor}_n^{\mathbb{Z}}(A,B) = 0$ *for* $n \geq 2$.

Remark Proposition 2.4.5 says that $\mathbb{Z}$ has flat dimension 1.

Proof A is the direct limit of its finitely generated subgroups A_α. This is a filtered colimit, by Proposition 2.4.2 we obtain

$$\mathrm{Tor}_i^{\mathbb{Z}}(A,B) \cong \varinjlim \mathrm{Tor}_i^{\mathbb{Z}}(A_\alpha, B).$$

But, the direct limit of torsion groups is again a torsion group, so we may assume that A is finitely generated, i.e.,

$$A \cong \mathbb{Z}^m \oplus \mathbb{Z}/n_1 \oplus \mathbb{Z}/n_2 \oplus \cdots \oplus \mathbb{Z}/n_r$$

for some integers $m, n_1, \ldots, n_r$. As $\mathbb{Z}^m$ is free (so is projective), $\mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}^m, \Box)$ vanishes for $i \geq 1$, and so we have

$$\mathrm{Tor}_i^{\mathbb{Z}}(A,B) \cong \mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n_1, B) \oplus \cdots \oplus \mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n_r, B).$$

The result follows from Lemma 2.4.6. □

Proposition 2.4.6

$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B)$ *is the torsion subgroup of* B *for every abelian group* B.

Proof We may write $\mathbb{Q}/\mathbb{Z}$ as a filtered colimit $\varinjlim \frac{1}{n}\mathbb{Z}/\mathbb{Z}$. By Proposition 2.4.2, we obtain

$$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B) \cong \varinjlim \mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) \cong \varinjlim B[n]$$

which is the torsion subgroup of B. □

Example 2.16 Let $R = \mathbb{Z}/m\mathbb{Z}$ and $A = \mathbb{Z}/d\mathbb{Z}$ with $d|m$. Writing $m = dd'$, then we have a periodic free resolution of $\mathbb{Z}/d\mathbb{Z}$ as follows

$$\cdots \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{d'} \mathbb{Z}/m\mathbb{Z} \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/d\mathbb{Z} \longrightarrow 0,$$

so that for any $\mathbb{Z}/m\mathbb{Z}$-module B, we have

$$\mathrm{Tor}_i^{\mathbb{Z}/m\mathbb{Z}}(\mathbb{Z}/d\mathbb{Z}, B) = \begin{cases} B/dB & i = 0, \\ B[d]/d'B & i > 0 \text{ odd}, \\ B[d']/dB & i > 0 \text{ even}. \end{cases}$$

For example, if $m = 8, d = 4$, then $d' = 2$ and

$$\mathrm{Tor}_i^{\mathbb{Z}/8\mathbb{Z}}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}) = \begin{cases} \mathbb{Z}/4\mathbb{Z} & i = 0 \\ (\mathbb{Z}/4\mathbb{Z})/(2\mathbb{Z}/4\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} & i > 0 \text{ odd}, \\ (2\mathbb{Z}/4\mathbb{Z})/0 \cong \mathbb{Z}/2\mathbb{Z} & i > 0 \text{ even}. \end{cases}$$

2.5 Ext

2.5.1 Ext^n

Let $R \in \mathbf{Rings}$, $A, B \in \mathbf{Mod}_R$, there are two ways to define $\mathrm{Ext}_R^n(A,B) \in \mathbf{Ab}$.

- $R^n\,\mathrm{Hom}_R(\Box, B)(A)$: take a projective resolution $P_\bullet \to A$ and set

$$\mathrm{Ext}_R^n(A,B) := H^n(\mathrm{Hom}(P_\bullet, B)).$$

- $R^n\operatorname{Hom}_R(A,\square)(B)$: take an injective resolution $B\to I^\bullet$ and set
$$\overline{\operatorname{Ext}}_R^n(A,B):=H^n(\operatorname{Hom}(A,I^\bullet)).$$

Theorem 2.5.1

We have a natural isomorphism $\operatorname{Ext}_R^n(A,B)\cong\overline{\operatorname{Ext}}_R^n(A,B)$.

Proof **We give a sketch of proof:**

Similar to the proof of Theorem 2.4.1: Choose a projective resolution $P_\bullet\to A$ and injective resolution $B\to I^\bullet$, consider $\operatorname{Hom}(P_\bullet,I^\bullet)$, we have induced maps:
$$\operatorname{Hom}(P_\bullet,B)\longrightarrow\operatorname{Hom}(P_\bullet,I^\bullet)\longleftarrow\operatorname{Hom}(A,I^\bullet).$$
Then show the induced maps of complexes
$$\operatorname{Tot}(\operatorname{Hom}(P_\bullet,B))=\operatorname{Hom}(P_\bullet,B)\longrightarrow\operatorname{Tot}(\operatorname{Hom}(P_\bullet,I^\bullet))\longleftarrow\operatorname{Tot}(\operatorname{Hom}(A,I^\bullet))=\operatorname{Hom}(A,I^\bullet)$$
are quasi isomorphisms by showing the mapping cones are acyclic. $\square$

Proposition 2.5.1

Let $A\in\mathbf{Mod}_R$, the following are equivalent:

(1) B is an injective R-module.

(2) $\operatorname{Hom}_R(\square,B)$ is an exact functor.

(3) $\operatorname{Ext}_R^i(A,B)$ vanishes $\forall i>0$ and $\forall A\in\mathbf{Mod}_R$.

(4) $\operatorname{Ext}_R^1(A,B)$ vanishes $\forall A\in\mathbf{Mod}_R$.

Proof (1) $\Leftrightarrow$ (2): Proposition 2.2.1.

(2) $\Rightarrow$ (3): For any $A\in\mathbf{Mod}_R$, choose a projective resolution $P_\bullet\to A\twoheadrightarrow 0$. Apply exact functor $\operatorname{Hom}_R(\square,B)$, we have exact sequence
$$0\longrightarrow\operatorname{Hom}(A,B)\longrightarrow\operatorname{Hom}(P_0,B)\longrightarrow\operatorname{Hom}(P_1,B)\longrightarrow\cdots.$$
So, for $i>1$, $\operatorname{Ext}_R^i(A,B)=H^i(\operatorname{Hom}(P_\bullet,B))=0$.

(3) $\Rightarrow$ (4): Clearly.

(4) $\Rightarrow$ (2): Let $0\to X\to Y\to Z\to 0$ be a short exact sequence, apply $\operatorname{Ext}(\square,B)$, we have long exact sequence
$$0\longrightarrow\operatorname{Hom}(Z,B)\longrightarrow\operatorname{Hom}(Y,B)\longrightarrow\operatorname{Hom}(X,B)\longrightarrow\operatorname{Ext}^1(Z,B)\longrightarrow\cdots.$$
Since $\operatorname{Ext}^1(Z,B)=0$, $0\to\operatorname{Hom}(Z,B)\to\operatorname{Hom}(Y,B)\to\operatorname{Hom}(X,B)\to 0$ is exact, we win. $\square$

Proposition 2.5.2

Let $B\in\mathbf{Mod}_R$, the following are equivalent:

(1) A is a projective R-module.

(2) $\operatorname{Hom}_R(A,\square)$ is an exact functor.

(3) $\operatorname{Ext}_R^i(A,B)$ vanishes $\forall i>0$ and $\forall B\in\mathbf{Mod}_R$.

(4) $\operatorname{Ext}_R^1(A,B)$ vanishes $\forall B\in\mathbf{Mod}_R$.

Proof Same as Proposition 2.5.1. $\square$

Proposition 2.5.3 (Ext^n **and direct sum/product)**

$\forall n \geq 0$, we have

(1) $\mathrm{Ext}_R^n\left(\bigoplus_i A_i, B\right) \cong \prod_i \mathrm{Ext}_R^n(A_i, B)$.

(2) $\mathrm{Ext}_R^n(A, \prod_i B_i) \cong \prod_i \mathrm{Ext}_R^n(A, B_i)$.

Proof If $P_{i,\bullet} \to A_i$ are projective resolutions, so is $\bigoplus_i P_{i,\bullet} \to \bigoplus_i A_i$, since $\mathrm{Hom}(\bigoplus_i P_{i,\bullet}, B) \cong \prod_i \mathrm{Hom}(P_{i,\bullet}, B)$, we get

$$\begin{aligned}
\mathrm{Ext}_R^n\left(\bigoplus_i A_i\right) &= H^n\left(\mathrm{Hom}\left(\bigoplus_i P_{i,\bullet}, B\right)\right) \\
&= H^n\left(\prod_i \mathrm{Hom}(P_{i,\bullet}, B)\right) \\
&= \prod_i H^n(\mathrm{Hom}(P_{i,\bullet}, B)) \\
&= \prod_i \mathrm{Ext}_R^n(A_i, B).
\end{aligned}$$

Similarly, if $B_i \to I_i^\bullet$ are injective resolutions, so is $\prod_i B_i \to \prod_i I_i^\bullet$, and we conclude as before. □

2.5.2 Ext^n for commutative rings

We assume R is commutative in this subsection. If A, B are R-modules, then so are $\mathrm{Hom}_R(A, B)$ and $\mathrm{Ext}_R^*(A, B)$.

Proposition 2.5.4

If $\mu : A \to A$ is given by multiplication by $r \in R$, then so are the induced endomorphisms $\mu^* : \mathrm{Ext}_R^*(A, B) \to \mathrm{Ext}_R^*(A, B)$. Similar statement if we replace A by B.

Proof Pick a projective resolution $P_\bullet \twoheadrightarrow A$. Comparison Theorem 2.2.2 tells us multiplication by r gives an R-module chain map

$$\widetilde{\mu}_\bullet : P_\bullet \longrightarrow P_\bullet.$$

The induced map $\mathrm{Hom}_R(P_n, B) \to \mathrm{Hom}_R(P_n, B)$ is also multiplication by r, because it sends $f \in \mathrm{Hom}_R(P_n, B)$ to $f \circ \widetilde{\mu}_n$ which takes $x \in P_n$ to $f(rx) = rf(x)$. Hence the map μ^* on the subquotient $\mathrm{Ext}_R^n(A, B)$ of $\mathrm{Hom}_R(P_n, B)$ is also multiplication by r.

If $v : B \to B$ is given by multiplication by $r \in R$, similar argument as above by using an injective resolution $B \to I^\bullet$, the induced endomorphisms $v^* : \mathrm{Ext}_R^*(A, B) \to \mathrm{Ext}_R^*(A, B)$ is given by multiplication by $r \in R$. □

Corollary 2.5.1

*If $R \in$ **CRings**, and A annihilated by $r \in R$ (i.e., A is an $R/(r)$-module), then $\mathrm{Ext}_R^*(A, B)$ are also annihilated by $r \in R$ for any R-module B (i.e., $\mathrm{Ext}_R^*(A, B)$ are $R/(r)$-modules).*

Proof Applying Proposition 2.5.4. □

Proposition 2.5.5

Assume R is Noetherian. Then for any finitely generated R-modules A, B, the R-modules $\operatorname{Ext}^*_R(A,B)$ are also finitely generated.

Remark **Geometric version:**

Suppose X is a locally Noetherian scheme, $\mathscr{F},\mathscr{G}$ are coherent sheaves on X, then $\mathscr{E}xt^*_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$ are coherent sheaves on X.

Proof Since A is finitely generated and R is Noetherian, we may find a projective resolution $P_\bullet \twoheadrightarrow A$, such that each P_n is projective and finitely generated. Claim that $\operatorname{Hom}_R(P_n, B)$ is also finitely generated. Since P_n is finitely generated, we have an exact sequence $R^m \to P_n \to 0$ for some m, by applying $\operatorname{Hom}(\square, B)$, we have

$$0 \longrightarrow \operatorname{Hom}_R(P_n, B) \longrightarrow \operatorname{Hom}_R(R^m, B)$$

exact. Note that $\operatorname{Hom}_R(R^m, B) \cong \operatorname{Hom}_R(R,B)^m \cong B^m$ and B^m is Noetherian (since B is f.g., and R is Noetherian), $\operatorname{Hom}_R(P_n, B)$ is finitely generated, since R is Noetherian, $\operatorname{Hom}_R(P_n, B)$ is Noetherian. Because $\operatorname{Ext}^n_R(A,B)$ is a subquotient of $\operatorname{Hom}_R(P_n,B)$, $\operatorname{Ext}^n_R(A,B)$ is finitely generated. $\square$

2.5.3 Computations of Ext^n

Lemma 2.5.1

$\operatorname{Ext}^n_{\mathbb{Z}}(A,B) = 0$ *for* $n \geq 2$ *and all abelian groups* A, B.

Proof Embed B in an injective abelian group I^0. Since I^0 is injective, by Proposition 2.2.4, I^0 is divisible. Let $I^1 = \operatorname{Coker}(B \to I^0)$, since I^1 is the quotient of I^0, I^1 is divisible. Since $\mathbb{Z}$ is PID, by Proposition 2.2.4 again, I^1 is an injective object. Therefore $B \to I^0 \to I^1 \to 0$ is an injective resolution of length 2, so, we have long exact sequence

$$\begin{tikzcd}
 & & \cdots \\
\operatorname{Ext}^2(A,I^0) \arrow[dr] & & \operatorname{Ext}^2(A,B) \arrow[ll] \\
\operatorname{Ext}^1(A,I^0) \arrow[dr] & \operatorname{Ext}^2(A,I^1) \arrow[uur] & \operatorname{Ext}^1(A,B) \arrow[ll, dotted] \\
\operatorname{Hom}(A,I^0) \arrow[dr] & \operatorname{Ext}^1(A,I^1) \arrow[uur] & \operatorname{Hom}(A,B) \arrow[ll, dotted] \\
 & \operatorname{Hom}(A,I^1) \arrow[uur] & \\
 & 0 \arrow[uur] &
\end{tikzcd}$$

since $\operatorname{Ext}^i(A,I^0) = \operatorname{Ext}^i(A,I^1) = 0$ for all $i \geq 1$, so $\operatorname{Ext}^n(A,B) = 0$ for $n \geq 2$. $\square$

Lemma 2.5.2

We have $\operatorname{Ext}^0_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) \cong B[n]$ *and* $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) \cong B/nB$.

Proof We have a projective resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

Then, we have a long exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, B) \arrow[r, "\times n"] \arrow[d, equal] & \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, B) \arrow[r, "\delta^0"] \arrow[d, equal] & \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) \arrow[r] & \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}, B) \arrow[r] \arrow[d, equal] & \cdots \\
& & B & B & & 0. &
\end{tikzcd}$$

So

$$\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) = \mathrm{Ker}(\times n) = B[n]$$

and

$$\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) = B/\,\mathrm{Ker}\,\delta = B/\,\mathrm{Im}(\times n) = B/nB,$$

we win. □

Remark One may realize that the result is dual to $\mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B)$ (c.f., Lemma 2.4.6). Later we will see the dual relation between Tor and Ext.

Example 2.17 Write $R = \mathbb{Z}/m\mathbb{Z}$ and $A = \mathbb{Z}/d\mathbb{Z}$ with $d \mid m$. Writing $m = dd'$, then

$$\cdots \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{d'} \mathbb{Z}/m\mathbb{Z} \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/d\mathbb{Z} \longrightarrow 0$$

is an infinite periodic injective resolution of A. Then $\forall B \in \mathbf{Mod}_{\mathbb{Z}/m\mathbb{Z}}$, we have

$$\mathrm{Ext}^i_{\mathbb{Z}/m\mathbb{Z}}(\mathbb{Z}/d\mathbb{Z}, B) \cong \begin{cases} B[d] & i = 0; \\ B[d']/dB & i > 0 \quad \text{odd}; \\ B[d]/d'B & i > 0 \quad \text{even}. \end{cases}$$

$\mathrm{Ext}^n_{\mathbb{Z}}(A, \mathbb{Z})$

Let A be a torsion abelian group (an abelian group in which every element has finite order), and recall that A^* denotes its Pontrjagin dual $\mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})$ (Definition 2.2.3).

Lemma 2.5.3

Let A be a torsion abelian group, then

$$\mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Z}) = 0, \quad \mathrm{Ext}^1_{\mathbb{Z}}(A, \mathbb{Z}) = A^*.$$

Proof In fact, the sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Q} \xrightarrow{\beta} \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

is an injective resolution of $\mathbb{Z}$. Then, we have long exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Z}) \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}) \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z}) \arrow[r] \arrow[d, equal] & \mathrm{Ext}^1_{\mathbb{Z}}(A, \mathbb{Z}) \arrow[r] & \mathrm{Ext}^1_{\mathbb{Z}}(A, \mathbb{Q}) \arrow[r] \arrow[d, equal] & \cdots \\
& & & A^* & & 0 &
\end{tikzcd}$$

Since A is a torsion abelian group, A does not have free part, we may assume that $A = \mathbb{Z}/m\mathbb{Z}$ for some n. Consider $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Q})$, by Lemma 2.5.2,

$$\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Q}) \cong \mathbb{Q}[n] = 0.$$

Hence by above long exact sequence, we have

$$\mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Z}) = 0, \quad \mathrm{Ext}^1_{\mathbb{Z}}(A, \mathbb{Z}) = A^*.$$

□

Proposition 2.5.6

If A is a finite torsion abelian group, then $A^{**} \cong A$.

Proof In fact, $A \hookrightarrow A^{**}$ (recall remark of Proposition 2.2.7). It suffices to show that $|A| = |A^{**}|$. Since A is a finite torsion free abelian group, we may assume that $A = \mathbb{Z}/n\mathbb{Z}$. By Lemma 2.5.3 and Lemma 2.5.2, we have $A^* = \operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}/n\mathbb{Z} = A$. So $|A^{**}| = |A^*| = |A|$, as we desired. □

Example 2.18 Let $\mathbb{Z}_{p^\infty} := \varinjlim (p^{-n}\mathbb{Z})/\mathbb{Z}$, then

$$\begin{aligned}\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) &= (\mathbb{Z}_{p^\infty})^* \\ &= \operatorname{Hom}_{\mathbb{Z}}(\varinjlim (p^{-n}\mathbb{Z})/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \\ &\cong \varprojlim \operatorname{Hom}_{\mathbb{Z}}((p^{-n}\mathbb{Z})/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}).\end{aligned}$$

Consider $\operatorname{Hom}_{\mathbb{Z}}((p^{-n}\mathbb{Z})/\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$, let $\varphi : (p^{-n}\mathbb{Z})/\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$, consider $\varphi\left(\frac{1}{p^n}\right)$. Note that $p^n \cdot \varphi\left(\frac{1}{p^n}\right) = 0$, so $\operatorname{ord}\left(\varphi\left(\frac{1}{p^n}\right)\right) \mid p^n$, hence $\operatorname{Hom}_{\mathbb{Z}}((p^{-n}\mathbb{Z})/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/p^n\mathbb{Z}$. Hence

$$\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) = (\mathbb{Z}_{p^\infty})^* \cong \varprojlim \mathbb{Z}/p^n\mathbb{Z} =: \widehat{\mathbb{Z}}_p.$$

where $\widehat{\mathbb{Z}}_p$ denotes the p-adic integral number.

2.6 Extensions

Although the material discussed below work for any abelian category $\mathscr{A}$, we will assume $\mathscr{A} = \mathbf{Mod}_R$. (We will drop the subscript R in the discussion below.)

Philosophy

In an Abelian category such as $\mathbf{Mod}_R$, the structure problem of module extensions is frequently studied. A long exact sequence of R-modules from A to C via n intermediate modules B_i $(0 \le i \le n-1)$:

$$0 \longrightarrow A \longrightarrow B_{n-1} \longrightarrow \cdots \longrightarrow B_1 \longrightarrow B_0 \longrightarrow C \longrightarrow 0,$$

is called an n**-fold extension** of C by A. These extensions can be classified up to congruence (i.e., in the sense of a commutative diagram involving morphisms between long exact sequences of length n with fixed endpoints A and C). Their congruence classes are in bijective correspondence with the elements of the additive group $\operatorname{Ext}^n(C, A)$.

2.6.1 Yoneda extension

In this subsection, we will analyze the meaning of Ext^1_R — namely, the problem of module extensions.

Definition 2.6.1 (Extension)

(1) An ***extension*** *ξ of A by B is an exact sequence $0 \to B \to X \to A \to 0$. Two extensions ξ and ξ' are equivalent if there is a commutative diagram*

$$\begin{tikzcd} \xi : & 0 \arrow[r] & B \arrow[r] \arrow[d, equal] & X \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\ \xi' : & 0 \arrow[r] & B \arrow[r] & X' \arrow[r] & A \arrow[r] & 0. \end{tikzcd} \tag{2.6}$$

Note that the morphism $X \to X'$ is automatically an isomorphism by Snake Lemma 1.5.4.

*(2) An extension is **split** if it is equivalent to*

$$0 \longrightarrow B \xrightarrow{\begin{pmatrix} 0 \\ \mathrm{id}_B \end{pmatrix}} A \oplus B \longrightarrow A \longrightarrow 0.$$

Lemma 2.6.1

If $\mathrm{Ext}^1_R(A, B) = 0$, then every extension of A by B is split.

Proof Given an exact sequence $0 \to B \xrightarrow{f} X \to A \to 0$, applying $\mathrm{Hom}(\Box, B)$ to it gives

$$0 \longrightarrow \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}(X, B) \xrightarrow{f^*} \mathrm{Hom}(B, B) \xrightarrow{\partial} \mathrm{Ext}^1_R(A, B) = 0.$$

Thus there exists a retraction $X \to B$, by Corollary 1.5.2 (S1), the extansion of A by B is split. $\square$

Remark Taking the construction of this lemma to heart, we see that the class $\mathrm{ch}([\xi]) := \partial\,(\mathrm{id}_B)$ in $\mathrm{Ext}^1(A, B)$ is an obstruction to ξ being split: ξ is split iff id_B lifts to $\mathrm{Hom}(X, B)$ iff the class $\mathrm{ch}([\xi]) \in \mathrm{Ext}^1(A, B)$ vanishes. Equivalent extensions have the same obstruction by naturality of the map ∂, so the obstruction $\mathrm{ch}([\xi])$ only depends on the equivalence class of ξ.

Definition 2.6.2 (Yoneda extension group, map of characteristic class)

Let $\mathrm{YExt}^1_R(A, B)$ be the set of equivalence classes of extensions of A by B. Define

$$\mathrm{ch} : \mathrm{YExt}^1_R(A, B) \longrightarrow \mathrm{Ext}^1_R(A, B), \quad [\xi] \longmapsto \partial(\mathrm{id}_B),$$

where $\partial : \mathrm{Hom}(B, B) \to \mathrm{Ext}^1_R(A, B)$ is the connection morphism when apply $\mathrm{Hom}(\Box, B)$ to ξ.

Remark Given ξ as above, we could alternatively apply $\mathrm{Hom}(A, \Box)$ to get

$$\partial' : \mathrm{Hom}_R(A, A) \longrightarrow \overline{\mathrm{Ext}}^1_R(A, B)$$

and define $\mathrm{ch}'([\xi])$ to be $\partial'(\mathrm{id}_A)$. You may check $\mathrm{ch}'([\xi])$ is the same thing as $\mathrm{ch}([\xi])$.

The map ch is well-defined. Indeed, consider the diagram (2.6) which is an equivalence; it induces a commutative diagram:

$$\begin{tikzcd} \mathrm{Hom}_R(X, B) \arrow[r] & \mathrm{Hom}_R(B, B) \arrow[r, "\partial"] \arrow[d, equal] & \mathrm{Ext}^1_R(A, B) \arrow[d, equal] \\ \mathrm{Hom}_R(X', B) \arrow[r] \arrow[u, "\sim"] & \mathrm{Hom}_R(B, B) \arrow[r, "\partial"] & \mathrm{Ext}^1_R(A, B) \end{tikzcd}$$

hence $\mathrm{ch}([\xi]) = \mathrm{ch}([\xi'])$.

Lemma 2.6.2

Consider a pushout diagram

$$\begin{tikzcd} Y \arrow[r, "j"] \arrow[d, "j'"'] & X \arrow[d] \\ X' \arrow[r, "i"] & X \sqcup_Y X' \end{tikzcd}$$

If j is injective, then so is i and

$$\mathrm{Coker}(Y \to X) \cong \mathrm{Coker}(X' \to X \sqcup_Y X').$$

Proof Recall that $X \sqcup_Y X' = X \oplus X' / \{(j(y), -j'(y)) : y \in Y\}$. So if $i(x') = 0$, i.e., $(0, x') = (j(y), -j'(y))$ for some $y \in Y$, then $y = 0$ (as j is injective), hence $x' = 0$.

We next show that $\mathrm{Coker}(j) \cong \mathrm{Coker}(i)$. Since i, j are injective, we may regard Y as a submodule of X and regard X' as a submodule of $X \sqcup_Y X'$. Hence

$$\mathrm{Coker}(i) \cong X \sqcup_Y X'/X' \cong X \oplus 0/(j(Y) \oplus 0) \cong X/Y \cong \mathrm{Coker}(j).$$

$\square$

Theorem 2.6.1 (Baer) *Given two R-modules A, B, the map of characteristic class*

$$\mathrm{ch} : \mathrm{YExt}^1_R(A, B) \xrightarrow{\sim} \mathrm{Ext}^1_R(A, B)$$

is a bijection which sends the split extensions to zero.

Proof Fix an exact sequence: $0 \to M \xrightarrow{j} P \to A \to 0$ with P projective. Applying $\mathrm{Hom}(\square, B)$ yields an exact sequence

$$\mathrm{Hom}(P, B) \xrightarrow{j^*} \mathrm{Hom}(M, B) \xrightarrow{\delta} \mathrm{Ext}^1_R(A, B) \longrightarrow 0.$$

Given $x \in \mathrm{Ext}^1(A, B)$, choose $\beta \in \mathrm{Hom}(M, B)$ with $\delta(\beta) = x$. Let X be the pushout of j and β, so that we obtain a commutative diagram with exact row (by Lemma 2.6.2):

$$\begin{tikzcd}
 & 0 \arrow[r] & M \arrow[r, "j"] \arrow[d, "\beta"'] & P \arrow[r, "e"] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\
\xi : & 0 \arrow[r] & B \arrow[r] & X \arrow[r] & A \arrow[r] & 0.
\end{tikzcd}$$

This induces the following commutative diagram:

$$\begin{tikzcd}
\mathrm{Hom}(P, B) \arrow[r] & \mathrm{Hom}(M, B) \arrow[r, "\delta"] & \mathrm{Ext}^1(A, B) \\
\mathrm{Hom}(X, B) \arrow[r] \arrow[u] & \mathrm{Hom}(B, B) \arrow[r, "\partial"] \arrow[u, "\beta^*"] & \mathrm{Ext}^1(A, B). \arrow[u, equal]
\end{tikzcd}$$

Then, we see that $\mathrm{ch}([\xi]) = \partial(\mathrm{id}_B) = \delta(\beta^*(\mathrm{id}_B)) = \delta(\beta) = x$, which means ch is surjective.

It remains to prove that ch is injective. Suppose $\mathrm{ch}([\xi]) = \mathrm{ch}([\xi'])$, where

$$\begin{tikzcd}
\xi : & 0 \arrow[r] & B \arrow[r] & X \arrow[r] & A \arrow[r] & 0 \\
\xi' : & 0 \arrow[r] & B \arrow[r] & X' \arrow[r] & A \arrow[r] & 0.
\end{tikzcd}$$

Let $0 \to M \xrightarrow{j} P \to A \to 0$ with P projective. Since P is projective, there exists $\alpha : P \to X$, with an induced morphism β, fitting in the following commutative diagram

$$\begin{tikzcd}
 & 0 \arrow[r] & M \arrow[r, "j"] \arrow[d, dashed, "\beta"] & P \arrow[r, "e"] \arrow[d, dashed, "\alpha"] & A \arrow[r] \arrow[d, equal] & 0 \\
\xi : & 0 \arrow[r] & B \arrow[r] & X \arrow[r] & A \arrow[r] & 0,
\end{tikzcd}$$

from which we get the following commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}(P, B) \arrow[r, "j^*"] & \mathrm{Hom}(M, B) \arrow[r, "\delta"] & \mathrm{Ext}^1(A, B) \arrow[r] & 0 \\
\mathrm{Hom}(X, B) \arrow[r] \arrow[u, "\alpha^*"] & \mathrm{Hom}(B, B) \arrow[r, "\partial"] \arrow[u, "\beta^*"] & \mathrm{Ext}^1(A, B). \arrow[u, equal] &
\end{tikzcd}$$

Then, we see that $\mathrm{ch}(\xi) := \partial(\mathrm{id}_B) = \delta(\beta)$. Similarly, we have a morphism $\alpha' : P \to X'$ (with induced β') and a commutative diagram

$$\begin{tikzcd} & 0 \arrow[r] & M \arrow[r, "j"] \arrow[d, dashed, "\beta'"] & P \arrow[r, "e"] \arrow[d, dashed, "\alpha'"] & A \arrow[r] \arrow[d, equal] & 0 \\ \xi' : & 0 \arrow[r] & B \arrow[r] & X' \arrow[r] & A \arrow[r] & 0. \end{tikzcd}$$

so that $\mathrm{ch}([\xi']) := \partial'(\mathrm{id}_B) = \delta(\beta')$. Since $\mathrm{ch}([\xi]) = \mathrm{ch}([\xi'])$ by assumption, we obtain $\delta(\beta) = \delta(\beta')$, i.e., $\beta' - \beta \in \operatorname{Ker} \delta$. Hence there exists $\gamma \in \mathrm{Hom}(P, B)$ such that

$$j^*(\gamma) = \gamma \circ j = \beta' - \beta.$$

Consider the following diagram.

$$\begin{tikzcd} \xi : & & 0 \arrow[rr] & & B \arrow[rr] & & X \arrow[rr] & & A \arrow[rr] & & 0 \\ & 0 \arrow[rr] & & M \arrow[ur] \arrow[rr, "j"] \arrow[d, "\beta"] & & P \arrow[ur] \arrow[rr] \arrow[d] & & A \arrow[ur] \arrow[rr] \arrow[d, equal] & & 0 & \\ & 0 \arrow[rr] & & B \arrow[rr] \arrow[uur, equal] & & B \sqcup_M P \arrow[rr] \arrow[uur, dashed] & & A \arrow[rr] \arrow[uur, equal] & & 0 & \end{tikzcd}$$

The universal property of pushout gives a morphism $B \sqcup_M P \to X$, and Fiver Lemma 1.5.3 gives the isomorphism $B \sqcup_{M,(j,\beta)} P \cong X$. Similarly, we have $X' \cong B \sqcup_{M,(j,\beta')} P$. Say

$$\begin{tikzcd} M \arrow[r, "j"] \arrow[d, "\beta"'] & P \arrow[d, "\alpha"] \\ B \arrow[r, "i"'] & X \end{tikzcd} \qquad \begin{tikzcd} M \arrow[r, "j"] \arrow[d, "\beta'"'] & P \arrow[d, "\alpha'"] \\ B \arrow[r, "i'"'] & X'. \end{tikzcd}$$

Set $\sigma = \alpha + i\gamma$, then

$$i \circ \beta' = i \circ (\beta + \gamma \circ j) = \alpha \circ j + i \circ \gamma \circ j = \sigma \circ j$$

as illustrated as follows:

$$\begin{tikzcd} M \arrow[r, "j"] \arrow[d, "\beta'"'] & P \arrow[d, "\alpha'"'] \arrow[ddr, bend left, "\sigma"] & \\ B \arrow[r, "i'"] \arrow[drr, bend right, "i"'] & X' \arrow[dr, dashed, "v"] & \\ & & X. \end{tikzcd}$$

By the universal property of pushout, there is an induced morphism $v : X' \to X$. Conversely, set $\sigma' = \alpha' - i' \circ \gamma$, then

$$i' \circ \beta = i' \circ (\beta' - \gamma \circ j) = i' \circ \beta' - i' \circ \gamma \circ j = \alpha' \circ j - i' \circ \gamma \circ j = \sigma' \circ j$$

as illustrated as follows:

$$\begin{tikzcd} M \arrow[r, "j"] \arrow[d, "\beta"'] & P \arrow[d, "\alpha"'] \arrow[ddr, bend left, "\sigma'"] & \\ B \arrow[r, "i"] \arrow[drr, bend right, "i'"'] & X \arrow[dr, dashed, "v'"] & \\ & & X'. \end{tikzcd}$$

By the universal property of pushout, there is an induced morphism $v' : X \to X'$. Hence $X \cong X'$, which implies that $[\xi] = [\xi']$.

Finally, if $\xi : 0 \to B \xrightarrow{f} X \to A \to 0$ split, by Corollary 1.5.2 (S1), there exists $g \in \mathrm{Hom}(X, B)$ such that $g \circ f = \mathrm{id}_B$. So we have $\mathrm{ch}([\xi]) = \partial(\mathrm{id}_B) = \partial(f^*(g)) = 0$, where $f^* : \mathrm{Hom}(X, B) \to \mathrm{Hom}(B, B)$. It follows that ch sends the split extensions to zero. $\square$

Baer sum

By Theorem 2.6.1, the abelian group structure on $\mathrm{Ext}^1(A, B)$ naturally translates to an abelian group structure on $\mathrm{YExt}^1(A, B)$. The precise definition is as follows.

Let $\xi : 0 \to B \xrightarrow{f} X \xrightarrow{g} A \to 0$ and $\xi' : 0 \to B \xrightarrow{f'} X' \xrightarrow{g'} A \to 0$ be two extensions of A by B. Let X'' be the pullback

$$\begin{tikzcd} X'' \arrow[r] \arrow[d] & X \arrow[d, "g"] \\ X' \arrow[r, "g'"] & A \end{tikzcd}$$

i.e., $X'' = \{(x, x') \in X \times X' : g(x) = g(x') \text{ in } A\}$. Then X'' contains the skew diagonal $\nabla := \{(f(b), -f'(b)) : b \in B\}$, let $Y = X''/\nabla$. Then we obtain an exact sequence

$$\varphi : \qquad 0 \longrightarrow B \longrightarrow Y \longrightarrow A \longrightarrow 0,$$

where the first map is $b \mapsto [(f(b), 0)] = [(0, f'(b))]$ and the second is $(x, x') \mapsto g(x) = g'(x')$.

Set $[\xi] +_b [\xi'] := [\varphi]$, and call it the **Baer sum** of $[\xi]$ and $[\xi']$ (first introduced by Baer in 1934, this year Grothendieck was 6 years old).

Corollary 2.6.1

The set $\mathrm{YExt}^1(A, B)$ under Baer sum, with zero being the class of the split extension. The map ch is an isomorphism of abelian groups.

Proof Pick $[\xi], [\xi'] \in \mathrm{YExt}^1_R(A, B)$, we want to show that $\mathrm{ch}([\xi] +_b [\xi']) = \mathrm{ch}([\xi]) + \mathrm{ch}([\xi'])$. Fix an exact sequence $0 \to M \xrightarrow{j} P \to A \to 0$ with P projective. As in the proof of Theorem 2.6.1, we have two commutative diagrams with exact rows:

$$\begin{tikzcd} & 0 \arrow[r] & M \arrow[r, "j"] \arrow[d, dashed, "\beta"'] & P \arrow[r, "e"] \arrow[d, dashed, "\alpha"'] & A \arrow[r] \arrow[d, equal] & 0 \\ \xi : & 0 \arrow[r] & B \arrow[r, "i"] & X \arrow[r] & A \arrow[r] & 0, \end{tikzcd}$$

$$\begin{tikzcd} & 0 \arrow[r] & M \arrow[r, "j"] \arrow[d, dashed, "\beta'"'] & P \arrow[r, "e"] \arrow[d, dashed, "\alpha'"'] & A \arrow[r] \arrow[d, equal] & 0 \\ \xi' : & 0 \arrow[r] & B \arrow[r, "i'"] & X' \arrow[r] & A \arrow[r] & 0. \end{tikzcd}$$

By applying $\mathrm{Hom}(\square, B)$, we have two commutative diagrams:

$$\begin{tikzcd}
\mathrm{Hom}(P,B) \arrow[r, "j^*"] & \mathrm{Hom}(M,B) \arrow[r, "\delta"] & \mathrm{Ext}^1(A,B) \arrow[r] \arrow[d, equal] & 0 \\
\mathrm{Hom}(X,B) \arrow[r] \arrow[u, "\alpha^*"] & \mathrm{Hom}(B,B) \arrow[r, "\partial"] \arrow[u, "\beta^*"] & \mathrm{Ext}^1(A,B) &
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{Hom}(P,B) \arrow[r, "j^*"] & \mathrm{Hom}(M,B) \arrow[r, "\delta"] & \mathrm{Ext}^1(A,B) \arrow[r] \arrow[d, equal] & 0 \\
\mathrm{Hom}(X',B) \arrow[r] \arrow[u, "\alpha'^*"] & \mathrm{Hom}(B,B) \arrow[r, "\partial'"] \arrow[u, "\beta'^*"] & \mathrm{Ext}^1(A,B) &
\end{tikzcd}$$

so we have $\mathrm{ch}([\xi]) + \mathrm{ch}([\xi']) = \partial(\mathrm{id}_B) + \partial'(\mathrm{id}_B) = \delta(\beta) + \delta(\beta') = \delta(\beta+\beta')$. Let Y be the pushout of j and $\beta+\beta'$, so we obtain a commutative diagram with exact row (by Lemma 2.6.2):

$$\begin{tikzcd}
& 0 \arrow[r] & M \arrow[r, "j"] \arrow[d, "\beta+\beta'"'] & P \arrow[r, "e"] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\
\varphi: & 0 \arrow[r] & B \arrow[r] & Y \arrow[r] & A \arrow[r] & 0.
\end{tikzcd}$$

It suffices to show that $[\varphi] = [\xi] +_b [\xi']$, i.e., to show $Y \cong (X \times_A X')/\nabla$. Say $BS := (X \times_A X')/\nabla$. Define $u : P \to BS$ by setting $u(p) = [(\alpha(p), \alpha'(p))]$ where $(\alpha(p), \alpha'(p)) \in X \times_A X'$. Define $v : B \to BS$ by setting $v(b) = [(i(b), 0)]$ where $(i(b), 0) \in X \times_A X'$. One check that $v \circ (\beta + \beta') = u \circ j$. So by the universal property of Y, there exists a unique morphism $Y \to BS$. Then by Five Lemma 1.5.3, $Y \cong BS$, we win. $\square$

Remark We can define $\mathrm{Ext}^1(A, B)$ in any abelian category $\mathscr{A}$, even if it has no projectives and no injectives, to be the set of equivalence classes of extensions under Baer sum (if indeed this is a set). The Freyd-Mitchell Embedding Theorem shows that $\mathrm{Ext}^1(A, B)$ is an abelian group-but one could also prove this fact directly.

2.6.2 Higher Yoneda Ext groups

Definition 2.6.3 (Higher Yoneda Ext group)

An element of the Yoneda $\mathrm{YExt}^n(A, B)$ is an equivalence class of exact sequence of the form

$$\xi: \quad 0 \to B \to X_n \to \cdots \to X_1 \to A \to 0.$$

*It is called **similar** to an extension ξ' if there is a commutative diagram*

$$\begin{tikzcd}
\xi: & 0 \arrow[r] & B \arrow[r] \arrow[d, equal] & X_n \arrow[r] \arrow[d] & \cdots \arrow[r] & X_1 \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\
\xi': & 0 \arrow[r] & B \arrow[r] & X_n' \arrow[r] & \cdots \arrow[r] & X_1' \arrow[r] & A \arrow[r] & 0.
\end{tikzcd}$$

*Two extensions ξ and ξ' are called **equivalent** if there exist $\xi = \xi^0, \xi^1, \ldots, \xi^m = \xi'$ for some m, such that ξ^i and ξ^{i+1} are similar, i.e., either $\xi^i \to \xi^{i+1}$ or $\xi^i \leftarrow \xi^{i+1}$.*

To define $[\xi] +_b [\xi']$ when $n \geq 2$, let $X_1'' := X_1 \times_A X_1'$, let $X_n'' := X_n \sqcup_B X_n'$. Then $[\xi] +_b [\xi']$ is the class of extension

$$0 \to B \to X_n'' \to X_{n-1} \oplus X_{n-1}' \to \cdots \to X_2 \oplus X_2' \to X_1'' \to A \to 0$$

Theorem 2.6.2 (Yoneda)

There is an isomorphism of abelian groups

$$\mathrm{YExt}^n(A,B) \xrightarrow{\sim} \mathrm{Ext}^n(A,B).$$

Proof **We give a sketch of proof:**

Choose $\varepsilon : P_\bullet \to A$ to be a projective resolution, then the Comparison Theorem yields a map from $P_\bullet$ to ξ, hence a diagram

$$\begin{tikzcd} & 0 \arrow[r] & M \arrow[r] \arrow[d, "\beta"] & P_{n-1} \arrow[r] \arrow[d] & \cdots \arrow[r] & P_0 \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\ \xi : & 0 \arrow[r] & B \arrow[r] & X_n \arrow[r] & \cdots \arrow[r] & X_0 \arrow[r] & A \arrow[r] & 0. \end{tikzcd}$$

By dimension shifting trick, there is an exact sequence

$$\mathrm{Hom}(P_{n-1},B) \longrightarrow \mathrm{Hom}(M,B) \xrightarrow{\partial} \mathrm{Ext}^n(A,B) \longrightarrow 0$$

and we define a map $\mathrm{ch} : \mathrm{YExt}^n(A,B) \to \mathrm{Ext}^n(A,B)$ by setting $\mathrm{ch}([\xi]) := \partial(\beta)$. Then we check that this is well-defined and is an isomorphism. $\square$

Yoneda products

Given $A, B, C \in \mathbf{Mod}_R$, we can define the **Yoneda product**

$$\mathrm{Ext}^m_R(B,C) \otimes \mathrm{Ext}^n_R(A,B) \longrightarrow \mathrm{Ext}^{m+n}_R(A,C)$$

as follows. Given extensions

$$\xi : \quad 0 \longrightarrow B \xrightarrow{\alpha} X_n \longrightarrow \cdots \longrightarrow X_1 \longrightarrow A \longrightarrow 0$$

and

$$\eta : \quad 0 \longrightarrow C \longrightarrow Y_m \longrightarrow \cdots \longrightarrow Y_1 \xrightarrow{\alpha} B \longrightarrow 0$$

we define the product of $\eta \otimes \xi$ to be the equivalence class of the extension

$$0 \longrightarrow C \longrightarrow Y_m \longrightarrow \cdots \longrightarrow Y_1 \xrightarrow{\alpha\circ\beta} X_n \longrightarrow \cdots \longrightarrow X_1 \longrightarrow A \longrightarrow A \longrightarrow 0.$$

Yoneda product is associative. As a result, $\mathrm{Ext}^*_R(A,A)$ is a graded ring. Moreover, for any R-module M, $\mathrm{Ext}^*_R(A,B)$ is a graded module over $\mathrm{Ext}^*_R(A,A)$.

2.7 Universal Coefficient Theorems

Künneth formula for tensor and Universal Coefficient Theorem for homology

There is a very useful formula for using the homology of a chain complex $P_\bullet$ to compute the homology of the complex $P_\bullet \otimes M$. Here is the most useful general formulation we can give:

Theorem 2.7.1 (Künneth formula)

Let $P_\bullet$ be a chain complex of flat right R-modules such that each submodule $d(P_n)$ of P_{n-1} is also flat. Then for every n and every left R-module M, there is an exact sequence

$$0 \longrightarrow H_n(P_\bullet) \otimes_R M \longrightarrow H_n(P_\bullet \otimes_R M) \longrightarrow \mathrm{Tor}^R_1(H_{n-1}(P_\bullet), M) \longrightarrow 0.$$

Proof Since $d(P_n)$ is flat, by Proposition 2.4.3, $\mathrm{Tor}_1^R(d(P_n), M) = 0$, then we get a short exact sequence

$$0 \longrightarrow Z_n \otimes_R M \longrightarrow P_n \otimes_R M \longrightarrow d(P_n) \otimes M \longrightarrow 0$$

for every n. These give a short exact sequence of chain complexes

$$0 \longrightarrow Z_\bullet \otimes M \longrightarrow P_\bullet \otimes M \longrightarrow d(P_\bullet) \otimes M \longrightarrow 0$$

with the differentials of $Z_\bullet$ and $d(P_\bullet)$ are zero. The homology sequence is

$$H_{n+1}(d(P_\bullet) \otimes M) \xrightarrow{\partial_{n+1}} H_n(Z_\bullet \otimes M) \longrightarrow H_n(P_\bullet \otimes M) \longrightarrow H_n(d(P_\bullet) \otimes M) \xrightarrow{\partial_n} H_{n-1}(Z_\bullet \otimes M), \tag{2.7}$$

with

$$H_i(Z_\bullet \otimes M) = Z_i \otimes M, \quad H_i(dP_\bullet \otimes M) = d(P_i) \otimes M.$$

Moreover, the morphism ∂ is just induced from the inclusion map $d(P_n) \subseteq Z_{n-1}$ (tensored with M). We deduce from (2.7) a short exact sequence for each n

$$0 \longrightarrow \mathrm{Coker}(\partial_{n+1}) \longrightarrow H_n(P_\bullet \otimes M) \longrightarrow \mathrm{Ker}(\partial_n) \longrightarrow 0. \tag{2.8}$$

On the other hand, for any n, we have two short exact sequences

$$0 \longrightarrow Z_n \longrightarrow P_n \longrightarrow d(P_n) \longrightarrow 0, \quad 0 \longrightarrow d(P_{n+1}) \longrightarrow Z_n \longrightarrow H_n(P) \longrightarrow 0.$$

The first one shows that Z_n is also flat (use Lemma 2.7.1 below), since P_n and $d(P_n)$ are both flat. Thus the second one is a flat resolution of $H_n(P_\bullet)$, and $\mathrm{Tor}_*(H_n(P), M)$ is the homology of the complex

$$0 \longrightarrow d(P_{n+1}) \otimes_R M \xrightarrow{\partial_{n+1}} Z_n \otimes_R M \longrightarrow 0,$$

hence $\mathrm{Ker}\,\partial_{n+1} = d(P_{n+1}) \otimes_R M = \mathrm{Tor}_1^R(H_n(P_\bullet), M)$ and $\mathrm{Coker}\,\partial_{n+1} = Z_n \otimes M = \mathrm{Tor}_0^R(H_n(P_\bullet), M) = H_n(P_\bullet) \otimes M$. By (2.8), we have short exact sequence

$$0 \longrightarrow H_n(P_\bullet) \otimes_R M \longrightarrow H_n(P_\bullet \otimes_R M) \longrightarrow \mathrm{Tor}_1^R(H_{n-1}(P_\bullet), M) \longrightarrow 0.$$

□

Lemma 2.7.1

Suppose that $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of R-modules. If M' and M'' are flat so is M. If M and M'' are flat so is M'.

Proof See [Sta25, Lemma 00HM]. □

Theorem 2.7.2 (Universal Theorem for Homology)

Let $P_\bullet$ be a chain complex of free abelian groups. Then for every n and every abelian group M the Künneth formula 2.7.1 splits noncanonically, yielding a direct sum decomposition

$$H_n(P_\bullet \otimes M) \cong (H_n(P_\bullet) \otimes M) \oplus \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(P_\bullet), M).$$

Remark This uses the key fact that $\mathbb{Z}$ has flat dimension 1.

Proof We shall use the well-known fact that every subgroup of a free abelian group is free abelian. Since $d(P_n)$ is a subgroup of P_{n+1}, it is free abelian. By Proposition 2.2.1, the exact sequence

$$0 \longrightarrow Z_n \longrightarrow P_n \longrightarrow d(P_n) \longrightarrow 0$$

splits, giving a noncanonical isomorphism $P_n \cong Z_n \oplus d(P_n)$. Applying $\otimes M$,

$$P_n \otimes M \cong (Z_n \otimes M) \oplus (d(P_n) \otimes M),$$

we see that $Z_n \otimes M$ is a direct summand of $P_n \otimes M$; moreover, $Z_n \otimes M$ is a direct summand of the intermediate group

$$\mathrm{Ker}(d_n \otimes \mathrm{id}_M : P_n \otimes M \to P_{n-1} \otimes M).$$

So for group morphism $f : Z_n \otimes M \to \mathrm{Ker}(d_n \otimes \mathrm{id}_M)$, there is a retraction $r : \mathrm{Ker}(d_n \otimes \mathrm{id}_M) \to Z_n \otimes M$ such that $r \circ f = \mathrm{id}$. Note that $\mathrm{Im}(d_{n+1} \otimes \mathrm{id}_M) \subseteq Z_n \otimes M$ and $\mathrm{Im}(d_{n+1} \otimes \mathrm{id}_M) \subseteq \mathrm{Ker}(d_n \otimes \mathrm{id}_M)$. Let $\overline{f} : \frac{Z_n \otimes M}{\mathrm{Im}(d_{n+1} \otimes \mathrm{id}_M)} \to \frac{\mathrm{Ker}(d_n \otimes \mathrm{id}_M)}{\mathrm{Im}(d_{n+1} \otimes \mathrm{id}_M)}$, r induces a retraction $\overline{r} : \frac{\mathrm{Ker}(d_n \otimes \mathrm{id}_M)}{\mathrm{Im}(d_{n+1} \otimes \mathrm{id}_M)} \to \frac{Z_n \otimes M}{\mathrm{Im}(d_{n+1} \otimes \mathrm{id}_M)}$, so $\frac{Z_n \otimes M}{\mathrm{Im}(d_{n+1} \otimes \mathrm{id}_M)} = H_n(P_\bullet) \otimes M$ is a direct summand of $H_n(P_\bullet \otimes M)$. Since P_n and $d(P_n)$ are flat, the Künneth formula tells us that the other summand is $\mathrm{Tor}_1^R(H_{n-1}(P_\bullet), M)$. □

Theorem 2.7.3 (Künneth formula for complexes)

Let $P_\bullet$ and $Q_\bullet$ be right and left R-module complexes, respectively. Recall that the tensor product complex $P_\bullet \otimes_R Q_\bullet$ is the complex whose degree n part is $\bigoplus_{i+j=n} P_i \otimes Q_j$ and whose differential is given by $d(a \otimes b) = (da) \otimes b + (-1)^i a \otimes (db)$ for $a \in P_i, b \in Q_j$. If P_n and dP_n are flat for each n, then there is an exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(P_\bullet) \otimes H_j(Q_\bullet) \longrightarrow H_n(P_\bullet \otimes_R Q_\bullet) \longrightarrow \bigoplus_{i+j=n-1} \mathrm{Tor}_1^R(H_i(P_\bullet), H_j(Q_\bullet)) \longrightarrow 0$$

for each n.

If $R = \mathbb{Z}$ and $P_\bullet$ is a complex of free abelian groups, this sequence is noncanonically split.

Proof Modify the proof given in Künneth formula 2.7.1 for $Q_\bullet = M$. □

Application: Universal Coefficient Theorem for Homology in topology

Let $S(X)$ denote the singular chain complex of a topological space X; each $S_n(X)$ is a free abelian group. If M is any abelian group, the homology of X with "coefficients" in M is

$$H_*(X; M) = H_*(S(X) \otimes M).$$

Writing $H_*(X)$ for $H_*(X; \mathbb{Z})$, the formula in this case becomes

$$H_n(X; M) \cong (H_n(X) \otimes M) \oplus \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(X), M).$$

This formula is often called the **Universal Coefficient Theorem in topology**.

If Y is another topological space, the Eilenberg-Zilber Theorem [Wik25] states that $H_*(X \times Y) \cong H_*(S(X) \otimes S(Y))$. Therefore the Künneth formula yields the "Künneth formula for cohomology":

$$H_n(X \times Y) \cong \left\{ \bigoplus_{i=0}^{n} H_i(X) \otimes H_{n-i}(Y) \right\} \oplus \left\{ \bigoplus_{i=1}^{n} \mathrm{Tor}_1^{\mathbb{Z}} \left(H_{i-1}(X), H_{n-i}(Y)\right) \right\}.$$

Künneth formula for Hom and Universal Coefficient Theorem for Cohomology

We now turn to the analogue of the Künneth formula for Hom in place of $\otimes$.

Theorem 2.7.4 (Universal Coefficient Theorem for Cohomology)

Let $P_\bullet$ be a chain complex of projective R-modules such that each dP_n is also projective. Then for every n and every R-module M, there is a (noncanonically) split exact sequence

$$0 \longrightarrow \mathrm{Ext}_R^1(H_{n-1}(P_\bullet), M) \longrightarrow H^n(\mathrm{Hom}_R(P_\bullet, M)) \longrightarrow \mathrm{Hom}_R(H_n(P_\bullet), M) \longrightarrow 0. \tag{2.9}$$

Proof Since dP_n is projective, by Proposition 2.2.1, there is a (noncanonical) isomorphism $P_n \cong Z_n \oplus dP_n$

for each n. Apply $\mathrm{Hom}_R(\square, M)$, we have $\mathrm{Hom}_R(P_n, M) \cong \mathrm{Hom}_R(Z_n, M) \times \mathrm{Hom}_R(dP_n, M)$. Therefore each sequence

$$0 \longrightarrow \mathrm{Hom}(dP_n, M) \longrightarrow \mathrm{Hom}(P_n, M) \longrightarrow \mathrm{Hom}(Z_n, M) \longrightarrow 0$$

is exact. We may now copy the proof of the Künneth formula 2.7.1 for $\otimes$, using $\mathrm{Hom}(\square, M)$ instead of $\square \otimes M$, to see that the sequence is indeed exact. We may copy the proof of the Universal Coefficient Theorem 2.7.2 for $\otimes$ in the same way to see that the sequence is split. $\square$

Application: Universal Coefficient Theorem for Cohomology in topology

The cohomology of a topological space X with "coefficients" in M is defined to be

$$H^*(X; M) := H^*(\mathrm{Hom}(S(X), M)).$$

In this case, the Universal Coefficient Theorem becomes

$$H^n(X; M) \cong \mathrm{Hom}(H_n(X), M) \oplus \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(X), M).$$

2.8 Derived functors of the inverse limit

Let I be a small category and $\mathscr{A}$ an abelian category. We will see in Proposition 2.8.1 that the functor category $\mathscr{A}^I$ has enough injectives, at least when $\mathscr{A}$ is complete and has enough injective. (For example, $\mathscr{A}$ could be $\mathbf{Ab}$, $\mathbf{Mod}_R$, or $\mathbf{Shv}(X)$.) Therefore we can define the right derived functors $R^n \lim_{i\in I}$ from $\mathscr{A}^I$ to $\mathscr{A}$.

We are most interested in the case in which $\mathscr{A}$ is $\mathbf{Ab}$ and I is the poset $\cdots \to 2 \to 1 \to 0$ of whole numbers in reverse order. We shall call the objects of $\mathbf{Ab}^I$ (countable) **towers of abelian groups**; they have the form

$$\{A_i\}: \qquad \cdots \longrightarrow A_2 \longrightarrow A_1 \longrightarrow A_0.$$

In this section we shall give the alternative construction $\varprojlim^1$ of $R^1 \varprojlim$ for countable towers due to Eilenberg and prove that $R^n \varprojlim = 0$ for $n \neq 0, 1$. This construction generalizes from $\mathbf{Ab}$ to other abelian categories that satisfy the following axiom, introduced by Grothendieck in [Gro57]:

(AB4) $\mathscr{A}$ is complete, and the product of any set of surjections is a surjection.

Explanation: If I is a discrete set, $\mathscr{A}^I$ is the product category $\prod_{i\in I} \mathscr{A}$ of indexed families of objects $\{A_i\}$ in $\mathscr{A}$. For $\{A_i\}$ in $\mathscr{A}^I$, $\lim_{i\in I} A_i$ is the product $\prod A_i$. Axiom (AB4) states that the left exact functor $\prod$ from $\mathscr{A}^I$ to $\mathscr{A}$ is exact for all discrete I. Axiom (AB4) fails ($\prod_{i=1}^{\infty}$ is not exact) for some important abelian categories, such as $\mathbf{Shv}(X)$. On the other hand, axiom (AB4) is satisfied by many abelian categories in which objects have underlying sets, such as $\mathbf{Ab}$, $\mathbf{Mod}_R$, and $C(\mathbf{Mod}_R)$.

Proposition 2.8.1

If $\mathscr{A}$ is complete (i.e., any finite product exists, hence any small limit exists) and has enough injectives, then $\mathscr{A}^I$ has enough injectives.

Proof The construction is similar to Theorem 2.2.5. Given $\{A_i\} \in \mathscr{A}^I$, we may associate the k-th coordinate $A_k \in \mathscr{A}$, which gives an exact functor from $\mathscr{A}^I \to \mathscr{A}$; more precisely, $\mathrm{ev}_k : \mathscr{A}^I \to \mathscr{A}$ given by $\{A_i\}_{i\in I} \mapsto A_k$

is exact. On the other hand, given $k \in I$ and $B \in \mathscr{A}$, we may define an object $k_*(B) \in \mathscr{A}^I$ by:

$$(k_*B)_i = \begin{cases} B & i \geq k \\ 0 & \text{otherwise} \end{cases}$$

where for $i \geq j \geq k$, the transition morphism $(k_*B)_i \to (k_*B)_j$ is just the identity map on B. Let $\eta \in \mathrm{Hom}_{\mathscr{A}^I}(\{A_i\}, k_*B)$, note that for $i > k$

$$\begin{tikzcd} A_i \arrow[r, "\eta_i"] \arrow[d] & (k_*B)_i \arrow[r, equal] \arrow[d, "\mathrm{id}_B"] & B \\ A_k \arrow[r, "\eta_k"'] \arrow[ur, dashed] & (k_*B)_k \arrow[r, equal] & B \end{tikzcd}$$

i.e., $\eta_i = \eta_k \circ (A_i \to A_k)$ and $\eta_j = 0 : A_j \to (k_*B)_j = 0$ for $j < k$, so η is determined by η_k, it follows that

$$\mathrm{Hom}_{\mathscr{A}^I}(\{A_i\}, k_*B) \cong \mathrm{Hom}_{\mathscr{A}}(A_k, B),$$

i.e., $\mathrm{ev}_k \dashv k_*$. Since ev_k is exact, by Proposition 2.2.10, k_* sends injective objects to injective objects. Thus for any injective object E in $\mathscr{A}$, k_*E is injective object in $\mathscr{A}^I$.

Now given $\{A_i\} \in \mathscr{A}^I$, choose a monomorphism $A_k \hookrightarrow E_k$ for each $k \in I$, with E_k injective object. Then k_*E_k is injective object, and so is $\prod_{k\in I} k_*E_k$. Moreover, by $\mathrm{ev}_k \dashv k_*$, $A_k \hookrightarrow E_k$ is corresponding to a morphism $f_k : \{A_i\} \to k_*E_k$ for each k, by the universal property of product, here is a morphism

$$f : \{A_i\} \longrightarrow \prod_{k\in I} k_*(E_k). \tag{2.10}$$

Finally, we need to check that (2.10) is a monomorphism. Let $v, w : \{B_i\} \to \{A_i\}$ be two morphism such that $f \circ v = f \circ w$, we want to show that $v = w$. It suffices to check $v_j = w_j$ for all $j \in I$. Consider the following diagram.

$$\begin{tikzcd} \{B_i\} \arrow[r, shift left, "v"] \arrow[r, shift right, "w"'] & \{A_i\} \arrow[r, "f"] \arrow[dr, "f_j"'] & \prod\limits_{k\in I} k_*(E_k) \arrow[d, "\mathrm{pr}_j"] \\ & & j_*(E_j) \end{tikzcd}$$

Since $f \circ v = f \circ w$, we have $\mathrm{pr}_j \circ f \circ v = \mathrm{pr}_j \circ f \circ w$, and thus $f_j \circ v = f_j \circ w$; in particular, $(f_j)_j \circ v_j = (f_j)_j \circ w_j$. Note that $(f_j)_j : A_j \hookrightarrow E_j$ is a monomorphism, $v_j = w_j$, as we desired. ☐

In the following, we will consider the case $\mathscr{A} = \mathbf{Ab}$ or more generally $\mathscr{A} = \mathbf{Mod}_R$, and I is the poset $\cdots \to 2 \to 1 \to 0$ of whole natural numbers $\mathbb{N}$. Recall the following Theorem 1.5.1.

Theorem

Let

$$0 \longrightarrow A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i \longrightarrow 0$$

be an exact sequence of directed inverse systems of abelian groups over $\mathbb{N}$*. If* $\{A_i\}$ *is Mittag-Leffler, then*

$$0 \longrightarrow \varprojlim A_i \longrightarrow \varprojlim B_i \longrightarrow \varprojlim C_i \longrightarrow 0$$

is exact.

This may be consider as that $R^1 \varprojlim A_i = 0$ for an inverse system satisfying the Mittag-Leffler condition. In fact, taking a monomorphism to an injective object, say $\{A_i\} \hookrightarrow \{B_i\}$, we obtain

$$0 \longrightarrow \varprojlim A_i \longrightarrow \varprojlim B_i \longrightarrow \varprojlim B_i/A_i \longrightarrow R^1 \varprojlim A_i \longrightarrow 0,$$

hence $R^1 \varprojlim A_i = 0$ by Theorem 1.5.1.

Definition 2.8.1

Given an inverse system $\{A_i\}$ in $\mathbf{Ab}^{\mathbb{N}}$, define the map

$$\Delta : \prod_{i=0}^{\infty} A_i \longrightarrow \prod_{i=0}^{\infty} A_i$$

by the formula

$$\Delta(\cdots, a_i, \cdots, a_0) = (\cdots, a_i - \overline{a_{i+1}}, \cdots, a_1 - \overline{a_2}, a_0 - \overline{a_1})$$

where $\overline{a_{i+1}}$ denote the image of $a_{i+1} \in A_{i+1}$ in A_i. The kernel of Δ is $\varprojlim A_i$ as is easily checked. We define $\varprojlim^1 A_i$ to be the cokernel of Δ, making $\varprojlim^1$ to be a functor $\mathbf{Ab}^{\mathbb{N}} \to \mathbf{Ab}$.
We also set

$$\varprojlim^0 A_i = \varprojlim A_i, \quad \varprojlim^n A_i = 0, \forall n \neq 0, 1.$$

Lemma 2.8.1

If all the maps $A_{i+1} \to A_i$ are onto, then $\varprojlim^1 A_i = 0$. Moreover $\varprojlim A_i \neq 0$ (unless every $A_i = 0$), because each of the natural projections $\varprojlim A_i \to A_j$ are onto.

Proof Given elements $b_i \in A_i$ for $i = 0, 1, \ldots$, and any $a_0 \in A_0$, inductively choose $a_{i+1} \in A_{i+1}$ to be a lift of $a_i - b_i \in A_i$. The map Δ sends $(\cdots, a_1, a_0)$ to $(\cdots, b_1, b_0)$, so Δ is onto, and thus $\operatorname{Coker} \Delta = 0$. If all the $b_i = 0$, then $(\cdots, a_1, a_0) \in \varprojlim A_i$. $\square$

Lemma 2.8.2

The functors $\{\varprojlim^n\}$ form a universal cohomological δ-functor. Hence we have $\varprojlim^n = R^n \varprojlim$ for any n.

Proof If $0 \to \{A_i\} \to \{B_i\} \to \{C_i\} \to 0$ is a short exact sequence of inverse systems, we may apply Snake Lemma to

$$\begin{tikzcd}
0 \arrow[r] & \prod_i A_i \arrow[r] \arrow[d, "\Delta"] & \prod_i B_i \arrow[r] \arrow[d, "\Delta"] & \prod_i C_i \arrow[r] \arrow[d, "\Delta"] & 0 \\
0 \arrow[r] & \prod_i A_i \arrow[r] & \prod_i B_i \arrow[r] & \prod_i C_i \arrow[r] & 0
\end{tikzcd}$$

to get the natural long exact sequence

$$0 \to \varprojlim^0 A_i \to \varprojlim^0 B_i \to \varprojlim^0 C_i \to \varprojlim^1 A_i \to \varprojlim^1 B_i \to \varprojlim^1 C_i \to 0 \to \cdots$$

By the universal property of limits, it is easy to check (2) in Definition 2.3.1. Hence $\{\varprojlim^n\}$ form a δ-functor.

Recall Theorem 2.3.1, we only need to check $\{\varprojlim^n\}$ are effaceable for $n \geq 1$, for which it suffices to check $\varprojlim^1$ vanishes on enough injective objects, namely the injective objects in (2.10). By construction, it suffices to show $\varprojlim^1(k_*E)$ vanishes, where E is an injective object in $\mathscr{A}$. The inverse system $k_*E =: \{E_i\}$ is simply

$$k_*E: \qquad 0 \longleftarrow \cdots \longleftarrow 0 \longleftarrow E \xleftarrow{\mathrm{id}} E \xleftarrow{\mathrm{id}} E \xleftarrow{\mathrm{id}} \cdots.$$

In particular, all the transition maps are surjective, by Lemma 2.8.1, $\Delta_{k_*E}: \prod E_i \to \prod E_i$ is surjective, so $\operatorname{Coker}\Delta_{k_*E} = 0$, i.e., $\varprojlim^1(k_*E)$ vanishes. $\square$

Remark If we replace $\mathbf{Ab}$ by $\mathscr{A} = \mathbf{Mod}_R, C(\mathbf{Mod}_R)$ or any abelian category $\mathscr{A}$ satisfying Grothendieck's axiom (AB4), the above proof goes through to show that $\varprojlim^1 = R^1\varprojlim^1$ and $R^n\varprojlim = 0$ for $n \neq 0, 1$ as functors on the category of towers in $\mathscr{A}$. However, the proof breaks down for other abelian categories.

Corollary 2.8.1

For $n \geq 2$, $R^n\varprojlim(\square)$ vanishes.

Example 2.19 (Calculate $\varprojlim^1 p^i\mathbb{Z}$) Set $A_0 = \mathbb{Z}$ and let $A_i = p^i\mathbb{Z}$ be the subgroup generated by p^i. Applying $\varprojlim$ to the short exact sequence of towers

$$0 \longrightarrow \{p^i\mathbb{Z}\} \longrightarrow \{\mathbb{Z}\} \longrightarrow \{\mathbb{Z}/p^i\mathbb{Z}\} \longrightarrow 0,$$

we have long exact sequence

$$\begin{array}{ccccccccccc} 0 \longrightarrow \varprojlim p^i\mathbb{Z} & \longrightarrow & \varprojlim \mathbb{Z} & \longrightarrow & \varprojlim_i \mathbb{Z}/p^i\mathbb{Z} & \longrightarrow & \varprojlim^1 p^i\mathbb{Z} & \longrightarrow & \varprojlim^1 \mathbb{Z} & \longrightarrow & \cdots \\ & & \| & & \| & & & & \| & & \\ & & \mathbb{Z} & & \widehat{\mathbb{Z}}_p & & & & 0, & & \end{array}$$

where $\widehat{\mathbb{Z}}_p = \varprojlim_i \mathbb{Z}/p^i\mathbb{Z}$ is the group of p-adic integers. Consider $\varprojlim p^i\mathbb{Z}$, by the construction of inverse limit, $\varprojlim p^i\mathbb{Z} = 0$. Then we have an exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \widehat{\mathbb{Z}}_p \longrightarrow \varprojlim^1 p^i\mathbb{Z} \longrightarrow 0,$$

it follows that $\varprojlim^1 p^i\mathbb{Z} \cong \widehat{\mathbb{Z}}_p/\mathbb{Z}$

Proposition 2.8.2

If $\{A_i\}$ satisfies the Mittag-Leffler condition 1.5.3, then

$$\varprojlim^1 A_i = 0.$$

Proof We first show the case that for each k, there exists a $j > k$ such that the map $A_j \to A_k$ is zero (we call this **trivial Mittag-Leffler condition**). Let $(\cdots, b_1, b_0) \in \prod_i A_i$, set $a_k = b_k + \overline{b_{k+1}} + \cdots + \overline{b_{j-1}}$ (note that $\overline{b_i} = 0$ for $i \geq j$), where $\overline{b_i}$ denotes the image of b_i in A_k. Then Δ maps $(\cdots, a_1, a_0)$ to $(\cdots, b_1, b_0)$. Thus Δ is onto and $\varprojlim^1 A_i = 0$.

We next show the general case, i.e., $\{A_i\}$ satisfies the Mittag-Leffler condition. Let $B_k \subseteq A_k$ be the image of $A_i \to A_k$ for large i. Say $f_{k,k+1}: A_{k+1} \to A_k$, then we have $f_{k,i} = f_{k,k+1} \circ f_{k+1,i}$, hence $f_{k,i}(A_i) = B_k = f_{k,k+1} \circ f_{k+1,i}(A_i) = f_{k,k+1}(B_{k+1})$, i.e., $f_{k,k+1}|_{B_{k+1}}: B_{k+1} \to B_k$ are all onto, by Lemma 2.8.1, $\varprojlim^1 B_i = 0$. By the construction, $\{A_k/B_k\}$ holds trivial Mittag-Leffler condition, so $\varprojlim^1 A_k/B_k = 0$. Applying $\varprojlim$ to the short exact sequence

$$0 \longrightarrow \{B_i\} \longrightarrow \{A_i\} \longrightarrow \{A_i/B_i\} \longrightarrow 0,$$

we have long exact sequence

$$\begin{tikzcd}
\cdots \arrow[r] & \varprojlim A_i/B_i \arrow[r] & \varprojlim^1 B_i \arrow[r] \arrow[d, equal] & \varprojlim^1 A_i \arrow[r] & \varprojlim^1 A_i/B_i \arrow[r] \arrow[d, equal] & \cdots \\
& & 0 & & 0, &
\end{tikzcd}$$

we see that $\varprojlim^1 A_i = 0$, as we desired. □

Theorem 2.8.1

Let $C_0^\bullet \leftarrow C_1^\bullet \leftarrow \cdots$ be a system of cochain complexes of abelian groups satisfying the Mittag-Leffler condition, and set $C^\bullet = \varprojlim C_i^\bullet$. Then there is an exact sequence for each q:

$$0 \longrightarrow \varprojlim^1 H^{q-1}(C_i^\bullet) \longrightarrow H^q(C^\bullet) \longrightarrow \varprojlim H^q(C_i^\bullet) \longrightarrow 0.$$

Proof We first note the following facts:

(a) Since $\varprojlim$ is left exact, if we have a morphism $\{A_i\} \xrightarrow{f_i} \{A_i'\}$, with limit $A \xrightarrow{f} A'$, then $\operatorname{Ker} f = \varprojlim \operatorname{Ker} f_i$.

(b) However, it is subtler for image: the exact sequence $0 \to \operatorname{Ker} f_i \to A_i \to \operatorname{Im} f_i \to 0$ implies

$$0 \longrightarrow \operatorname{Ker} f \longrightarrow A \longrightarrow \varprojlim \operatorname{Im} f_i \longrightarrow \varprojlim^1 \operatorname{Ker} f_i \longrightarrow \varprojlim^1 A_i \longrightarrow \varprojlim^1 \operatorname{Im} f_i \longrightarrow 0$$

In the case $\{A_i\}$ satisfies the Mittag-Leffler condition, we have $\varprojlim^1 A_i = 0$, hence by (a) we obtain an exact sequence

$$0 \longrightarrow \operatorname{Ker} f \longrightarrow A \longrightarrow \varprojlim \operatorname{Im} f_i \longrightarrow \varprojlim^1 \operatorname{Ker} f_i \longrightarrow 0 \longrightarrow \varprojlim^1 \operatorname{Im} f_i \longrightarrow 0.$$

Note that

$$\operatorname{Im}(A \to \varprojlim \operatorname{Im} f_i) \cong A/\operatorname{Ker}(A \to \varprojlim \operatorname{Im} f_i) \cong A/\operatorname{Im}(\operatorname{Ker} f \to A) \cong A/\operatorname{Ker} f \cong \operatorname{Im} f,$$

then we obtain

$$0 \longrightarrow \operatorname{Im} f \longrightarrow \varprojlim \operatorname{Im} f_i \longrightarrow \varprojlim^1 \operatorname{Ker} f_i \longrightarrow 0, \qquad \varprojlim^1 \operatorname{Im} f_i = 0.$$

Now we prove the theorem. Define $B_i^\bullet \subseteq Z_i^\bullet \subseteq C_i^\bullet$ to be the cocycles and coboundaries, and similarly $B^\bullet \subseteq Z^\bullet \subseteq C^\bullet$. Then we have a SES for each q

$$0 \longrightarrow Z_i^{q-1} \longrightarrow C_i^{q-1} \longrightarrow B_i^q \longrightarrow 0.$$

By (a) above, $Z^q = \varprojlim Z_i^q$; by (b) above, we have $\varprojlim^1 B_i^q = 0$ and SES

$$0 \longrightarrow B^q \longrightarrow \varprojlim B_i^q \longrightarrow \varprojlim^1 Z_i^{q-1} \longrightarrow 0. \tag{2.11}$$

Hence, we obtain $(\varprojlim B_i^q)/B^q \cong \varprojlim^1 Z_i^{q-1}$.

On the other hand, by applying $\varprojlim$ to the SES $0 \to B_i^q \to Z_i^q \to H^q(C_i^\bullet) \to 0$ we have

$$\begin{tikzcd}
0 \arrow[r] & \varprojlim B_i^q \arrow[r] & \varprojlim Z_i^q \arrow[r] \arrow[d, equal] & \varprojlim H^q(C_i^\bullet) \arrow[r] & \varprojlim^1 B_i^q \arrow[r] \arrow[d, equal] & \varprojlim^1 Z_i^q \arrow[r] & \varprojlim^1 H^q(C_i^\bullet) \arrow[r] & 0 \\
& & Z^q & & 0, & & &
\end{tikzcd} \tag{2.12}$$

hence, we obtain $Z^q/\varprojlim B_i^q \cong \varprojlim^1 H^q(C_i^\bullet)$ and $\varprojlim^1 Z_i^q \cong \varprojlim^1 H^q(C_i^\bullet)$.

The result then follows: indeed, C^q has a filtration

$$0 \subseteq B^q \subseteq \varprojlim B_i^q \subseteq Z^q \subseteq C^q,$$

then there is a SES

$$0 \longrightarrow (\varprojlim B_i^q)/B^q \longrightarrow Z^q/B^q \longrightarrow Z^q/\varprojlim B_i^q \longrightarrow 0.$$

Recall that $Z_q/\varprojlim B_i^q \cong \varprojlim^1 H^q(C_i^\bullet)$ and $(\varprojlim B_i^q)/B^q \cong \varprojlim^1 Z_i^{q-1} \cong \varprojlim^1 H^{q-1}(C_i^\bullet)$, we have SES

$$0 \longrightarrow \varprojlim^1 H^{q-1}(C_i^\bullet) \longrightarrow H^q(C^\bullet) \longrightarrow \varprojlim H^q(C_i^\bullet) \longrightarrow 0,$$

we win. □

Corollary 2.8.2

Let A be a left R-module that is the union of submodules: $A_0 \subseteq \cdots \subseteq A_i \subseteq A_{i+1} \subseteq \cdots$. Then, for every R-module B and every q, the sequence

$$0 \longrightarrow \varprojlim^1 \operatorname{Ext}_R^{q-1}(A_i, B) \longrightarrow \operatorname{Ext}_R^q(A, B) \longrightarrow \varprojlim \operatorname{Ext}_R^q(A_i, B) \longrightarrow 0 \qquad (2.13)$$

is exact.

Proof Choose an injective resolution $B \hookrightarrow E^\bullet$; then we obtain a system of cochain complexes

$$\cdots \longrightarrow \operatorname{Hom}_R(A_{i+1}, E^\bullet) \longrightarrow \operatorname{Hom}_R(A_i, E^\bullet) \longrightarrow \cdots \longrightarrow \operatorname{Hom}_R(A_0, E^\bullet). \qquad (2.14)$$

The Mittag-Leffler condition is satisfied because each E^q is injective, so that $\operatorname{Hom}(A_{i+1}, E^q) \to \operatorname{Hom}(A_i, E^q)$ is surjective. Since the cohomology of $\operatorname{Hom}(A, E^\bullet)$ (resp. $\operatorname{Hom}(A_i, E^\bullet)$) computes $\operatorname{Ext}^q(A, B)$ (resp. $\operatorname{Ext}^q(A_i, B)$), the result follows from Theorem 2.8.1. □

Remark In Corollary 2.8.2, we need the morphisms $A_i \to A_{i+1}$ to be injective so that the inverse system (2.14) satisfies the Mittag-Leffler condition.

More generally, say $A = \varinjlim_i A_i$. For $i \leq k$, write

$$B_{i,k} = \operatorname{Im}(A_i \to A_k), \quad Z_{i,k} = \operatorname{Ker}(A_i \to A_k)$$

Then since each E^n (for n fixed) is injective, we see that the image of

$$\operatorname{Hom}_R(A_k, E^n) \longrightarrow \operatorname{Hom}_R(A_i, E^n)$$

is equal to $\operatorname{Hom}_R(B_{i,k}, E^n)$, hence $\operatorname{Hom}_R(A_i, E^n)$ satisfies Mittag-Leffler condition if and only if the submodule $\operatorname{Hom}_R(B_{i,k}, E^n)$ stabilize. On the other hand, there is an exact sequence

$$0 \longrightarrow \operatorname{Hom}_R(B_{i,k}, E^n) \longrightarrow \operatorname{Hom}_R(A_i, E^n) \longrightarrow \operatorname{Hom}_R(Z_{i,k}, E^n) \longrightarrow 0$$

so the latter condition is equivalent to the quotient modules $\operatorname{Hom}_R(Z_{i,k}, E^n)$ being stable. But, $\{Z_{i,k}, k \geq i\}$ form an increasing sequence of submodules of A_i, the condition holds **if**, for example, each A_i is a Noetherian R-module.

Chapter 3 Spectral Sequences

3.1 Motivating example

Before giving the precise definition of a spectral sequence, we look at some basic examples showing that spectral sequences are ubiquitous.

3.1.1 Example: Non-acyclic resolutions

Let us work in the category $\mathbf{Mod}_R$. When we define $\mathrm{Ext}^i_R(A, B)$, we choose first an injective resolution $B \to I^\bullet$, and calculate the cohomology of $\mathrm{Hom}_R(A, I^\bullet)$. One may ask the following question:

Question

What happen if we start with a resolution $B \to I^\bullet$ with I^i not necessarily injective?

Let us look at some special cases.

(a) Suppose given a resolution $0 \to B \to I^0 \xrightarrow{d^0} I^1 \to 0$. Then the long exact sequence gives

$$0 \longrightarrow \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}(A, I^0) \xrightarrow{d^{00}} \mathrm{Hom}(A, I^1) \longrightarrow$$

$$\longrightarrow \mathrm{Ext}^1(A, B) \longrightarrow \mathrm{Ext}^1(A, I^0) \xrightarrow{d^{01}} \mathrm{Ext}^1(A, I^1) \longrightarrow \cdots .$$

In particular, if we denote $d^{0q} : \mathrm{Ext}^q(A, I^0) \to \mathrm{Ext}^q(A, I^1)$, then there is an exact sequence

$$0 \longrightarrow \mathrm{Coker}(d^{0,q-1}) \longrightarrow \mathrm{Ext}^q(A, B) \longrightarrow \mathrm{Ker}(d^{0,q}) \longrightarrow 0,$$

where it is easy to check that $\mathrm{Coker}(d^{0,q-1}) \cong \mathrm{Ker}(\mathrm{Ext}^q(A, B) \to \mathrm{Ext}^q(A, I^0))$.

On way to graphically draw it as follows: put at (p, q) coordinate the module $\mathrm{Ext}^q(A, I^p)$ (with 0 for $p \neq 0, 1$ or $q < 0$):

$$\begin{array}{ccc}
\cdots & & \cdots \\
\mathrm{Ext}^1(A, I^0) & \xrightarrow{d^{01}} & \mathrm{Ext}^1(A, I^1) \\
\mathrm{Hom}(A, I^0) & \xrightarrow{d^{00}} & \mathrm{Hom}(A, I^1)
\end{array}$$

Each row $\mathrm{Ext}^q(A, I^\bullet)$ is a complex induced from $I^\bullet$, and the cohomology are related to $\mathrm{Ext}^i(A, B)$ via

$$0 \longrightarrow H^1(\mathrm{Ext}^{q-1}(A, I^\bullet)) \longrightarrow \mathrm{Ext}^q(A, B) \longrightarrow H^0(\mathrm{Ext}^q(A, I^\bullet)) \longrightarrow 0.$$

(It is easy to see that $H^0(\mathrm{Ext}^q(A, I^\bullet)) \cong \mathrm{Ker}(d^{0,q})$ and $H^1(\mathrm{Ext}^{q-1}(A, I^\bullet)) \cong \mathrm{Coker}(d^{0,q-1})$.)

(b) Suppose given a resolution $0 \to B \xrightarrow{\varepsilon} I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \to 0$. Then we may split it as

$$0 \to B \xrightarrow{\varepsilon} I^0 \xrightarrow{\eta} M \to 0, \qquad 0 \to M \xrightarrow{\iota} I^1 \to I^2 \to 0$$

(where $M = \operatorname{Ker} d^1$) which induce respectively

$$0 \longrightarrow \operatorname{Hom}(A,B) \longrightarrow \operatorname{Hom}(A,I^0) \xrightarrow{\eta_*} \operatorname{Hom}(A,M) \longrightarrow$$

$$\longrightarrow \operatorname{Ext}^1(A,B) \xrightarrow{\varepsilon^1} \operatorname{Ext}^1(A,I^0) \xrightarrow{\eta^1} \operatorname{Ext}^1(A,M) \longrightarrow \cdots,$$

$$0 \longrightarrow \operatorname{Hom}(A,M) \xrightarrow{\iota_*} \operatorname{Hom}(A,I^1) \xrightarrow{d^{10}} \operatorname{Hom}(A,I^2) \longrightarrow$$

$$\longrightarrow \operatorname{Ext}^1(A,M) \xrightarrow{\iota^1} \operatorname{Ext}^1(A,I^1) \longrightarrow \operatorname{Ext}^1(A,I^2) \longrightarrow \cdots.$$

We draw the picture as follows:

$$\begin{tikzcd}
\cdots & \cdots & \cdots \\
\operatorname{Ext}^1(A,I^0) \arrow[r, "d^{01}"] & \operatorname{Ext}^1(A,I^1) \arrow[r, "d^{11}"] & \operatorname{Ext}^1(A,I^2) \\
\operatorname{Hom}(A,I^0) \arrow[r, "d^{00}"] & \operatorname{Hom}(A,I^1) \arrow[r, "d^{10}"] & \operatorname{Hom}(A,I^2)
\end{tikzcd}$$

We check that:

- In fact, $d^{00} = \iota_* \circ \eta_*$, and ι_* is injective, so $\operatorname{Im}(d^{00}) = \operatorname{Im}\eta_* \cong \operatorname{Ker}(\operatorname{Hom}(A,M) \to \operatorname{Ext}^1(A,B))$. Hence

$$\begin{aligned}
H^1(\operatorname{Hom}(A,I^\bullet)) &= \operatorname{Ker}(d^{10}) / \operatorname{Im}(d^{00}) = \operatorname{Hom}(A,M)/\operatorname{Im}(\eta_*) \\
&\cong \operatorname{Hom}(A,M)/\operatorname{Ker}(\operatorname{Hom}(A,M) \to \operatorname{Ext}^1(A,B)) \\
&\cong \operatorname{Im}(\operatorname{Hom}(A,M) \to \operatorname{Ext}^1(A,B)) \\
&\cong \operatorname{Ker}(\varepsilon^1).
\end{aligned}$$

- Since $d^0 : I^0 \to I^1$ factor through M and $\operatorname{Ext}^1(A,\square)$ is a functor, we have $d^{01} = \iota^1 \circ \eta^1$, so there is an inclusion $\operatorname{Im}\varepsilon^1 = \operatorname{Ker}\eta^1 \subseteq \operatorname{Ker} d^{01}$ (**warning:** in general not equality). Rather, note that

$$\begin{aligned}
\operatorname{Ker}\iota^1 &= \operatorname{Im}(\operatorname{Hom}(A,I^2) \to \operatorname{Ext}^1(A,M)) \\
&= \operatorname{Hom}(A,I^2)/\operatorname{Ker}(\operatorname{Hom}(A,I^2) \to \operatorname{Ext}^1(A,M)) \\
&= \operatorname{Hom}(A,I^2)/\operatorname{Im} d^{10} \\
&= \operatorname{Coker} d^{10},
\end{aligned}$$

there exists a map $d_2^{01} : \operatorname{Ker} d^{01} \to \operatorname{Coker} d^{10}$ fitting in the diagram below

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Ker} d^{01} \arrow[r] \arrow[d, dashed, "d_2^{01}"'] & \operatorname{Ext}^1(A,I^0) \arrow[r, "d^{01}"] \arrow[d, "\eta^1"'] & \operatorname{Ext}^1(A,I^1) \arrow[d, equal] \\
0 \arrow[r] & \operatorname{Coker} d^{10} \arrow[r] & \operatorname{Ext}^1(A,M) \arrow[r, "\iota^1"] & \operatorname{Ext}^1(A,I^1)
\end{tikzcd}$$

and crucially $\operatorname{Ker} d_2^{01} = \operatorname{Ker}\eta^1 = \operatorname{Im}\varepsilon^1$ by Snake Lemma. Moreover, we have $H^0(\operatorname{Ext}^1(A,I^\bullet)) = \operatorname{Ker} d^{01}$ and $H^1(\operatorname{Ext}^0(A,I^\bullet)) = \operatorname{Coker} d^{10}$.

Summary:

Setting $E_2^{pq} := H^p(\mathrm{Ext}^q(A, I^\bullet))$, we obtain again a family of objects indexed by (p, q):

$$\begin{tikzcd}
\cdots & \cdots & \cdots \\
E_2^{01} \arrow[rrd, dashed, "d_2^{01}"] & E_2^{11} & E_2^{21} \\
E_2^{00} & E_2^{10} & E_2^{20}
\end{tikzcd}$$

together with an extra (but not obvious) morphism $d_2^{01} : E_2^{01} \to E_2^{20}$ such that there is a short exact sequence (recall $E_2^{10} = H^1(\mathrm{Hom}(A, I^\bullet)) = \mathrm{Ker}\,\varepsilon^1$, we have proved above)

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Ker}\,\varepsilon^1 \arrow[r] \arrow[d, equal] & \mathrm{Ext}^1(A, B) \arrow[r] & \mathrm{Im}\,\varepsilon^1 \arrow[r] \arrow[d, equal] & 0 \\
& E_2^{10} & & \mathrm{Ker}\,d_2^{01} &
\end{tikzcd}$$

Note that if $I^\bullet$ is an ijective resolution, then all $E_2^{pq} = 0$ for $q \geq 1$, hence the above sequence degenerates to $E_2^{10} \cong \mathrm{Ext}^1(A, B)$ as usual.

Remark The argument goes through for any left exact covariant functor in place of $\mathrm{Hom}_R(A, \square)$.

3.1.2 Example: Universal Coefficients Theorem

Question

Give a (not necessarily acyclic) complex $I^\bullet$ of injective objects, can we compute $H^n(\mathrm{Hom}(A, I^\bullet))$ using $\mathrm{Ext}^i(A, H^i(I^\bullet))$?

Remark If $I^\bullet$ is acyclic, then it is an injective resolution of $H^0(I^\bullet)$ and we have $H^n(\mathrm{Hom}(A, I^\bullet))$ using $\mathrm{Ext}^n(A, H^0(I^\bullet))$.

Recall the Universal Coefficient Theorem for Cohomology.

Theorem 3.1.1 (Universal Coefficient Theorem for Cohomology)

Let $I^\bullet$ be a cochain complex of injective R-modules such that each $d(I^n)$ is also injective. Then for every n and every R-module A, there is a (noncanonically) split exact sequence

$$0 \longrightarrow \mathrm{Ext}_R^1(A, H^{n-1}(I^\bullet)) \longrightarrow H^n(\mathrm{Hom}_R(A, I^\bullet)) \longrightarrow \mathrm{Hom}_R(A, H^n(I^\bullet)) \longrightarrow 0.$$

Let's drop the assumption that "each $d(I^n)$ is injective", and assume simply that $I^\bullet = (0 \to I^0 \to I^1 \to 0)$. Then we obtain two short exact sequences:

$$0 \to H^0(I^\bullet) \to I^0 \to B^0 \to 0, \qquad 0 \to B^0 \to I^1 \to H^1(I^\bullet) \to 0,$$

where $B^0 = d(I^0)$. Applying $\mathrm{Hom}(A, \square)$ we obtain (using the injectivity of I^0, I^1):

$$\begin{gathered}
0 \longrightarrow \mathrm{Hom}(A, H^0(I^\bullet)) \longrightarrow \mathrm{Hom}(A, I^0) \xrightarrow{\alpha} \mathrm{Hom}(A, B^0) \longrightarrow \mathrm{Ext}^1(A, H^0(I^\bullet)) \longrightarrow 0, \\
0 \longrightarrow \mathrm{Ext}^1(A, B^0) \xrightarrow{\sim} \mathrm{Ext}^2(A, H^0(I^\bullet)) \longrightarrow 0
\end{gathered} \tag{3.1}$$

and

$$0 \longrightarrow \mathrm{Hom}(A, B^0) \xrightarrow{\beta} \mathrm{Hom}(A, I^1) \longrightarrow \mathrm{Hom}(A, H^1(I^\bullet)) \xrightarrow{\gamma} \mathrm{Ext}^1(A, B^0) \longrightarrow 0.$$

Since β is injective, by Lemma 3.1.1 below, we have an isomorphism

$$\mathrm{Hom}(A, H^0(I^\bullet)) = \mathrm{Ker}\,\alpha \cong \mathrm{Ker}\,\beta\alpha = H^0(\mathrm{Hom}(A, I^\bullet)),$$

and also $0 \to \mathrm{Coker}(\alpha) \to \mathrm{Coker}(\beta\alpha) \to \mathrm{Coker}(\beta) \to 0$ which translates to

$$0 \longrightarrow \mathrm{Ext}^1(A, H^0(I^\bullet)) \longrightarrow H^1(\mathrm{Hom}(A, I^\bullet)) \longrightarrow \mathrm{Ker}(\gamma) \longrightarrow 0. \tag{3.2}$$

Note that we wish to view γ as $\mathrm{Hom}(A, H^1(I^\bullet)) \xrightarrow{\gamma} \mathrm{Ext}^1(A, B^0) \cong \mathrm{Ext}^2(A, H^0(I^\bullet))$ by (3.1), because we want an expression using only $H^i(I^\bullet)$ (not B^i!). Thus, we arrive at the following situation (again!)

$$\begin{tikzcd}
\cdots & \cdots & \cdots \\
E_2^{01} \arrow[drr, dashed, "d_2^{01} := \gamma"] & E_2^{11} & E_2^{21} \\
E_2^{00} & E_2^{10} & E_2^{20}
\end{tikzcd}$$

where $E_2^{pq} := \mathrm{Ext}_R^p(A, H^q(I^\bullet))$. With this notation, we have a long exact sequence from (3.2)

$$0 \longrightarrow \mathrm{Ext}^1(A, H^0(I^\bullet)) \longrightarrow H^1(\mathrm{Hom}(A, I^\bullet)) \longrightarrow E_2^{01} \xrightarrow{\gamma} E_2^{20} \longrightarrow 0. \tag{3.3}$$

Remark that, under the assumption "$d(I^0) = B^0$ is injective object", we have $E_2^{20} = 0$, which is compatible with the Universal Coefficient Theorem 3.1.1.

Lemma 3.1.1

Let $\alpha : A \to B$, $\beta : B \to C$ be morphisms in abelian category $\mathscr{A}$. Then we have a long exact sequence

$$0 \to \mathrm{Ker}(\alpha) \to \mathrm{Ker}(\beta\alpha) \to \mathrm{Ker}(\beta) \to \mathrm{Coker}(\alpha) \to \mathrm{Coker}(\beta\alpha) \to \mathrm{Coker}(\beta) \to 0.$$

Proof Consider the following diagram

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r, "\begin{pmatrix} \mathrm{id}_A \\ 0 \end{pmatrix}"] \arrow[d, "\alpha"'] & A \oplus B \arrow[r, "(0, \mathrm{id}_B)"] \arrow[d, "\varphi"] & B \arrow[r] \arrow[d, "\beta"] & 0 \\
0 \arrow[r] & B \arrow[r, "\begin{pmatrix} \mathrm{id}_A \\ 0 \end{pmatrix}"'] & B \oplus C \arrow[r, "(0, \mathrm{id}_C)"'] & C \arrow[r] & 0
\end{tikzcd} \tag{3.4}$$

where $\varphi := \begin{pmatrix} \alpha & -\mathrm{id}_B \\ 0 & \beta \end{pmatrix}$, on check that above diagram commutes. By Snake Lemma, we have long exact sequence

$$0 \to \mathrm{Ker}(\alpha) \to \mathrm{Ker}(\varphi) \to \mathrm{Ker}(\beta) \to \mathrm{Coker}(\alpha) \to \mathrm{Coker}(\varphi) \to \mathrm{Coker}(\beta) \to 0.$$

By the construction of φ, it is easy to see that $\mathrm{Ker}\,\beta\alpha \cong \mathrm{Ker}\,\varphi$ and $\mathrm{Coker}\,\beta\alpha \cong \mathrm{Coker}\,\varphi$ (use Freyd-Mitchell Embedding Theorem 1.5.2), we are done. $\square$

3.2 Definition: Spectral sequence

Setting

Let $\mathscr{A}$ be an abelian category.

Spectral sequence

Definition 3.2.1 (Spectral sequence)

*A **cohomological spectral sequence** in $\mathscr{A}$ starting on page $a \geq 0$ consisting of the following data:*

(a) an object $E_r^{p,q}$ of $\mathscr{A}$ for every $p, q \in \mathbb{Z}$ and $r \geq a$;

(b) a morphism $d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}$ for $p, q \in \mathbb{Z}$ and $r \geq a$ such that $d_r^{p+r,q-r+1} d^{p,q} = 0$;

(c) an isomorphism $\alpha_{r+1}^{p,q} : E_{r+1}^{p,q} \xrightarrow{\sim} \operatorname{Ker}(d_r^{p,q}) / \operatorname{Im}(d_r^{p-r,q+r-1})$ for $p, q \in \mathbb{Z}$ and $r \geq a$.

Note that $E_{r+1}^{p,q}$ is a subquotient of $E_r^{p,q}$. The total degree of $E_r^{p,q}$ is $n = p + q$. Each differential $d_r^{p,q}$ increases the total degree by one.

A morphism $f : E \to E'$ between two spectral sequence is a family of maps $f_r^{p,q} : E_r^{p,q} \to E_r'^{p,q}$ with $d_r' f_r = f_r d_r$. With this, spectral sequences form a category.

The the sequences of arrows giving the spectral sequence would look like this:

$$
\begin{tikzcd}
(0,1) & & & \\
\bullet \arrow[u, "\text{page 0}"] \arrow[r, "\text{page 1}"] \arrow[drr, "\text{page 2}"] \arrow[ddrrr, "\text{page 3}"] & (1,0) & & \\
 & & (2,-1) & \\
 & & & (3,-2)
\end{tikzcd}
$$

Example 3.1 A first quadrant spectral sequence is one with $E_r^{p,q} = 0$, unless $p \geq 0$ and $q \geq 0$. For fixed p, q, $E_r^{p,q} = E_{r+1}^{p,q}$ for large r, i.e., $r > \max\{p, q+1\}$. For example (we only consider the position $(0,0)$): At page E_0:

$$
\begin{tikzcd}
0 & E_0^{0,1} & E_0^{1,1} \\
0 \arrow[u] & E_0^{0,0} \arrow[u, "d_0^{0,0}"] & E_0^{1,0} \arrow[u, "d_0^{1,0}"] \\
0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u]
\end{tikzcd}
$$

At page E_1:

$$
\begin{tikzcd}
0 \arrow[r] & ? \arrow[r] & ? \\
0 \arrow[r] & \operatorname{Ker} d_0^{0,0} \arrow[r, "d_1^{0,0}"] & \operatorname{Ker} d_0^{1,0} \\
0 \arrow[r] & 0 \arrow[r] & 0
\end{tikzcd}
$$

At page E_2:

$$\begin{tikzcd}
0 \arrow[drr] & 0 & ? & ? & ? \\
0 & 0 & \operatorname{Ker} d_1^{0,0} \arrow[drr] & ? & ? \\
0 & 0 & 0 & 0 & 0
\end{tikzcd}$$

Hence $E_r^{0,0} = E_{r+1}^{0,0}$ for all $r \geq 2$.

Example 3.2 Let $A, B \in \mathscr{A}$ (with enough projective objects and also enough injective objects). Choose a projective resolution $P_\bullet \to A$ and an injective resolution $B \to I^\bullet$, we form the double complexes $\operatorname{Hom}(P_\bullet, I^\bullet)$. Set $a = 0$ and $E_0^{p,q} = \operatorname{Hom}(P_q, I^p)$, we obtain the E_0 page of the spectral sequence $(E_0^{p,q} \to E_0^{p,q+1})$

$$\begin{tikzcd}
\vdots & \vdots & \vdots \\
E_0^{01} \arrow[u] & E_0^{11} \arrow[u] & E_0^{21} \arrow[u] \\
E_0^{00} \arrow[u] & E_0^{10} \arrow[u] & E_0^{20} \arrow[u]
\end{tikzcd}$$

Taking the cohomology of each column, we obtain $E_1^{p,q} = \operatorname{Ext}^q(A, I^p)$ since $P_\bullet$ is a projective resolution of A, and the E_1 page is as follows:

$$\begin{tikzcd}
\cdots \arrow[r] & \cdots \arrow[r] & \cdots \\
E_1^{01} \arrow[r] & E_1^{11} \arrow[r] & \cdots \\
E_1^{00} \arrow[r] & E_1^{10} \arrow[r] & \cdots
\end{tikzcd}$$

Again taking the cohomology, we obtain the E_2 page:

$$\begin{tikzcd}
\cdots & \cdots & \cdots \\
E_2^{01} \arrow[drr, dashed, "d_2^{01}"] & E_2^{11} & E_2^{21} \\
E_2^{00} & E_2^{10} & E_2^{20}
\end{tikzcd}$$

E_3 page:

$$\begin{tikzcd}
\cdots & \cdots & \cdots & \cdots \\
E_3^{02} \arrow[ddrrr, dashed, "d_3^{0,2}"] & E_3^{12} & E_3^{22} & E_3^{32} \\
E_3^{01} & E_3^{11} & E_3^{21} & E_3^{31} \\
E_3^{00} & E_3^{10} & E_3^{20} & E_3^{30}
\end{tikzcd}$$

E_∞-terms

By definition, each $E_{r+1}^{p,q}$ is a subquotient of $E_r^{p,q}$, hence inductively a subquotient of $E_a^{p,q}$. Thus, there is a sequence of subobjects $B_r^{p,q}$, $Z_r^{p,q}$ of $E_a^{p,q}$ for all $r \geq a$:

$$0 = B_a^{p,q} \subseteq B_{a+1}^{p,q} \subseteq \cdots \subseteq B_r^{p,q} \subseteq \cdots \subseteq Z_r^{p,q} \subseteq \cdots \subseteq Z_{a+1}^{p,q} \subseteq Z_a^{p,q} = E_a^{p,q}$$

such that $E_r^{p,q} \cong Z_r^{p,q} / B_r^{p,q}$.

Definition 3.2.2 (E_∞-terms)

We define infinite coboundaries

$$B_\infty^{p,q} := \bigcup_{r=a}^{\infty} B_r^{p,q} = \operatorname{colim}_r B_r^{p,q},$$

infinite cocycles

$$Z_\infty^{p,q} := \bigcap_{r=a}^{\infty} Z_r^{p,q} = \lim_r Z_r^{p,q},$$

and E_∞-terms

$$E_\infty^{p,q} := Z_\infty^{p,q} / B_\infty^{p,q}.$$

Remark One may also define a derived E_∞ term:

$$RE_\infty^{p,q} = \varprojlim{}^1 Z_r^{p,q}.$$

Lemma 3.2.1

Let $f : E \to E'$ be a morphism of spectral sequences, such that for some r, $f_r^{p,q} : E_r^{p,q} \to E_r'^{p,q}$ is an isomorphism for all $p, q \in \mathbb{Z}$. Then $f_\infty^{p,q} : E_\infty^{p,q} \to E_\infty'^{p,q}$ is an isomorphism.

Proof Consider the following diagram

$$\begin{tikzcd}
0 \arrow[r] & Z_r^{p,q} \arrow[r, hook] \arrow[d] & E_r^{p,q} \arrow[r, "d_r^{p,q}"] \arrow[d, "\sim"'] & E_r^{p+r,q-r+1} \arrow[d, "\sim"] \\
0 \arrow[r] & Z_r'^{p,q} \arrow[r, hook] & E_r'^{p,q} \arrow[r, "d_r'^{p,q}"] & E_r'^{p+r,q-r+1}
\end{tikzcd}$$

by Five Lemma, $Z_r^{p,q} \cong Z_r'^{p,q}$. Since $f_r^{p,q}$ is isomorphism and $d_r' f_r = f_r d_r$, we have $B_r^{p,q} \cong B_r'^{p,q}$, so $E_{r+1}^{p,q} \cong E_{r+1}'^{p,q}$ is an isomorphism. Inductively, $E_k^{p,q} \cong E_k'^{p,q}$ for all $k \geq r$, hence $E_\infty^{p,q} \cong E_\infty'^{p,q}$. □

By construction, we have inclusions for any r:

$$B_r^{p,q} \subseteq B_\infty^{p,q} \subseteq Z_\infty^{p,q} \subseteq Z_r^{p,q}.$$

Definition 3.2.3 (Regular, coregular, and biregular)

*(1) A spectral sequence is **regular** if and only if for all (p, q) there exists an $r = r(p, q)$ such that $Z_r^{p,q} = Z_{r+1}^{p,q} = \cdots$, i.e., $Z_\infty^{p,q} = Z_r^{p,q}$ for large r.*

*(2) A spectral sequence is **coregular** if and only if for all (p, q) there exists an $r = r(p, q)$ such that $B_r^{p,q} = B_{r+1}^{p,q} = \cdots$, i.e., $B_\infty^{p,q} = B_r^{p,q}$ for large r.*

*(3) A spectral sequence is **biregular**, if both regular and coregular.*

Convergence

Definition 3.2.4 (Filtration)

*Let $A \in \mathrm{obj}(\mathscr{A})$. A **filtration** (or decreasing filtration) of A is a sequence of subobjects of A*

$$\cdots \supseteq F^0(A) \supseteq F^1(A) \supseteq \cdots \supseteq F^p(A) \supseteq \cdots.$$

*If they exist, we write $\bigcap_p F^p(A)$ for the intersection (or limit) and $\bigcup_p F^p(A)$ for the union (or colimit). We say that the filtration is **separated** if $\bigcap_p F^p(A) = 0$ and **exhaustive** if $\bigcup_p F^p(A) = A$.*
*We say that the filtration is **bounded below** if there exists $p \in \mathbb{Z}$ with $F^p(A) = 0$ (trivially implies separated), and **bounded above** if there exists $p \in \mathbb{Z}$ with $F^p(A) = A$ (trivially implies exhaustive). It is called **bounded** if both bounded above and below.*

Definition 3.2.5 (Converge)

Let $E = \{E_r^{p,q}\}$ be a spectral sequence and $H = \{H_n\}_{n\in\mathbb{Z}}$ with $H^n \in \mathscr{A}$.

*(1) We say **the spectral sequence weakly converges to given objects** H^n **of** $\mathscr{A}$ if each H^n has a decreasing filtration*

$$H^n \supseteq \cdots \supseteq F^pH^n \supseteq F^{p+1}H^n \supseteq \cdots$$

together with isomorphisms

$$\beta^{p,q} : E_\infty^{p,q} \xrightarrow{\sim} F^pH^{p+q}/F^{p+1}H^{p+q}$$

for all $p, q \in \mathbb{Z}$, $p + q = n$. Note that we don't require the filtration on H^n to be separated or exhaustive.

*(2) We say that **the spectral sequence** $\{E_r^{p,q}\}$ **abuts to** H^n if it weakly converges to H^n and the filtration on H^n is separated and exhaustive.*

*(3) We say that **the spectral sequence converge to** H^n if it abuts to H^n, it is regular, and $H^n = \varprojlim_p (H^n/F^pH^n)$ for each n. Denote it by $E_r^{p,q} \Rightarrow H^{p+q}$.*

Remark

(i) Every weakly convergent spectral sequence abuts to $\bigcup F^pH^n/\bigcap F^pH^n$.

(ii) In some literatures, abutment in our sense is called convergence, while convergence in our sense is called strongly convergence.

(iii) The same spectral sequence may converge to two different objects H^*.

Proof We give a sketch of proof to show (i). Suppose spectral sequence $\{E_r^{p,q}\}$ weakly converges to H^n, then we have a filtration

$$H^n \supseteq \cdots \supseteq F^pH^n \supseteq F^{p+1}H^n \supseteq \cdots,$$

such that

$$E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}H^{p+q}.$$

Let $U^n = \bigcup_p F^pH^n$, $I^n = \bigcap_p F^pH^n$, and $K^n = U^n/I^n$, then we have a filtration of K^n,

$$K^n \supseteq \cdots \supseteq F^pH^n/I^n \supseteq F^{p+1}H^n/I^n \supseteq \cdots,$$

denote $F^pK^n := F^pH^n/I^n$. We first show this filtration is separated. Consider $\bigcap_p F^pK^n$,

$$\bigcap_p F^pK^n = \bigcap_p \left(\frac{F^pH^n}{I^n}\right) = \frac{\bigcap_p F^pH^n}{I^n} = I^n/I^n = 0,$$

so filtration $\{F^pK^n\}$ is separated. We next show this filtration is exhaustive. Consider $\bigcup_p F^pK^n$,

$$\bigcup_p F^pK^n = \bigcup_p \left(\frac{F^pH^n}{I^n}\right) = \frac{\bigcup_p F^pH^n}{I^n} = U^n/I^n = K^n,$$

so $\{F^pK^n\}$ is exhaustive. Note that

$$\frac{F^pH^{p+q}/I^n}{F^{p+1}H^{p+q}/I^n} \cong \frac{F^pH^{p+q}}{F^{p+1}H^{p+q}} \cong E_\infty^{p,q},$$

which implies $\{E_r^{p,q}\}$ weakly converges to K^n. Hence every weakly convergent spectral sequence abuts to $\bigcup F^pH^n/\bigcap F^pH^n$. □

Definition 3.2.6 (Degenerates)

We say that the spectral sequence ***degenerates*** *at page r if, for any $s \geq r$, the differential d_s is zero. This implies that $E_r \cong E_{r+1} \cong E_{r+2} \cong \cdots$. In particular, it implies that E_r is isomorphic to E_∞.*

Bounded spectral sequence

Definition 3.2.7 (Bounded spectral sequence)

(1) A spectral sequence is said to be ***bounded*** *if there are only finitely many non-zero terms of each total degree n in $E_a^{p,q}$ ($p+q=n$). This implies the same statement for $E_r^{p,q}$ for all $r \geq a$.*

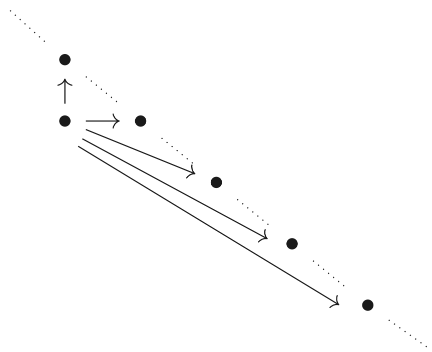
(2) A spectral sequence is said to be ***bounded below*** *if for each n there is an integer $s = s(n)$ such that the terms $E_a^{p,q}$ of total degree n vanish for all $p > s$.*

Remark Bounded spectral sequences are clearly bounded below.

For a bounded spectral sequence:

- For each p, q there is an r_0 such that $E_r^{p,q} = E_{r_0}^{p,q}$, for all $r \geq r_0$, and consequently $E_\infty^{p,q} = E_r^{p,q}$ for large r.

  Proof This is clear, you may consider the following graph:

  □

- If $E_r^{p,q}$ weakly converges to H^n, then the filtration on H^n is finite, meaning that F^pH^n stabilizes when p goes to ∞ or $-\infty$.

  Proof Suppose bounded spectral sequence $\{E_r^{p,q}\}$ started at page a. Since $\{E_r^{p,q}\}$ weakly converges to H^n, there exists a filtration $\{F^pH^n\}$ of H^n such that

  $$E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}H^{p+q}.$$

  Recall that $E_\infty^{p,q}$ is subquotient of $E_a^{p,q}$, if p goes to ∞ or $-\infty$, we have $E_\infty^{p,q} = 0$, which implies that $F^pH^{p+q} = F^{p+1}H^{p+q}$ for $p \to \infty$ or $p \to -\infty$. It follows that $\{F^pH^n\}$ stabilizes. □

- $\{E_r^{p,q}\}$ converges to H^* whenever $\{E_r^{p,q}\}$ abuts to H^*.

  Proof It suffices to show that $\{E_r^{p,q}\}$ is regular and $H^n = \varprojlim_p (H^n/F^pH^n)$ for each n. $\{E_r^{p,q}\}$ is regular given by the definition of bounded. Recall that the filtration is stable, for p large enough, $F^pH^n = 0$, hence $H^n = \varprojlim_p (H^n/F^pH^n)$. $\square$

Examples

Example 3.3 A first quadrant spectral sequence is bounded. If it converges to H^n, then each H^n has a finite filtration of length $n+1$:

$$H^n = F^0H^n \supseteq F^1H^n \supseteq \cdots \supseteq F^nH^n \supseteq F^{n+1}H^n = 0.$$

Since the spectral sequence lives in the first quadrant, each $E_{r+1}^{0,n}$ is a subobject of $E_r^{0,n}$, hence $E_\infty^{0,n}$ is a subobject of $E_a^{0,n}$. On the other hand, $E_\infty^{0,n}$ is isomorphic to H^n/F^1H^n, hence is a quotient of H^n. The composite morphisms

$$H^n \twoheadrightarrow E_\infty^{0,n} \hookrightarrow E_a^{0,n}$$

is called **edge morphisms**. Similarly, we have edge morphisms

$$E_a^{n,0} \twoheadrightarrow E_\infty^{n,0} \hookrightarrow H^n.$$

Example 3.4 (2 columns) Suppose E is a spectral sequence with $E_2^{p,q} = 0$ except $p = 0, 1$, then the spectral sequence degenerates at page 2 and for each total degree n, only two terms $E_2^{1,n-1}, E_2^{0,n}$ are possibly nonzero. If it abuts to H^*, then $E_\infty^{p,n-p} = F^pH^n/F^{p+1}H^n$. If $p \geq 2$ or $p \leq -1$, $E_\infty^{p,n-p} = 0$, so $F^pH^n = F^{p+1}H^n$. If $p = 0$, then $E_\infty^{0,n} = E_2^{0,n} = F^0H^n/F^1H^n$. If $p = 1$, then $E_\infty^{1,n-1} = E_2^{1,n-1} = F^1H^n/F^2H^n$. Since the filtration on H^n is separated and exhaustive, we obtained that $F^0H^n = H^n$ and $F^2H^n = 0$, so we have $E_2^{0,n} = H^n/E_2^{1,n-1}$, i.e., there is an exact sequence

$$0 \longrightarrow E_2^{1,n-1} \longrightarrow H^n \longrightarrow E_2^{0,n} \longrightarrow 0.$$

Example 3.5 Suppose E is a spectral sequence with $E_2^{p,q} = 0$ except $q = 0, 1$. At page 2:

$$\begin{tikzcd}
0 \arrow[drr] & 0 \arrow[drr] & 0 \arrow[drr] & & & & & \\
& \cdots \arrow[drr] & E_2^{-1,1} \arrow[drr] & E_2^{0,1} \arrow[drr] & E_2^{1,1} \arrow[drr] & E_2^{2,1} & \cdots & \\
& \cdots & E_2^{-1,0} & E_2^{0,0} \arrow[drr] & E_2^{1,0} \arrow[drr] & E_2^{2,0} \arrow[drr] & \cdots & \\
& & & & & 0 & 0 & 0
\end{tikzcd}$$

and at page 3:

$$\begin{tikzcd}
0 \arrow[rrrdd] & 0 \arrow[rrrdd] & 0 \arrow[rrrdd] & & & & & & & \\
& 0 \arrow[rrrdd] & 0 \arrow[rrrdd] & 0 \arrow[rrrdd] & & & & & & \\
& & \cdots & E_3^{-1,1} \arrow[rrrdd] & E_3^{0,1} \arrow[rrrdd] & E_3^{1,1} \arrow[rrrdd] & E_3^{2,1} & \cdots & & \\
& & \cdots & E_3^{-1,0} & E_3^{0,0} \arrow[rrrdd] & E_3^{1,0} \arrow[rrrdd] & E_3^{2,0} \arrow[rrrdd] & \cdots & & \\
& & & & & & 0 & 0 & 0 & \\
& & & & & & & 0 & 0 & 0
\end{tikzcd}$$

The spectral sequence degenerates at $r = 3$ and if it abuts to H^*, we have exact sequences for each n:

$$0 \longrightarrow E_3^{n,0} \longrightarrow H^n \longrightarrow E_3^{n-1,1} \longrightarrow 0.$$

But, by definition, $E_3^{n,0}$ is just the cokernel of $d_2^{n-1,1}$ and $E_3^{n-1,1}$ is the kernel of $d_2^{n-1,1}$, hence there is a long exact sequence:

$$\cdots \longrightarrow H^{n-1} \longrightarrow E_2^{n-2,1} \xrightarrow{d_2^{n-2,1}} E_2^{n,0} \longrightarrow H^n \longrightarrow E_2^{n-1,1} \xrightarrow{d_2^{n-1,1}} E_2^{n+1,0} \longrightarrow H^{n+1} \longrightarrow \cdots .$$

Lemma 3.2.2

Let $\{E_r^{p,q}\}_{r\geq 2}$ be a first quadrant spectral sequence converges to H^n. Then there exists an exact sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \longrightarrow E_2^{2,0} \longrightarrow H^2,$$

*called **the exact sequence of low degree terms**.*

Proof At E_2 page:

$$\begin{tikzcd}
0 \arrow[rrd] & & & & & & \\
& 0 \arrow[rrd] & E_2^{0,1} \arrow[rrd] & E_2^{1,1} & E_2^{2,1} & & \\
& & E_2^{0,0} & E_2^{1,0} \arrow[rrd] & E_2^{2,0} \arrow[rrd] & & \\
& & & & & 0 & 0
\end{tikzcd}$$

Hence $E_\infty^{1,0} = E_2^{1,0}$. Recall Example 3.3 , we have filtration $H^1 = F^0H^1 \supseteq F^1H^1 \supseteq F^2H^1 = 0$, so $E_\infty^{0,1} = F^0H^1/F^1H^1 = H^1/F^1H^1 = H^1/E_\infty^{1,0} = H^1/E_2^{1,0}$. Hence there is an exact sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_\infty^{0,1} \longrightarrow 0.$$

Now consider page E_3:

$$\begin{tikzcd}
0 \arrow[ddrrr] & 0 \arrow[ddrrr] & 0 \arrow[ddrrr] & & & & & & \\
0 \arrow[ddrrr] & 0 \arrow[ddrrr] & 0 \arrow[ddrrr] & & & & & & \\
& & & E_3^{0,1} \arrow[ddrrr] & E_3^{1,1} \arrow[ddrrr] & E_3^{2,1} \arrow[ddrrr] & & & \\
& & & E_3^{0,0} \arrow[ddrrr] & E_3^{1,0} \arrow[ddrrr] & E_3^{2,0} \arrow[ddrrr] & & & \\
& & & & & & 0 & 0 & 0 \\
& & & & & & 0 & 0 & 0
\end{tikzcd}$$

We have $E_\infty^{0,1} = E_3^{0,1}$ and $E_\infty^{2,0} = E_3^{2,0}$. By Example 3.3 , we have filtration $H^2 = F^0H^2 \supseteq F^1H^2 \supseteq F^2H^2 \supseteq F^3H^2 = 0$. Hence

$$E_\infty^{2,0} = F^2H^2/F^3H^2 = F^2H^2 = H^2 = E_3^{2,0}.$$

Note that $E_3^{0,1} = \operatorname{Ker} d_2^{0,1}$ and $E_3^{2,0} = \operatorname{Coker} d_2^{0,1}$, we obtain an exact sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \xrightarrow{d_2^{0,1}} E_2^{2,0} \longrightarrow H^2,$$

as we desired. □

3.3 The spectral sequence of a filtered complex

Setting

We work on abelian category $\mathscr{A}$. Let $C^\bullet$ be a complex in $\mathscr{A}$, with a decreasing filtration $\{F^pC^\bullet\}_{p\in\mathbb{Z}}$ of $C^\bullet$; the connecting morphism $\partial : C^n \to C^{n+1}$ satisfies

$$\partial(F^pC^n) \subseteq F^pC^{n+1}, \forall p \in \mathbb{Z},$$

so that ∂ induces morphisms

$$F^pC^n/F^{p+1}C^n \to F^pC^{n+1}/F^{p+1}C^{n+1}.$$

The aim is to construct a spectral sequence whose E_0 page consists of $E_0^{p,q} := F^pC^{p+q}/F^{p+1}C^{p+q}$:

$$\begin{tikzcd}
\vdots & \vdots \\
E_0^{p,q+1} \arrow[u] & E_0^{p+1,q+1} \arrow[u] \\
E_0^{p,q} \arrow[u, "\partial"] & E_0^{p+1,q} \arrow[u, "\partial"]
\end{tikzcd}$$

Construct spectral sequences

Consider

$$\begin{array}{ccccc} C^{p+q-1} & \xrightarrow{\partial} & C^{p+q} & \xrightarrow{\partial} & C^{p+q+1} \\ \cup\!| & & \cup\!| & & \cup\!| \\ F^{p-r+1}C^{p+q-1} & & F^pC^{p+q} & & F^{p+r}C^{p+q+1} \end{array}$$

For $r \geq 0$, set

$$A_r^{p,q} = F^pC^{p+q} \cap \partial^{-1}(F^{p+r}C^{p+q+1}), \qquad D_r^{p,q} = F^pC^{p+q} \cap \partial(F^{p-r+1}C^{p+q-1}).$$

Since $F^{p-r+1}C^{p+q-1} \subseteq F^{p-r}C^{p+q-1}$, we have

$$D_0^{p,q} \subseteq \cdots \subseteq D_r^{p,q} \subseteq \cdots \subseteq \operatorname{Im}\partial \cap F^pC^{p+q}.$$

By $0 \subseteq \cdots \subseteq F^{p+r}C^{p+q+1} \subseteq F^{p+r-1}C^{p+q+1} \subseteq \cdots$, we have

$$\operatorname{Ker}\partial \cap F^pC^{p+q} \subseteq \cdots \subseteq A_r^{p,q} \subseteq \cdots \subseteq A_0^{p,q} = F^pC^{p+q}.$$

So we have

$$D_0^{p,q} \subseteq \cdots \subseteq D_r^{p,q} \subseteq \cdots \subseteq \operatorname{Im}\partial \cap F^pC^{p+q} \subseteq \operatorname{Ker}\partial \cap F^pC^{p+q} \subseteq \cdots \subseteq A_r^{p,q} \subseteq \cdots \subseteq A_0^{p,q} = F^pC^{p+q}.$$

Note *Set*

$$Z_r^{p,q} := \frac{A_r^{p,q} + F^{p+1}C^{p+q}}{F^{p+1}C^{p+q}} \subseteq B_r^{p,q} := \frac{D_r^{p,q} + F^{p+1}C^{p+q}}{F^{p+1}C^{p+q}}, \quad E_r^{p,q} := Z_r^{p,q}/B_r^{p,q},$$

then

$$0 = B_0^{p,q} \subseteq \cdots \subseteq B_r^{p,q} \subseteq \cdots \subseteq Z_r^{p,q} \subseteq \cdots \subseteq Z_0^{p,q} = E_0^{p,q}.$$

Lemma 3.3.1

(1) $D_r^{p,q} = \partial(A_{r-1}^{p-r+1,q+r-2})$*, thus* $B_r^{p,q} = \partial(Z_{r-1}^{p-r+1,q+r-2})$.

(2) $A_r^{p,q} \cap F^{p+1}C^{p+q} = A_{r-1}^{p+1,q-1}$*, thus* $Z_r^{p,q} = A_r^{p,q}/A_{r-1}^{p+1,q-1}$.

(3) $D_r^{p,q} \cap F^{p+1}C^{p+q} = D_{r+1}^{p+1,q-1}$*, thus*

$$B_r^{p,q} = D_r^{p,q}/D_{r+1}^{p+1,q-1} = \partial(A_{r-1}^{p-r+1,q+r-2})/\partial(A_r^{p-r+1,q+r-2}).$$

(4) $\partial : A_r^{p,q} \to D_{r+1}^{p+r,q-r+1}$ *induces an isomorphism*

$$Z_r^{p,q}/Z_{r+1}^{p,q} \cong B_{r+1}^{p+r,q-r+1}/B_r^{p+r,q-r+1}.$$

Proof

(1) We have $\partial(A \cap A') \subseteq \partial(A) \cap \partial(A')$, hence

$\partial(A_{r-1}^{p-r+1,q+r-2}) \subseteq \partial(F^{p-r+1}C^{p+q-1}) \cap \partial(\partial^{-1}(F^pC^{p+q})) \subseteq \partial(F^{p-r+1}C^{p+q-1}) \cap F^pC^{p+q} = D_r^{p,q}.$

Conversely, if $\partial(y) \in F^pC^{p+q}$ for some $y \in F^{p-r+1}C^{p+q-1}$, then $y \in A_{r-1}^{p-r+1,q+r-2}$, and thus $\partial(y) \in \partial(A_{r-1}^{p-r+1,q+r-2})$. Hence, $D_r^{p,q} = \partial(A_{r-1}^{p-r+1,q+r-2})$. Thus, $B_r^{p,q} = \partial(Z_{r-1}^{p-r+1,q+r-2})$.

(2) By definition, clearly, $A_r^{p,q} \cap F^{p+1}C^{p+q} = A_{r-1}^{p+1,q-1}$. Now, we have

$$A_r^{p,q}/A_{r-1}^{p+1,q-1} = \frac{A_r^{p,q}}{A_r^{p,q} \cap F^{p+1}C^{p+q}} = \frac{A_r^{p,q} + F^{p+1}C^{p+q}}{F^{p+1}C^{p+q}} = Z_r^{p,q}.$$

(3) By definition, $D_r^{p,q} \cap F^{p+1}C^{p+q} = D_{r+1}^{p+1,q-1}$ is clearly, and

$$D_r^{p,q}/D_{r+1}^{p+1,q-1} = \frac{D_r^{p,q}}{D_r^{p,q} \cap F^{p+1}C^{p+q}} = \frac{D_r^{p,q} + F^{p+1}C^{p+q}}{F^{p+1}C^{p+q}} = B_r^{p,q}.$$

By part (1), we also have

$$B_r^{p,q} = \partial(A_{r-1}^{p-r+1,q+r-2})/\partial(A_r^{p-r+1,q+r-2}).$$

(4) By part (2), we have

$$Z_r^{p,q}/Z_{r+1}^{p,q} \cong \frac{A_r^{p,q}}{A_{r-1}^{p+1,q-1}} \Big/ \frac{A_{r+1}^{p,q}}{A_r^{p+1,q-1}} \cong \frac{A_r^{p,q}}{A_{r+1}^{p,q} + A_{r-1}^{p+1,q-1}}.$$

On the other hand, by part (3),

$$\frac{B_{r+1}^{p+r,q-r+1}}{B_r^{p+r,q-r+1}} = \frac{\partial(A_r^{p,q})}{\partial(A_{r+1}^{p,q})} \Big/ \frac{\partial(A_{r-1}^{p+1,q-1})}{\partial(A_r^{p+1,q-1})} \cong \frac{\partial(A_r^{p,q})}{\partial(A_{r+1}^{p,q}) + \partial(A_{r-1}^{p+1,q-1})}.$$

Consider $\partial|_{A_r^{p,q}} : A_r^{p,q} \to \frac{\partial(A_r^{p,q})}{\partial(A_{r+1}^{p,q})+\partial(A_{r-1}^{p+1,q-1})}$, clearly, $\operatorname{Ker} \partial|_{A_r^{p,q}} = A_{r+1}^{p,q} + A_{r-1}^{p+1,q-1}$, so

$$Z_r^{p,q}/Z_{r+1}^{p,q} \cong B_{r+1}^{p+r,q-r+1}/B_r^{p+r,q-r+1}.$$

□

Note *It remain to construct:*

(a) $d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}$ *such that* $d_r \circ d_r = 0$.

(b) $E_{r+1}^{p,q} \cong \operatorname{Ker} d_r^{p,q} / \operatorname{Im} d_r^{p-r,q+r-1}$.

By construction, $\partial : A_r^{p,q} \to A_r^{p+r,q-r+1}$ induces

$$d_r^{p,q} : E_r^{p,q} \twoheadrightarrow \frac{Z_r^{p,q}}{Z_{r+1}^{p,q}} \xrightarrow[\text{Lem. 3.3.1}]{\sim} \frac{B_{r+1}^{p+r,q-r+1}}{B_r^{p+r,q-r+1}} \hookrightarrow \frac{Z_r^{p+r,q-r+1}}{B_r^{p+r,q-r+1}} = E_r^{p+r,q-r+1}.$$

Then $\partial \circ \partial = 0$ induces $d_r^{p+r,q-r+1} \circ d_r^{p,q} = 0$. And

$$\operatorname{Ker} d_r^{p,q} = Z_{r+1}^{p,q}/B_r^{p,q}, \qquad \operatorname{Im} d_r^{p,q} = B_{r+1}^{p+r,q-r+1}/B_r^{p+r,q-r+1}$$

induces

$$\operatorname{Ker} d_r^{p,q} / \operatorname{Im} d_r^{p-r,q+r-1} \cong \frac{Z_{r+1}^{p,q}}{B_r^{p,q}} \Big/ \frac{B_{r+1}^{p,q}}{B_r^{p,q}} \cong \frac{Z_{r+1}^{p,q}}{B_{r+1}^{p,q}} = E_{r+1}^{p,q}.$$

E_∞-terms

Let

$$Z_\infty^{p,q} := \bigcap_r Z_r^{p,q}, \quad B_\infty^{p,q} := \bigcup_r B_r^{p,q}, \quad \text{and } E_\infty^{p,q} := Z_r^{p,q}/B_r^{p,q}.$$

Assume the filtration $\{F^pC^\bullet\}$ is separated and exhaustive (Definition 3.2.4), then

$$A_\infty^{p,q} := \bigcap_r A_r^{p,q} = \operatorname{Ker} \partial \cap F^pC^{p+q}, \qquad D_\infty^{p,q} = \bigcup_r D_r^{p,q} = \operatorname{Im} \partial \cap F^pC^{p+q}.$$

Convergence

Since $F^pC^\bullet$ is a subcomplex of $C^\bullet$, this induces map between cohomology groups:

$$H^{p+q}(F^pC^\bullet) \longrightarrow H^{p+q}(C^\bullet).$$

Denote the image by

$$F^pH^{p+q}(C^\bullet) := \frac{\operatorname{Ker} \partial \cap F^pC^{p+q} + \operatorname{Im} \partial}{\operatorname{Im} \partial} \cong \frac{A_\infty^{p,q}}{A_\infty^{p,q} \cap \operatorname{Im} \partial} \cong \frac{A_\infty^{p,q}}{D_\infty^{p,q}},$$

this defines a filtration on $H^{p+q}(C^\bullet)$, hence (we obtain a associated graded object)

$$\operatorname{gr}^p H^{p+q}(C^\bullet) := \frac{F^pH^{p+q}(C^\bullet)}{F^{p+1}H^{p+q}(C^\bullet)} \cong \frac{A_\infty^{p,q}/D_\infty^{p,q}}{A_\infty^{p+1,q-1}/D_\infty^{p+1,q-1}} \cong \frac{A_\infty^{p,q}/A_\infty^{p+1,q-1}}{D_\infty^{p,q}/D_\infty^{p+1,q-1}},$$

where the last isomorphism comes from the relation $D_\infty^{p,q} \cap A_\infty^{p+1,q-1} = D_\infty^{p+1,q-1}$ and Lemma 3.3.2 below.

Lemma 3.3.2

Given $A, B, C, D \in \mathrm{obj}(\mathscr{A})$, such that

$$\begin{tikzcd} D \arrow[r, hook] \arrow[d, hook] & C \arrow[d, hook] \\ B \arrow[r, hook] & A \end{tikzcd}$$

and $D = B \cap C$. Then we have an isomorphism

$$\frac{A/B}{C/D} \cong \frac{A/C}{B/D}.$$

Proof Since $B \cap C \hookrightarrow B, C \hookrightarrow A$ induces $C/B \cap C \hookrightarrow A/B$. Consider the following diagram.

$$\begin{tikzcd} 0 \arrow[r] & D \arrow[r, hook] \arrow[d, hook] & C \arrow[r, two heads] \arrow[d, hook] & C/D \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & B \arrow[r, hook] & A \arrow[r, two heads] & A/B \arrow[r] & 0 \end{tikzcd}$$

By Snake Lemma we have exact sequence

$$0 \longrightarrow B/D \longrightarrow A/C \longrightarrow \frac{A/B}{C/D} \longrightarrow 0,$$

which implies that

$$\frac{A/B}{C/D} \cong \frac{A/C}{B/D}.$$

$\square$

By Lemma 3.3.1, we have $Z_r^{p,q} = A_r^{p,q}/A_{r-1}^{p+1,q-1}$, so there is an exact sequence

$$0 \longrightarrow A_{r-1}^{p+1,q-1} \longrightarrow A_r^{p,q} \longrightarrow Z_r^{p,q} \longrightarrow 0.$$

Recall that $\lim \square$ is left exact, we have

$$\begin{tikzcd} 0 \arrow[r] & \lim_r A_{r-1}^{p+1,q-1} \arrow[r] \arrow[d, equal] & \lim_r A_r^{p,q} \arrow[r] \arrow[d, equal] & \lim_r Z_r^{p,q} \arrow[d, equal] \\ & \bigcap_r A_{r-1}^{p+1,q-1} = A_\infty^{p+1,q-1} & \bigcap_r A_r^{p,q} = A_\infty^{p,q} & \bigcap_r Z_r^{p,q} = Z_\infty^{p,q} \end{tikzcd}$$

So $A_\infty^{p,q}/A_\infty^{p+1,q-1} \cong \mathrm{Im}(A_\infty^{p,q} \to Z_\infty^{p,q})$. By Lemma 3.3.1, we have $B_r^{p,q} = D_r^{p,q}/D_{r+1}^{p+1,q-1}$, recall that taking small filtered colimit is exact, we have

$$B_\infty^{p,q} = \bigcup_r B_r^{p,q} = \mathrm{colim}_r B_r^{p,q} = \frac{\mathrm{colim}_r D_r^{p,q}}{\mathrm{colim}_r D_{r+1}^{p+1,q-1}} = D_\infty^{p,q}/D_\infty^{p+1,q-1}.$$

Hence

$$\mathrm{gr}^p H^{p+q}(C^\bullet) = \mathrm{Im}(A_\infty^{p,q} \to Z_\infty^{p,q})/B_\infty^{p,q}. \tag{3.5}$$

We next give a key observation:

Lemma 3.3.3

If the filtration $\{F^pC^\bullet\}$ is bounded below, (3.5) implies:

$$\mathrm{gr}^p H^{p+q}(C^\bullet) = Z_\infty^{p,q}/B_\infty^{p,q} = E_\infty^{p,q}.$$

Proof By Lemma 3.3.1, we have $Z_r^{p,q} = A_r^{p,q}/A_{r-1}^{p+1,q-1}$, then we have short exact sequence

$$0 \longrightarrow A_{r-1}^{p+1,q-1} \longrightarrow A_r^{p,q} \longrightarrow Z_r^{p,q} \longrightarrow 0.$$

Apply $\varprojlim_r \square$, we have

$$0 \longrightarrow \varprojlim_r A_{r-1}^{p+1,q-1} \longrightarrow \varprojlim_r A_r^{p,q} \longrightarrow \varprojlim_r Z_r^{p,q} \longrightarrow \varprojlim_r^1 A_{r-1}^{p+1,q-1}.$$

We claim that $\varprojlim_r^1 A_{r-1}^{p+1,q-1} = 0$. Since $\{F^pC^\bullet\}$ is bounded below, for large p, $F^pC^\bullet = 0$, recall that $A_r^{p,q} = F^pC^{p+q} \cap \partial^{-1}(F^{p+r}C^{p+q+1})$, $A_r^{p,q}$ is stable for large p, in particular, $A_{r-1}^{p+1,q-1}$ is stable. Hence $\{A_{r-1}^{p+1,q-1}\}_r$ holds Mittag-Leffler condition, by Proposition 2.8.2, $\varprojlim_r^1 A_{r-1}^{p+1,q-1} = 0$. It follows that $\mathrm{Im}(A_\infty^{p,q} \to Z_\infty^{p,q}) = Z_\infty^{p,q}$, we win. $\square$

In summary, we obtain the following theorem:

Theorem 3.3.1

Suppose that the filtration $\{F^pC^\bullet\}$ is bounded below and exhaustive, then the associated spectral sequence $\{E_r^{p,q}\}_{r\geq 0}$, with $E_0^{p,q} = F^pC^{p+q}/F^{p+1}C^{p+q}$, is bounded below (Definition 3.2.7) and converges to $H^*(C^\bullet)$:

$$E_1^{p,q} := H^{p+q}(F^pC^\bullet/F^{p+1}C^\bullet) \Rightarrow H^{p+q}(C^\bullet).$$

Proof $\{F^pC^\bullet\}$ is bounded below implies that $\{E_r^{p,q}\}_{r\geq 0}$ is bounded below. $\{F^pC^\bullet\}$ is separated and exhaustive since it is bounded below and exhaustive, then $\{F^pH^{p+q}(C^\bullet)\}$ is separated and exhaustive. Separatedness of $\{F^pH^{p+q}(C^\bullet)\}$ implies $H^{p+q}(C^\bullet) = \varprojlim_p (H^{p+q}(C^\bullet)/F^pH^{p+q}(C^\bullet))$ for each $p, q \in \mathbb{Z}$. By the proof in Lemma 3.3.3, $\{A_r^{p,q}\}$ is stable, then so is $\{Z_r^{p,q}\}$, and therefore regular. The weak convergence is given by our key observation Lemma 3.3.3:

$$E_\infty^{p,q} = Z_\infty^{p,q}/B_\infty^{p,q} = \mathrm{gr}^p\, H^{p+q}(C^\bullet) = \frac{F^pH^{p+q}(C^\bullet)}{F^{p+1}H^{p+q}(C^\bullet)}.$$

Hence $\{E_r^{p,q}\}_{r\geq 0}$ converges to $H^*(C^\bullet)$. $\square$

Example 3.6 (First quadrant filtration) Suppose that $C^i = 0$ for $i < 0$ and that the filtration on C^n for $n \geq 0$ has the following form

$$C^m = F^0C^n \supseteq \cdots \supseteq F^nC^n \supseteq F^{n+1}C^n = 0.$$

That is, $F^jC^n = 0$ for $j > 0$ and $F^jC^n = C^n$ for $j < 0$. Such a filtration is called **canonically bounded**. This means that when we place the filtered pieces of $C^\bullet$ in a diagram, everything essentially lives in the first quadrant:

$$\begin{tikzcd}
F^0C^3 & & & \\
F^0C^2 \arrow[u] & F^1C^3 \arrow[ul, dotted] & & \\
F^0C^1 \arrow[u] & F^1C^2 \arrow[u] \arrow[ul, dotted] & F^2C^3 \arrow[ul, dotted] & \\
F^0C^0 \arrow[u] & F^1C^1 \arrow[u] \arrow[ul, dotted] & F^2C^2 \arrow[u] \arrow[ul, dotted] & F^3C^3 \arrow[ul, dotted]
\end{tikzcd}$$

Then the associated spectral sequence converges (Theorem 3.3.1).

Remark The reason to put the objects in such a way is that we may view it as the "page -1" of the spectral sequence (see the dotted arrows).

3.4 The spectral sequence of a double complex

Let $C^{\bullet,\bullet} \in C^2(\mathscr{A})$ be a double complex:

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \\
\cdots \arrow[r] & C^{i-1,j+1} \arrow[r] \arrow[u] & C^{i,j+1} \arrow[r] \arrow[u] & C^{i+1,j+1} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & C^{i-1,j} \arrow[r] \arrow[u] & C^{i,j} \arrow[r, "\mathrm{d_I}^{i,j}"] \arrow[u, "\mathrm{d_{II}}^{i,j}"] & C^{i+1,j} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & C^{i-1,j-1} \arrow[r] \arrow[u] & C^{i,j-1} \arrow[r] \arrow[u] & C^{i+1,j-1} \arrow[r] \arrow[u] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}$$

Reset

$$\mathrm{H_I}^{i,j}(C^{\bullet,\bullet}) := \operatorname{Ker} \mathrm{d_{II}}^{i,j} / \operatorname{Im} \mathrm{d_{II}}^{i,j-1}, \qquad \mathrm{H_{II}}^{i,j}(C^{\bullet,\bullet}) := \operatorname{Ker} \mathrm{d_I}^{i,j} / \operatorname{Im} \mathrm{d_I}^{i-1,j}.$$

This gives complexes $\{(\mathrm{H_I}^{\bullet,j}(C^{\bullet,\bullet}), \mathrm{d_I}^{\bullet,j})\}$ and $\{(\mathrm{H_{II}}^{i,\bullet}(C^{\bullet,\bullet}), \mathrm{d_{II}}^{i,\bullet})\}$. This yields at each position two cohomology objects $H^p(\mathrm{H_I}^{\bullet,q}(C^{\bullet,\bullet}))$ and $H^q(\mathrm{H_{II}}^{p,\bullet}(C^{\bullet,\bullet}))$. In general, there are not necessarily equal, but we shall see that they "converge" to the same limit in some sense.

Recall the total complex $\operatorname{Tot}(C^{\bullet,\bullet})$ with

$$\operatorname{Tot}(C^{\bullet,\bullet})^n := \operatorname{Tot}_{\oplus}(C^{\bullet,\bullet})^n := \bigoplus_{i+j=n} C^{i,j}, \qquad d|_{C^{i,j}} = \mathrm{d_I}^{i,j} + (-1)^i \mathrm{d_{II}}^{i,j}.$$

There are two natural filtration on $\operatorname{Tot}(C^{\bullet,\bullet})$:

$$\mathrm{F_I}^p \operatorname{Tot}(C^{\bullet,\bullet})^n := \bigoplus_{r\geq p} C^{r,n-r}, \qquad \mathrm{F_{II}}^p \operatorname{Tot}(C^{\bullet,\bullet})^n := \bigoplus_{r\geq p} C^{n-r,r},$$

both are separated and exhaustive.

Remark We may also consider $\operatorname{Tot}^{\prod}(C^{\bullet,\bullet})$ and define filtrations on it. However, the filtrations are not exhaustive in general.

By construction in §3.3, we obtain two spectral sequence ${}'E$ and ${}''E$ corresponding to the filtrations $\{\mathrm{F_I}^p \operatorname{Tot}(C^{\bullet,\bullet})^\bullet\}_{p\in\mathbb{Z}}$ and $\{\mathrm{F_{II}}^p \operatorname{Tot}(C^{\bullet,\bullet})^\bullet\}_{p\in\mathbb{Z}}$ respectively. The following lemma computes the first pages of ${}'E$ and ${}''E$:

Lemma 3.4.1

(1) For $p, q \in \mathbb{Z}$, there exists canonical isomorphisms:

$${}'E_0^{p,q} \xrightarrow{\sim} C^{p,q}, \qquad {}''E_0^{p,q} \xrightarrow{\sim} C^{q,p}$$

such that the diagram commutes:

$$\begin{tikzcd}
{}'E_0^{p,q+1} \arrow[r, "\sim"] & C^{p,q+1} \\
{}'E_0^{p,q} \arrow[r, "\sim"] \arrow[u, "{}'d_0^{p,q}"] & C^{p,q} \arrow[u, "(-1)^p \mathrm{d_{II}}^{p,q}"']
\end{tikzcd} \qquad \begin{tikzcd}
{}''E_0^{p,q+1} \arrow[r, "\sim"] & C^{q+1,p} \\
{}''E_0^{p,q} \arrow[r, "\sim"] \arrow[u, "{}''d_0^{p,q}"] & C^{q,p} \arrow[u, "\mathrm{d_I}^{q,p}"']
\end{tikzcd}$$

(2) For $p, q \in \mathbb{Z}$, there are canonical isomorphisms

$${}'E_1^{p,q} \xrightarrow{\sim} \mathrm{H_I}^{p,q}(C^{\bullet,\bullet}), \qquad {}''E_1^{p,q} \xrightarrow{\sim} \mathrm{H_{II}}^{q,p}(C^{\bullet,\bullet})$$

making the following diagrams commutes

$$\begin{tikzcd} {'E_1^{p,q}} \arrow[r, "{'d_1^{p,q}}"] \arrow[d, "\sim"'] & {'E_1^{p+1,q}} \arrow[d, "\sim"] \\ {\mathrm{H_I}^{p,q}(C^{\bullet,\bullet})} \arrow[r, "{\mathrm{d_I}^{p,q}}"] & {\mathrm{H_I}^{p+1,q}(C^{\bullet,\bullet})} \end{tikzcd} \qquad \begin{tikzcd} {''E_1^{p,q}} \arrow[r, "{''d_1^{p,q}}"] \arrow[d, "\sim"'] & {''E_1^{p+1,q}} \arrow[d, "\sim"] \\ {\mathrm{H_{II}}^{q,p}(C^{\bullet,\bullet})} \arrow[r, "{(-1)^q \mathrm{d_{II}}^{q,p}}"] & {\mathrm{H_{II}}^{q,p+1}(C^{\bullet,\bullet})} \end{tikzcd}$$

(3) For $p, q \in \mathbb{Z}$, there exists canonical isomorphisms:

$$'E_2^{p,q} \xrightarrow{\sim} H^p(\mathrm{H_I}^{\bullet,q}(C^{\bullet,\bullet})), \qquad ''E_2^{p,q} \xrightarrow{\sim} H^p(\mathrm{H_{II}}^{q,\bullet}(C^{\bullet,\bullet})).$$

Proof Direct check, it is easy. □

Convergence

Theorem 3.4.1

If $C^{p,q} = 0$ for (p, q) in the forth quadrant, then the first filtration is bounded below and exhaustive, and we have a convergent spectral sequence

$$'E_2^{p,q} = H^p(\mathrm{H_I}^{\bullet,q}(C^{\bullet,\bullet})) \Rightarrow H^{p+q}(\mathrm{Tot}(C^{\bullet,\bullet})).$$

If $C^{p,q} = 0$ for (p, q) in the second quadrant, then the second filtration is bounded below and exhaustive, and we have a convergent spectral sequence

$$''E_2^{p,q} = H^p(\mathrm{H_{II}}^{q,\bullet}(C^{\bullet,\bullet})) \Rightarrow H^{p+q}(\mathrm{Tot}(C^{\bullet,\bullet})).$$

Remark In the case $C^\bullet$ is a right half-plane complex, $'E$ is right half-plane while $''E$ is upper half-plane.

Proof The two filtrations are always exhaustive and the bounded below is also clear under the condition imposed. We conclude by Theorem 3.3.1. □

Corollary 3.4.1

If $C^{\bullet,\bullet}$ is a first quadrant complex, then both the filtrations $\mathrm{F_I}^p$ and $\mathrm{F_{II}}^p$ on $\mathrm{Tot}(C^{\bullet,\bullet})$ are bounded, and we have convergent spectral sequences

$$'E_2^{p,q} = H^p(\mathrm{H_I}^{\bullet,q}(C^{\bullet,\bullet})) \Rightarrow H^{p+q}(\mathrm{Tot}(C^{\bullet,\bullet})), \quad ''E_2^{p,q} = H^p(\mathrm{H_{II}}^{q,\bullet}(C^{\bullet,\bullet})) \Rightarrow H^{p+q}(\mathrm{Tot}(C^{\bullet,\bullet})).$$

Application: Künneth spectral sequence

We first introduce the following definition.

Definition 3.4.1 (Collapse)

*A spectral sequence **collapses** at E_r ($r \geq 2$) if there is exactly one non-zero row or column in $\{E_r^{p,q}\}$. This implies that all connecting morphisms are zero, hence if $E_r^{p,q}$ converges to H^n, then $H^n \cong E_r^{p,q}$, the unique term in the row (or column) with $p + q = n$.*

Theorem 3.4.2

Let $P_\bullet$ be a bounded below chain complex of projective R-modules (i.e., $P_n = 0$ for $n \ll 0$). Then for any R-module M, there is a convergent spectral sequence

$$E_2^{p,q} = \mathrm{Ext}_R^p(H_q(P_\bullet), M) \Rightarrow H^{p+q}(\mathrm{Hom}_R(P_\bullet, M)).$$

Proof Choose an injective resolution $M \hookrightarrow I^\bullet$ and consider the double complex $C^{p,q} := \mathrm{Hom}_R(P_q, I^p)$. Since $C^{p,q}$ is (essentially) in the first quadrant, both spectral sequence $'E$ and $''E$ converge to $H^n(\mathrm{Tot}(C^{\bullet,\bullet}))$ by Corollary 3.4.1.

About $'E$: since I^p is injective, $\mathrm{Hom}(\square, I^p)$ is exact, we have $\mathrm{H_I}^{p,q}(C^{\bullet,\bullet}) \cong \mathrm{Hom}_R(H_q(P_\bullet), I^p)$, so $\mathrm{H_I}^{\bullet,q}(C^{\bullet,\bullet}) \cong \mathrm{Hom}_R(H_q(P_\bullet), I^\bullet)$, hence

$$'E_2^{p,q} = H^p(\mathrm{Hom}_R(H_q(P_\bullet), I^\bullet)) \cong \mathrm{Ext}^p(H_q(P_\bullet), M),$$

where last isomorphism since $M \hookrightarrow I^\bullet$ is an injective resolution.

About $''E$: it also converges to $H^n(\mathrm{Tot}(C^{\bullet,\bullet}))$. Since $M \hookrightarrow I^\bullet$ is an injective resolution and each P_n is projective (i.e., the functor $\mathrm{Hom}(P_n, \square)$ is exact), we obtain

$$\mathrm{H_{II}}^{q,p}(C^{\bullet,\bullet}) = \begin{cases} \mathrm{Hom}(P_p, M) & q = 0 \\ 0 & q > 0, \end{cases}$$

so that $''E_2^{p,q} = H^p(\mathrm{H_{II}}^{q,\bullet}(C^{\bullet,\bullet})) = 0$ except for $q = 0$ in which case

$$''E_2^{p,0} = H^p(\mathrm{Hom}(P_\bullet, M)).$$

Consequently, the spectral sequence $''E$ collapses yielding

$$``E_2^{n,0} = H^n(\mathrm{Hom}(P_\bullet, M)) = H^n(\mathrm{Tot}(C^{\bullet,\bullet})).$$

The proof is complete. $\square$

Remark Note that, in the Universal coefficient theorem, the chain complex $P_\bullet$ is not required to be bounded below, but with another condition that each $d(P_n)$ is projective. It can be proved via a spectral sequence argument as above, because the spectral sequences concentrate on a vertical (or horizontal) strip, hence are bounded (below and above).

Now we give the definition of a homological spectral sequence:

Definition 3.4.2

A homological spectral sequence (starting with E^a) in $\mathscr{A}$ consists of the following data:

(a) objects $\{E^r_{p,q}\}$, for all $r \geq a, p, q \in \mathbb{Z}$;

(b) differentials $d^r_{p,q} : E^r_{p,q} \to E^r_{p-r,q+r-1}$ satisfying $d^r_{p,q} \circ d^r_{p+r,q-r+1} = 0$;

(c) isomorphism $E^r_{p,q} \cong \mathrm{Ker}(d^r_{p+r,q-r+1})$.

Similarly, we have a Künneth homological spectral sequence for Tor:

Theorem 3.4.3

Let $P_\bullet$ be a bounded below complex of flat R-modules and M an R-module. Then there is a boundedly converging spectral sequence

$$E^2_{p,q} = \mathrm{Tor}^R_p(H_q(P_\bullet), M) \Rightarrow H_{p+q}(P \otimes_R M).$$

3.5 Cartan-Eilenberg resolution

Definition 3.5.1 (Cartan-Eilenberg resolution)

Given a complex $C^\bullet$ in $\mathscr{A}$, an injective resolution of $C^\bullet$ is a commutative diagram

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \\
\cdots \arrow[r] & I^{p,1} \arrow[r] \arrow[u] & I^{p+1,1} \arrow[r] \arrow[u] & I^{p+2,1} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & I^{p,0} \arrow[r] \arrow[u] & I^{p+1,0} \arrow[r] \arrow[u] & I^{p+2,0} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & C^{p} \arrow[r] \arrow[u] & C^{p+1} \arrow[r] \arrow[u] & C^{p+2} \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

in which the rows are complexes and each column is an injective resolution. That is, $C^p \hookrightarrow I^{p,\bullet}$ is an injective resolution in $\mathscr{A}$. For each $p \in \mathbb{Z}$, we have complexes

$$0 \longrightarrow Z^p(C^\bullet) \longrightarrow Z^p(I^{\bullet,0}) \longrightarrow Z^p(I^{\bullet,1}) \longrightarrow \cdots$$

$$0 \longrightarrow B^p(C^\bullet) \longrightarrow B^p(I^{\bullet,0}) \longrightarrow B^p(I^{\bullet,1}) \longrightarrow \cdots$$

$$0 \longrightarrow H^p(C^\bullet) \longrightarrow H^p(I^{\bullet,0}) \longrightarrow H^p(I^{\bullet,1}) \longrightarrow \cdots$$

and we say the injective resolution is a ***Cartan-Eilenberg resolution*** *if for each $p \in \mathbb{Z}$ these complexes are all injective resolutions.*

Lemma 3.5.1

Assume $\mathscr{A}$ has enough injective objects, then for every $C^\bullet \in C(\mathscr{A})$, it has a Cartan-Eilenberg resolution. In particular, if $C^\bullet$ is acyclic, we may choose a Cartan-Eilenberg resolution $I^{\bullet,\bullet}$ with each row acyclic.

Proof For any $p \in \mathbb{Z}$, we have short exact sequences:

$$0 \longrightarrow Z^p(C^\bullet) \longrightarrow C^p \longrightarrow B^{p+1}(C^\bullet) \longrightarrow 0,$$

$$0 \longrightarrow B^p(C^\bullet) \longrightarrow Z^p(C^\bullet) \longrightarrow H^p(C^\bullet) \longrightarrow 0.$$

For any $p \in \mathbb{Z}$, choose injective resolutions: $H^p(C^\bullet) \hookrightarrow H^{p,\bullet}$ and $B^p(C^\bullet) \hookrightarrow B^{p,\bullet}$. The Horseshoe Lemma implies that there exists an injective resolution $Z^{p,\bullet}$ of $Z^p(C^\bullet)$ fitting into a short exact sequence. Using the Horseshoe Lemma again, we have an injective resolution $C^p \hookrightarrow I^{p,\bullet}$. For any $q \in \mathbb{Z}$, let $d_{p,q} : I^{p,q} \to I^{p+1,q}$ be the composition:

$$I^{p,q} \longrightarrow B^{p+1,q} \longrightarrow Z^{p+1,q} \longrightarrow I^{p+1,q}.$$

Then $I^{\bullet,\bullet}$ is a Cartan-Eilenberg resolution of $C^\bullet$.

If $C^\bullet$ is acyclic, i.e., $H^p(C^\bullet) = 0, \forall p \in \mathbb{Z}$, we set 0 to be the injective resolution of $H^p(C^\bullet)$ in the above construction. Then the resolution of $C^\bullet$ has acyclic rows. $\square$

Theorem 3.5.1

Let $B \in \mathbf{Mod}_R$ and $B \hookrightarrow C^\bullet$ be a resolution (not necessarily injective). Then $\forall A \in \mathbf{Mod}_R$, there is a convergent spectral sequence:

$$E_1^{p,q} = \mathrm{Ext}_R^q(A, C^q) \Rightarrow \mathrm{Ext}_R^{p+q}(A, B).$$

Proof Take a Cartan-Eilenberg resolution $C^\bullet \to I^{\bullet,\bullet}$ such that $\mathrm{H_I}^{p,q}(I^{\bullet,\bullet}) = 0$ for any $p \geq 1$. For $p = 0$, $\mathrm{H_I}^{0,\bullet}(I^{\bullet,\bullet})$ is an injective resolution of $B = H^0(C^\bullet)$. Let $F = \mathrm{Hom}_R(A, \square)$ and consider $F(I^{\bullet,\bullet})$ with the two associated spectral sequences $'E$ and $''E$. Then

$$'E_1^{p,q} = \mathrm{Ext}_R^q(A, C^p) \Rightarrow H^{p+q}(\mathrm{Tot}(F(I^{\bullet,\bullet}))).$$

Also from the construction of Cartan-Eilenberg resolution, we know $\mathrm{H_I}^{p,q}(I^{\bullet,\bullet}) = 0$, recall that $I^{\bullet,\bullet}$ is given by Horseshoe Lemma, so $I^{p,q} = Z^{p,q} \oplus B^{p+1,q}$, hence $F(I^{p,q}) = F(Z^{p,q}) \oplus F(B^{p+1,q})$. Hence for $p \geq 1$, $F(I^{\bullet,\bullet})$ has exact rows. By Lemma 3.4.1,

$$''E_1^{p,q} = 0, \forall q \geq 1 \qquad \text{and} \qquad ''E_1^{p,0} = \mathrm{H_I}^{0,p}(F(I^{\bullet,\bullet})).$$

Thus

$$''E_2^{p,q} = \begin{cases} \mathrm{Ext}_R^p(A, B), & q = 0 \\ 0, & q \geq 1 \end{cases}$$

i.e., $''E$ collapses. And

$$H^n(\mathrm{Tot}(F(I^{\bullet,\bullet}))) \cong'' E_2^{n,0}.$$

$\square$

Application: Right hyper-derived functor

Note *Assume $\mathscr{A}$ has enough injectives, let $C^+(\mathscr{A}) = \{A^\bullet : A^n = 0, n \ll 0\}$.*

Lemma 3.5.2

For any $A^\bullet$ in $C^+(\mathscr{A})$, there exists $I^\bullet$ in $C^+(\mathscr{A})$ with each I^n injective and quasi-isomorphism: $A^\bullet \to I^\bullet$. Moreover, for every $J^\bullet \in C^+(\mathscr{A})$ with each J^n injective and $A^\bullet \to J^\bullet$ quasi-isomorphism, then $I^\bullet \sim J^\bullet$ is homotopy equivalent.

Proof We may assume that $A^n = 0$ for all $n < 0$. By Lemma 3.5.1, we may take a Cartan-Eilenberg resolution $A^\bullet \to I^{\bullet,\bullet}$. Since each $I^{p,q}$ is injective object, $\mathrm{Tot}(I^{\bullet,\bullet})^{p+q}$ is injective object for each $p, q \in \mathbb{Z}$. We want to show that $A^\bullet \to \mathrm{Tot}(I^{\bullet,\bullet})$ is a quasi-isomorphism. Recall Lemma 3.4.1, ${}'E_0$ page is

$$\begin{tikzcd}
\vdots & \vdots & \vdots \\
I^{0,2} \arrow[u] & I^{1,2} \arrow[u] & I^{2,2} \arrow[u] \\
I^{0,1} \arrow[u] & I^{1,1} \arrow[u] & I^{2,1} \arrow[u] \\
I^{0,0} \arrow[u] & I^{1,0} \arrow[u] & I^{2,0} \arrow[u] \\
0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u]
\end{tikzcd}$$

${}'E_1$ page is:

$$\begin{tikzcd}
0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots \\
0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots \\
H^0(I^{0,\bullet}) \arrow[r] \arrow[d, equal] & H^0(I^{1,\bullet}) \arrow[r] \arrow[d, equal] & H^0(I^{2,\bullet}) \arrow[r] \arrow[d, equal] & \cdots \\
A^0 & A^1 & A^2 &
\end{tikzcd}$$

and ${}'E_2$ page is:

$$\begin{tikzcd}
0 \arrow[drr] & 0 & 0 & \cdots \\
0 \arrow[drr] & 0 & 0 & \cdots \\
H^0(A^\bullet) \arrow[drr] & H^1(A^\bullet) & H^2(A^\bullet) & \cdots \\
0 & 0 & 0 & \cdots
\end{tikzcd}$$

Hence ${}'E$ collapses at ${}'E_2$, recall Corollary 3.4.1, ${}'E_2^{p,q} \Rightarrow H^{p+q}(\mathrm{Tot}(I^\bullet))$, we have ${}'E_2^{n,0} = H^n(A^\bullet) \cong H^n(\mathrm{Tot}(I^\bullet))$.

The proof of homotopy equivalent is omitted here. $\square$

Definition 3.5.2 (Right hyper-derived functor)

*Let $F : \mathscr{A} \to \mathscr{B}$ be a left exact functor and $\mathscr{A}$ has enough injective objects. For any $A^\bullet \in C^+(\mathscr{A})$, we define $\mathbb{R}^n F(A^\bullet) := H^n(F(I^\bullet))$ where $I^\bullet \in C^+(\mathscr{A})$ with each I^n injective and $A^\bullet \to I^\bullet$ quasi-isomorphism. $(\mathbb{R}^n F)_{n\geq 0}$ is called the **right hyper-derived functor** of F.*

So we get a δ-functor from $C^+(\mathscr{A})$ to $\mathscr{B}$. In particular, if $A \in \mathscr{A} \hookrightarrow C^+(\mathscr{A})$, then $\mathbb{R}^n F(A) = R^n F(A), \forall n \geq 0$.

Corollary 3.5.1

Let $A^\bullet \in C^+(\mathscr{A})$. Then there is a convergent spectral sequences:

$$'E_1^{p,q} = R^qF(A^p) \Rightarrow \mathbb{R}^{p+q}F(A^\bullet), \qquad ''E_2^{p,q} = R^pF(H^q(A^\bullet)) \Rightarrow \mathbb{R}^{p+q}F(A^\bullet).$$

In particular, $'E_2^{p,q} = H^p(R^qF(A^\bullet))$.

Proof Take a Cartan-Eilenberg resolution $A^\bullet \to I^{\bullet,\bullet}$, by Lemma 3.5.2, $A^\bullet \to \mathrm{Tot}(I^{\bullet,\bullet})$ is a quasi-isomrophism. Recall Definition 3.5.2, we have $\mathbb{R}^nF(A^\bullet) = H^n(F\,\mathrm{Tot}(I^{\bullet,\bullet}))$.

By Corollary 3.4.1, we have

$$'E_2^{p,q} = H^p(\mathrm{H_I}^{\bullet,q}(FI^{\bullet,\bullet})) \Rightarrow H^{p+q}(F\,\mathrm{Tot}(I^{\bullet,\bullet})) = \mathbb{R}^{p+q}F(A^\bullet)$$

and

$$''E_2^{p,q} = H^p(\mathrm{H_{II}}^{q,\bullet}(FI^{\bullet,\bullet})) \Rightarrow H^{p+q}(F\,\mathrm{Tot}(I^{\bullet,\bullet})) = \mathbb{R}^{p+q}F(A^\bullet).$$

Note that $\mathrm{H_I}^{p,q}(FI^{\bullet,\bullet}) = R^qF(A^p)$, we have $'E_1^{p,q} = R^qF(A^p)$, so $'E_2^{p,q} = H^p(R^qF(A^\bullet))$. We next calculate $''E$. At $''E_0$ page:

$$\begin{array}{ccc}
\vdots & \vdots & \vdots \\
\uparrow & \uparrow & \uparrow \\
FI^{2,0} & FI^{2,1} & FI^{2,2} \\
\uparrow & \uparrow & \uparrow \\
FI^{1,0} & FI^{1,1} & FI^{1,2} \\
\uparrow & \uparrow & \uparrow \\
FI^{0,0} & FI^{0,1} & FI^{0,2}
\end{array}$$

$''E_1$ page:

$$\begin{array}{ccccccc}
H^2(FI^{\bullet,0}) & \longrightarrow & H^2(FI^{\bullet,1}) & \longrightarrow & H^2(FI^{\bullet,2}) & \longrightarrow & \cdots \\
H^1(FI^{\bullet,0}) & \longrightarrow & H^1(FI^{\bullet,1}) & \longrightarrow & H^1(FI^{\bullet,2}) & \longrightarrow & \cdots \\
H^0(FI^{\bullet,0}) & \longrightarrow & H^0(FI^{\bullet,1}) & \longrightarrow & H^0(FI^{\bullet,2}) & \longrightarrow & \cdots
\end{array}$$

Since $I^{\bullet,\bullet}$ is Cartan-Eilenberg resolution, $I^{p,q} = Z^{p,q} \oplus B^{p+1,q}$, so we have $H^q(FI^{\bullet,p}) = FH^q(I^{\bullet,p})$, hence $''E_1$ page is:

$$\begin{array}{ccccccc}
FH^2(I^{\bullet,0}) & \longrightarrow & FH^2(I^{\bullet,1}) & \longrightarrow & FH^2(I^{\bullet,2}) & \longrightarrow & \cdots \\
FH^1(I^{\bullet,0}) & \longrightarrow & FH^1(I^{\bullet,1}) & \longrightarrow & FH^1(I^{\bullet,2}) & \longrightarrow & \cdots \\
FH^0(I^{\bullet,0}) & \longrightarrow & FH^0(I^{\bullet,1}) & \longrightarrow & FH^0(I^{\bullet,2}) & \longrightarrow & \cdots
\end{array}$$

Recall that $H^p(A^\bullet) \hookrightarrow H^p(I^{\bullet,0}) \to H^p(I^{\bullet,1}) \to \cdots$ is injective resolution (Definition 3.5.1), $''E_2$ page is:

$$\begin{tikzcd}
R^0F(H^2(A^\bullet)) \arrow[drr] & R^1F(H^2(A^\bullet)) & R^2F(H^2(A^\bullet)) \\
R^0F(H^1(A^\bullet)) \arrow[drr] & R^1F(H^1(A^\bullet)) & R^2F(H^1(A^\bullet)) \\
R^0F(H^0(A^\bullet)) & R^1F(H^0(A^\bullet)) & R^2F(H^0(A^\bullet))
\end{tikzcd}$$

So

$$''E_2^{p,q} = R^pF(H^q(A^\bullet)) \Rightarrow \mathbb{R}^{p+q}F(A^\bullet).$$

□

3.6 The Grothendieck spectral sequence

Definition 3.6.1 (Complete, cocomplete)

*(1) A category $\mathscr{C}$ is called **small-complete** (usually just complete) if all small diagrams in $\mathscr{C}$ has limits in $\mathscr{C}$, that is, if every functor $F : \mathscr{I} \to \mathscr{C}$ from a small category $\mathscr{I}$ to $\mathscr{C}$ has a limit.*

*(2) A category $\mathscr{C}$ is called **small-cocomplete** (usually just complete) if all small diagrams in $\mathscr{C}$ has colimits in $\mathscr{C}$, that is, if every functor $F : \mathscr{I} \to \mathscr{C}$ from a small category $\mathscr{I}$ to $\mathscr{C}$ has a colimit.*

Theorem 3.6.1 (Grothendieck spectral sequence)

Let $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{B} \to \mathscr{C}$ be additive left exact functors between abelian categories where $\mathscr{A}, \mathscr{B}$ have enough injective objects and $\mathscr{C}$ is cocomplex. Suppose that F sends injective objects to G-acyclics (i.e., $R^iG(\square) = 0$ for $i \geq 1$), then $\forall A \in \mathscr{A}$, there is converging first quadrant spectral sequence E starting on page zero, such that

$$E_2^{p,q} = R^pG(R^qF(A)) \Rightarrow R^{p+q}(GF)(A).$$

The exact sequence of low terms is

$$0 \longrightarrow R^1G(FA) \longrightarrow R^1(GF)A \longrightarrow G(R^1F(A)) \longrightarrow R^2G(FA) \longrightarrow R^2(GF)(A).$$

Proof Let $A \in \operatorname{obj}(\mathscr{A})$, and $A \hookrightarrow C^\bullet$ be injective resolution of A, let $I^{\bullet,\bullet}$ be a Cartan-Eilenberg resolution of the complex $F(C^\bullet)$ (with $I^{p,q} = 0$ unless $p, q \geq 0$). That is, we have a have a commutative diagram with exact columns:

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \\
0 \arrow[r] & I^{0,1} \arrow[r] \arrow[u] & I^{1,1} \arrow[r] \arrow[u] & I^{2,1} \arrow[r] \arrow[u] & \cdots \\
0 \arrow[r] & I^{0,0} \arrow[r] \arrow[u] & I^{1,0} \arrow[r] \arrow[u] & I^{2,0} \arrow[r] \arrow[u] & \cdots \\
0 \arrow[r] & F(C^0) \arrow[r] \arrow[u] & F(C^1) \arrow[r] \arrow[u] & F(C^2) \arrow[r] \arrow[u] & \cdots \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

We can apply the results in §3.4 to the double complex $GI^{\bullet,\bullet}$ to obtain two canonical filtrations

$$\mathrm{F_I}^p(\mathrm{Tot}(GI^{\bullet,\bullet}))^n = \bigoplus_{r\geq p} G(I^{r,n-r}), \qquad \mathrm{F_{II}}^p(\mathrm{Tot}(GI))^n = \bigoplus_{r\geq p} G(I^{n-r,r}),$$

and spectral sequence ${}'E_r^{p,q},{}'' E_r^{p,q}$ converging to $H^{p+q}(\mathrm{Tot}(GI^{\bullet,\bullet}))$ by Corollary 3.4.1 (since these are both biregular first quadrant spectral sequence). We have a canonical isomorphism

$${}'E_2^{p,q} \cong H^p(\mathrm{H_I}^{\bullet,q}(GI^{\bullet,\bullet})) \cong H^p(R^qG(FC^{\bullet})).$$

But $C^{\bullet}$ is a complex of injective objects and F sends injective objects to G-acyclics by assumption, so for $q > 0$ the complex $R^pG(FC^{\bullet})$ is zero, and for $q = 0$ it is canonically isomorphic to $(GF)(C^{\bullet})$. In other words, we have

$${}'E_2^{p,q} = \begin{cases} 0 & q > 0 \\ R^p(GF)(A) & q = 0. \end{cases}$$

i.e., this spectral sequence collapses, hence

$$R^n(GF)(A) \cong H^n(\mathrm{Tot}(GI^{\bullet,\bullet})). \tag{3.6}$$

Now we turn to the second spectral sequence ${}''E$. By Lemma 3.4.1 (3), we know ${}''E_2^{p,q} = H^p(\mathrm{H_{II}}^{q,\bullet}(GI^{\bullet,\bullet}))$.

Claim: there is a canonical isomorphism of complexes: $\mathrm{H_{II}}^{q,\bullet}(GI^{\bullet,\bullet}) \cong G(\mathrm{H_{II}}^{q,\bullet}(I^{\bullet,\bullet}))$.

Proof Since the resolution $I^{\bullet,\bullet}$ of $F(C^{\bullet})$ is Cartan-Eilenberg, we have exact sequences

$$0 \longrightarrow Z^{p,q} \longrightarrow I^{p,q} \longrightarrow B^{p+1,q} \longrightarrow 0$$

$$0 \longrightarrow B^{p,q} \longrightarrow Z^{p,q} \longrightarrow \mathrm{H_{II}}^{p,q}(I^{\bullet,\bullet}) \longrightarrow 0$$

which must be split exact because all the objects are injective. It follows that the images under G of those sequences remain exact, so that $\mathrm{H_{II}}^{q,\bullet}(GI^{\bullet,\bullet}) \cong G(\mathrm{H_{II}}^{q,\bullet}(I^{\bullet,\bullet}))$, as we desired. □

On the other hand, since $I^{\bullet}$ is CE-resolution of $F(C^{\bullet})$, $\mathrm{H_{II}}^{q,\bullet}(I^{\bullet,\bullet})$ is an injective resolution of $H^q(FC^{\bullet}) \cong R^qF(A)$:

$$0 \longrightarrow H^p(FC^{\bullet}) \longrightarrow \mathrm{H_{II}}^{p,0}(I^{\bullet,\bullet}) \longrightarrow \mathrm{H_{II}}^{p,1}(I^{\bullet,\bullet}) \longrightarrow \cdots.$$

Thus, there is a canonical isomorphism

$$H^pG(\mathrm{H_{II}}^{q,\bullet}(I^{\bullet,\bullet})) \cong R^pG(R^qF(A)).$$

Hence,

$${}''E_2^{p,q} = H^p(\mathrm{H_{II}}^{q,\bullet}(GI^{\bullet,\bullet})) \cong H^pG(\mathrm{H_{II}}^{q,\bullet}(I^{\bullet,\bullet})) \cong R^pG(R^qF(A)) \Rightarrow H^n(\mathrm{Tot}(GI^{\bullet,\bullet})) \cong R^n(GF)(A),$$

we win.

The exact sequence of low terms is given by Lemma 3.2.2. □

3.7 Examples of spectral sequences

3.7.1 Base change spectral sequences

Theorem 3.7.1 (Base change spectral sequences)

Let $f : R \to S$ be a ring map. Then there is a first quadrant cohomological spectral sequence

$$E_2^{p,q} = \operatorname{Ext}_S^p(A, \operatorname{Ext}_R^q(S, B)) \Rightarrow \operatorname{Ext}_R^{p+q}(A, B) \tag{3.7}$$

for every S-module A and R-module B. In particular, if S is projective as an R-module, then

$$\operatorname{Ext}_S^n(A, \operatorname{Hom}_R(S, B)) \cong \operatorname{Ext}_R^n(A, B).$$

Proof This is an example of Grothendieck's spectral sequences, by setting

- **categories:** $\mathscr{A} = \mathbf{Mod}_R$, $\mathscr{B} = \mathbf{Mod}_S$, $\mathscr{C} = \mathbf{Ab}$;
- **functors:** $F = \operatorname{Hom}_R(S, \square)$, $G = \operatorname{Hom}_S(A, \square)$ so that

$$G \circ F = \operatorname{Hom}_S(A, \operatorname{Hom}_R(S, \square)) \cong \operatorname{Hom}_R(A \otimes_S S, \square) = \operatorname{Hom}_R(A, \square).$$

Recall Proposition 2.2.10 and the proof of Theorem 2.2.3, F admits a left adjoint which means F is exact, hence it sends injectives to injectives.

In particular, if S is projective as an R-module, it is easy to see that E collapses at E_2, then

$$\operatorname{Ext}_S^n(A, \operatorname{Hom}_R(S, B)) \cong \operatorname{Ext}_R^n(A, B).$$

$\square$

Here is a generalization of the base change spectral sequence.

Theorem 3.7.2 (Base change spectral sequence)

Assume that A is a left S-module and B is an (R, S)-module, satisfying $\operatorname{Ext}_R^i(A, \operatorname{Hom}_S(B, E)) = 0$ for all $i \geq 1$ whenever E is an injective S-module. Then there is a first quadrant spectral sequence

$$\operatorname{Ext}_S^p\left(A, \operatorname{Ext}_R^q(B, C)\right) \Rightarrow \operatorname{Ext}_R^n\left(B \otimes_S A, C\right).$$

In particular, if B is projective as B-module, then

$$\operatorname{Ext}_S^n\left(A, \operatorname{Hom}_R(B, C)\right) \cong \operatorname{Ext}_R^n\left(B \otimes_S A, C\right).$$

Proof Similar proof. The condition on vanishing of Ext^i ensures that $F := \operatorname{Hom}_S(B, \square)$ sends injective S-modules to acyclic R-modules. $\square$

Theorem 3.7.3

Let $f : R \to S$ be a ring map. Then there is a first quadrant homological spectral sequence

$$E_{p,q}^2 = \operatorname{Tor}_p^S(\operatorname{Tor}_q^R(A, S), B) \Rightarrow \operatorname{Tor}_{p+q}^R(A, B).$$

for every R-module A and S-module B. In particular, if S is a flat R-module, then

$$\operatorname{Tor}_n^S(A \otimes_R S, B) \cong \operatorname{Tor}_n^R(A, B).$$

Proof This is again an example of Grothendieck spectral sequences, by taking

- **categories:** $\mathscr{A} = \mathbf{Mod}_R$, $\mathscr{B} = \mathbf{Mod}_S$, $\mathscr{C} = \mathbf{Ab}$;
- **functors:** $F = \square \otimes_R S$, $G = \square \otimes_S B$.

$\square$

3.7.2 Double tower of inverse limits

Let $(A_{ij})_{i,j\in\mathbb{N}}$ be an inverse system indexed by $\mathbb{N}\times\mathbb{N}$. The following result concerns $R^n \varprojlim_{(i,j)} A_{ij}$.

Theorem 3.7.4

We have $\varprojlim_{(i,j)} A_{ij}$, *a short exact sequence*

$$0 \longrightarrow R^1 \varprojlim_i (\varprojlim_j A_{ij}) \longrightarrow R^1 \varprojlim_{(i,j)} A_{ij} \longrightarrow \varprojlim_i (R^1 \varprojlim_j A_{ij}) \longrightarrow 0,$$

$$R^2 \varprojlim_{(i,j)} A_{ij} = R^1 \varprojlim_i (R^1 \varprojlim_j A_{ij})$$

and $R^n \varprojlim_{(i,j)} = 0$ *for* $n \geq 3$.

Proof First, we have a Grothendieck spectral sequence

$$E_2^{p,q} = R^p \varprojlim_i (R^q \varprojlim_j A_{ij}) \Rightarrow R^{p+q} \varprojlim_{(i,j)} A_{ij}.$$

Then we use the fact that $R^p \varprojlim = R^q \varprojlim = 0$ for $p, q \neq 0, 1$. □

3.7.3 The Leray spectral sequence

Definition 3.7.1 (Flasque sheaves)

Suppose X is a topological space, we say a sheaf $\mathscr{F}$ on X is flasque if for any subsets $V \subseteq U$, the restriction $\mathscr{F}(U) \to \mathscr{F}(V)$ is onto.

Proposition 3.7.1

Let $\mathscr{F}$ be a flasque sheaf on a topological space X. Then $H^i(X, \mathscr{F}) = 0$ for any $i > 0$. In other words, a flasque sheaf is $\Gamma(X, \square)$-acyclic.

Proof See [Har13, Prop. 2.5., page 208]. □

Lemma 3.7.1 (Injectives are flasque)

Let X be a ringed space. Let $V \subseteq U \subseteq X$ be open subspaces. For any injective $\mathscr{O}_X$-module $\mathscr{I}$ the restriction mapping $\mathscr{I}(U) \to \mathscr{I}(V)$ is surjective.

Proof See [Sta25, Lemma 01EA]. □

Lemma 3.7.2

Let $f : X \to Y$ be a continuous map of topological spaces and $\mathscr{F}$ is an injective object in $\mathbf{Shv}(X)$, *then $f_*\mathscr{F}$ is flasque.*

Proof Use the definition of flasque and Lemma 3.7.1, we get the consequence immediately. □

Theorem 3.7.5 (Leray spectral sequence)

Let $f : X \to Y$ be a continuous map of topological spaces and $\mathscr{F}$ a sheaf of abelian groups on X. Then there is a biregular first quadrant spectral sequence

$$E_2^{p,q} = H^p(Y, R^q f_*(\mathscr{F})) \Rightarrow H^{p+q}(X, \mathcal{F}).$$

Proof This is the Grouthendieck spectral sequence, by taking

- **categories:** $\mathscr{A} = \mathbf{Shv}(X)$, $\mathscr{B} = \mathbf{Shv}(Y)$, $\mathscr{C} = \mathbf{Ab}$;
- **functors:** $F = f_*$, $G = \Gamma(Y, \square)$.

F sends injectives to flasque sheaves is given by Lemma 3.7.2. $\square$

3.7.4 Cěch spectral sequence

Let X be a topological space and $\mathfrak{U} := \{U_i\}_{i\in I}$ a covering of X. This induces a left exact additive functor

$$\check{H}^0(\mathfrak{U}, \square) : \mathbf{PreShv}(X) \longrightarrow \mathbf{Ab}, \qquad \mathscr{F} \longmapsto \operatorname{Ker}\left(\prod_{i\in I} \mathscr{F}(U_i) \to \prod_{i,j\in U} \mathscr{F}(U_i \cap U_j)\right).$$

The right derived functor is the Cěch cohomology $\check{H}^p(\mathfrak{U}, \square)$, $p > 0$. We have **the Cěch spectral sequence**:

$$\forall \mathscr{F} \in \mathbf{Shv}(X), \qquad E_2^{p,q} = \check{H}^p(\mathfrak{U}, \mathscr{H}^q(\mathscr{F})) \Rightarrow H^{p+q}(X, \mathscr{F}),$$

where $\mathscr{H}^q(\mathscr{F}) \in \mathbf{PreShv}(X)$ is the presheaf $V \mapsto H^q(V, \mathscr{F})$.

This is the Grothendieck spectral sequence with categories

$$\mathscr{A} = \mathbf{Shv}(X), \quad \mathscr{B} = \mathbf{PreShv}(X), \quad \mathscr{C} = \mathbf{Ab}$$

and functors

$$F = \iota, \quad G = \check{H}^0(\mathfrak{U}, \square), \quad G \circ F = \Gamma(X, \square)$$

where ι is the inclusion functor. Notice that ι admits a left adjoint, the sheafification functor $\square^{\mathrm{sh}}$, which is exact, then ι sends injective objects to injective objects.

We can also write E_1 explicitly:

$$E_1^{p,q} = \prod_{i_0\cdots i_p \in I} H^q(U_{i_0} \cap \cdots \cap U_{i_p}, \mathscr{F}).$$

Then if $\mathfrak{U}$ is $\mathscr{F}$-acyclic, i.e., $H^q(U_{i_0} \cap \cdots \cap U_{i_p}, \mathscr{F}) = 0$, $\forall i_0 \cdots i_p \in I$ and $q > 0$, then $E_1^{p,q} = 0$, $\forall q > 0$. Then $E_2^{p,q} = 0$ for $q > 0$ and

$$E_2^{p,0} = \check{H}^p(\mathfrak{U}, \mathscr{F}) = H^p(C^\bullet(\mathfrak{U}, \mathscr{F})) \xrightarrow{\sim} H^p(X, \mathscr{F}),$$

where $C^\bullet(\mathfrak{U}, \mathscr{F})$ is the Cěch complex associated to $\mathfrak{U}$ and $\mathscr{F}$.

3.7.5 Hodge-to-de Rham spectral sequence

Let X be a (smooth) complex manifold, Ω_X^p be the sheaf of holomorphic differential p-forms on X, and $H^n(X, \underline{\mathbb{C}})$ be its sheaf cohomology with complex coefficients.

Theorem 3.7.6 (Hodge-to-de Rham spectral sequence)

We have the Hodge-to-de Rham spectral sequence:

$$E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H_{\mathrm{dR}}^{p+q}(X).$$

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